

# The memory and purity cost of immediate delivery

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## Abstract

How much memory does a device need to apply one quantum channel  $n$  times, if each output must leave before the next input arrives? We count the qubits the device keeps, the qubits it exchanges for fresh ones, and the purity it is given. Error is measured on the whole experiment, against an adaptive observer.

Immediate release has a price. For the qutrit Werner–Holevo channel, vanishing error requires  $\log_2 3$  bits of purity per use, a known rate; we add a bound at finite error. If all outputs may leave together,  $O(\log n)$  bits suffice in total at polynomially small error, refining a known zero rate. At error  $n^{-2}$  the change comes when outputs wait in batches of about  $\log n$ .

Purity separates cheap from expensive repeated use along one set of channels, the closure of those that a finite, maximally mixed bath implements exactly: outside it, linear purity is known to be needed at small error, hence linear memory when fresh qubits are maximally mixed. We prove a converse for memory, keeping maximally mixed fresh qubits and the optimal exchange rate from here on. A channel in the closure with a flat probe (an input that leaves the minimal environment maximally mixed) can be served with sublinear memory at any error that is not exponentially small. For channels built from finitely many group relations, memory at fixed error is at most logarithmic, divergent but sublinear, or linear, and some are linear exactly when some finitely presented group is not hyperlinear.

At high precision, the memory of  $n$  uses of such a channel equals, up to squared logarithms, the cost of the best single use that never leaves the channel’s support. That cost is the bath entropy plus  $n$  times the entropy one use adds to the bath. For Houghton’s group, devices acting through the group need and attain memory  $n^{1/3}$  up to logarithmic factors. Whether the single-use law holds at fixed error is open.

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# 1 Introduction

A quantum channel is usually described by what it does to one input. Suppose instead that a device must apply the same channel  $n$  times, and that each output must leave before the next input arrives. With a fresh pure environment for every use, the device needs almost no memory. But fresh pure qubits are a resource [72], and a device given little purity must keep what it cannot afford to throw away. How much must it keep, and what decides the answer?

We count three resources. The memory  $Q$  is the number of qubits the device keeps between uses. The exchange  $J$  is the number of qubits it hands over for good and replaces with fresh ones. The purity of a register is its number of qubits minus its entropy [49];  $P$  is the purity of the initial memory and  $C$  the total purity of the fresh qubits. Every reset, purifier and random seed the device keeps is counted. Unless we say otherwise, fresh qubits are maximally mixed, so  $C = 0$ . The observer may keep a reference and may choose each input after seeing earlier outputs. The error is the largest half trace distance, over all such observers, between this whole experiment and  $n$  independent uses of the channel. A device that releases each output before the next input arrives is a *causal service*.

The paper has three messages. Immediate release has a price, which a single qutrit channel already shows. One closed set of channels, those that a finite maximally mixed bath can imitate, separates cheap from expensive repeated use. Purity is known to mark this set, and we show that memory marks it too, for channels with a flat probe. And for channels built from group relations, at high precision, the memory of  $n$  uses is governed by a single use.

**The price of immediacy.** Consider the Werner–Holevo channel [83] on  $3 \times 3$  matrices,

$$\Theta(X) = \frac{1}{2}(\text{Tr}(X)I - X^T), \quad H = \log_2 3.$$

Up to a change of basis it is the spin-one channel that Landau and Streater used to show that a unital channel need not be a mixture of unitaries [58].

**Theorem A** (Theorems 3.1 and 3.3). *Every causal service of  $\Theta$  whose error tends to zero needs supplied purity  $P + C \geq (H - o(1))n$  [27, Theorem 3.2(c) and Proposition F.1(c)], and exact causal services attain  $P + C = nH + O(\sqrt{n})$ . If all  $n$  outputs may be released together,  $\Theta^{\otimes n}$  has an implementation with error  $1/\text{poly}(n)$  and supplied purity  $O(\log n)$ , refining the rate zero of [29].*

The rate  $H$  is the purity cost of an extreme channel, from earlier work [27]. What is added here is a bound at finite error and the comparison with collective delivery. The reason for the rate is rigidity. The Choi state of  $\Theta$  lives on a three-dimensional antisymmetric subspace, where tracing out the output acts as  $\Theta$  itself, which has a bounded inverse. So a nearly correct use cannot correlate its output and reference with a purification of the memory, and each use must deposit  $\log_2 3$  fresh bits of entropy there. Collective delivery avoids this by holding all inputs until the end, so its memory is linear.

Letting outputs wait interpolates between the two. At error  $n^{-2}$ , with sublinear memory and linear exchange, batches much shorter than  $\log n$  keep the full rate, while batches much longer than  $\log n$  have rate zero (Theorem 3.4). Rate zero for batches much longer than  $\log n$  holds for every unital channel (Theorem 3.5). The same logarithmic threshold appears for a closed apparatus delivering an extreme ququart Schur channel [25, Theorems 7 and 8]; we carry it to open services.

The collective bound holds for every unital channel (Lemma A.3, which extends [25, Theorem 1]). At the level of rates it is a theorem of Faist, Berta and Brandão. With trivial Hamiltonians, thermal operations are the noisy operations, work is counted in pure qubits, and unital channels have zero thermodynamic capacity [29], [30, Equation (4.9) and Theorem 7.1].

The lemma adds the explicit  $O(\log n)$  at polynomially small error. Classical randomness cannot explain it. The channel  $\Theta$  is not factorizable [43, Example 3.1], so its tensor powers stay a fixed distance from mixtures of unitaries [43, Section 6]. This refuted the asymptotic quantum Birkhoff conjecture of Smolin, Verstraete and Winter [79]; see also [75].

**The dividing line.** A *finite tracial factorization* of a channel is an implementation by one unitary acting on the input and a finite bath that starts maximally mixed. It uses no purity, and on a maximally mixed input it produces no bath entropy. Horodecki, Horodecki and Oppenheim introduced these implementations as noisy operations [49], and Gour et al. define noisy operations as their closure [37]. Let  $\overline{\mathcal{F}}$  be this closure. By Haagerup and Musat,  $\overline{\mathcal{F}}$  consists of the factorizable channels that admit an ancilla embeddable in an ultrapower of the hyperfinite  $\text{II}_1$  factor [44, Theorem 3.6], and it contains every factorizable channel exactly when Connes' embedding problem has a positive answer [44, Theorem 3.7]. Distance is the normalized Choi distance  $d_J$ , half the trace distance between Choi states.

A channel of Choi rank  $r$  has a *flat probe* if some input, with a reference, leaves its minimal environment maximally mixed; equivalently, its entropy exchange [73] can reach  $\log_2 r$ . For such a channel in  $\overline{\mathcal{F}}$ , about  $\frac{1}{2} \log r$  exchanged qubits per use is the least that permits sublinear memory as the error tends to zero [27, Theorem 3.4]. From here on, unless we say otherwise, the exchange is at most  $\lfloor (n-1) \log r/2 \rfloor$  and errors are at most  $\frac{1}{16}$ .

**Theorem B** (Theorem 4.1). *Let  $\Phi$  be a channel on  $M_d$ .*

- (a) *If  $\Phi \in \overline{\mathcal{F}}$  has a flat probe of rank  $r \geq 2$ , then for every error sequence  $\epsilon_n > 0$  with  $\log(1/\epsilon_n) = o(n)$  there are causal services with error at most  $\epsilon_n$  and memory  $Q = o(n)$ .*
- (b) [27, Theorem 3.2(b)] *If  $\Phi$  is at normalized Choi distance  $\Delta > 0$  from  $\overline{\mathcal{F}}$ , every causal service with imports between uses and error  $\epsilon$  has, at any exchange,  $P + C \geq n(\Delta - \epsilon)_+^2 / (2 \ln 2)$ . With maximally mixed imports its memory is therefore linear whenever  $\epsilon < \Delta$ .*

Part (a) sharpens [27, Corollary 3.3(c)], which gives sublinear memory at an error that tends to zero at no prescribed rate. The mechanism of part (b) is local tracialization [27, Lemma 5.4]: a use that raises the bath entropy only slightly, on a maximally mixed input, is close in Choi distance to a mixture of finite tracial factorizations (Lemma 4.5). A channel a fixed distance from  $\overline{\mathcal{F}}$  must therefore produce bath entropy at a positive rate, and supplied purity pays for it. The qutrit channel  $\Theta$  lies at a positive distance from  $\overline{\mathcal{F}}$ , computed by Haagerup and Musat [44, Theorem 5.6 and Corollary 5.7] and restated in Choi distance in Proposition 3.2.

The question of reusing one device without resets goes back to Rybár and Ziman [72]. In their model one fixed unitary acts on a finite memory and each input, and a channel is repeatable if every use, on uncorrelated inputs, induces exactly that channel. They showed that random-unitary channels are repeatable and that nonunital channels are not repeatable with a finite memory. They left the other unital channels open [72, Section IV]. Their entropy count [72, Equation (3.7)] is the prototype of the purity bookkeeping behind Theorem B(b).

Proposition 4.6 answers their question: some unital channels that are not random-unitary are repeatable, and some unital channels are not. The positive half rests on known results. A Schur channel of Haagerup and Musat [43, Example 3.3] is unital and not random-unitary. Lie, Son, Boes, Ng and Wilming showed that it has a catalytic dilation, in which the memory state never changes [60, Proposition 15 and Appendix C]. Such a device repeats its channel, here with one memory qubit. For the negative half we use the same bookkeeping. A device that repeats  $\Phi$  for  $n$  uses with memory dimension  $M$  has  $\log M \geq n \text{dist}_J(\Phi, \overline{\mathcal{F}})^2 / (2 \ln 2)$ . A channel that one finite memory repeats forever is a finite mixture of finite tracial factorizations.

The qutrit channel  $\Theta$  is unital, but repeating it  $n$  times needs  $\log M \geq n \log 3$ , which their  $n$ -cell device attains.

Rybár and Ziman also proposed devices whose memory starts maximally mixed, which are our finite tracial factorizations, followed by a reset to the maximally mixed state. A reused reset device receives inputs correlated with its own memory, and repeatability says nothing about such inputs. Theorem B(a) instead replaces about  $\frac{1}{2} \log r$  memory qubits per use by fresh maximally mixed ones, under joint error and with memory that may grow sublinearly. It can be read as an open-system, quantitative form of their proposal. Lie, Son, Boes, Ng and Wilming call the finite mixtures of finite tracial factorizations strongly factorizable [60]; their closure is  $\overline{\mathcal{F}}$ .

Exact factorizability and membership in  $\overline{\mathcal{F}}$  are different questions. Haagerup and Musat's equivalence, combined with  $\text{MIP}^* = \text{RE}$  [53], implies nonconstructively that in some finite dimension an exactly factorizable channel lies outside  $\overline{\mathcal{F}}$ ; Musat records this for a Schur channel [65, Corollary 3.11]. Our companion paper [63] gives explicit rational examples, conditional on recent preprints. Their distance from  $\overline{\mathcal{F}}$  is explicit but astronomically small (Proposition 5.3). In the other direction, some mixtures of two unitaries lie in  $\overline{\mathcal{F}}$  but have no finite tracial factorization [65, Proposition 3.6]. Musat and Rørdam give channels in  $\overline{\mathcal{F}}$  with no finite-dimensional ancilla at all, in every dimension  $d \geq 11$  [66, Theorem 4.1]; Mori lowers this to  $d \geq 8$  [65, Theorem 2.5].

**Three regimes for group gadgets.** A *canonical gadget* (Section 2.2) is a channel built from a finite list of relations in a group, by the construction of [28, Section 9]. Its Kraus operators are indexed by finitely many group elements, the *labels*, and it is factorizable through the group von Neumann algebra. Write  $P_\Phi$  for the group presented by the gadget's own relations. For gadgets the dividing line refines into three regimes at every fixed error (Theorem 5.1 and Table 2). Memory is at most logarithmic when a finite quotient of  $P_\Phi$  separates the labels; the gadget then has an exact finite tracial factorization [27, Theorem 8.6]. Memory diverges but stays sublinear when the labels are separated only by unitaries in a tracial ultrapower of the hyperfinite factor. Otherwise it is linear. The linear regime contains a gadget exactly when some finitely presented group is not hyperlinear (Proposition 5.2).

Slofstra and Vidick read the same three-way split off hyperlinear profiles [78, Section 2]. For linear-system games it governs the entanglement needed for near-perfect play [19, 76, 78]. There the dimension diverges as the error tends to zero; here memory diverges with the number of uses at fixed error.

**One use governs many.** Realize one use of a gadget  $\Phi$  by a unitary acting on the input and a bath in state  $\rho$ . Over  $n$  uses the bath entropy  $S = S(\rho)$  is paid once, and the entropy  $a$  that one use adds to the bath (with maximally mixed input) is paid  $n$  times. So the natural cost of the use is  $S + na$ . Let the *profile*  $Z_\Phi(n)$  be the least such cost over uses whose output stays exactly in the span of the gadget's Kraus operators and which tell the labels apart to a fixed coarse accuracy (Definition 6.1). The bath state may be coherent.

**Theorem C** (Theorem 6.5). *Let  $\Phi$  be a canonical gadget with Choi rank  $r \geq 2$ , and let  $\epsilon \leq 1/[n \log_2^2(en)]$ . The least memory  $Q$  of a causal service of  $\Phi$  with error  $\epsilon$  satisfies*

$$c(1 + Z_\Phi(n)) \leq 1 + Q \leq C \log_2^2(n/\epsilon)(1 + Z_\Phi(n)),$$

*with constants that depend only on the dimension and the Choi rank.*

At fixed error the lower bound is a second profile,  $L_\Phi^*(n)$ , which also allows each use a leakage of order  $1/n$  outside the Kraus span (Definition 2.4). Whether  $Z_\Phi(n)$  and  $L_\Phi^*(n)$  agree up to polylogarithmic factors is the main open problem (Problem 9.1).

The profiles make intermediate scales visible. For the configuration-lamp gadget of [28, Section 11], memory is  $n^{1/2+o(1)}$  [28, Theorem 11.1(b)], and the fixed-error profile  $L_\Phi^*$  matches it (Corollary 7.9). For Houghton’s channel, devices whose uses act through Houghton’s group, with negligible discard, need memory  $n^{1/3}$  up to logarithmic factors, and such devices attain it (Theorem 8.7). For arbitrary devices,  $n^{1/3}$  up to logarithms is an upper bound [27, Corollary 6.7], and the exponent is open.

**Related work.** Memory channels [57, 14] and collision models [18] describe repeated interactions with an environment. Sequential protocols are described by quantum combs [15] and strategies [42]; our error uses their adaptive discrimination setting. The memory needed to realize a given comb has been minimized when discarding is free [8, 9]; here discarding costs exchange and every retained qubit counts. Programmable processors store a channel in a program register [68, 47]; here the channel is fixed and the memory serves its repetition. Process tensors describe the multi-time response to interventions and detect environmental memory [70]. Here the target process has independent uses, and the memory belongs to its resource-constrained implementation. Quantum statistical complexity minimizes simulator entropy for prescribed output statistics [40], not for a channel on arbitrary entangled inputs. Mendl and Wolf analyze the mixed-unitary structure of unital and covariant channels [62]; Haagerup and Musat determined which members of their qutrit family are factorizable [44, Corollary 5.7], the input to Proposition 3.2. On the operator-algebra side, a unital channel is factorizable exactly when its iterates are compressions of the iterates of one trace-preserving automorphism of a larger algebra, equivalently when it has a Markov dilation [43, Theorem 4.4]; Haagerup and Musat credit the first equivalence to Kümmerer. These are iterates of the channel, not independent uses.

Thermodynamics prices processes in work. Faist and Renner give the single-shot work cost of a process on a specified input [33]. Faist, Dupuis, Oppenheim and Renner bound the work cost of a logical process on a fixed input by a smooth max-entropy of the discarded information given the output [31]. Both allow every Gibbs-preserving map for free; with trivial Hamiltonians these are the unital channels, so  $\Theta$  costs nothing there. Gibbs-preserving maps can outperform thermal operations on single instances [32], but for i.i.d. implementations the two cost the same [29], [30, Theorem 7.1]. Theorem A shows that the gap reopens, at the full rate  $H$ , when each output must leave at once. Faist, Berta and Brandão asked how their costs change under distinguishability measures suited to memory effects [29]. For trivial Hamiltonians the answer depends on the delivery contract: the qutrit lower bound needs only fresh entangled inputs chosen in advance, so the jump comes from immediate release, not from the stronger measure.

Catalysis is the closest neighbour. Exact return of a catalyst’s marginal can coexist with correlations: entropy and rank govern unitary state conversion [84], and free energy governs approximate correlated-catalytic conversion under Gibbs-preserving operations [74]. Lie and Jeong show that every catalytic channel is factorizable, so that not every unital channel is catalytic, and treat uncorrelatedness as a consumable randomness resource [59]. Boes et al. show that a quantum source of randomness of dimension  $\sqrt{d}$  suffices for every noisy state transition on a  $d$ -dimensional system [10], a halving of the same kind as the exchange rate  $\frac{1}{2} \log r$ . Son et al. show that exact catalyst return near one input implies return on every input [80]. Returning a marginal is weaker than simulating a full adaptive experiment; here retained catalysts count as memory and every import is independent and charged. Channel resource theories quantify processes under specified free operations [38], thermodynamic ones included [30]; our additional constraint is the chronological memory and exchange budget.

**Relation to earlier papers.** The Houghton paper [26] supplies the far-commutator area bound and the finite models of Houghton’s group; the presentation itself is Johnson’s, as used in [27, Section 4]. The causal-memory paper [27] supplies the model, the step from a service to a single good use, the service at the optimal exchange rate, and the purity bounds behind Theorem A and Theorem B(b). The configuration-lamps paper [28] supplies the gadget construction and the lamp channel’s memory laws. The rational-channel paper [63] supplies exactly factorizable channels at an explicit distance from  $\mathcal{F}$ , and the group certificate behind our canonical example in regime (iii), conditional on recent preprints [82, 1, 81]. Closed repeated use [24], the no-reset setting of [72], concerns one unchanging bath with no exchange. Finite numerical checks of the constants are in the accompanying `verify` directory; they support but do not replace the proofs.

**Organization.** Section 2 specifies the model. The qutrit separation comes first, followed by the dividing line and the one-use memory bounds. The remaining sections develop examples and the open comparison. Each main statement links to its proof.

The appendices fall into three groups. Appendices A, B, C, E, F and J prove the results of Sections 2–7: the qutrit and batching results, the closure law and repeatability, the gadget classification, the memory law with its static comparisons, and the one-round lower bound. Appendix N proves the Houghton results of Section 8, which also use the device of Theorem H.18 and the cap of Proposition I.4. The rest is specialized. Appendix D bounds a mixture profile for phased monomial dynamics. Appendices G, H and I, with proofs in Appendices K, L, M and S, develop flat-source, conserved-bias and spatial constructions and their limits. Appendices O–R extend the Houghton analysis; Section 8 summarizes their results.

## 2 Model and profiles

**Conventions.** Logarithms are to base 2 and  $\ln$  is natural. We write  $\tau_m = \text{Tr}/m$  for the normalized trace on  $M_m$ ,  $\|X\|_F$  for the Hilbert–Schmidt norm,  $\|X\|_2 = \tau(X^*X)^{1/2}$  for its normalized version,  $\|X\|$  for the operator norm, and  $h_2$  for the binary entropy. For a channel  $\Psi$  on  $M_d$ ,  $\mathcal{J}(\Psi) = (\Psi \otimes \text{id})(|\Omega\rangle\langle\Omega|)$  with  $|\Omega\rangle = d^{-1/2} \sum_i |ii\rangle$  is its normalized Choi state,  $d_\diamond(\Psi, \Psi') = \frac{1}{2} \|\Psi - \Psi'\|_\diamond$ , and  $d_J(\Psi, \Psi') = \frac{1}{2} \|\mathcal{J}(\Psi) - \mathcal{J}(\Psi')\|_1$  is the *normalized Choi distance*, and  $\text{dist}_J(\Psi, \mathcal{S}) = \inf_{\Psi' \in \mathcal{S}} d_J(\Psi, \Psi')$  for a set  $\mathcal{S}$  of channels.

Throughout,  $\Phi$  is a channel on  $M_d$  with linearly independent Kraus operators  $A_1, \dots, A_r$ ,  $r \geq 2$ . So  $r$  is the Choi rank of  $\Phi$ , and the support projection  $\Pi_\Phi$  of  $\mathcal{J}(\Phi)$  projects onto the span of the vectorized Kraus operators. A *flat probe* is a pure state  $\psi_*$  of the input and a reference whose output  $(\Phi \otimes \text{id})(\psi_*)$  is flat of rank  $r$ ; equivalently, its input marginal  $\rho_*$  satisfies  $\text{Tr}(\rho_* A_i^* A_j) = \delta_{ij}/r$ .

### 2.1 Rounds

**Definition 2.1** (Rounds). A *round* of  $\Phi$  is a pair  $(U, \rho)$  of a unitary  $U$  on  $\mathbb{C}^d \otimes \mathbb{C}^M$  and a state  $\rho$  on  $\mathbb{C}^M$ , the *bath*. It has the channel and the bath map

$$\Psi(X) = \text{Tr}_M U(X \otimes \rho) U^*, \quad T(\rho) = \text{Tr}_d U(\tau_d \otimes \rho) U^*,$$

and four numbers: its *distance*  $u = d_\diamond(\Psi, \Phi)$ , its *support leakage*  $\ell = \text{Tr}[(I - \Pi_\Phi)\mathcal{J}(\Psi)]$ , its *entropy production*  $a = S(T(\rho)) - S(\rho)$ , and its *bath entropy*  $S = S(\rho)$ . The round is *flat* if  $\rho = P_K/k$  is the normalized projection onto a subspace  $K$  of dimension  $k$ , has *zero leakage* if  $\ell = 0$ , and is *exact* if  $\ell = 0$  and  $u = 0$ .

The map  $T$  is unital and trace preserving, so  $a \geq 0$ . A round is one use of a device seen from inside: the bath is the memory,  $\rho$  is its state when the input arrives, and  $T(\rho)$  is the memory after a use with maximally mixed input. Leakage measures how far the round leaves the Kraus span of  $\Phi$ ; entropy production measures how much the memory is stirred. The letter  $a$  always denotes entropy production; group generators that are traditionally called  $a$  and  $b$  are written  $\mathbf{a}, \mathbf{b}$ .

## 2.2 Canonical untwisted anchored gadgets

**Definition 2.2** (Canonical untwisted anchored gadget). Choose a group  $G$ , a finite generating set  $S$ , and a finite list  $R$  of words in  $S^{\pm 1}$ , of length at least two, equal to 1 in  $G$ . The list need not present  $G$ . Let the nonempty *symbols* consist of the  $2|S|$  formal signed letters and one tagged prefix  $(i, j)$  for each word  $r_i = \sigma_{i,1} \cdots \sigma_{i,m_i} \in R$  and  $2 \leq j < m_i$ . Tags distinguish physical positions even when two prefixes name the same word or group element. Write  $p_{i,1} = \sigma_{i,1}$ ,  $p_{i,m_i} = \emptyset$ , and let  $p_{i,j}$  denote the tagged prefix for  $2 \leq j < m_i$ . The triangle list is  $(p_{i,j-1}, \sigma_{i,j}, p_{i,j})$ ,  $2 \leq j \leq m_i$ , together with  $(s, s^{-1}, \emptyset)$ ,  $s \in S$ . Use exactly the scalar coefficients and block placement in (1) below, with a singleton diagonal slot for the identity and for each nonempty symbol.

The adjective *canonical* refers to this fixed table and its regular-trace channel. *Untwisted* means that triangle products are ordinary products: no projective phase or cocycle enters a label relation. The minus sign in each two-by-two block is part of the fixed table. *Anchored* means that every distinct group label has a coefficient-1 diagonal slot whose row and column have no other slot. We also call these channels *canonical gadgets*. If  $s = |S|$ ,  $r_0 = |R|$  and  $\Lambda = \sum_i m_i$ , the table has  $2s + \Lambda - 2r_0$  nonempty symbols and  $\Lambda - r_0 + s$  triangles, hence  $d = 1 + 4s + 3\Lambda - 4r_0$ .

This is the construction of [28, Section 9]. Write  $[w]$  for the element of  $G$  represented by a symbol, so every triangle has  $[x][y] = [z]$ . Assign arbitrary unitaries  $\lambda(w)$  to the nonempty symbols and set  $\lambda(\emptyset) = I$ . The physical *gadget unitary* is

$$U_\lambda = \text{diag}(I, (\lambda(w))_w) \oplus \bigoplus_{(x,y,z)} \frac{1}{\sqrt{2}} \begin{pmatrix} I & \lambda(y) \\ \lambda(x) & -\lambda(x)\lambda(y) \end{pmatrix}, \quad (1)$$

which is unitary for every unitary  $\lambda$ . The gadget channel evaluates the symbols in the left regular representation and traces out the group von Neumann algebra:  $\Phi_G(X) = (\text{id} \otimes \tau)[U_\lambda(X \otimes 1)U_\lambda^*]$ . Writing  $U_\lambda = \sum_a c_a E_a \otimes \lambda([w_a])$  over the *slots*  $a$  (nonzero entries), the group elements  $[w_a]$  are the *labels*,  $\mathcal{L}$  is their set, and

$$\Phi_G(X) = \sum_{g \in \mathcal{L}} A_g X A_g^*, \quad A_g = \sum_{[w_a]=g} c_a E_a.$$

So every gadget channel is factorizable by construction, through  $M_d \otimes L(G)$  with the trace  $\tau_d \otimes \tau$  and the unitary  $U_\lambda^*$ , in the sense of Anantharaman-Delaroche [2], in the unitary form of [43, Theorem 2.2]. The only question is whether it lies in the closure  $\overline{\mathcal{F}}$  of finite tracial factorizations (Section 2.3). The  $A_g$  have disjoint supports, so  $r = |\mathcal{L}|$  is the Choi rank. Every label has an *anchor*: a diagonal slot  $z_g$  with coefficient 1 whose row and column contain no other slot. Hence the *label weights*  $\mu_g = \|A_g\|_F^2/d$  satisfy  $\mu_g \geq 1/d$  and  $\sum_g \mu_g = 1$ , and half of a maximally entangled state on the  $r$  anchors is a flat probe of rank  $r$  [28, Section 9]. The labels are the actual distinct elements  $\mathcal{L} \subseteq G$ , rather than the words naming them. In particular equal words in  $G$  share a Kraus operator. The finite *presented label group* is

$$P_\Phi = \langle v_g \ (g \in \mathcal{L}) \mid v_1 = 1, \ v_{[x]}v_{[y]} = v_{[z]} \text{ for every triangle } (x, y, z) \rangle.$$

It is finitely presented and has the surjective homomorphism  $q : P_\Phi \rightarrow \langle \mathcal{L} \rangle \leq G$ ,  $q(v_g) = g$ . This quotient need not be an isomorphism. All representation criteria below concern  $P_\Phi$  and its named label set. A representation *separates the labels* if it distinguishes  $v_g$  and  $v_h$  whenever  $g \neq h$  in  $\mathcal{L}$ .

A *label chunk*  $E$  is a finite subset of the presented label group containing these named labels, with partial multiplication  $xy = z$  whenever  $x, y, z \in E$  and that equality holds. Its *sofic profile*  $\sigma_E(s)$ ,  $s > 1$ , is the least  $N$  for which some map  $f : E \rightarrow \text{Sym}_N$  with  $f(1) = \text{id}$  has normalized Hamming multiplicative defect at most  $1/s$  on defined products and distance at least  $1 - 1/s$  between distinct elements, and  $+\infty$  if there is none [21, Definitions 3.2 and 3.4]. In Cornulier's terms such an  $f$  is a  $(1 - s^{-1})$ -expansive  $s^{-1}$ -morphism. In source-specific applications we specify whether  $E$  is instead the image chunk in  $G$ .

**Example: a two-element group.** Take  $G = \mathbb{Z}/2$  with generator  $\mathbf{a}$ ,  $S = \{\mathbf{a}\}$  and  $R = \{\mathbf{a}^2\}$ , so  $s = r_0 = 1$  and  $\Lambda = 2$ . The relator  $\mathbf{a}\mathbf{a}$  has no tagged prefix, so the nonempty symbols are the two signed letters  $\mathbf{a}, \mathbf{a}^{-1}$ . The triangles are  $(\mathbf{a}, \mathbf{a}, \emptyset)$  from the relator and  $(\mathbf{a}, \mathbf{a}^{-1}, \emptyset)$ . Hence  $d = 1 + 2 + 2 \cdot 2 = 7$ . In the regular representation  $\lambda(\mathbf{a}) = \lambda(\mathbf{a}^{-1}) = u$  with  $u^2 = 1$ , and (1) reads

$$U_\lambda = \text{diag}(1, u, u) \oplus \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & u \\ u & -1 \end{pmatrix} \oplus \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & u \\ u & -1 \end{pmatrix}.$$

The labels are 1 and  $\mathbf{a}$ , so  $r = 2$ . With the basis ordered as in this display, the Kraus operators are

$$A_1 = E_{11} + \frac{1}{\sqrt{2}}(E_{44} - E_{55} + E_{66} - E_{77}), \quad A_{\mathbf{a}} = E_{22} + E_{33} + \frac{1}{\sqrt{2}}(E_{45} + E_{54} + E_{67} + E_{76}),$$

and  $A_1^* A_1 + A_{\mathbf{a}}^* A_{\mathbf{a}} = I_7$ . The slots  $(1, 1)$  and  $(2, 2)$  are anchors, so half of a maximally entangled state on  $\text{span}\{|1\rangle, |2\rangle\}$  is a flat probe. The presented label group  $P_\Phi$  is  $\mathbb{Z}/2$  itself.

**Example: the Houghton gadget.** Take Johnson's presentation of  $H_3$  in Section 8. The relator  $\alpha^2$  contributes the single triangle  $(\alpha, \alpha, \emptyset)$ ; the other relators contribute their successive prefixes in the same way. Its table has 33 nonempty symbols, 35 triangles,  $d = 104$  and  $r = 26$  actual labels. In particular 1 and  $\gamma = \alpha\alpha\alpha^{-1}$  have separate singleton anchors. The latter is a three-cycle in the action on the rays. This supplies both a physical channel and the finite presentation  $P_{\Phi_H}$ ; here the full Johnson relators recover  $H_3$  after eliminating prefix generators. Other gadgets use relations that hold in the group without presenting it. The configuration-lamp gadget is one: its presented label group is only known to map onto the group generated by its labels.

**Permutation rounds.** Let  $f : \mathcal{L} \rightarrow \text{Sym}(\Omega)$  be any map with  $f(1) = \text{id}$  on a finite set  $\Omega$ , and let  $U_f$  be the gadget unitary with  $\lambda(w)$  the permutation matrix of  $f([w])$ ; it is unitary although no relation need hold [28, proof of Proposition 12.1]. With a diagonal bath  $\rho = \text{diag}(p)$  this is a *permutation round*. A point  $\omega$  is *triangle-good* if  $f([x])f([y])\omega = f([z])\omega$  for every triangle, labels  $g \neq h$  *collide* at  $\omega$  if  $f(g)\omega = f(h)\omega$ , and the round is *free* if no labels collide on the points it uses. At a triangle-good point every slot of label  $g$  sends  $\omega$  to  $f(g)\omega$ , so  $U_f(\xi \otimes |\omega\rangle) = \sum_g A_g \xi \otimes |f(g)\omega\rangle$ . For a bath supported on triangle-good points,  $S = H(p)$  and  $a = H(\mathcal{T}p) - H(p)$  with  $\mathcal{T}p = \sum_g \mu_g f(g)_* p$ . The regular action of a group in which the gadget relations hold and the labels are distinct gives free permutation rounds with every bias.

**Lemma 2.3** (Leakage of permutation rounds). *In a permutation round with distance  $u$  and leakage  $\ell$ , the pure-point round at  $\omega$  has leakage 0 if  $\omega$  is triangle-good, whatever the collisions,*

and at least  $1/(4d)$  otherwise; hence the bath gives mass at most  $4dl$  to triangle-bad points. Moreover, for labels  $g \neq h$ , the bath gives mass at most  $2u$  to the points where  $g$  and  $h$  collide.

### 2.3 Causal services

We use the model of [27, Definition 2.3]. It extends the no-reset setting of Rybár and Ziman [72], in which one fixed unitary acts on a finite memory and each use must induce the channel exactly on uncorrelated inputs. Here the unitaries may depend on the round, memory qubits may be exchanged, inputs may be adaptive and entangled, and the error is approximate and joint. The joint criterion matters: one fair bath bit controlling  $I$  or  $Z$  on every qubit repeats dephasing in their sense, but its joint error is at least  $1 - 2^{1-n}$  [24, Proposition 8.1]. A *causal service* of  $\Phi$  for  $n$  rounds has a memory of  $Q$  qubits in a fixed initial state  $\beta_0$  and, in each round, a fixed unitary on the input and the memory, after which the output is released; after each round but the last, a prescribed set of memory qubits is exported for good and replaced by imported qubits in a fixed state, independent of everything else. An observer chooses the inputs adaptively, possibly entangled with quantum references of its own, and the *error* is the largest half trace distance between the observer's final state and its final state with  $n$  independent uses of  $\Phi$ . The resources are the memory  $Q$ , the *exchange*  $J$  (the number of exported qubits) and the *purity*  $P + C$ , with  $P = Q - S(\beta_0)$  and  $C$  the sum over the imports of their number of qubits minus their entropy. Every retained qubit counts.

The prescribed exchange takes place *between rounds*; this is the between-round service contract used throughout, except for the explicit timing extension in Theorem 3.1. A *public service* has its unitaries, schedule and imported states described to the observer, and no uncounted runtime seed is supplied. Randomness in the specified mixed source states is allowed and counted in the purity. The description does not reveal the samples drawn from such sources. A *chronological compiler* turns a one-round response family into such a service, releasing each output before the next round and charging every workspace and clock qubit. In the language of quantum strategies, the observer error is one half of the full strategy discrimination norm: equal-prior discrimination succeeds with probability  $1/2 + \Delta/2$  when our error is  $\Delta$  [41, Definition 1 and Section 3.3].

A channel has a *finite tracial factorization* if  $\Psi(X) = (\text{id} \otimes \tau_M)[U(X \otimes I_M)U^*]$  for a unitary  $U$  on  $\mathbb{C}^d \otimes \mathbb{C}^M$ ; the bath dimension is  $M$ . Write  $\mathcal{F}_m$  for the channels with a finite tracial factorization of bath dimension at most  $m$ , and  $\overline{\mathcal{F}}$  for the closure of their union. Finite tracial factorizations are the noisy operations of [49], and  $\overline{\mathcal{F}}$  is the closed set of noisy operations of [37]. The finite mixtures of finite tracial factorizations are the strongly factorizable channels of [60], and  $\overline{\mathcal{F}}$  is their closure. By a theorem of Haagerup and Musat,  $\overline{\mathcal{F}}$  consists of the factorizable channels, in the unitary form of [43, Theorem 2.2], that admit an exact factorization through  $M_d \otimes N$  for some tracial von Neumann algebra  $N$  with a trace-preserving embedding into an ultrapower of the hyperfinite  $\text{II}_1$  factor, and whether it contains every factorizable channel is equivalent to the Connes embedding problem [44, Theorems 3.6 and 3.7]. Gadget channels of hyperlinear groups lie in the closure: the regular group trace has matrix approximants, which realize the fixed table. This uses the theorem that the group von Neumann algebra of a countable group embeds in  $\mathcal{R}^\omega$  exactly when the group is hyperlinear [69, Theorem 8.5]; Pestov credits it to Kirchberg [56] and Rădulescu [71, Proposition 2.5], among others. Haagerup–Musat's characterization applies [44, Theorem 3.6]. More generally the criterion is separation of the finite labels of  $P_\Phi$  in  $U(\mathcal{R}^\omega)$ , as stated in Section 5. For a gadget channel in the closure,  $\frac{1}{2} \log r$  qubits per use is the optimal exchange rate [27, Propositions 3.1 and 9.1]. We work at this rate: a service is at the *rank rate* if  $J \leq \frac{n}{2} \log r$ , and  $Q^*(n, \epsilon)$  denotes the least memory of a rank-rate service of  $\Phi$  for  $n$  rounds with error at most  $\epsilon$ , with any purity.

We write  $Q_{\text{flat}}^*(n, \epsilon)$  when imports are required to be flat ( $C = 0$ ), with the sharper schedule  $J \leq \lfloor (n-1) \log r/2 \rfloor$ . The explicit upper constructions meet this contract. Initial purity is always charged as  $P = Q - S(\beta_0)$  and may be as large as the retained memory.

| Symbol                                  | Meaning                                                                                             |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------|
| $Q, J, P, C$                            | Retained qubits, replacement slots counted once, initial entropy deficit, imported entropy deficit  |
| $u, \epsilon$                           | One-round half-diamond error; complete adaptive-experiment half trace distance                      |
| $S, a, \ell$                            | Source entropy; Choi-input bath entropy production; normalized-Choi support leakage                 |
| $L_{\Phi}^*, Z_{\Phi}, U_{\Phi}$        | Leaky entropy cost; exact-support entropy cost; exact rectangular rank cost                         |
| $\mathcal{F}_m, \overline{\mathcal{F}}$ | Channels with a finite tracial factorization of bath dimension at most $m$ ; closure of their union |
| $D_{\text{mix}, \Phi}$                  | Approximation profile using finite mixtures of finite tracial factorizations                        |
| $B_g, \mu_g$                            | Response isometries; fixed Choi weights $\ A_g\ _F^2/d$                                             |

Table 1: Notation used across the main results. Entropies are in bits.

## 2.4 Profiles

**Definition 2.4** (Two-scale profiles). For  $n \geq 1$ ,  $u \geq 0$  and  $\ell_0 \geq 0$  put

$$L_{\Phi}(n; u, \ell_0) = \inf \{S(\rho) + n a : (U, \rho) \text{ a round of } \Phi \text{ with distance } \leq u \text{ and leakage } \leq \ell_0\},$$

over rounds with any finite bath, and  $L_{\Phi}(n, \epsilon) = L_{\Phi}(n; \epsilon, \frac{32}{15}\epsilon/n)$ . The *two-scale profile* is

$$L^*(n) = L_{\Phi}(n; \frac{1}{16}, \frac{2}{15n}).$$

For a class  $\mathcal{C}$  of rounds,  $L^{\mathcal{C}}(n; u, \ell_0)$  is the same infimum over rounds of  $\mathcal{C}$ .

The distance  $\frac{1}{16}$  and the leakage  $\frac{2}{15n}$  are the values that Theorem 7.3 obtains from a service with error at most  $\frac{1}{16}$ , using the extraction theorem with  $\lambda = 2$ . The bath entropy and the entropy production enter at different scales: the constraint on leakage is read at  $1/n$ , while entropy production is charged  $n$  times.

*Remark 2.5* (An upper comparison with the hyperlinear profile). Let  $E$  be a label chunk in  $P_{\Phi}$ . In the convention of Slofstra–Vidick [78, Definition 7.1],  $\eta(E; \delta, e)$  is the least unitary dimension with normalized Frobenius multiplicative defect at most  $e$  and pair separation at least  $\delta$ . Their conditions (defects on all products in  $E$ , separation of all distinct pairs,  $\phi(1) = I$ ) are stronger than the triangle and label conditions used here, so the bound below applies. For a canonical untwisted anchored gadget, constants  $t_{\Phi} \in \mathbb{N}$  and  $c_{\Phi} > 0$  give

$$L_{\Phi}^*(n) \leq L_{\Phi}^{\text{flat}}(n; 1/16, 2/(15n)) \leq t_{\Phi} \log_2(4\eta(E; 1, c_{\Phi}/\sqrt{n})).$$

The constructed round is a finite tracial factorization, so its production is zero. Fixed amplification pays only the displayed logarithmic dimension cost; Appendix F.8 proves the comparison. This is a one-sided flat-round bound, rather than an equivalence with sofic profiles or with biased or rectangular source costs.

The profile of [27, Remark 5.3] reads a round under the single budget  $\ell + a \leq v$ . At the rank rate the budget that a service supplies is of order  $(Q + \log n)/n$ , which is far above the error scale  $1/n$  for leakage and is exactly the cost scale for entropy production; reading both quantities at one scale is what the two-scale profile avoids.

**Lemma 2.6** (Monotonicity).  $L_{\Phi}(n; u, \ell_0)$  is nonincreasing in  $u$  and in  $\ell_0$ , and  $L_{\Phi}(n, \epsilon) \geq L^*(n)$  for  $0 < \epsilon \leq \frac{1}{16}$ . If  $0 < c_1 \leq c$  and  $K \geq c/c_1$ , then  $L_{\Phi}(n; u, c_1/n) \leq L_{\Phi}(Kn; u, c/(Kn))$ .

So an upper bound proved at leakage  $c_1/n$  instead of  $c/n$  costs a constant rescaling of  $n$ , which is harmless for profiles that grow at most polynomially.

**Proposition 2.7** (The linear ceiling). *Every channel  $\Phi$  with Choi rank  $r$  has, for all  $n \geq 1$ , a causal service with error 0, exchange  $J = 0$  and memory  $Q = n\lceil \log r \rceil$ . In particular  $Q^*(n, \epsilon) \leq n\lceil \log r \rceil$ .*

This is the  $n$ -cell construction of Rybár and Ziman [72, Section III], with minimal pure Stinespring environments.

The same device read as a round shows  $L^*(n) \leq n \log r$ : the Stinespring dilation with a pure environment of dimension  $r$  is an exact round with  $S = 0$  and  $a \leq \log r$ . So both sides of the law lie between 0 and  $n\lceil \log r \rceil$ . The profile is sublinear only through rounds with  $a = o(1)$ , and a round with  $a \geq 1/\text{polylog}(n)$  already costs at least  $n/\text{polylog}(n)$ .

## 2.5 Dilations

**Definition 2.8** (Rectangular dilations [28, Theorem 12.2]). Maps  $B_1, \dots, B_r : \mathbb{C}^k \rightarrow \mathbb{C}^{k'}$ ,  $k \leq k'$ , form an *exact rectangular dilation* of  $\Phi$  if  $W = \sum_i A_i \otimes B_i$  is an isometry,  $\text{Tr}(B_i^* B_j) = k \delta_{ij}$  and  $\sum_i B_i^* B_i = r I_k$ . Its *room* is  $\log(k'/k)$ , and

$$U_\Phi(n) = \inf \{ \log k + n \log(k'/k) \}$$

over exact rectangular dilations.

For a gadget channel, maps  $B_g$  indexed by the labels form a *gadget family* if they satisfy the identities of Lemma 2.9(a). Its *flat Gram* is the matrix  $(\tau_k(B_g^* B_h))_{g,h}$ .

**Lemma 2.9** (Gadget dilations). (a) *For a gadget channel and maps  $B_g : \mathbb{C}^k \rightarrow \mathbb{C}^{k'}$ ,  $W = \sum_g A_g \otimes B_g$  is an isometry if and only if every  $B_g$  is an isometry and  $B_1^* B_{[y]} = B_{[x]}^* B_{[z]}$  for every triangle  $(x, y, z)$ . Then  $\sum_g B_g^* B_g = r I$ , and replacing every  $B_g$  by  $V B_g$  with  $V$  unitary changes nothing; so one may take  $B_1$  to be the inclusion of  $\mathbb{C}^k$ .*

(b) *If  $\Phi$  has a flat probe, then  $W^* W = I$  implies  $\sum_i B_i^* B_i = r I_k$ .*

(c) *An exact rectangular dilation gives an exact round with bath dimension  $k'$ ,  $S = \log k$  and  $a \leq \log(k'/k)$ . Hence  $L_\Phi(n; 0, 0) \leq U_\Phi(n)$ .*

**Lemma 2.10** (Zero-leakage rounds are exact on their support). *Let  $(U, \rho)$  be a round of  $\Phi$  with  $\ell = 0$ , and  $K = \text{supp } \rho$ . There are operators  $X_i$  on  $\mathbb{C}^M$  with  $U(\xi \otimes \kappa) = \sum_i A_i \xi \otimes X_i \kappa$  for all  $\xi \in \mathbb{C}^d$  and  $\kappa \in K$ . Thus  $W = \sum_i A_i \otimes X_i|_K$  is an isometry and  $\Psi(X) = \sum_{i,j} \text{Tr}(\rho X_j^* X_i) A_i X A_j^*$ . With a flat probe,  $\sum_i X_i^* X_i = r I$  on  $K$ . For a gadget channel the restrictions  $X_g|_K$  form a gadget family, and its weighted Gram  $G_{gh} = \text{Tr}(\rho X_g^* X_h)$  satisfies  $|G_{gh}| \leq 2u$  for  $g \neq h$ .*

The expansion of  $U$  is the finite form of the coefficient expansion in [43, Theorem 2.2].

So a flat round with zero leakage is a gadget family with exact identities whose flat Gram is off the identity by at most  $2u$ , and at  $u = 0$  it is an exact rectangular dilation. What separates a zero-leakage round from a rectangular dilation is only the bias of its bath and the size of its output space. Lemmas 2.3–2.10 and Proposition 2.7 are proved in Appendix J.

## 3 A sharp qutrit supplied-purity rate

The general nonclosure converse measures distance to a tracial closure. For the qutrit Werner–Holevo channel [83] a stronger converse measures an entire bit rate. Put

$$\Theta(X) = \frac{1}{2}(\text{Tr}(X)I - X^T), \quad H = \log_2 3, \quad g_D(t) = 2t \log_2 D + (1+t)h_2\left(\frac{t}{1+t}\right).$$

Its normalized Choi state is the flat state on the three-dimensional antisymmetric subspace. In this section only, we also permit prescribed, independent imports and permanent exports during a round. Their timing and states are fixed independently of the observer and device state, and their purity is counted in  $P + C$ . Elsewhere exchange happens only between rounds.

**Theorem 3.1** (Qutrit purity, at arbitrary exchange timing). *Every immediate-output service of  $\Theta$  with complete-experiment error at most  $\epsilon$  satisfies*

$$P + C \geq n [H - g_9(\min\{1, 10\sqrt{\epsilon}\})]_+.$$

*There are exact services with*

$$Q = O(\sqrt{n}), \quad P + C = nH + O(\sqrt{n} + \log n), \quad J = nH + O(\sqrt{n}).$$

*Consequently the optimal asymptotic supplied-purity rate is  $H$  when the error tends to zero, even under the enlarged timing contract.*

The attained rate equals the purity supplied by one pure three-dimensional Stinespring environment per round. The lower bound comes from rigidity (Lemma A.1). The Choi state of  $\Theta$  is the flat state on the three-dimensional antisymmetric subspace of reference and output, so a round that implements  $\Theta$  nearly correctly keeps them nearly inside that subspace. There, tracing out the output acts as  $\Theta$  itself, which is invertible with stabilized inverse norm at most 9. The reference starts independent of a purification of the bath, so reference and output stay nearly independent of it too. Each use must then raise the bath entropy by nearly  $\log_2 3$  fresh bits, and these gains add up to at most the supplied purity. The argument holds for every bath dimension and for memories correlated with earlier outputs. It is a vanishing-error statement: at coarse error its continuity bound may be vacuous. The  $O(\sqrt{n})$  terms of the upper bound come from packing. The pure qutrit environments of  $\lceil \sqrt{n} \rceil$  consecutive rounds share one bank of  $\lceil \sqrt{n} \rceil H$  qubits, which is exported when full. Memory holds one bank, and rounding loses less than one qubit per bank.

Theorem 3.1 is the finite-error form of a law that holds for every channel. For any  $\Phi$ , the least supplied-purity rate at vanishing error is the purity cost  $\kappa(\Phi)$  of closed repeated use [24, Definition 1.3], and with memory  $o(n)$  the achievable exchange and purity rates  $(j, p)$  fill the region  $p \geq \kappa(\Phi)$ ,  $j \geq p$ ,  $2j - p \geq H(\Phi)$ , where  $H(\Phi)$  is the largest entropy of a complementary output [27, Theorems 3.2 and 3.4]. The map  $\Theta$  is an extreme point of the channels on  $M_3$ , so  $\kappa(\Theta) = H(\Theta) = H$  [27, Proposition F.1(c)], and the banks above sit at the corner  $j = p = H$ , where the least purity and the least exchange are reached together. This section adds the explicit finite-error bound, the enlarged timing contract, exact services with memory  $O(\sqrt{n})$ , and the comparison with collective and batched delivery below. The cost  $\kappa$  vanishes exactly on  $\overline{\mathcal{F}}$  [27, Theorem 3.2], so gadgets in the first two regimes of Theorem 5.1 have  $\kappa = 0$  and those in the linear regime have  $\kappa > 0$ .

**Proposition 3.2** (The static qutrit gap; Haagerup–Musat [44, Theorem 5.6 and Corollary 5.7]). *In normalized Choi distance,*

$$\text{dist}_J(\Theta, \overline{\mathcal{F}}) = 2/27.$$

*A closest point is  $(2/27)W_3^+ + (25/27)W_3^-$ , where  $W_3^- = \Theta$  and  $W_3^+(X) = (\text{Tr}(X)I + X^T)/4$ .*

As in [44, Section 4 and the proof of Theorem 5.6(2)], Werner twirling reduces the question to the line  $\lambda W_3^+ + (1 - \lambda)W_3^-$ , whose normalized Choi distance from  $\Theta$  is  $\lambda$ . Haagerup–Musat’s factorizability threshold places the closest factorizable point at  $\lambda = 2/27$  [44, Theorem 5.6 and Corollary 5.7], and that point has a finite tracial factorization with a three-dimensional bath [44, Theorem 5.6(1)]. The proposition gives an explicit example for Theorem 4.1(b). The sharper rate theorem uses antisymmetric-support rigidity instead. Appendix A proves both statements.

**Theorem 3.3** (Collective putrit implementation). *For  $n \geq 1$  and  $0 < \epsilon \leq 1/2$ , there is one unitary implementation of  $\Theta^{\otimes n}$  with all outputs released together, half-diamond error at most  $\epsilon$ , and*

$$\begin{aligned} Q_{\text{bath}} &\leq nH/2 + 2\log_2(1/\epsilon) + 2\log_2(n+1) + O(1), \\ P &\leq 2\log_2(1/\epsilon) + 2\log_2(n+1) + O(1), \quad J = C = 0, \\ Q_{\text{peak}} &\leq 2n + Q_{\text{bath}} + O(1). \end{aligned}$$

Here  $Q_{\text{bath}}$  counts the fixed auxiliary bath and  $Q_{\text{peak}}$  also counts accepted input/output payload registers held until delivery. Subexponential inverse error gives supplied-purity rate zero with linear peak retained memory.

The proof applies Lemma A.3, which holds for every unital channel. For a fixed unital channel  $\Psi$ , it gives an implementation of  $\Psi^{\otimes n}$  with all outputs released together, error  $1/\text{poly}(n)$  and supplied purity  $O(\log n)$ . At the level of rates this is the zero thermodynamic capacity of unital channels with trivial Hamiltonians [29], [30, Equation (4.9) and Theorem 7.1]. The lemma adds the explicit  $O(\log n)$  term, and Theorem 3.1 shows that immediate delivery raises the rate to  $H$ .

The collective construction holds the inputs until delivery, so its memory is linear. The inputs are chosen by the observer, so they are not independent imports, and releasing them as outputs is not replacement exchange. Purity is therefore  $P = Q_{\text{bath}} - S(\beta_{\text{bath}})$ , not  $Q_{\text{peak}} - S(\beta_{\text{bath}})$ . The same convention applies to batching with sequential input arrivals.

**Theorem 3.4** (Logarithmic batching threshold). *Fix contiguous batch lengths  $1 \leq b_n = o(n)$ , complete adaptive error  $n^{-2}$ ,  $Q_{\text{peak}} = o(n)$  and  $J = O(n)$ . Let  $R[b]$  be the infimum achievable  $\limsup(P + C)/n$ , and put  $x = \limsup_{n \rightarrow \infty} \log_2 n/b_n \in [0, \infty]$ . Then*

$$H \min\{1, x/\log_2 9\} \leq R[b] \leq \min\{H, 6x\}.$$

In particular  $R[b] = H$  for  $b_n = o(\log n)$ , and  $R[b] = 0$  if and only if  $b_n/\log n \rightarrow \infty$ . More generally  $R[b] = H$  whenever  $x \geq \log_2 9$ , including  $b_n \sim c \log_2 n$  with  $c \leq 1/\log_2 9$ . The remaining part of the critical logarithmic curve is not determined. In the superlogarithmic regime the zero-rate construction also has  $J = nH/2 + o(n)$ ; at critical logarithmic batching only  $J = O(n)$  is asserted.

The same logarithmic threshold appears for a closed apparatus serving a ququart Schur channel on a delivery schedule [25, Theorems 7 and 8]. Theorem 3.4 carries it to open services with exchange. Its upper half holds for every unital channel, with  $H$  replaced by  $\log_2 r$ .

**Theorem 3.5** (Batching any unital channel). *Let  $\Psi$  be a unital channel on  $M_d$  with Choi rank  $r$ . Under the contract of Theorem 3.4, let  $R_\Psi[b]$  be the infimum achievable  $\limsup(P + C)/n$  for services of  $\Psi$ , with  $x$  as there. Then*

$$R_\Psi[b] \leq \min\{\log_2 r, 6x\}.$$

In particular  $R_\Psi[b] = 0$  if  $b_n/\log n \rightarrow \infty$ , and the zero-rate construction then has  $J = \frac{n}{2} \log_2 r + o(n)$ . With all outputs released together, rate zero is the zero thermodynamic capacity of unital channels [29], [30, Equation (4.9) and Theorem 7.1].

For  $\Theta$ ,  $r = 3$  and this is the upper half of Theorem 3.4. For a closed apparatus serving one ququart Schur channel, independent delayed blocks give purity  $o(n)$  under superlogarithmic delay [25, Theorem 8]. The lower half has no analogue for all unital channels. A channel with a finite tracial factorization of bath dimension  $M$ , such as an equal mixture of  $M$  unitaries, has exact immediate services with  $P + C = O(\sqrt{n})$ ,  $Q = O(\sqrt{n})$  and  $J = n \log_2 M + O(\sqrt{n})$ . Each round uses its own maximally mixed bath, and the baths are packed in banks as in the proof of Theorem 3.1, so each bank costs less than one qubit of purity.

## 4 A qualitative closure law

Recall the sets  $\mathcal{F}_m$  and their closure  $\overline{\mathcal{F}}$  from Section 2.3, and the normalized Choi distance  $d_J$ . Put

$$D_{\text{mix},\Phi}(\delta) = \min\{m : d_J(\Phi, \Psi) \leq \delta, \quad \Psi \in \text{conv } \mathcal{F}_m\}.$$

The corresponding profile for single finite tracial factorizations is  $D_{J,\Phi}(\delta) = \min\{m : d_J(\Phi, \Psi) \leq \delta, \quad \Psi \in \mathcal{F}_m\}$ . Infinite profile values are allowed. Finite mixtures and single factorizations have the same closure (Lemma 4.2; see also [65, Proposition 3.8(ii)]), but their exact bath dimensions need not agree [65, Propositions 3.6 and 3.8(i)].

**Theorem 4.1** (Qualitative closure law). *Let  $\Phi$  be fixed on  $M_d$ , with a flat probe of rank  $r \geq 2$ .*

- (a) *If  $\Phi \in \overline{\mathcal{F}}$  and  $0 < \epsilon_n \leq 1/16$  satisfies  $\log(1/\epsilon_n) = o(n)$ , there are services of the exact target  $\Phi$  with error at most  $\epsilon_n$ ,  $Q = o(n)$ ,  $C = 0$ ,  $P \leq Q$ , and  $J \leq \lfloor (n-1) \log r/2 \rfloor$ .*
- (b) ([27, Theorem 3.2(b)]) *For any channel on  $M_d$ , not necessarily with a flat probe, let  $\Delta = \text{dist}_J(\Phi, \overline{\mathcal{F}}) > 0$ . Every between-round service with error at most  $\epsilon$ , at any exchange, satisfies*

$$P + C \geq \frac{n(\Delta - \epsilon)_+^2}{2 \ln 2}, \quad Q + C \geq \frac{n(\Delta - \epsilon)_+^2}{2 \ln 2}.$$

*In particular, flat imports and fixed  $\epsilon < \Delta$  force  $Q = \Omega_{\Phi,\epsilon}(n)$ .*

Part (b) is the purity floor of [27, Theorem 3.2(b)], restated for completeness. Part (a) sharpens [27, Corollary 3.3(c)], which gives sublinear memory at a vanishing error with no prescribed rate, to every subexponential error with maximally mixed imports. The operational mechanism is local tracialization: a round producing  $a$  bits of bath entropy is within  $d_J \leq \sqrt{2 \ln 2} a$  of a mixture of finite tracial factorizations (Lemma 4.5). Small production therefore supplies a finite-bath approximant. A channel a positive distance from the closure must produce entropy on a positive fraction of rounds, which is paid for by supplied purity. This includes exactly factorizable channels outside the closure: an infinite-dimensional tracial realization alone does not provide finite memory.

The theorem is qualitative. Part (a) uses the rank-rate exchange budget and allows initial purity up to the memory; part (b) holds at any exchange, and its error threshold is the distance  $\Delta$  of the given channel.

### 4.1 From a finite approximant to an exact flat source

**Lemma 4.2** (Mixtures to one maximally mixed bath). *A channel in  $\text{conv } \mathcal{F}_m$  can be approximated at half-diamond error at most  $\gamma$  by a single finite tracial factorization with bath dimension  $N = \lceil (d^4 + 1)m/\gamma \rceil$ , for  $0 < \gamma \leq 1$ .*

The proof is in Appendix B.

**Lemma 4.3** (Positive-Gram repair for a finite tracial approximant). *For a fixed channel  $\Phi$  with a flat probe, a finite tracial factorization with bath dimension  $k$  and Choi distance at most  $\delta$  from  $\Phi$  yields an exact rectangular dilation of  $\Phi$  with source rank  $k_1$ , output dimension at most  $rk_1$ , and flat-round entropy production  $a_1$  satisfying*

$$\log k_1 \leq \log k + O_\Phi(\log(1/\delta)), \quad a_1 \leq C_\Phi \sqrt{\delta} \log(e/\delta)$$

*for sufficiently small  $\delta$ . The source is flat and its Gram is exactly  $I_r$ .*

The proof is in Appendix B.

## 4.2 The quantitative upper bound

**Theorem 4.4** (Profile-controlled exact-target upper). *For a fixed flat-probe channel there are  $C_\Phi, \delta_\Phi > 0$  such that, for  $n \geq 2$  and  $0 < \epsilon < 1$ ,*

$$\begin{aligned} Q^*(n, \epsilon) \leq C_\Phi \inf_{0 < \delta \leq \delta_\Phi} \{ & \log(D_{\text{mix}, \Phi}(\delta) + 1) + \log(1/\delta) \\ & + n\sqrt{\delta} \log^2(e/\delta) + \log(n/\epsilon) \log(e/\delta) \\ & + \log(2 + \log(D_{\text{mix}, \Phi}(\delta) + 1)) \}. \end{aligned}$$

The service has  $C = 0$ ,  $P \leq Q$  and  $J \leq \lfloor (n-1) \log r/2 \rfloor$ .

The proof is in Appendix B.

## 4.3 The nonclosure direction

**Lemma 4.5** (Local tracialization). *For a round  $(U, \rho)$  with maximally mixed system input and entropy production  $a$ , its channel is within Choi distance  $\sqrt{2 \ln 2} a$  of a finite mixture of finite tracial factorizations, each with bath dimension at most that of  $\rho$ .*

This is [27, Lemma 5.4]; Appendix B gives a short proof.

## 4.4 Repeatability without resets

Rybár and Ziman's device [72, Sections II and III] is one unitary  $U$  on  $\mathbb{C}^d \otimes \mathbb{C}^M$  and one memory state  $\xi_1$  on  $\mathbb{C}^M$ . There is no reset and no exchange. Uncorrelated inputs  $\rho_1, \rho_2, \dots$  arrive one at a time. The memory before use  $j$  is  $\xi_j$ , with  $\xi_{j+1} = \text{Tr}_d U(\rho_j \otimes \xi_j) U^*$ , and use  $j$  induces the channel  $X \mapsto \text{Tr}_M U(X \otimes \xi_j) U^*$ . The device *repeats*  $\Phi$  for  $n$  uses if this channel is  $\Phi$  for every  $j \leq n$  and every choice of uncorrelated inputs. The channel is *n-repeatable with memory dimension  $M$*  if some such device repeats it for  $n$  uses, and *repeatable* if one finite device repeats it for every  $n$ . A device is a *catalytic dilation* if  $\text{Tr}_d U(\rho \otimes \xi_1) U^* = \xi_1$  for every input  $\rho$  [60, Definition 2]. Its memory state never changes, so it repeats its channel for every  $n$ .

**Proposition 4.6** (Repeatability without resets). *Let  $\Phi$  be a channel on  $M_d$ .*

(a) *If a device with memory dimension  $M$  and initial state  $\xi_1$  repeats  $\Phi$  for  $n$  uses, then*

$$\log M - S(\xi_1) \geq \frac{n \text{dist}_J(\Phi, \text{conv } \mathcal{F}_M)^2}{2 \ln 2} \geq \frac{n \text{dist}_J(\Phi, \overline{\mathcal{F}})^2}{2 \ln 2}.$$

(b) *If  $\Phi$  is  $n$ -repeatable with memory dimension  $M$  for every  $n$ , then  $\Phi = \sum_k p_k \Psi_k$  with  $p_k > 0$ ,  $\Psi_k \in \mathcal{F}_{m_k}$  and  $\sum_k m_k \leq M$ . In particular  $\Phi$  is unital and  $\Phi \in \text{conv } \mathcal{F}_M \subseteq \overline{\mathcal{F}}$ . For a device whose memory state is unchanged by a maximally mixed input, this is the characterization of Lie, Son, Boes, Ng and Wilming [60, Proposition 4 and Equation (14)].*

(c) *Every device that repeats the qutrit channel  $\Theta$  of Section 3 for  $n$  uses has  $\log M \geq S(\xi_1) + n \log 3$ . Rybár and Ziman's  $n$ -cell device [72, Section III] with pure three-dimensional cells has  $M = 3^n$  and attains equality. Hence  $\Theta$ , which is unital and not random-unitary [58], is not repeatable.*

(d) *(Haagerup and Musat [43, Example 3.3]; Lie, Son, Boes, Ng and Wilming [60, Proposition 15].) Put  $v_0 = (1, 0, 0)$  and  $v_j = 5^{-1/2}(1, 2 \cos \frac{2\pi j}{5}, 2 \sin \frac{2\pi j}{5})$  for  $1 \leq j \leq 5$ , and  $B_{kl} = v_k \cdot v_l$ . The Schur channel  $T_B(X) = B \circ X$  on  $M_6$  is unital and not random-unitary. With  $w_k = v_{k,1}\sigma_x + v_{k,2}\sigma_y + v_{k,3}\sigma_z$ , the unitary  $U = \sum_{k=0}^5 |k\rangle\langle k| \otimes w_k$  and the*

memory state  $\xi_1 = I/2$  form a catalytic dilation of  $T_B$ . So  $T_B$  is repeatable with one memory qubit.

Part (d) answers Rybár and Ziman’s question [72, Section IV]: a unital channel that is not random-unitary can be repeated without resets. The channel and the proof that it is not random-unitary are Haagerup and Musat’s; the matrix  $B$  is that of [43, Example 3.3], whose (2, 6) entry is printed there as  $-5^{-1/2}$  although the decomposition given there makes it  $+5^{-1/2}$ . Lie, Son, Boes, Ng and Wilming gave a catalytic dilation of every real Schur channel, with memory dimension  $2^{\text{rank } B}$  [60, Proposition 15], and used this channel to show that catalytic channels need not be random-unitary [60, Appendix C]. Part (d) adds that a catalytic dilation repeats its channel, and that for this  $B$  the three Pauli matrices give one memory qubit. Part (b) places every repeatable channel among the finite mixtures of finite tracial factorizations. These are the strongly factorizable channels of [60], and they are dense in  $\overline{\mathcal{F}}$ . Lie and Jeong showed that catalytic channels are factorizable [59]. Part (c) shows that unitality is not enough in dimension three, although it is for qubits [72, Section IV]. Part (b) also rules out the channels in  $\overline{\mathcal{F}}$  that need an infinite-dimensional ancilla [66, Theorem 4.1]. The proof, in Appendix B, uses only maximally mixed inputs and single-use marginals, so the outputs may be correlated through the memory.

## 5 Three regimes for anchored gadgets

The closure law distinguishes sublinear from linear memory. In the gadget class, exact support and the anchors also distinguish bounded one-round cost from cost that diverges. The mechanism is compactness at bounded *entropy*, followed by support-preserving conversion of a finite mixture to one maximally mixed bath. Slofstra and Vidick read the same three-way split (bounded, finite but divergent, infinite) off hyperlinear profiles [78, Section 2]; for sofic profiles of chunks, boundedness means that the chunk embeds in a finite group [21, Fact 3.5]. For a linear-system game that can be won with probability approaching one, the entanglement needed to win with probability  $1 - \epsilon$  is governed by such a profile [78, Corollaries 5.2 and 5.3]. It stays bounded exactly when a finite-dimensional perfect strategy exists [20, 19], and it is unbounded for some games [76, 77]. There the dimension diverges as the error tends to zero; here memory diverges with the number of uses at fixed error, the leakage scale  $1/n$  playing the role of the defect (Remark 2.5). Channel profiles correspond to group profiles by [27, Theorem 9.2].

**Theorem 5.1** (Gadget trichotomy). *For a canonical untwisted anchored gadget, exactly one of the following holds.*

- (i)  $\Phi$  has an exact finite tracial factorization. Equivalently, a finite-dimensional unitary representation of  $P_\Phi$  separates all labels, or a finite quotient of  $P_\Phi$  distinguishes every named label. Then  $\sup_n L_\Phi^*(n) < \infty$  and  $Q_{\text{flat}}^*(n, \epsilon) = O_\Phi(\log(n/\epsilon))$  [27, Theorem 8.6].
- (ii)  $\Phi \in \overline{\mathcal{F}}$  but has no exact finite tracial factorization. Equivalently, the named labels are separated by a homomorphism  $P_\Phi \rightarrow U(\mathcal{R}^\omega)$  but by no finite quotient. Then  $L_\Phi^*(n) \rightarrow \infty$ ,  $L_\Phi^*(n) = o(n)$ , and  $Q^*(n, \epsilon_n) \rightarrow \infty$  for every  $0 < \epsilon_n \leq 1/16$ . If also  $\log(1/\epsilon_n) = o(n)$ , flat-import rank-rate services have  $Q = o(n)$ .
- (iii)  $\Phi \notin \overline{\mathcal{F}}$ . Equivalently, some distinct named labels are identified by every homomorphism  $P_\Phi \rightarrow U(\mathcal{R}^\omega)$ . Then  $L_\Phi^*(n) = \Theta_\Phi(n)$  and  $Q^*(n, \epsilon_n) = \Theta_\Phi(n)$  for every error sequence  $0 < \epsilon_n \leq 1/16$ .

All upper constructions have  $C = 0$  and  $P \leq Q$ .

In (i), [27, Theorem 8.6] allows exchange  $\lfloor n \log r/2 \rfloor$ . Skipping exports of at most  $\lceil \log r/2 \rceil$  qubits, and holding from the start a maximally mixed qubit in place of each skipped import, meets the budget  $\lfloor (n-1) \log r/2 \rfloor$  with the same error and  $O_r(1)$  more memory.

Here  $\mathcal{R}^\omega$  is a tracial ultrapower of the hyperfinite  $\text{II}_1$  factor. Separation means operator-distinct images; it does not require a map to preserve the regular group trace. Padding by a trivial representation and tensor amplification turn separation of the finite label set into the needed vanishing trace overlaps (compare [78, Lemma 2.2 and the proof of Lemma 2.6]). Lemma C.3 gives this conversion. Finite-dimensional separation gives a finite quotient by residual finiteness of the finitely generated matrix image. These criteria concern the presented label group, not every element of its source-group image.

In the middle regime memory diverges, possibly no faster than  $\log n$ , and the theorem gives no rate there. In the linear regime the error threshold is the fixed value  $1/16$ , whereas the general converse of Theorem 4.1 needs error below the channel's distance from the closure.

**Why bounded entropy is enough.** Take rounds with bounded source entropy and production  $a_j \rightarrow 0$ . Bounded entropy keeps the sorted source spectra from spreading over ever more eigenvalues, so they converge. A round moves weight between bath eigenvalues through a doubly stochastic matrix, and small production forces the spectrum after a use to match the spectrum before it, up to  $\sqrt{2 \ln 2 a_j}$  in  $\ell^1$ . In the limit the round maps each eigenspace of the limiting source spectrum unitarily onto itself. So the limiting channel is a finite mixture of finite tracial factorizations (Lemma C.1). This mixture conclusion holds for general channels. Anchors turn an exact-support mixture close to the canonical gadget into a separating representation.

Let  $B = \sup_n L_\Phi^*(n)$ . Let  $R_{\text{sep}}$  be the least  $\log m$  for which a representation  $\pi$  of  $P_\Phi$  on  $\mathbb{C}^m$  satisfies  $|\tau_m(\pi(v_g)^* \pi(v_h))| \leq 1/2$  for all distinct labels  $g \neq h$ . Then

$$R_{\text{sep}} \leq 16(B+1) + \log_2(33(d^4+1)), \quad B \leq \lceil \log_2(8r(r-1)) \rceil R_{\text{sep}}.$$

The exact finite tracial factorization is obtained by Malcev residual finiteness of the finitely generated matrix image and its finite quotient's regular representation [67]. No bound on that quotient's order follows from the displayed representation bound. Appendix C proves the classification.

## 5.1 Known examples

The placements in regimes (i) and (ii) follow from results in this paper; regime (iii) is discussed after Proposition 5.2. The two-element example has  $P_\Phi = \mathbb{Z}/2$ . A gadget from the lamplighter  $C_2 \wr \mathbb{Z}$  has labels separated by finite quotients (Section 7). The Houghton gadget has  $P_{\Phi_H} \cong H_3$  (Section 2.2). Every finite-dimensional representation of  $H_3$  identifies a three-cycle label with the label 1 (Remark 8.1), so regime (i) fails. Since  $H_3$  is amenable, it is hyperlinear, so it embeds in  $U(\mathcal{R}^\omega)$  [69] and the labels are separated: this is regime (ii). For the configuration-lamp gadget, Corollary 7.9 gives  $L_\Phi^*(n) \rightarrow \infty$  and  $Q^*(n, \epsilon) = o(n)$  at fixed error, which excludes regimes (i) and (iii).

**Proposition 5.2** (Regime (iii) and non-hyperlinear groups). *The following are equivalent.*

- (a) *Some canonical untwisted anchored gadget lies in regime (iii) of Theorem 5.1.*
- (b) *Some finitely presented group  $P$  has an element  $\zeta \neq 1$  such that  $\pi(\zeta) = 1$  for every homomorphism  $\pi : P \rightarrow U(\mathcal{R}^\omega)$ .*
- (c) *Some finitely presented group is not hyperlinear.*

| Regime                                                     | Criterion on $P_\Phi$ ; memory                                                                         | Known examples                                                                                                                                                                                                                                                           |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i) exact finite tracial factorization                     | A finite quotient separates the labels; memory $O(\log(n/\epsilon))$ .                                 | Labels separated in a finite, residually finite or maximally almost periodic group satisfying the relations (Corollary 7.8): the two-element example of Section 2.2, the lamplighter $C_2 \wr \mathbb{Z}$ .                                                              |
| (ii) in $\overline{\mathcal{F}}$ , no finite factorization | Labels separated in $U(\mathcal{R}^\omega)$ but by no finite quotient; memory diverges and is $o(n)$ . | Houghton's group $H_3$ ; the configuration-lamp group $\Gamma$ .                                                                                                                                                                                                         |
| (iii) outside $\overline{\mathcal{F}}$                     | Two labels identified by every homomorphism into $U(\mathcal{R}^\omega)$ ; memory $\Theta(n)$ .        | Known only conditionally: the canonical gadget $\Phi_{\text{cert}}$ of the group certificate of [63], at an explicit distance from $\overline{\mathcal{F}}$ (Proposition 5.5). Nonempty exactly when some finitely presented group is not hyperlinear (Proposition 5.2). |

Table 2: The three regimes of Theorem 5.1, at fixed error at most 1/16.

For (a) $\Rightarrow$ (b), take  $P = P_\Phi$  and  $\zeta = v_g^{-1}v_h$  for two identified labels. For (b) $\Rightarrow$ (a), add the freely trivial word  $ww^{-1}$ , where  $w$  represents  $\zeta$ , to a presentation of  $P$ ; this makes  $\zeta$  a label with its own anchor. Appendix C.5 gives the proof, including the standard step (c) $\Rightarrow$ (b).

**Which gadgets are known outside the closure?** Our companion paper [63] gives explicit rational channels outside  $\overline{\mathcal{F}}$ , at explicit distances. Each factorizes exactly through a group von Neumann algebra. Its distances are the normalized Choi and half-diamond distances of Section 2, and its closure of the noisy operations is  $\overline{\mathcal{F}}$ , so its bounds apply here unchanged. Write  $\text{dist}_\diamond(\Phi, \overline{\mathcal{F}}) = \inf_{\Psi \in \overline{\mathcal{F}}} d_\diamond(\Phi, \Psi)$ .

**Proposition 5.3** (Mghirbi–Douglas [63, Theorems I and III]). *Put  $\delta_{\text{WZ}} = 2^{-2^{1122000}}$ , the error tolerance of Wang and Zhi [82, (2.20)], and assume the theorems of [82].*

- (a) *A unital channel  $\Phi_{12}$  on twelve qubits, with Kraus entries in  $\{0, 1, \pm 3/5, 4/5\}$  and Choi rank 949, has  $\text{dist}_J(\Phi_{12}, \overline{\mathcal{F}}) \geq 2^{-50}\delta_{\text{WZ}}^2$  and  $\text{dist}_\diamond(\Phi_{12}, \overline{\mathcal{F}}) \geq 2^{-50}\delta_{\text{WZ}}^2$ .*
- (b) *The fifteen-qubit channel  $\Phi_{15}$  and its reductions  $\Phi_1, \Phi_2$ , on  $M_d$  with  $d = 31\,981$  and  $31\,982$ , have Choi rank 11 230 and both distances at least  $2^{-41}\delta_{\text{WZ}}^2$ .*
- (c) *The fully referenced channel  $\Phi_{\text{full}}$  on  $M_{44\,086}$ , with a singleton anchor for each of its 11 230 labels, has  $\text{dist}_J(\Phi_{\text{full}}, \overline{\mathcal{F}}) \geq 2^{-36}\delta_{\text{WZ}}^2$  and  $\text{dist}_\diamond(\Phi_{\text{full}}, \overline{\mathcal{F}}) \geq 2^{-21}\delta_{\text{WZ}}^2$ .*

The constants are astronomically small: already  $2^{1122000}$  has 337 756 decimal digits. They come from the rigidity estimate, not from the channels. Theorem 4.1(b) turns them into explicit purity and memory bounds, as [63, Section 6] notes.

**Corollary 5.4** (Theorem 4.1(b) with [63, Theorem III]). *Every between-round service of  $\Phi_{\text{full}}$  with error at most  $\epsilon$  satisfies*

$$P + C \geq \frac{n(2^{-36}\delta_{\text{WZ}}^2 - \epsilon)_+^2}{2 \ln 2}, \quad Q + C \geq \frac{n(2^{-36}\delta_{\text{WZ}}^2 - \epsilon)_+^2}{2 \ln 2}.$$

*With maximally mixed imports and  $\epsilon \leq 2^{-37}\delta_{\text{WZ}}^2$ , this gives  $Q \geq 2^{-75}\delta_{\text{WZ}}^4 n$ .*

The memory is linear, but the bound reaches one qubit only at  $n = 2^{75} \delta_{\text{WZ}}^{-4}$ . It replaces a compactness argument by a number; it does not describe an experiment.

None of these channels is a canonical gadget, so Theorem 5.1 does not apply to them. Their blocks use  $3/5$  and  $4/5$  where (1) uses  $1/\sqrt{2}$ . Equal triangles share one block, while the canonical table gives every tagged prefix its own. Most labels have no anchor.  $\Phi_{12}$ , the output of the anchored-letter compiler [63, Definition 3.1], has singletons only for the identity, the letters and some further labels: 364 of its 949 labels.  $\Phi_2$  and  $\Phi_{15}$  have singletons only for the identity and the witness, and  $\Phi_1$  only for the identity.  $\Phi_{\text{full}}$  anchors every label but keeps the  $3/5, 4/5$  blocks.

A canonical example comes from the same group construction. Take the presentation of [63, Section 7], with 43 generators and 4417 relators of length at most 19, and its group image  $\Gamma$ . One generator is the witness  $j$ , and  $j \neq 1$  in  $\Gamma$ . Let  $\Phi_{\text{cert}}$  be the canonical gadget of  $\Gamma$ , these generators and these relators. It acts on  $M_d$  with  $d = 108\,364$ , and the letter  $j$  has its own singleton anchor. Every homomorphism from the presented group into a tracial matrix ultraproduct kills  $j$  [64, Theorem 1]. The presented group is countable, so a label-separating homomorphism into  $U(\mathcal{R}^\omega)$  would give one into a tracial matrix ultraproduct: the von Neumann algebra generated by its image embeds trace-preservingly into one. That would contradict the certificate, so Lemma C.3 places  $\Phi_{\text{cert}}$  outside  $\overline{\mathcal{F}}$ , in regime (iii).

The fully referenced bound of [63] makes this example explicit.

**Proposition 5.5** (An explicit canonical gadget outside the closure). *Assume the rigidity theorem [63, Theorem D.4], which rests on [82]. Then*

$$\begin{aligned} \text{dist}_J(\Phi_{\text{cert}}, \overline{\mathcal{F}}) &\geq \frac{\delta_{\text{WZ}}^2}{4096 \cdot 19^2 \cdot 108\,364} > 2^{-38} \delta_{\text{WZ}}^2, \\ \text{dist}_\diamond(\Phi_{\text{cert}}, \overline{\mathcal{F}}) &\geq \frac{\delta_{\text{WZ}}^2}{4096 \cdot 19^2} > 2^{-21} \delta_{\text{WZ}}^2. \end{aligned}$$

*Proof.* By [63, Theorem D.4], unitaries whose relator defects are at most  $\delta_{\text{WZ}}$  have  $\|v_j - I\|_2 < 2^{-17}$ , in every matrix dimension. So  $\delta_{\text{WZ}}$  is a relator tolerance in the sense of [63, Lemma 7.3]. That lemma, from [64, Theorem 6], gives the displayed bounds, with  $\ell_* = 19$ , for channels built from  $3/5, 4/5$  blocks with a singleton for every label. Its proof, with  $k$  as there, uses the coefficients only twice. Each block entry  $c$  has  $|c| \leq 4/5$ , which keeps the bath block within  $3k$  of  $c$  times its label unitary. Each block has  $ab \geq 12/25$ , which turns the row-orthogonality defect,  $13k$  or  $(12 + \sqrt{2})k$ , into a triangle defect below  $32k$ . The canonical coefficients  $a = b = 1/\sqrt{2}$  satisfy both. The canonical table contains every prefix triangle and every inverse triangle, every label has an anchor, and the anchors of 1 and  $j$  are distinct. The proof uses one anchor per label and treats blocks one at a time, so tagged repeats change nothing.  $\square$

So  $\Phi_{\text{cert}}$  is a canonical gadget in regime (iii) at an explicit distance. Theorem 4.1(b) gives  $P + C \geq n(2^{-38} \delta_{\text{WZ}}^2 - \epsilon)_+^2 / (2 \ln 2)$  for its between-round services, and Theorem 5.1(iii) gives linear memory at every error up to  $1/16$ , with a constant from compactness.

These results are conditional. The certificate rests on Alekseev–Liu–Thom, Theorem B, and Thom, Theorem 1.2 [1, 81]. The explicit distances rest on the rigidity theorems of Wang and Zhi [82], whose proofs follow steps of those two papers; [63, Section 1.4] lists which statements use them. All three are recent unrefereed preprints at the time of writing. Exact factorizability is unconditional. By contrast  $\text{MIP}^* = \text{RE}$  and Haagerup–Musat imply nonconstructively an exactly factorizable nonclosure channel in some dimension, recorded for a Schur channel in [65, Corollary 3.11], but do not by themselves supply an explicit group or canonical gadget.

## 6 The exact entropic profile and the quantitative memory law

An exact support certificate specifies what a round does on its source, without requiring a flat source or global multiplication of rectangular responses. That is enough for a chronological compiler. It is also the right static object to extract from high-precision services.

**Definition 6.1** (Exact-support entropy profile). For a canonical untwisted anchored gadget, take finite spaces  $K, L$ , a state  $\rho$  on  $K$ , and isometries  $B_g : K \rightarrow L$  satisfying

$$B_1^* B_{[y]} = B_{[x]}^* B_{[z]}, \quad \max_{g \neq h} |\text{Tr}(\rho B_g^* B_h)| \leq 1/2,$$

the first identity for every triangle  $(x, y, z)$  of the table, as in Lemma 2.9(a). Put  $\sigma = \sum_g \mu_g B_g \rho B_g^*$  and  $a = S(\sigma) - S(\rho)$ . Define, for  $t \geq 0$ ,

$$Z_\Phi(t) = \inf\{S(\rho) + ta\}$$

over these families. One may restrict  $K$  to  $\text{supp } \rho$  and  $L$  to  $\text{span}_g B_g K$ , so  $\dim L \leq r \dim K$ .

The family extends to a physical unitary round, and  $a$  is its actual Choi-input production. It is nonnegative because the isometric output ensemble has the source spectrum. Exact support does not mean exact target error: the coarse trace gap is amplified when designing a service.

**Proposition 6.2** (Finite-matrix form and profile structure). *For source dimension  $k$ , an equivalent optimization uses a positive block matrix  $Y = [Y_{gh}]$  and a state  $\rho$ , with  $Y_{gg} = I$ ,  $Y_{1[y]} = Y_{[x][z]}$  for every triangle  $(x, y, z)$ , and  $|\text{Tr } \rho Y_{gh}| \leq 1/2$  for  $g \neq h$ . Writing  $D = \text{diag}(\sqrt{\mu_g})$ , the state*

$$\omega = (D \otimes \sqrt{\rho}) Y (D \otimes \sqrt{\rho})$$

*has source marginal  $\rho$  and  $a = S(\omega) - S(\rho)$ . The function  $Z_\Phi$  is nondecreasing, concave and subadditive,  $Z_\Phi(0) = 0$ ,  $Z_\Phi(t) \leq t \log r$ , and  $Z_\Phi(ct) \leq c Z_\Phi(t)$  for  $c \geq 1$ . Moreover, for  $t_r = \lceil \log_2(8(r-1)) \rceil$ ,*

$$L_\Phi^*(n) \leq t_r Z_\Phi(n).$$

The spectrum identity follows from the two products of the row operator  $[\sqrt{\mu_g} B_g \sqrt{\rho}]_g$ . Nonnegativity is a property of this ensemble; quantum conditional entropy is not positive in general. Infima of the affine costs  $S + ta$  give concavity. Fixed tensor amplification preserves the untwisted identities and powers the scalar trace overlaps. Its output is the *mixture of powered outputs*, not the tensor power of their mixture. Appendix F proves these assertions.

**Theorem 6.3** (Coherent exact-support compiler). *Every family in Definition 6.1 gives, for  $n \geq 1$  and  $0 < \epsilon \leq 1$ , a fixed public service of the canonical target with complete adaptive error at most  $\epsilon$  and*

$$Q \leq C_{d,r} \log_2^2 \left( \frac{128(n+1)}{\epsilon} \right) (1 + S + na), \quad J \leq \lfloor (n-1) \log r / 2 \rfloor, \quad C = 0, \quad P \leq Q.$$

*The constant is independent of source dimension, spectrum and response geometry.*

The construction replaces the biased source by maximally mixed registers. Each eigenvalue of the source, and of the output state, is coded as a block whose dimension is proportional to it, inside sectors of geometrically increasing dimension; this is the standard embedding of nonuniform states into flat ones from thermodynamic resource theory [50]. The sector index then acts as an energy, and one use changes it by the logarithm of the ratio of an output eigenvalue to a source eigenvalue. Tensor amplification bounds the moment generating

function of these changes as a matrix over pairs of labels, which covers coherent sources. Excess dimension is exported and replaced by maximally mixed qubits. One induction over the rounds bounds both the chance of leaving the allowed range of sectors and the error of the whole adaptive experiment. The device never measures the source or postselects. Appendix E gives the construction and its resource count.

**Theorem 6.4** (Any-exchange necessity and high precision). *Every between-round  $n$ -round service of the canonical gadget, at any exchange, with error  $0 < \epsilon \leq 1/16$  satisfies*

$$Z_\Phi(n) \leq K_{d,r} [1 + Q + C + \sqrt{n\epsilon} \log_2(en/\epsilon)].$$

*If  $\epsilon \leq 1/[n \log_2^2(en)]$ , this reduces to  $1 + Z_\Phi(n) \leq K'_{d,r}(1 + Q + C)$ . There is no exchange restriction. With flat imports, Theorem 6.3 gives the matching rank-rate upper up to the displayed squared logarithm.*

Extraction chooses a round with leakage  $O(\epsilon/n)$  and production paid by  $(P + C)/n$ . Biased first-order repair makes support exact at additional cost  $O(\sqrt{n\epsilon} \log(en/\epsilon))$ . Keeping the experiment error is essential: setting it to  $1/16$  first loses the high-precision conclusion. Its proof is independent of the compiler (Appendix F).

**Theorem 6.5** (Memory from one-use entropy). *Fix a canonical untwisted anchored gadget on  $M_d$  with Choi rank  $r \geq 2$ . Write  $y = \log_2(128(n+1)/\epsilon)$ , where  $0 < \epsilon \leq 1/16$ . Constants depend only on  $d, r$ .*

- (a) *Every service with exchange  $J \leq n \log_2 r/2$  has  $Q \geq c_{d,r} L_\Phi^*(n) - C_{d,r} \log_2(en)$ .*
- (b) *A service exists with maximally mixed imports, initial purity  $P \leq Q$ , exchange  $J \leq \lfloor (n-1) \log_2 r/2 \rfloor$  and  $Q \leq C_{d,r} y^2(1 + Z_\Phi(n))$ .*
- (c) *At precision  $\epsilon \leq 1/[n \log_2^2(en)]$ ,*

$$c_{d,r}(1 + Z_\Phi(n)) \leq 1 + Q_{\text{flat}}^*(n, \epsilon) \leq C_{d,r} y^2(1 + Z_\Phi(n)),$$

*where  $Q_{\text{flat}}^*$  minimizes memory under the flat-import contract in (b).*

The converse uses Theorem 7.3; the construction and high-precision comparison use Theorems 6.3 and 6.4. At fixed error the explicit converse correction grows with  $n$ , which is why part (c) states its precision hypothesis.

**Theorem 6.6** (Horizon envelope). *For a fixed threshold  $N_0(d, r)$ , put*

$$E_\Phi(n) = \inf_{N \geq \max(n, N_0)} \left[ L_\Phi^*(N) + \frac{n}{\sqrt{N}} \log_2(eN) \right],$$

*where  $N$  is integer. Then  $Z_\Phi(n) \leq C_{d,r}(1 + E_\Phi(n))$  and*

$$Q_{\text{flat}}^*(n, \epsilon) \leq C_{d,r} \min \left\{ n, y^2(1 + E_\Phi(n)) \right\}, \quad y = \log_2(128(n+1)/\epsilon).$$

*No regularity of  $L_\Phi^*$  is assumed.*

Choose a round at the design horizon  $N$ , repair its leakage at that scale, and serve  $n \leq N$  rounds. At polynomial inverse error,  $L_\Phi^*(N) = N^{\beta+o(1)}$  gives the generic upper exponent  $\beta/(\beta + 1/2)$  for  $0 < \beta < 1/2$  and  $\beta$  for  $\beta \geq 1/2$ . The lower exponent is  $\beta$ . Polylogarithmic profiles give polylogarithmic services and subpolynomial profiles give subpolynomial services, but the envelope does not close the interval below  $1/2$ .

**The missing theorem.** A universal all-accuracy rank-rate upper bound on  $Q_{\text{flat}}^*$  through  $L_{\Phi}^*(n)$ , with flat imports, is equivalent, up to finite logarithmic powers, to

$$Z_{\Phi}(n) \leq K_{\Phi} \log_2^p(en)(1 + L_{\Phi}^*(n)) \quad (2)$$

for some finite  $p$ . Sufficiency is Theorem 6.3; necessity applies Theorem 6.4 at  $\epsilon_n = 1/[16n \log_2^2(en)]$ . This is a comparison of optimized static costs. It permits choosing an exact family wholly different from the original leaky round. Requiring a close repair of every round is stronger and can fail.

**Other descriptions of the same cost.** Appendix F gives two equivalent descriptions of  $Z_{\Phi}$ . First, a family whose source-weighted trace defect is at most  $c_r/n$  can be repaired to an exact one, at a cost bounded by a constant multiple of its own cost plus  $\log(en)$  (equation (23)), without filtering the source. Second, an optimization over ensembles of square unitaries, penalized by entropy, matches  $Z_{\Phi}$  within constants. Neither proves (2): the ensemble's entropy gain need not be that of a physical round, and ordinary leakage controls the trace defect only at the square-root scale.

**A sufficient probability certificate.** There is one useful setting in which support probability can be used directly. Let  $\Lambda$  be the positive bath leakage operator, normalized so that  $\ell = \text{Tr}(\rho\Lambda)/d$ . If the round has distance  $u \leq 1/16$  and the source has mass  $1 - O(1/n)$  on  $\ker \Lambda$ , then for all sufficiently large  $n$ , even without commutation between the source and that kernel projection,

$$Z_{\Phi}(n) \leq C_{d,r} \log_2(en)(1 + S + na).$$

Appendix F proves the dimension-free source projection estimate and its coherent permutation corollary. Small mean leakage alone does not imply this hypothesis: the quartic example there has an empty exact kernel and nevertheless a cheap, different exact finite tracial factorization.

## 7 Extracting a one-round memory lower bound

### 7.1 Two scales in one round

The lower half rests on a single observation about the proof of sequential extraction [27, Theorem 5.2]: it bounds the leakage and the entropy production of the extracted round separately, and the two bounds live at different scales. Leakage is a probability of failing a support test, so the error budget  $\epsilon$  controls its sum over the rounds; entropy production is an increase of memory entropy, so the purity budget controls its sum.

**Theorem 7.1** (Two-scale extraction). *Let a causal service of  $\Phi$  for  $n$  rounds have memory  $Q$ , error  $\epsilon < \frac{1}{2}$  and purity  $P + C \leq B$ , and put  $B' = B + h_2(\epsilon) + \epsilon \log n$ . For every  $\lambda > 1$  some round  $t$  gives a round  $(U_t, \rho_t)$  of  $\Phi$ , where  $U_t$  is the unitary of round  $t$  on the input and the  $Q$  memory qubits and  $\rho_t$  is a memory state, with bath dimension  $2^Q$  and*

$$d_{\diamond}(\Psi_t, \Phi) \leq \epsilon, \quad \ell_t \leq \frac{\lambda \epsilon}{n(1 - \epsilon)}, \quad a_t \leq \frac{\lambda}{\lambda - 1} \cdot \frac{B'}{n(1 - \epsilon)}.$$

The state  $\rho_t$  is the memory state conditioned on an observer passing the support tests of rounds  $1, \dots, t-1$ . The joint budget  $\ell_t + a_t \leq (B + \epsilon + h_2(\epsilon) + \epsilon \log n)/(n(1 - \epsilon))$  of [27, Theorem 5.2] follows from the two budgets established in its proof; the split is what is new. At the rank rate the purity is paid for with memory.

**Proposition 7.2** (Purity budget [28, Proposition 13.1]). *If  $\Phi$  has a flat probe of rank  $r$ , every causal service of  $\Phi$  for  $n$  rounds with error  $\epsilon \leq \frac{1}{16}$  and exchange  $J \leq \frac{n}{2} \log r$  has  $P + C < 3Q + 6$ .*

**Theorem 7.3** (The memory lower bound). *Let  $\Phi$  have a flat probe of rank  $r$ . A rank-rate service with  $n \geq 1$  rounds and error  $\epsilon \leq 1/16$  has a static round of bath dimension  $2^Q$ , distance at most  $\epsilon$ , leakage at most  $32\epsilon/(15n)$ , and  $na \leq CQ + C' \log_2(en)$ , for universal constants. Consequently, for universal  $c, C'' > 0$ ,*

$$Q \geq cL_\Phi(n, \epsilon) - C'' \log_2(en) \geq cL_\Phi^*(n) - C'' \log_2(en).$$

*At arbitrary exchange, Theorem 7.1 retains the separate leakage and supplied-purity budgets. The detailed inequalities are in Theorem J.1.*

The profile does not depend on  $\epsilon$ : one function  $L^*$  bounds the memory at every error up to  $\frac{1}{16}$ . Under the single budget of [27, Theorem 5.2] the leakage would be read at the purity scale  $(Q + \log n)/n$ ; for Houghton's channel that caps the method at  $n^{1/4}$ , while the two-scale profile can reach  $n^{1/3}$  (Remark 8.11 and Proposition 8.10).

*Remark 7.4* (The extracted round is a round to be streamed). The round of Theorem 7.3 has an arbitrary bath state on  $Q$  qubits, distance at most  $\frac{1}{16}$ , leakage at most  $\frac{2}{15n}$ ,  $S \leq Q$  and  $na = O(Q + \log(en))$ . It belongs to the class of rounds over which  $L^*(n)$  is defined, at the same scales. So an upper bound  $Q^*(n, \epsilon) \leq K(\log(n/\epsilon))^\kappa(1 + L^*(n))$  follows if all these rounds stream in the following sense; no rescaling of  $\epsilon$  is needed to match the two halves.

**Definition 7.5** (Streaming). A class  $\mathcal{C}$  of rounds of  $\Phi$  *streams* if there are  $K, \kappa, n_0$  such that for all  $n \geq n_0$  and  $0 < \epsilon \leq \frac{1}{16}$ , every round of  $\mathcal{C}$  with distance at most  $\frac{1}{16}$  and leakage at most  $\frac{2}{15n}$  gives a rank-rate service of  $\Phi$  for  $n$  rounds with error at most  $\epsilon$  and memory

$$Q \leq K(\log(n/\epsilon))^\kappa(1 + S + na).$$

It streams *up to an additive term*  $E(n)$  if the same holds with  $S + na + E(n)$  in place of  $S + na$ .

If the class of all rounds streams, then  $Q^*(n, \epsilon) \leq K(\log(n/\epsilon))^\kappa(1 + L^*(n) + 1)$ , which together with Theorem 7.3 is the two-scale law. Conversely, if the law holds then every class streams, since the service need not come from the round. Rounds with  $a \geq (\log(n/\epsilon))^{-\kappa}$  always stream with  $K \geq \lceil \log r \rceil$ : for them the linear ceiling of Proposition 2.7 gives  $n \lceil \log r \rceil \leq \lceil \log r \rceil (\log(n/\epsilon))^\kappa na$ .

## 7.2 The rank form

The upper half of [28, Theorem 12.2] streams exact rectangular dilations at the cost  $\log k + n \log(k'/k)$ , which is the cost of the round they give (Lemma 2.9(c)) with the entropy production replaced by the room. Its device reads only the flat Gram of the family, so it tolerates label traces that are merely bounded away from one.

**Theorem 7.6** (Robust streaming). *Let  $\Phi$  have a flat probe of rank  $r$ .*

- (a) *Let  $W = \sum_i A_i \otimes B_i : \mathbb{C}^d \otimes \mathbb{C}^k \rightarrow \mathbb{C}^d \otimes \mathbb{C}^{k'}$  be an isometry, with no condition on the traces, and put  $G_{ij} = \tau_k(B_i^* B_j)$ . For  $n \geq 2$ ,  $0 < \epsilon < 1$  and  $n \|G - I\| \leq 1$  there is a causal service of  $\Phi$  for  $n$  rounds with  $C = 0$ ,  $J = \lfloor \frac{n-1}{2} \log r \rfloor$ ,  $P \leq n \log(k'/k) + 2$ ,*

$$Q \leq \log k + n \log(k'/k) + O_r(\log(n/\epsilon) + \log(2 + \log k)), \quad \text{error} \leq e^2 \epsilon + \frac{n}{2} \|G - I\|.$$

- (b) For a gadget channel, if  $(B_g)$  is a gadget family then so is  $(B_g^{\otimes t})$ , with room  $t \log(k'/k)$ , dimension  $k^t$  and flat Gram entries  $\tau_k(B_g^* B_h)^t$ .
- (c) For a gadget channel, let  $(B_g)$  be a gadget family with room  $\delta$  and flat Gram off-diagonals at most  $c < 1$ , and put  $t = \lceil \log(2rn/\epsilon) / \log(1/c) \rceil$ . Then, for  $n \geq 2$ , there is a rank-rate service with error at most  $(e^2 + 1)\epsilon$ ,  $C = 0$ ,  $P \leq nt\delta + 2$  and  $Q \leq t \log k + nt\delta + O_r(\log(n/\epsilon) + \log(2 + t \log k))$ .

**Corollary 7.7** (The rank-form sandwich). *If  $\Phi$  has a flat probe, then for  $n \geq 2$  and  $0 < \epsilon \leq \frac{1}{16}$ ,*

$$cL_\Phi(n, \epsilon) - C \log_2(en) \leq Q^*(n, \epsilon) \leq U_\Phi(n) + K_r(\log(n/\epsilon) + \log(2 + U_\Phi(n))),$$

and  $L_\Phi(n, \epsilon) \leq U_\Phi(n)$ .

The gap between the two sides is a gap between Shannon and rank entropy, and between a biased and a flat bath. When the labels are separated in finite dimension it closes trivially.

**Corollary 7.8** (Labels separated in finite dimension). *Let  $\Phi$  be a gadget channel. If some finite-dimensional unitary representation of  $P_\Phi$  separates the labels, then  $Q^*(n, \epsilon) = O_\Phi(\log(n/\epsilon))$ .*

This covers every gadget channel whose labels are distinct in  $\mathbb{Z}^k$ , in a residually finite group or in a maximally almost periodic group in which the relations hold. Houghton's channel is not of this kind: every finite-dimensional unitary representation of  $H_3$  is trivial on the finitary alternating group, by Jordan's theorem on finite subgroups of  $U(D)$  [55], so it identifies a label that is a three-cycle with the label 1 (Remark 8.1).

### 7.3 The configuration-lamp class

The concrete group in [28, Section 3] is the permutational wreath product

$$\Gamma = S_3 \wr_V A = \left( \bigoplus_{v \in V} S_3 \right) \rtimes A, \quad A = V \rtimes (\text{GL}_{\text{fin}}(Y) H_3), \quad V = \mathbb{F}_2^{(Y)}, \quad Y = \{1, 2, 3\} \times \mathbb{N}.$$

Here  $\text{GL}_{\text{fin}}(Y)$  consists of the linear maps of  $V$  acting as the identity off finitely many coordinates, and  $H_3$  permutes the coordinates through its action on  $Y$ . The affine group  $A$  acts on  $V$  by translations and these linear maps. A lamp  $S_3$  sits at each point of  $V$ , a finite binary *configuration*, and  $A$  permutes the lamps. The group  $\Gamma$  is finitely presented, generated by five elements, and elementary amenable [28, Theorem A]. This is the configuration-lamp group and channel throughout this paper. It differs from the usual binary lamplighter  $C_2 \wr \mathbb{Z}$ , whose finite quotients separate finite label sets (reduce the shift and lamp locations modulo a sufficiently large integer). A canonical gadget from that usual lamplighter lies in the logarithmic upper branch.

For the channels of [28] the two sides already meet. Let  $\Phi$  be a gadget channel that satisfies the hypotheses of [28, Theorem 10.1]; its generators include the transpositions  $\mathbf{a}, \mathbf{b}$  of one lamp  $S_3$ . Suppose that for every  $k \geq 1$  there are  $2^k$  transport words  $u_1, \dots, u_{2^k}$  of length at most  $L_k$  whose *pairwise areas* are at most  $A_k$ : each commutator  $[u_i x u_i^{-1}, u_{i'} y u_{i'}^{-1}]$  with  $i \neq i'$  and  $x, y \in \{\mathbf{a}, \mathbf{b}\}$  is a product of at most  $A_k$  conjugates of relators and their inverses. Suppose that  $A_k + L_k + 1 \leq K k^\kappa$ , as in [28, Corollary 10.2(a)], and that the chunk  $E$  of labels in  $\Gamma$  used by these transport words has sofic profile  $\log \sigma_E(s) \leq C_E s (\log s)^\lambda$ .

**Corollary 7.9** (The configuration-lamp class). *For such a channel there are  $c, C, K, n_0$  with, for  $n \geq n_0$  and  $0 < \epsilon \leq \frac{1}{16}$ ,*

$$L^*(n) \geq \frac{c\sqrt{n}}{(\log n)^\kappa}, \quad U_\Phi(n) \leq C\sqrt{n}(\log n)^\lambda, \quad Q^*(n, \epsilon) \leq K(\log(n/\epsilon))^{\kappa+\lambda+1}(1+L^*(n)).$$

So the two-scale law holds. For the configuration-lamp channel of [28, Section 11],  $L^*(n)$ ,  $Q^*(n, \epsilon)$  and  $U_\Phi(n)$  are all  $n^{1/2+o(1)}$ ; for  $Q^*$  this is [28, Theorem 11.1(b) and Corollary 12.3] at  $\epsilon = \frac{1}{16}$ , and for smaller fixed  $\epsilon$  the upper bound comes from [28, Theorem 12.2], as in the proof of Corollary 12.3 there.

The exponent  $\frac{1}{2}$  of [28, Theorem 11.1(b)] is therefore not a feature of the method: it is the two-scale profile of the channel, which balances a bath of entropy  $S$  against entropy production of order  $1/S$ . Theorems 7.1–7.3, Theorem 7.6 and Corollaries 7.7–7.9 are proved in Appendix J.

This square-root memory statement allows initial purity up to the memory. With a purity budget  $B$ ,  $\log n \leq B \leq c_0 n$ , the same channel needs memory of order  $n/B$  up to logarithmic factors, matched by exact services [28, Theorem 11.1(a)]; at logarithmic purity this is nearly linear.

## 8 Houghton's channel

Houghton's group  $H_3$  [51] is finitely presented [13]. Let  $\Phi_H$  be the gadget channel of Johnson's presentation of  $H_3$  [54],

$$H_3 = \langle \mathbf{a}, \mathbf{b}, \alpha \mid \alpha^2, (\alpha \mathbf{a} \alpha \mathbf{a}^{-1})^3, [\alpha, \mathbf{a}^2 \alpha \mathbf{a}^{-2}], \mathbf{a} \mathbf{b} \mathbf{a}^{-1} \mathbf{b}^{-1} \alpha^{-1}, \mathbf{a} \alpha \mathbf{a}^{-1} \mathbf{b} \alpha^{-1} \mathbf{b}^{-1} \rangle,$$

as in [27, Section 4]:  $d = 104$  and  $r = 26$  labels. Houghton's group has a chunk whose sofic profile [21, Definition 3.4] is not bounded by any polynomial [26, Theorem 1.1 and Corollary 1.2]. Through the far-commutator area bound [26, Theorem 3.2], the one-round theorem of [27] gives memory at least  $c n^{1/(2 \log 107+2)}$  at the rank rate, an exponent of about 0.06459, and devices reach  $n^{1/3}$  up to logarithms [27, Corollary 6.7] (Remark 8.14). This section shows that the exponent is  $\frac{1}{3}$  for devices whose rounds act through the group itself, by two independent lower bounds, and that for general devices both the two-scale law and the exponent  $\frac{1}{3}$  come down to one lower bound on  $L_{\Phi_H}^*$ .

*Remark 8.1* (No finite-dimensional separating representation). Every finite-dimensional unitary representation of  $H_3$  is trivial on the finitary alternating group: for  $n \geq 5$  the simple group  $\text{Alt}(n)$  embeds in  $H_3$ , and a faithful image in  $U(D)$  would be a finite subgroup with no nontrivial abelian normal subgroup, whose order Jordan's theorem bounds in terms of  $D$  [55]. The label  $\gamma = [\alpha \mathbf{a} \alpha \mathbf{a}^{-1}]$  is a three-cycle, so every finite-dimensional representation identifies it with the label 1, and Corollary 7.8 does not apply. By Lemma H.5, no zero-leakage round of  $\Phi_H$  at distance below  $\frac{1}{2}$  is square.

### 8.1 Genuine rounds

**Definition 8.2** (Representation rounds). Let  $\pi$  be a unitary representation of the free group on  $\mathbf{a}, \mathbf{b}, \alpha$  on a Hilbert space  $H$ , and  $U_\pi$  the gadget unitary of  $\Phi_H$  with each symbol  $w$  evaluated as  $\pi(w)$  and the lower right slot of a triangle  $(x, y, z)$  as  $-\pi(x)\pi(y)$ . A *representation round* is a round  $(U, \rho)$  in which  $\rho$  is a finite-rank state on  $H$  with support  $K$ , and  $U$  is a unitary on  $\mathbb{C}^d \otimes V$  that extends  $U_\pi$  on  $\mathbb{C}^d \otimes K$ , for a finite-dimensional  $V \subseteq H$  containing  $K$  and  $\pi(w)K$  for every slot word  $w$ . It is *genuine* if  $\pi$  factors through  $H_3$ .

Every  $H_3$ -set with a finitely supported law gives a genuine round, and the regular action gives exact ones with every bias. For  $\xi = \sqrt{\rho}$  and an operator  $W$  we use the displacement  $E(W) = \|(W - I)\xi\|_F$  and the centrality defect  $\chi(W) = \|W\xi - \xi W\|_F$  of [28, Appendix C.1]. Put  $\gamma = \alpha \mathbf{a} \alpha \mathbf{a}^{-1}$  and, for  $i \geq 0$ , the *copies*  $c_i = \pi(\mathbf{a}^{3i} \gamma \mathbf{a}^{-3i})$  and  $x_i = \pi(\mathbf{a}^{3i} \alpha \mathbf{a}^{-3i})$ , with  $\Delta_i = E(c_i)$ ,  $\theta_i = \chi(c_i)$  and  $\eta_i = \chi(x_i)$ . In  $H_3$  these are a three-cycle and a transposition on

the triple of consecutive points of the line through the first two rays shifted by  $3i$ , so the copies generate commuting copies of  $S_3$ .

**Lemma 8.3** (Exact local sector bound). *Let  $c_i, x_i$  ( $0 \leq i < N$ ) be unitaries on a Hilbert space with  $c_i^3 = x_i^2 = I$ ,  $x_i c_i x_i = c_i^{-1}$ , and  $c_i, x_i$  commuting with  $c_k$  for  $k < i$ . For every state  $\rho$  of finite rank,*

$$S(\rho) \geq \sum_i \left[ \frac{\Delta_i^2}{3} - \frac{\eta_i^2}{2 \ln 2} - h_2\left(\frac{\theta_i^2}{3}\right) - \frac{\theta_i^2}{3} \right].$$

Each copy pays only its own centrality. In the weighted sector bound of [28, Theorem C.6] the copies commute only approximately and the bound carries row sums over all pairs; here they commute exactly and the row sums disappear. Equality holds for the flat state on a tensor product of standard representations of  $S_3$ .

**Lemma 8.4** (Centrality from entropy production). *In every representation round,  $\chi(\pi(w)) \leq 2\sqrt{d a \ln 2}$  for every symbol  $w$ .*

**Theorem 8.5** (Genuine rounds cost  $n^{1/3}$ ). *A genuine round of  $\Phi_H$  at distance at most  $1/8$  has*

$$S(\rho) + 1 \geq c_H a^{-1/2}, \quad S(\rho) + n a + 1 \geq c'_H n^{1/3},$$

for fixed positive constants. More generally, a representation round obeys the same entropy mechanism up to the number  $N_{\text{ex}}$  of copies satisfying the exact local sector relations: with  $\delta_{\text{cent}} = \max\{\chi(\pi(a)), \chi(\pi(\alpha))\}$ ,

$$S(\rho) + 1 \geq c_H \min\{N_{\text{ex}}, \delta_{\text{cent}}^{-1}\}, \quad \delta_{\text{cent}} \leq 2\sqrt{104 \ln 2 a}.$$

The explicit constants are in Theorem N.1.

In particular no genuine round at distance  $\frac{1}{8}$  has  $a = 0$ , and a round with  $S + n a \leq n^{1/3-\delta}$  is not a representation round in which the copies up to distance of order  $a^{-1/2}$  commute exactly.

## 8.2 Genuine devices

A round unitary  $U$  is *genuine* on a memory subspace  $V$  if there are a genuine representation  $\pi$  on  $H$ , an isometry  $J : V \rightarrow H$  and an isometry  $R$  from  $\text{span}\{\pi(w)JV\}$  into the memory with  $U(\xi \otimes v) = (I \otimes R)U_\pi(\xi \otimes Jv)$ ; memory unitaries before and after the round are absorbed in  $J$  and  $R$ . A causal service of  $\Phi_H$  is *genuine with discard*  $\delta$  if every round unitary  $U_t$  is genuine on some  $V_t$  and, for the observer that tests the support in every round (the observer of Theorem 7.1), the memory has mass at most  $(1 - \epsilon)\delta$  outside  $V_t$  at the start of every round  $t$ . Devices that store points of an  $H_3$ -set and apply the gadget to them, with any processing of the bath between rounds (permutations of points and slots, block twirls, re-folds, exports), are genuine on their windows.

**Theorem 8.6** (Genuine services). *For  $n \geq n_0$ , every genuine service of  $\Phi_H$  with error  $\epsilon \leq \frac{1}{16}$ ,  $J \leq \frac{n}{2} \log 26$  and discard at most  $1/n^2$  has  $Q \geq c_H n^{1/3}$  for a fixed  $c_H > 0$ . The zero-discard and small-discard constants are preserved in Theorem N.2.*

**Theorem 8.7** (The Houghton exponent on genuine devices). *Let  $Q_{\text{gen}}^*(n, \epsilon)$  be the least memory of a genuine rank-rate service of  $\Phi_H$  with discard at most  $1/n^2$  and error at most  $\epsilon$ . For  $0 < \epsilon \leq \frac{1}{16}$  and  $n \geq n_0(\epsilon)$ ,*

$$c_H n^{1/3} \leq Q_{\text{gen}}^*(n, \epsilon) \leq C_H n^{1/3} (\log n)^{2/3} (\log(n/\epsilon))^{1/3}.$$

The upper bound is the device of Theorem H.18 for  $q = 3$ , which re-folds the translation part of the regular action and carries finite symmetric groups as fibres. The lower bound reads the group through a sector bound. A second, independent route reads it through its growth.

**Corollary 8.8** (The growth route). *Every genuine round of  $\Phi_H$  given by the regular action of  $H_3$ , with a bath that is diagonal in the basis of group elements, has  $S + na \geq cn^{1/3} - C$ .*

The constants are worse than those of Theorem 8.5, and the corollary covers only classical baths on the regular action; but it uses nothing about  $S_3$  or sector bounds, only that  $H_3$  has exponential growth, which follows from Chou's theorem [17, Theorem 3.2] because  $H_3$  is elementary amenable and not virtually nilpotent (Remark 8.1: it is not residually finite). Both routes place the regular action at  $n^{1/3}$ , and the device of Theorem H.18 attains it.

**Proposition 8.9** (Genuine rounds are streamed). *There is  $K$  such that for  $n \geq 2$  and  $0 < \epsilon \leq \frac{1}{16}$ , every genuine round of  $\Phi_H$  at distance at most  $\frac{1}{8}$  satisfies  $Q^*(n, \epsilon) \leq K \log^2(n/\epsilon)(1 + S + na)$ .*

So genuine rounds stream, and biasing a genuine action cannot separate the two halves of the law.

### 8.3 Smoothed one-round realizations

Appendix O studies an explicit family of smoothed rounds of  $\Phi_H$ , built from the corner links of [27, Section 4.2] with a clock and several tensor copies of a fermionic register. Their static cost is of order  $n^{1/3}$  for symmetric sources, for a moderate number of copies and for sources near the stationary algebra, while for unrestricted sources with positive production the proved lower bound is only  $n^{1/4}$  (Theorems Q.3 and Q.4, and Table 3 in Appendix O).

### 8.4 General devices

**Proposition 8.10** (The two-scale ceiling; compare [27, Proposition 4.14]). *For every  $\epsilon \in (0, \frac{1}{16}]$  there is  $C(\epsilon)$  with  $L_{\Phi_H}(n, \epsilon) \leq C(\epsilon)n^{1/3}$ ; in particular  $L_{\Phi_H}^*(n) \leq Cn^{1/3}$ .*

So no lower bound obtained from the two-scale profile through Theorem 7.3 can exceed  $n^{1/3}$ , the exponent of the genuine devices.

*Remark 8.11* (The single budget). Under the single budget  $\ell + a \leq t_n$  of [27, Theorem 5.2] with the purity budget  $P + C < 3Q + 6$ , the same smoothed rounds cap the one-round method at  $O(n^{1/4})$ : the leakage is then read at the purity scale  $(Q + \log n)/n$  instead of the error scale  $1/n$ . The two-scale profile removes this cap.

**Theorem 8.12** (What the law and the exponent reduce to). *(a) If  $L_{\Phi_H}^*(n) \geq n^{1/3}/p(n)$  for a polylogarithm  $p$ , then the two-scale law holds for  $\Phi_H$ , and  $Q_{\Phi_H}^*(n, \epsilon) = n^{1/3} \cdot \text{polylog}(n/\epsilon)$  at every fixed  $\epsilon \leq \frac{1}{16}$ .*

*(b) Conversely, if the two-scale law holds for  $\Phi_H$  and  $Q_{\Phi_H}^*(n, \epsilon) = n^{1/3+o(1)}$ , then  $L_{\Phi_H}^*(n) \geq n^{1/3-o(1)}$ . So if  $L_{\Phi_H}^*(n) \leq n^{1/3-\delta}$  for some  $\delta > 0$  and infinitely many  $n$ , the law or the exponent  $\frac{1}{3}$  fails for  $\Phi_H$ .*

*(c) Rounds of  $\Phi_H$  at distance at most  $\frac{1}{8}$  with  $S + na < c_H n^{1/3} - C_H$  are not genuine, for fixed positive constants.*

**Conjecture 8.13** (The Houghton exponent). *At every fixed error  $\epsilon \leq \frac{1}{16}$ ,  $Q_{\Phi_H}^*(n, \epsilon) = n^{1/3+o(1)}$ .*

By Theorem 8.12, the conjecture and the law for  $\Phi_H$  follow together from the single lower bound  $L_{\Phi_H}^*(n) \geq n^{1/3}/\text{polylog}(n)$ , which says that the ceiling of Proposition 8.10 is attained. A round below this scale cannot be genuine. If it is a permutation round, it cannot come from the regular action (Corollary 8.8). These exclusions do not identify a collision mechanism or rule out other finite approximate realizations.

*Remark 8.14* (General devices). For general devices, the one-round theorem [27, Theorem 5.5] with Theorem 7.1 and Proposition 7.2 gives  $Q_{\Phi_H}^*(n, \epsilon) \geq c n^{\beta_H/(1+\beta_H)}$  for  $n \geq n_0$ , with  $\beta_H = 1/(2 \log 107 + 1)$ , so  $\beta_H/(1 + \beta_H) = 1/(2 \log 107 + 2) \approx 0.06459$ ; the device of Theorem 8.7 gives  $Q_{\Phi_H}^*(n, \epsilon) \leq C n^{1/3} (\log n)^{2/3} (\log(n/\epsilon))^{1/3}$  for  $n \geq n_0(\epsilon)$ . This improves the dependence on  $\epsilon$  in the bound  $O((n/\epsilon)^{1/3} \log^{4/3}(2n/\epsilon))$  for general devices [27, Corollary 6.7]. The one-round method cannot close this gap: it certifies at most  $n^{1/6}$ , whatever area bound is used (Proposition I.4). The proofs of this section are in Appendix N.

## 9 What remains open

The high-precision law leaves a quantitative fixed-error question. The following four problems separate it from the sharper Houghton rate, entropy-rate service and autonomous local realization.

**Problem 9.1** (The concentration conjecture). For every canonical untwisted anchored gadget  $\Phi$ , do constants  $C_\Phi < \infty$  and  $p < \infty$  exist such that, for all  $n \geq 1$ ,

$$Z_\Phi(n) \leq C_\Phi \log_2^p(en)(1 + L_\Phi^*(n))?$$

A positive answer would complete the universal memory comparison at all accuracies (Section 6).

**Matrix form.** By Proposition 6.2,  $Z_\Phi(n)$  minimizes  $S(\rho) + na$  over exact solutions: a state  $\rho$  and a positive block matrix  $Y = [Y_{gh}]$  with  $Y_{gg} = I$ ,  $Y_{1[y]} = Y_{[x][z]}$  for every triangle  $(x, y, z)$ , and  $|\text{Tr } \rho Y_{gh}| \leq 1/2$  for  $g \neq h$ . The profile  $L_\Phi^*(n)$  minimizes the same cost over near-solutions: rounds that may leave the Kraus span, with probability at most  $2/(15n)$  when the bath is in state  $\rho$  and the input is maximally mixed. The defect is thus weighted by the source  $\rho$ , not by the normalized trace. The conjecture asks whether every cheap  $\rho$ -weighted near-solution can be traded for an exact solution, of any size and with any source, at a polylogarithmic increase in cost. The exact solution need not be close to the near-solution.

**Relation to group stability.** Stability asks whether almost-homomorphisms of a group are close to homomorphisms. In the Frobenius norm, cohomology vanishing gives stability and yields finitely presented groups that are not approximated by unitary matrices [23]. In the normalized Hilbert–Schmidt metric, infinite hyperlinear groups with property (T) are not stable, and Becker and Lubotzky proposed a flexible notion that allows the space to be enlarged before correcting [6, Section 4.4]. The conjecture is flexible in the same way. It differs in two respects: the defect is weighted by the source state  $\rho$  rather than by the normalized trace, and the price is the entropy cost  $S + na$  rather than a distance or a dimension. No reduction between the two questions is known.

**Where the conjecture is open.** The conjecture holds in regimes (i) and (iii) of Theorem 5.1. In regime (i), an exact finite tracial factorization is an exact solution with zero production, so  $Z_\Phi$  is bounded. In regime (iii),  $L_\Phi^*(n)$  is linear, while  $Z_\Phi(n) \leq n \log r$ . In regime (ii) the proved repair gives  $Z(n) \leq C[1 + L^*(n) + \sqrt{n} \log n]$ , together with the sharper

horizon envelope, so matching holds above the square-root scale up to logarithms. The open range is regime (ii) below  $\sqrt{n}$ . Concentration on an exact kernel and weighted affine trace defect  $O(1/n)$  are sufficient mechanisms; mean support leakage  $O(1/n)$  alone implies neither. Exact rectangular frames can satisfy the local gadget identities without extending to global group representations, so representation repair addresses a stronger question.

The first concrete target is Houghton's channel  $\Phi_H$ . It lies in regime (ii), and  $L_{\Phi_H}^*(n) \leq Cn^{1/3}$  (Proposition 8.10), below the scale where the proved repair gives matching bounds.

**A permutation subcase.** For permutation rounds, triangle-bad points have mass  $O(1/n)$  by Lemma 2.3. One may ask for an exact replacement whose entropy and production remain within logarithmic factors of the original cost. Collisions and the entropy cost of the good-point restriction still need control. Houghton's  $H_3$  is not P-stable: it is amenable, hence sofic, and not residually finite, whereas a sofic P-stable group must be residually finite [36], [3, Theorem 7.2(ii)], [7, discussion after Proposition 8.5]. So no general nearby repair by permutation representations exists for  $H_3$ . This leaves open whether the optimized costs separate, and whether unrestricted quantum devices outperform permutation ones.

**The flat subcase.** A useful special case asks whether every coarse-correct flat round with leakage  $O(1/n)$  admits an exact family of cost  $\text{polylog}(n)(1 + S + na)$ , allowing a larger bath and fixed coarse-error slack. Rounds with exactly zero production, of any source spectrum, reduce to a finite tracial factorization at entropy-scale dimension with a multiplicative leakage loss (Appendix F). Positive production remains outside that reduction. The second-order obstruction of Appendix G concerns twisted identities, while this conjecture concerns untwisted optimized costs.

**Problem 9.2** (All-family Houghton sharpness). Is  $L_{\Phi_H}^*(n) \geq n^{1/3}/\text{polylog}(n)$ ? More weakly, is  $Q_{\Phi_H}^*(n, \epsilon) \geq n^{1/3-o(1)}$  at fixed  $\epsilon \leq 1/16$ ?

A positive answer would show that the cube-root constructions remain optimal beyond genuine group devices. Table 3 in Appendix O separates the operational theorem from the static bounds for explicit smoothed rounds. Those bounds reach the cube-root scale for a moderate number of tensor copies and for stationary sources. For general sources with positive production they lie between the quarter- and cube-root scales. Small production has not been shown to force a source near the stationary algebra, and a uniform spectral estimate for the killed Laplacian of Appendix Q is open. Other rounds, and mixtures over clock sizes with positive production, remain beyond these bounds. The one-round operational method has its stated  $n^{1/6}$  ceiling; a lower bound for  $Z$  would need a separate all-family argument.

**Problem 9.3** (Low-purity and nonflat-probe entropy-rate service). For  $\Phi \in \overline{\mathcal{F}}$  and  $\log(1/\epsilon_n) = o(n)$ , can every flat-probe target be served with maximally mixed imports,  $J \leq \lfloor (n-1) \log r/2 \rfloor$ ,  $Q = o(n)$  and  $P = O(\log(n/\epsilon_n))$ ? For arbitrary nonflat probes, does sublinear service with maximally mixed imports hold at exchange rate  $H(\Phi)/2$ , where  $H(\Phi)$  is the maximum entropy of a complementary output?

These questions test whether sublinear memory can coexist with small initial purity, and whether rank can be replaced by complementary entropy in the general compiler. The principal upper constructions charge  $P \leq Q$ ; the nonflat rank-rate compiler does not give a half-complementary-entropy import threshold. With unrestricted imported purity the any-exchange converse controls  $Q + C$ : a pure replacer can recycle one retained qubit using one fresh pure qubit after each round.

**Spatial and spectral refinements.** For devices that move points along a Schreier graph through nested sets with balanced boundaries (the collar devices of Section H), and for the configuration-lamp devices, the sharp memory rates are open, independently of the general compiler. The rank-based profile  $U_\Phi$  and the entropy-based profiles may also differ: source entropy charges the occupied spectrum, while  $U_\Phi$  charges the logarithm of the ambient rank. How far apart they can be is open.

By an *exact stationary compression* we mean the following. Start a finite-dimensional probe coupled to a spin reservoir in its tracial product state, evolve by a fixed finite-range interaction, and trace out the reservoir. The reduced channels  $\Psi_t$  converge as  $t \rightarrow \infty$  to  $\Phi$  on every probe input, including a reference (equivalently, in diamond norm at fixed probe dimension). The target is this exact limit, rather than one selected finite-time approximation. All controllers and their initial states must belong to the specified dynamics.

**Problem 9.4** (An expensive autonomous local stationary limit). Is a channel with a superpolylogarithmic finite-mixture profile the exact stationary compression of finite-range quantum spin dynamics in a tracial product reservoir?

Such a construction would connect finite-memory approximation cost to an autonomous local reservoir. Known local-dynamics arguments exclude only narrow classes, so autonomous local realization remains open. This construction question is independent of the concentration conjecture.

**Repeatability without resets.** Proposition 4.6, with the catalytic dilation of Lie, Son, Boes, Ng and Wilming, answers Rybár and Ziman’s question [72]: some unital channels that are not random-unitary are repeatable, and some unital channels are not. What remains is to identify the repeatable channels. Catalytic channels are repeatable, and repeatable channels are finite mixtures of finite tracial factorizations. Lie, Son, Boes, Ng and Wilming show that the catalytic channels form a strict subset of these mixtures [60, Appendix C], so at least one of the two inclusions is strict. We do not know which. A repeatable device must reproduce the channel at every use, but its memory state may move; a catalytic device keeps it fixed. Haagerup, Musat and Ruskai’s channels  $\Phi_{\alpha,\beta}$  on  $M_3$  are extreme among unital channels for  $|\alpha|^2 \notin \{0, \frac{1}{2}, 1\}$ , are not unitary, and have finite tracial factorizations with one bath qubit [45]. They are not catalytic [60, Appendix C]. If one of them is repeatable, the catalytic channels are a strict subset of the repeatable ones. If one of them is not, the repeatable channels are a strict subset of the finite mixtures.

**AI assistance.** Generative AI was used in mathematical exploration, proof checking, code development and writing. The authors are responsible for the content.

## A Antisymmetric rigidity and purity accounting

### A.1 A dimension-free local estimate

**Lemma A.1** (Antisymmetric support decouples the source purifier). *Let a finite unitary round have source  $\beta$ , channel  $\Psi$ , and  $\delta = d_J(\Psi, \Theta)$ . Purify the source by  $F$ , feed a Bell pair on  $RA$ , and denote the final RYF marginal by  $\chi$ . Then*

$$\|\chi - \mathcal{J}(\Theta) \otimes \beta_F\|_1 \leq 20\sqrt{\delta}, \quad a \geq H - g_9(\min\{1, 10\sqrt{\delta}\}).$$

*Proof.* Choose the antisymmetric isometry  $W : \mathbb{C}^3 \rightarrow RY$ , sending the three coordinate vectors to the normalized alternating pairs, with signs chosen so its partial-trace channel is  $\Theta$ . Put  $P = WW^*$ . Since  $\mathcal{J}(\Theta) = P/3$ , support loss  $\text{Tr}((I - P)\chi) \leq \delta$ . Gentle projection gives  $\|\chi - P\chi P\|_1 \leq 2\sqrt{\delta}$ . The untouched marginal is  $\chi_{RF} = \tau_3 \otimes \beta_F$ . Thus, for the subnormalized  $\zeta = W^*P\chi PW$ ,

$$\|(\Theta \otimes \text{id}_F)(\zeta) - \tau_3 \otimes \beta_F\|_1 \leq 2\sqrt{\delta}.$$

The inverse linear map is  $\Theta^{-1}(Y) = \text{Tr}(Y)I - 2Y^T$ . The first summand has stabilized trace norm 3, and transposition on  $M_3$  has stabilized trace norm 3, so the inverse has norm at most 9. This also acts on subnormalized and nonpositive differences. Since  $\Theta(\tau_3) = \tau_3$ , it follows that  $\|\zeta - \tau_3 \otimes \beta_F\|_1 \leq 18\sqrt{\delta}$ . Re-embedding by  $W$  and adding the gentle error [85, Lemma 9] proves the first claim.

The global state on  $RYB'F$  is pure. Therefore the actual global bath gain is  $S(B') - S(\beta) = H(RY|F)_\chi$ . Conditional-entropy continuity on the fixed dimension  $\dim(RY) = 9$  gives the second claim [86, Lemma 2]; its comparison state has conditional entropy  $\log_2 3$ . No bound on  $\dim F$  is used.  $\square$

### A.2 Exports, imports and exact banks

*Proof of Theorem 3.1.* Send fresh Bell inputs and retain every output and reference. Each round's marginal Choi error is at most  $\epsilon$ , even when the memory is correlated with past outputs. For an intra-round exchange schedule, place every prescribed independent import at the mathematical beginning and retain every export to the mathematical end. Extend each operation by identities on unused imports and retired exports. This gives one finite unitary round with the same reduced channel. It is bookkeeping, not a smaller-memory implementation.

Let active dimensions before and after be  $b, b'$ , the product import dimension be  $a_0$ , and the product export dimension be  $g$ . Then  $ba_0 = b'g$ . If active marginals are  $\eta, \eta'$ , and the independent import state is  $\sigma$ , subadditivity bounds the round's global gain by

$$\begin{aligned} a_{\text{glob}} &\leq S(\eta') + \log_2 g - S(\eta) - S(\sigma) \\ &= F(\eta) + c - F(\eta'), \quad F(\eta) = \log_2 b - S(\eta), \quad c = \log_2 a_0 - S(\sigma). \end{aligned}$$

This uses only unconditional marginals and remains valid with past correlations. Summing telescopes to  $\sum_t a_{\text{glob},t} \leq P + C$ , because the final deficit is nonnegative. Each gain is nonnegative for the reconstructed round with maximally mixed system input. Lemma A.1 proves the lower, including its positive part. Correlated imports would require their joint deficit to be charged; adaptive state selection during a round is not included.

For the exact upper, let  $b = \lceil \sqrt{n} \rceil$  and  $q = \lceil bH \rceil$ . Encode  $b$  pure qutrit Stinespring cells in  $q$  qubits. Apply the cell isometry at its round and never touch that cell again until the bank is exported. For every full bank except the last, export all  $q$  qubits and import a pure bank.

Unused cells in the final bank stay pure; no accuracy error occurs. A reversible schedule clock costs  $\lceil \log_2(n+1) \rceil$  pure qubits, and a fixed wire encoding costs  $O(1)$ . With  $B = \lceil n/b \rceil$ ,

$$Q = q + O(\log n), \quad P + C = Bq + O(\log n), \quad J = (B - 1)q.$$

Since  $Bb = n + O(b)$  and  $B = O(\sqrt{n})$ , these are the claimed ledgers. Unitary extensions preserve the encoded bank and its unused padding. They realize independent channels for every adaptive observer.  $\square$

*Proof of Proposition 3.2.* Werner twirling fixes  $\Theta$ , contracts Choi trace distance and preserves factorizability: each conjugation preserves it, and convex mixtures and limits preserve the relevant closure. The twirled channel is  $T_\lambda = \lambda W_3^+ + (1 - \lambda)W_3^-$ . Its two normalized Choi states occupy orthogonal symmetric and antisymmetric subspaces, so  $d_J(T_\lambda, \Theta) = \lambda$ . Haagerup–Musat’s twirl calculation gives  $\lambda \geq 2/27$  for every factorizable twirl, and an exact factorization at equality through  $M_3 \otimes M_3$  [44, Lemmas 5.4 and 5.5, Theorem 5.6 and Corollary 5.7]. The threshold persists under limits. This proves both bounds for the closure  $\overline{\mathcal{F}}$ . The published full cb distance is  $4/27$  (half-diamond distance  $2/27$ ); no generic conversion of that norm into Choi distance is needed.  $\square$

### A.3 A lower bound for delayed batches

Suppose contiguous batches contain  $b_n = o(n)$  qutrit inputs and release their outputs together, with complete error at most  $n^{-2}$ . Within and between batches allow the prescribed independent exchange just accounted for. Define

$$K_n = \left\lfloor \frac{\log_2 n - 3 \log_2 \log_2(n+2)}{\log_2 9} \right\rfloor.$$

The following assertion concerns sufficiently large  $n$ , so  $K_n \geq 1$ .

**Proposition A.2** (Batch probes). *Every such service satisfies*

$$\frac{P + C}{n} \geq (H - o(1)) \min\{1, K_n/b_n\}.$$

*In particular  $b_n = o(\log n)$  forces asymptotic supplied-purity rate  $H$ , and the exact banks above attain it with  $Q_{\text{peak}} = o(n)$ ,  $J = O(n)$ . For arbitrary  $b_n$ , if  $R[b]$  is the infimum achievable  $\limsup(P + C)/n$  under these resource constraints and  $x = \limsup_{n \rightarrow \infty} \log_2 n/b_n \in [0, \infty]$ , then*

$$R[b] \geq H \min\{1, x/\log_2 9\}.$$

*Proof.* In a full batch probe  $m = \min(b_n, K_n)$  inputs with Bell pairs and set the others to  $\tau_3$ . Regard the unused inputs as an analytical bath, and retain their outputs at the mathematical end. By subadditivity and their maximum entropy, its gain is at most the actual bath gain:

$$S(B'Y_0) - S(B) - (b_n - m)H \leq S(B') - S(B).$$

Tensor the alternating isometry  $m$  times. Support loss is at most  $n^{-2}$ , and the stabilized inverse bound is  $9^m$ . The proof of Lemma A.1 now gives half trace distance at most  $t_n = (1 + 9^m)/n$  from the decoupled target state. Consequently the physical bath gain is at least  $mH - g_{9^m}(t_n)$ . By the definition of  $K_n$ ,  $t_n \leq n^{-1} + (\log_2(n+2))^{-3}$ . Since  $\log_2(9^m) = m \log_2 9$ ,  $g_{9^m}(t_n) = o(m)$  uniformly for  $1 \leq m \leq K_n$ . The leftover batch has fewer than  $b_n = o(n)$  rounds and nonnegative global gain, so it can be dropped from the lower sum.

There are  $\lfloor n/b_n \rfloor$  full batches. The deficit ledger therefore gives the displayed bound. Write  $z_n = \log_2 n/b_n$ . Since  $K_n/\log_2 n \rightarrow 1/\log_2 9$ , the bounded factors obey

$$\limsup_{n \rightarrow \infty} \min\{1, K_n/b_n\} = \min\{1, (\limsup_{n \rightarrow \infty} z_n)/\log_2 9\}.$$

For finite  $\limsup z_n$  this follows by continuity and a subsequence attaining the limsup; for infinite limsup the same subsequence makes the capped expression tend to one. The uniform  $o(1)$  and the discarded fraction  $b_n/n = o(1)$  vanish before taking the limsup. The lower holds for every admissible service sequence and hence for their infimum  $R[b]$ . This proves the stated bound.  $\square$

#### A.4 A counted collective Gaussian implementation

The following finite matrix argument controls both the isometry and the implemented channel. A small input Gram alone would not control the channel's Kraus coefficient Gram.

**Lemma A.3** (Rectangular Gaussian packing, extending [25, Theorem 1]). *Let  $\Psi$  be a unital trace-preserving channel on  $M_D$  with  $r$  Kraus operators. For  $0 < \delta \leq 1/2$  it has a finite unitary implementation of half-diamond error at most  $\delta$  with a flat rank- $k$  source in a dyadic  $m$ -dimensional bath, where*

$$\begin{aligned} \log_2 k &= \lceil \tfrac{1}{2} \log_2 r \rceil, \\ \log_2 m &\leq \tfrac{1}{2} \log_2 r + 2 \log_2(1/\delta) + 2 \log_2(1 + \ln(8Dk)) + O(1). \end{aligned}$$

*Its supplied purity is  $\log_2(m/k)$ ; no random seed is supplied.*

*Proof.* Write  $\Psi(X) = \sum_{\alpha} A_{\alpha} X A_{\alpha}^*$  and let  $G_{\alpha}$  be independent  $m \times k$  complex Gaussian matrices with entry variance  $1/m$ . Put  $W = \sum_{\alpha} A_{\alpha} \otimes G_{\alpha}$ ,  $Z = W^* W - I$ . We first justify the dimension-uniform moment estimate

$$\|\mathbb{E} Z^{2p}\| \leq (2p)!(k/m)^p, \quad k \leq m.$$

Wick expansion of a  $q$ th power and cancellation of fixed points under centering give

$$\mathbb{E} Z^q = \sum_{\substack{\pi \in S_q \\ \pi \text{ has no fixed point}}} m^{c(\pi)-q} k^{c(\gamma\pi)-1} T_{\pi} \otimes I_k, \quad \gamma = (1 \ 2 \ \cdots \ q).$$

Here  $c$  counts cycles. Row-index loops supply the  $m$  powers, column-index loops the  $k$  powers; the one external column pair accounts for the exponent minus one. Removing a fixed point contracts  $\sum_{\alpha} A_{\alpha}^* A_{\alpha} = I$  and leaves the other loop factors unchanged, exactly matching the subtraction of  $I$ . The contracted system word is  $T_{\pi} = \sum_{\alpha_1, \dots, \alpha_q} A_{\alpha_1}^* A_{\alpha_{\pi(1)}} \cdots A_{\alpha_q}^* A_{\alpha_{\pi(q)}}$ .

Each  $T_{\pi}$  is a contraction, including crossing pairings. Read its word from right to left. At a label's first occurrence create that label with either isometry  $v \mapsto \sum_{\alpha} A_{\alpha} v \otimes |\alpha\rangle$  or  $v \mapsto \sum_{\alpha} A_{\alpha}^* v \otimes |\alpha\rangle$ . Both are isometries, respectively by trace preservation and unitality. At its second occurrence erase it by the adjoint of its creation map. Permuting the open label registers is unitary. Each label appears once starred and once unstarred, so this construction gives exactly the contracted word and bounds its norm by one. For a fixed-point-free permutation  $c(\pi) \leq q/2$ , and  $c(\pi) + c(\gamma\pi) \leq q + 1$ . The latter follows from the triangle inequality for transposition length  $|\sigma| = q - c(\sigma)$ . With  $a = q - c(\pi) \geq q/2$  the coefficient is at most  $(k/m)^a \leq (k/m)^{q/2}$ . There are at most  $q!$  terms, proving the moment bound.

Choose  $k = 2^{\lceil (\log_2 r)/2 \rceil}$ ,  $p = \lceil \ln(8Dk) \rceil$ ,  $\eta = \delta/2$ , and take  $m$  to be the least power of two at least  $1024kp^2/\eta^2$ . Since  $Z$  is Hermitian, trace Markov gives

$$\Pr\{\|Z\| > \eta\} \leq Dk \left( \frac{4p^2 k}{m\eta^2} \right)^p < 1/4.$$

Also  $C_{\alpha\beta} = k^{-1} \text{Tr}(G_\alpha^* G_\beta)$  is the covariance matrix of  $mk$  independent standard Gaussian samples divided by  $mk$ . For a fixed unit vector its quadratic form is the mean of  $mk$  exponential variables; exponential Markov bounds the two tails. A  $1/4$ -net of the complex unit sphere, of size at most  $9^{2r}$ , gives

$$\Pr\{\|C - I\| > \eta\} \leq 2 \cdot 9^{2r} \exp(-mk\eta^2/32) < 1/4.$$

The chosen parameters suffice since  $k^2 \geq r$ . Hence one realization satisfies both bounds. Fix it as hardware.

Before normalization, tracing  $W(X \otimes I_k/k)W^*$  implements the Kraus coefficient matrix  $C^T$ . Its distance from  $I$  is the same as that of  $C$ . The coefficient perturbation has diamond norm at most  $\eta$ : use the Stinespring isometry of  $\Psi$  and trace duality with  $C - I$ . Polar normalization  $V = W(W^*W)^{-1/2}$  gives an isometry with  $\|V - W\| \leq \eta$ ,  $\|W\| \leq \sqrt{1 + \eta}$ . Left-right multiplication bounds its additional diamond error by  $3\eta$ . The total half-diamond error is at most  $2\eta = \delta$ . Complete  $V : \mathbb{C}^D \otimes \mathbb{C}^k \rightarrow \mathbb{C}^D \otimes \mathbb{C}^m$  to a unitary on the bath  $\mathbb{C}^m$ , initialized uniformly on the embedded  $\mathbb{C}^k$ . This source has entropy  $\log_2 k$ , giving the stated ledger. All probabilities prove finite existence; neither normalization nor Gaussian randomness is an online operation or postselection.  $\square$

For  $\Theta^{\otimes b}$ ,  $D = r = 3^b$  exactly. Thus the lemma yields

$$\begin{aligned} Q_{\text{bath},b} &\leq bH/2 + 2\log_2(1/\delta) + 2\log_2(b+1) + O(1), \\ P_b &\leq 2\log_2(1/\delta) + 2\log_2(b+1) + O(1). \end{aligned} \tag{3}$$

The  $b$  accepted system registers are held until the batch output; they use at most  $2b$  encoded qubits and are counted in  $Q_{\text{peak}}$ . They are observer payload, rather than part of the fixed auxiliary source whose entropy deficit is  $P_b$ . They leave as the system output, so their release does not add to  $J$ . With  $b = n$ , the displayed bath ledger and this custody count prove Theorem 3.3.

*Proof of Theorem 3.4.* The lower is Proposition A.2; exact banks give  $R[b] \leq H$  for every allowed batch sequence. For  $b_n \geq \log_2 n$ , put  $B = \lceil n/b_n \rceil$  and  $\delta = n^{-2}/B$ . Renew the whole collective bath independently between batches. The adaptive hybrid costs at most  $B\delta = n^{-2}$ , since there are no retained bank correlations after its permanent export. From (3) and  $Bb_n \leq n + b_n$ ,

$$\begin{aligned} P_b &\leq 4\log_2 n + 2\log_2 B + 2\log_2(b_n + 1) + O(1) \leq 6\log_2 n + O(1), \\ Q_{\text{bath},b} &\leq b_n H/2 + O(\log n). \end{aligned}$$

Add a pure reversible clock of  $O(\log n)$  qubits, charged once. A shorter final batch uses its own smaller  $k, m$  and flat inactive padding in the common bank; that padding costs zero purity, not an uncharged pure register. The payload bank can accept inputs in sequence and be transformed in place by the collective unitary. Hence  $Q_{\text{peak}} \leq (2 + H/2)b_n + O(\log n) = o(n)$  and  $J = O(n)$ . The supplied rate is at most  $6\log_2 n/b_n + o(1)$  in this range. For smaller  $b_n$  use exact banks; the pointwise minimum with exact banks is  $\min\{H, 6z_n\} + o(1)$  (when  $z_n > 1$ , exact banks already achieve the minimum). The same limsup identity gives  $R[b] \leq \min\{H, 6x\}$ , including  $x = \infty$ . If  $b_n/\log n \rightarrow \infty$ , renewing the  $O(\log n)$  overhead  $O(n/b_n)$  times costs  $o(n)$  exchange, and the leftover batch changes it by  $O(b_n + \log n) = o(n)$ . Hence  $J = nH/2 + o(n)$ . This last near-minimal exchange assertion is not made at critical logarithmic batching. The lower is positive exactly when  $x > 0$ . For the nonnegative sequence  $z_n$ ,  $x = 0$  is equivalent to  $b_n/\log_2 n \rightarrow \infty$ , which proves the zero-rate criterion. If  $b_n = o(\log n)$  then  $x = \infty$ , so the corrected lower is  $H$  and matches exact banks.  $\square$

*Proof of Theorem 3.5.* Write  $z_n = \log_2 n / b_n$ , so  $x = \limsup_n z_n$ .

*Exact banks.* Fix a Stinespring isometry of  $\Psi$  with a pure environment of dimension  $r$ . The banks in the proof of Theorem 3.1, with  $r$ -dimensional cells and  $q = \lceil \sqrt{n} \log_2 r \rceil$ , give exact services with  $Q = O(\sqrt{n})$ ,  $P + C = n \log_2 r + O(\sqrt{n})$  and  $J = O(n)$ . Holding each output until its batch ends keeps the error zero and adds at most  $b_n \lceil \log_2 d \rceil$  payload qubits to  $Q_{\text{peak}}$ .

*Collective batches.* Let  $n \geq 2$ ,  $B = \lceil n/b_n \rceil$  and  $\delta = n^{-2}/B$ . A batch of  $b' \leq b_n$  inputs uses Lemma A.3 for  $\Psi^{\otimes b'}$ , which is unital with  $D = d^{b'}$  and Choi rank  $r^{b'} \leq d^{2b'}$ . Thus  $k \leq 2d^{b'}$  and  $1 + \ln(8Dk) \leq 4(b_n + 1)(1 + \ln d)$ . Since  $B(b_n + 1) \leq 3n + 1$ , the source purity obeys

$$\log_2(m/k) \leq 4 \log_2 n + 2 \log_2(B(b_n + 1)) + 2 \log_2(1 + \ln d) + O(1) \leq s_n,$$

where  $s_n = 6 \log_2 n + 2 \log_2(1 + \ln d) + c$  and  $c$  is an absolute constant. Also  $\log_2 m \leq \frac{b_n}{2} \log_2 r + 1 + s_n$  for  $b' = b_n$ .

The memory holds a payload bank of  $b_n \lceil \log_2 d \rceil$  qubits, a bath bank of  $\log_2 m$  qubits sized for  $b' = b_n$ , and a pure reversible clock of  $\lceil \log_2(n + 1) \rceil$  qubits. The inputs of a batch enter the payload bank as they arrive. After the last one, the collective unitary acts on the payload and bath banks in place, and the outputs are released. Between batches the bath bank is exported, and a fresh one is imported in the lemma's source state:  $\log_2 k$  maximally mixed qubits and  $\log_2(m/k)$  pure ones. A shorter final batch uses its own smaller  $k$  and  $m$ . Its unused bath qubits are imported maximally mixed and are never touched, so they cost no purity.

No output is released inside a batch. So the inputs of a batch form one joint input, entangled with the observer's reference and with all earlier outputs. The batch applies a channel within half-diamond distance  $\delta$  of  $\Psi^{\otimes b'}$ . Its bath is then exported for good, the next batch starts from a fresh independent import, and the clock is pure and never meets an input. Replace the batch channels by  $\Psi^{\otimes b'}$  one at a time. At each step the state entering the batch is unchanged and everything after it is a fixed channel, so the observer's final state moves by at most  $\delta$ . The complete adaptive error is at most  $B\delta = n^{-2}$ .

The first bath and the clock give  $P \leq s_n + \log_2 n + 2$ , and the  $B - 1$  imports give  $C \leq (B - 1)s_n$ . With  $B - 1 \leq n/b_n$ ,

$$\begin{aligned} P + C &\leq B s_n + \log_2 n + 2, & J &= (B - 1) \log_2 m \leq \frac{n}{2} \log_2 r + \frac{(1 + s_n)n}{b_n}, \\ Q_{\text{peak}} &\leq (\lceil \log_2 d \rceil + \frac{1}{2} \log_2 r) b_n + s_n + \log_2 n + 3. \end{aligned}$$

Hence  $(P + C)/n \leq 6z_n + (s_n - 6 \log_2 n)/b_n + (s_n + \log_2 n + 2)/n$ .

*Choice.* For each  $n$ , use collective batches if  $6z_n < \log_2 r$ , and exact banks otherwise. In the first case  $b_n > 6 \log_2 n / \log_2 r$ . The two error terms are then  $O_d(1/\log n)$ ,  $J \leq \frac{3}{2} n \log_2 r + o(n)$ , and  $Q_{\text{peak}} = O(b_n + \log n) = o(n)$ . In both cases the contract holds and

$$\frac{P + C}{n} \leq \min\{\log_2 r, 6z_n\} + o(1).$$

The map  $t \mapsto \min\{\log_2 r, 6t\}$  is continuous and nondecreasing on  $[0, \infty]$ , so the limsup of the right side is  $\min\{\log_2 r, 6x\}$ . This proves the bound, including  $x = \infty$ .

If  $r = 1$ , exact banks have  $J = 0$ . If  $r \geq 2$  and  $b_n / \log n \rightarrow \infty$ , then  $z_n \rightarrow 0$  and collective batches are used for all large  $n$ . Since  $\log_2 m \geq \log_2 k \geq \frac{b_n}{2} \log_2 r$  and  $B - 1 \geq n/b_n - 1$ , also  $J \geq \frac{n}{2} \log_2 r - \frac{b_n}{2} \log_2 r$ . Hence  $J = \frac{n}{2} \log_2 r + o(n)$ .  $\square$

## B Proofs of the closure law

This appendix proves the finite-approximation upper and the local tracialization converse stated in Section 4. All bath, workspace and exchange counts refer to the service contract of Section 2.

*Proof of Lemma 4.2.* Carathéodory's theorem reduces the mixture to  $L \leq d^4 + 1$  terms with weights  $p_i$  and bath dimensions  $m_i \leq m$ . Take  $\lfloor Np_i/m_i \rfloor$  copies of the  $i$ th bath block and fill the remaining dimension by identity blocks; this is the block-diagonal construction in the proof of [44, Theorem 3.6]. The realized weights  $q_i = m_i \lfloor Np_i/m_i \rfloor / N$  obey  $q_i \leq p_i$  and  $1 - \sum_i q_i \leq Lm/N \leq \gamma$ . Both the removed mixture and the filler are channels, giving the stated half-diamond bound.  $\square$

*Proof of Lemma 4.3.* Choose Hilbert–Schmidt orthogonal minimal Kraus operators  $K_i$ , with  $\text{Tr } K_i^* K_j / d = \mu_i \delta_{ij}$  and  $\mu_i > 0$ . The flat-probe identity is unchanged by this unitary Kraus change. The unital completely positive map

$$\mathcal{A}(Y) = \sum_{i,j} K_i^* K_j \otimes Y_{ij}$$

has a fixed Hermiticity-preserving linear right inverse  $\mathcal{R}$  on its range. Its amplified trace norm is bounded independently of  $k$ : writing  $\mathcal{R}(H) = \sum_j \text{Tr}(F_j H) R_j$  in fixed dual bases, trace duality bounds each amplified coefficient extraction by  $\|F_j\| \|H\|_1$ , and hence the sum by a fixed constant.

Project the approximant unitary as  $U = G + L$ ,  $G = \sum_i K_i \otimes X_i$ , with the system coefficients of  $L$  orthogonal to the Kraus span. Its leakage is  $\ell = \|L\|_F^2 / (dk) \leq \delta$ . For  $Y^0 = [X_i^* X_j] \geq 0$ , let  $H_0 = I - G^* G$  and  $E = (\mathcal{R} \otimes \text{id})(H_0)$ . Here  $H_0$  is in the amplified range, and

$$\frac{\|H_0\|_1}{k} \leq d(2\sqrt{\ell} + \ell), \quad \frac{\|E\|_1}{k} = O_\Phi(\sqrt{\ell}).$$

The first inequality follows from  $U^* U = I$  and Hilbert–Schmidt Cauchy–Schwarz. Let  $Z = r \text{Tr}_{\mathbb{C}^r} E_- \geq 0$ . The positive pinching inequality gives  $E_- \leq I_r \otimes Z$ , so

$$Y_1 = Y^0 + E + I_r \otimes Z \geq 0, \quad \mathcal{A}(Y_1) = I_d \otimes (I_k + Z).$$

For  $C = (I + Z)^{-1/2}$ , the common congruence  $Y = (I_r \otimes C) Y_1 (I_r \otimes C)$  is positive and  $\mathcal{A}(Y) = I$ . Factoring  $Y$  gives maps  $B_i : \mathbb{C}^k \rightarrow \mathbb{C}^{rk}$  with  $B_i^* B_j = Y_{ij}$ , and  $W = \sum_i K_i \otimes B_i$  is a genuine isometry, completed to a unitary on the enlarged bath.

We must bound its entropy without paying  $\sqrt{\ell} \log k$ . Let  $D_\mu = \text{diag}(\sqrt{\mu_i}) \otimes I_k$ . The output spectrum is that of  $\Omega = D_\mu Y D_\mu / k$ , whose bath-reference marginal is  $I_k / k$ . It splits into two positive components,

$$\Omega_{\text{old}} = D_\mu (I \otimes C) Y^0 (I \otimes C) D_\mu / k, \quad \Omega_{\text{corr}} = D_\mu (I \otimes C) (E + I \otimes Z) (I \otimes C) D_\mu / k.$$

The old component factors through the *original square bath*, so its rank is at most  $k$ ; the correction has rank at most  $rk$ . Also  $\text{Tr}_d H_0 = \text{Tr}_d L^* L$ , by orthogonality. Thus, with  $b = \text{Tr } \Omega_{\text{corr}}$ ,

$$0 \leq b \leq \ell + \text{Tr } Z / k = O_\Phi(\sqrt{\ell}), \quad S(\Omega) \leq h_2(b) + (1 - b) \log k + b \log(rk).$$

The repaired entropy production is therefore at most  $h_2(b) + b \log r$ . This argument would fail for a general rank- $k$  source in ambient dimension  $M$ : its old component could have rank  $M$ , leaving an extra  $(1 - b) \log(M/k)$ .

The scalar Gram  $\bar{G}_{ij} = \text{Tr } B_i^* B_j / k$  obeys  $\|\bar{G} - I\| = O_\Phi(\sqrt{\delta})$  and  $\mathcal{A}(\bar{G}) = I_d$ . Indeed the original scalar coefficients are bounded fixed linear extractions of the Choi state; the correction has trace norm  $O_\Phi(k\sqrt{\ell})$ , and the common contraction changes scalar entries by the same order. The  $X_i$  have fixed operator-norm bounds because they are fixed coefficient extractions of a unitary.

Replicate the repaired source to the least positive multiple  $K$  of  $k$  with  $K \geq \delta^{-1/2}$ , without changing its normalized Gram or production. Thus  $K \leq k + \delta^{-1/2}$ . Put  $N = \max(1, \lceil K\|\bar{G} - I\| \rceil)$  and

$$C_0 = I_r - \frac{K}{N}(\bar{G} - I_r) \geq 0, \quad \mathcal{A}(C_0) = I_d.$$

If  $C_0 = [\langle c_i, c_j \rangle]$ , the maps  $c_i \otimes I_N$  give a correction isometry, with flat-source output entropy  $\log N + S(\sum_i \mu_i |c_i\rangle\langle c_i|) \leq \log N + \log r$ . The orthogonal direct sum has source rank  $k_1 = K + N$ , exact scalar Gram  $(K\bar{G} + NC_0)/(K + N) = I_r$ , and production

$$a_1 \leq (1 - p)a' + p \log r, \quad p = N/(K + N) = O_\Phi(\sqrt{\delta}).$$

The mixing entropy cancels against  $\log(K + N)$ . Replication and the added dimension give the logarithmic size bound.  $\square$

*Proof of Theorem 4.4.* Lemma 4.2 turns a mixture approximant at Choi accuracy  $\delta$  into a single finite tracial factorization at accuracy  $2\delta$ , with  $\log k \leq \log(D_{\text{mix},\Phi}(\delta) + 1) + \log(1/\delta) + O_d(1)$ . Lemma 4.3 gives an exact dilation with source rank  $k_1$  and production  $a_1 \leq C_\Phi \sqrt{\delta} \log(e/\delta)$ .

The path compiler of Theorem G.4 applies to this dilation without gadget identities. For clarity, the hypothesis check is as follows. The flat probe applied to  $W^*W = I$  gives  $\sum_i B_i^* B_i = rI$ , so  $\|B_i\|^2 \leq r$ . The flat output bath state is at most  $I/k_1$ , since  $WW^*$  is a projection. Its top- $k_1$  tail  $w$  is at most  $a_1 \ln 2$  by the proof of Lemma G.5, which needs only that spectral ceiling, not individual isometries  $B_i$ . Divide the output into at most  $r + 1$  rank- $k_1$  pieces. Their matrices  $M^{(b)} \geq 0$  sum to  $I_r$  and induce effects  $V^*(I \otimes M^{(b)})V$ , with norm at most  $\text{Tr } M^{(b)}$ . Consequently

$$\beta = \sum_{b \geq 2} w(b) \leq a_1 \ln 2 / \mu_{\min}.$$

Concentration in Lemma G.1(a) uses only  $\|B_i\|^2 \leq r$ . The path-type recursion uses positivity and  $\sum_b M^{(b)} = I_r$ . Inflow is bounded by  $\sum_i C_i^{(b)*} C_i^{(b)} \leq rI$ , and export uses Lemma G.2(a). These are exactly the invariants proved in Appendix K; no group relation or disjoint Kraus support is consumed.

Choose  $\bar{\beta} = C_\Phi \sqrt{\delta} \log(e/\delta) \leq 1/4$  and  $\theta = 1/\log(1/\bar{\beta})$ . Then  $u' \leq O_r(\bar{\beta})$  and  $\log \kappa_\theta / \theta \leq O_r(\bar{\beta} \log(1/\bar{\beta}))$  by the positive-piece Hölder bound in the proof of Theorem G.6. Insert these in Theorem G.4. The target is exact before compilation:  $\delta$  contributes to resources, not an  $n\delta$  operational error. Every path register, port and workspace is part of  $Q$ ; the fixed choices of twirls require no runtime random resource.  $\square$

*Proof of Theorem 4.1(a).* For fixed  $\delta > 0$  the profile is finite. Divide the preceding bound by  $n$  and let  $n \rightarrow \infty$ : the profile and fixed logarithmic terms vanish, and  $\log(n/\epsilon_n)/n \rightarrow 0$ . Then let  $\delta \downarrow 0$ . Explicitly choose  $\delta_j \downarrow 0$  so that  $C_\Phi \sqrt{\delta_j} \log^2(e/\delta_j) \leq 1/j$ . For each  $j$  choose increasing thresholds  $N_j$  beyond which the remaining terms divided by  $n$  are at most  $1/j$ , using the fixed finite value  $D_{\text{mix},\Phi}(\delta_j)$  and the given error sequence. Use  $\delta_j$  for  $N_j \leq n < N_{j+1}$ . This slow diagonal choice gives  $Q/n \leq 2/j \rightarrow 0$  and  $P/n \leq Q/n \rightarrow 0$ . It asserts no uniform rate of approximation and no logarithmic purity bound.  $\square$

*Proof of Lemma 4.5.* Let  $p_i, q_j$  be the eigenvalues of  $\rho, T(\rho)$  and  $w_{ji} = d^{-1} \sum_{s,t} |\langle s, j | U | t, i \rangle|^2$  in their eigenbases. This is doubly stochastic and  $q = wp$ . The laws  $P_{ji} = w_{ji} p_i$ ,  $R_{ji} = w_{ji} q_j$  satisfy  $D(P \| R) = S(q) - S(p) = a$ . Pinsker gives  $\sum_{ji} w_{ji} |p_i - q_j| \leq \sqrt{2 \ln 2} a$ . Zeros follow by restriction or continuity.

Sort both spectra decreasingly, matching their eigenvectors by index. The optimal one-dimensional spectral matching gives  $\sum_j |p_j - q_j| \leq \sum_{ji} w_{ji} |p_i - q_j|$ . Therefore  $\sum_{ji} w_{ji} |p_i - p_j| \leq 2 \sum_{ji} w_{ji} |p_i - q_j|$ . For each  $k$ , compress  $U$  from the top- $k$  input bath subspace to the top- $k$  output bath subspace, identified with  $\mathbb{C}^k$ . Its square compression  $A = VH$  is a contraction; choose a unitary polar extension  $V$ . The unitaries  $V(H \pm i\sqrt{I - H^2})$  average to the channel  $X \mapsto AXA^* + BXB^*$ ,  $B = V\sqrt{I - H^2}$ , before the flat bath is traced out. The original flat-source channel is the compressed channel plus a positive outgoing term; the repaired channel is the same compressed channel plus a positive replacement term. Both terms have normalized Choi trace  $e_k = k^{-1} \sum_{i \leq k, j > k} w_{ji}$ , so their Choi distance is at most  $e_k$ . Now use the layer-cake decomposition  $\rho = \sum_k k(p_k - p_{k+1})P_k/k$ , with  $p_{M+1} = 0$ . Its weights sum to one, and the total repair error is at most

$$\sum_k (p_k - p_{k+1}) \sum_{i \leq k, j > k} w_{ji} = \sum_{ji} w_{ji} (p_i - p_j)_+ = \frac{1}{2} \sum_{ji} w_{ji} |p_i - p_j|.$$

The last equality uses double stochasticity. The preceding transport bound completes the proof, with finitely many components that are finite tracial factorizations and bath caps at most  $M$ .  $\square$

*Proof of Theorem 4.1(b).* Use an observer sending fresh Bell inputs and retaining every output and reference, without adaptation. Even if the memory is correlated with previous outputs, each round's unconditional channel is the round determined by its current memory marginal. Its Choi marginal is within  $\epsilon$  of the target. Its production  $a_t \geq 0$  obeys  $\sum_t a_t \leq B := P + C$ : the deficit  $Q - S(\rho_t)$  decreases by  $a_t$  at a round, and a replacement raises it by at most the imported deficit. Lemma 4.5, averaging the repaired channels, and Jensen give

$$\Delta \leq \epsilon + \sqrt{2 \ln 2} B/n.$$

It gives  $B \geq n(\Delta - \epsilon)_+^2 / (2 \ln 2)$ ;  $P \leq Q$  proves (b). The boundary ledger remains valid with correlations: exporting  $E$  from  $AE$  and importing an independent state  $\beta$  raises the memory deficit by  $c + S(E|A) - \log \dim E \leq c$ , where  $c$  is the imported deficit.

An additional converse, [27, Corollary 3.3(b)], remains useful when imported purity is large. Let  $H(\Phi)$  be the largest entropy of a minimal complementary output over system inputs. Use a *different* observer sending independent purifications of an entropy-maximizing input. The ideal retained entropy is  $nH(\Phi)$ ; continuity on dimension at most  $d^{2n}$  gives actual entropy at least  $nH(\Phi) - 2\epsilon n \log d - 1$ . Purifying the device's resources gives an upper bound  $2(Q + J) - B$ : the complementary system is the final memory, permanent exports and inaccessible resource purifiers. Both observers test the *same fixed resource ledger*, so at  $J \leq nH(\Phi)/2$ ,  $2Q \geq B - 2\epsilon n \log d - 1$ . Combining the inequalities gives, at this entropy-rate exchange budget,

$$Q \geq n \left( \frac{(\Delta - \epsilon)_+^2}{4 \ln 2} - \epsilon \log d \right) - \frac{1}{2}.$$

This proof uses imports only between rounds, and the theorem is stated for that case.  $\square$

*Proof of Proposition 4.6.* (a) Feed the maximally mixed state  $\tau_d$  at every use. These inputs are uncorrelated, so the device repeats  $\Phi$  on them. Use  $j$  is then the round  $(U, \xi_j)$  of

Definition 2.1. Its channel is  $\Phi$ , its bath map  $T$  gives  $\xi_{j+1} = T(\xi_j)$ , and its entropy production is  $a_j = S(\xi_{j+1}) - S(\xi_j) \geq 0$ . The sum telescopes:

$$\sum_{j=1}^n a_j = S(\xi_{n+1}) - S(\xi_1) \leq \log M - S(\xi_1).$$

Choose  $j$  with  $a_j \leq (\log M - S(\xi_1))/n$ . Lemma 4.5 gives  $\Psi \in \text{conv } \mathcal{F}_M$  with  $d_J(\Phi, \Psi) \leq \sqrt{2 \ln 2} a_j$ . This is the first inequality. The second follows from  $\text{conv } \mathcal{F}_M \subseteq \bar{\mathcal{F}}$  (Lemma 4.2). With Bell inputs in place of  $\tau_d$  the memory marginals are the same, and each reference and output pair is exactly in the state  $\mathcal{J}(\Phi)$ . No joint state of several outputs is used.

(b) For each  $n$ , the proof of part (a) gives a round  $(U_n, \rho_n)$  on  $\mathbb{C}^d \otimes \mathbb{C}^M$  with channel  $\Phi$  and entropy production at most  $(\log M)/n$ . Unitaries and states on these fixed spaces form compact sets, and the channel and the production of a round depend continuously on it. A limit point  $(U, \rho)$  is therefore a round with channel  $\Phi$  and production 0. Let  $\omega = U(\tau_d \otimes \rho)U^*$ . Then  $S(\omega) = \log d + S(\rho)$ , the marginals of  $\omega$  are  $\Phi(\tau_d)$  and  $T(\rho)$ , and  $S(T(\rho)) = S(\rho)$ . Subadditivity gives  $S(\Phi(\tau_d)) \geq \log d$ . Hence  $\Phi(\tau_d) = \tau_d$ , subadditivity is an equality, and  $\omega = \tau_d \otimes T(\rho)$ , that is,  $U(I \otimes \rho)U^* = I \otimes T(\rho)$ . So  $T(\rho) = W\rho W^*$  for a unitary  $W$ , and  $U' = (I \otimes W^*)U$  has the same channel and commutes with  $I \otimes \rho$ . Write  $\rho = \sum_k \lambda_k P_k$  with distinct eigenvalues  $\lambda_k > 0$  and  $m_k = \text{rank } P_k$ . Then  $U'$  restricts to a unitary  $U_k$  on  $\mathbb{C}^d \otimes P_k \mathbb{C}^M$ . Identifying  $P_k \mathbb{C}^M$  with  $\mathbb{C}^{m_k}$ ,

$$\Phi(X) = \sum_k \lambda_k \text{Tr}_{P_k \mathbb{C}^M} U_k(X \otimes P_k)U_k^* = \sum_k \lambda_k m_k (\text{id} \otimes \tau_{m_k})[U_k(X \otimes I)U_k^*].$$

The weights  $p_k = \lambda_k m_k$  sum to 1, and  $\sum_k m_k = \text{rank } \rho \leq M$ .

(c) Use the inputs of part (a). Each use is a round with channel  $\Theta$ , so Lemma A.1 with  $\delta = 0$  gives  $a_j \geq \log 3$ . The telescoping sum gives  $n \log 3 \leq \log M - S(\xi_1)$ . In the  $n$ -cell device each cell holds the pure environment of a three-dimensional Stinespring dilation of  $\Theta$ . Use  $j$  acts on the input and the first cell and then shifts the cells cyclically, as in Proposition 2.7. Each use  $j \leq n$  meets a fresh pure cell, so the device repeats  $\Theta$  for  $n$  uses, with  $M = 3^n$  and  $S(\xi_1) = 0$ . Letting  $n \rightarrow \infty$  shows that no finite device repeats  $\Theta$ .

(d) The  $v_k$  are unit vectors, so the  $w_k$  are Hermitian unitaries with  $\tau_2(w_k w_l) = v_k \cdot v_l = B_{kl}$ . Thus  $U$  is unitary,  $B \geq 0$  and  $B_{kk} = 1$ , so  $T_B$  is a unital channel. For every input  $\rho$ ,

$$\text{Tr}_{\mathbb{C}^6} U(\rho \otimes \frac{I}{2})U^* = \sum_k \rho_{kk} w_k \frac{I}{2} w_k = \frac{I}{2}, \quad \text{Tr}_{\mathbb{C}^2} U(\rho \otimes \frac{I}{2})U^* = \sum_{k,l} \rho_{kl} \tau_2(w_k w_l) |k\rangle\langle l| = T_B(\rho).$$

This is the catalytic dilation. Haagerup and Musat proved that  $T_B$  is not random-unitary [43, Example 3.3]; here is a direct check. If  $T_B = \sum_i p_i \text{Ad } V_i$ , then  $T_B(|k\rangle\langle k|) = |k\rangle\langle k|$  forces every  $V_i$  to be diagonal. So  $B = \sum_i p_i z_i z_i^*$  with unimodular vectors  $z_i$ , each in the range of  $B$  because  $p_i z_i z_i^* \leq B$ . Then  $z_{i,k} = v_k \cdot c$  for some  $c = x + iy \in \mathbb{C}^3$ , so  $v_k^T (xx^T + yy^T - I)v_k = 0$  for every  $k$ . A quadratic form vanishing at  $v_1, \dots, v_5$  vanishes on the circle through them, since a nonzero trigonometric polynomial of degree at most two has at most four zeros; so it is a multiple of  $-4x_1^2 + x_2^2 + x_3^2$ , and vanishing at  $v_0$  makes it zero. Hence  $xx^T + yy^T = I$ , which is impossible since the left side has rank at most two.  $\square$

## C Entropy compactness and the gadget classification

This appendix first establishes a general-channel compactness statement. The later steps use the singleton anchors and untwisted table explicitly. In particular, the finite-mixture conclusion in the first lemma must not be replaced by a single-factor conclusion for an arbitrary channel.

## C.1 Spectral tightness and finite limiting layers

**Lemma C.1** (Entropy compactness). *Let rounds on a fixed  $M_d$  have channels  $\Psi_j \rightarrow \Theta$ , bounded source entropy, and production  $a_j \rightarrow 0$ . Then  $\Theta$  is a finite convex mixture of finite tracial factorizations.*

*Proof.* Diagonalize source and output bath states in separate decreasing eigenbases, with spectra  $p_{ji}, q_{jl}$ . Write the system blocks of the unitary as  $U_{j,li}$  and put  $w_{j,li} = \text{Tr}(U_{j,li}^* U_{j,li})/d$ . Unitarity makes  $w_j$  doubly stochastic and  $q_j = w_j p_j$ . The probability arrays  $w_{j,li} p_{ji}$  and  $w_{j,li} q_{jl}$  have relative entropy  $a_j$  in bits. Pinsker gives

$$\delta_j := \sum_{li} w_{j,li} |p_{ji} - q_{jl}| \leq \sqrt{2 \ln 2 a_j} \rightarrow 0.$$

Sorted matching minimizes absolute-distance transport. Indeed, integrate threshold indicators: any doubly stochastic coupling mismatches at least the difference between the numbers of eigenvalues above each threshold. Hence  $\sum_i |p_{ji} - q_{jl}| \leq \delta_j$ .

Sorted eigenvalues obey  $p_{ji} \leq 1/i$ . If the entropy bound is  $B$ , then  $\sum_{i>K} p_{ji} \leq B/\log_2(K+1)$ ; the same holds for  $q_j$  with a bounded entropy bound. After adjoining zero coordinates, diagonal subsequence extraction gives pointwise spectral limits, and these tail bounds upgrade convergence to  $\ell^1$ . Both limits are the same probability list  $p$ , with  $S(p) \leq \liminf_j S(p_j) \leq B$ .

All blocks  $U_{j,li}$  are contractions. A further diagonal subsequence gives limits  $U_{li}$ . If  $p_i \neq p_l$ , the transport bound forces  $U_{li} = 0$ . Fix  $p_i = \lambda > 0$  and choose  $K$  with  $1/(K+1) \leq \lambda/4$ . Eventually  $p_{ji} \geq \lambda/2$ , so

$$\sum_{l>K} w_{j,li} \leq 4\delta_j/\lambda \rightarrow 0.$$

The analogous row estimate follows from  $q_{jl} \geq \lambda/2$ . Thus no positive-weight column or row escapes to infinity. The set  $I_\lambda = \{i : p_i = \lambda\}$  is finite, of size  $k_\lambda \leq 1/\lambda$ . In the unitary column identities, terms outside this set vanish: finite unequal-eigenvalue terms vanish entrywise, tails vanish in squared block norm, and cross-column tails vanish by Cauchy–Schwarz. Therefore  $V_\lambda = [U_{li}]_{l,i \in I_\lambda}$  is a finite square isometry, hence unitary.

Let  $\Theta_\lambda$  be the channel of this finite tracial factorization. Splitting the source at a finite index cutoff makes its tail channel mass uniformly small. On each retained positive-weight column the preceding no-escape estimate permits taking the blockwise limit. This proves

$$\Theta = \sum_{\lambda>0} k_\lambda \lambda \Theta_\lambda, \quad \sum_{\lambda>0} k_\lambda \lambda = 1.$$

It remains to reduce a countable barycenter to a finite one. In a finite-dimensional real space, the barycenter of bounded atoms with strictly positive weights lies in their convex hull. To see this, if the barycenter lies on a relative boundary of the closed convex hull, a supporting functional has its maximum at the barycenter. Every positive-weight atom must attain that maximum; restrict to its lower-dimensional affine hyperplane. Repeat until the barycenter is in relative interior. A small simplex around it can then be approximated by vertices in the convex hull, still containing the barycenter. Carathéodory elimination leaves at most  $d^4 + 1$  of the original Choi atoms. Their dimensions are finite, proving the lemma.  $\square$

## C.2 Exact support, rational weights and separation

**Lemma C.2** (Support-preserving copies). *Let  $\Theta = \sum_i p_i \Theta_i$  be a finite mixture of finite tracial factorizations of bath dimensions  $m_i \leq m$ . For  $0 < \gamma < 1$ , it has an approximant  $\Gamma$  with a finite tracial factorization of bath dimension at most  $\lceil (d^4 + 1)m/\gamma \rceil$ , half-diamond*

error less than  $\gamma$ , and  $\mathcal{J}(\Gamma) \leq \mathcal{J}(\Theta)/(1 - \gamma)$ . In particular, a common exact Choi support is preserved.

*Proof.* Reduce to  $L \leq d^4 + 1$  original atoms. Put  $N = \lceil (d^4 + 1)m/\gamma \rceil$ ,  $c_i = \lfloor Np_i/m_i \rfloor$  and  $N' = \sum_i c_i m_i$ . The missing fraction  $\delta = (N - N')/N$  is less than  $\gamma$ , so  $N' > 0$ . Take the block-diagonal unitary with  $c_i$  copies of each factor, on a bath of dimension  $N'$ , as in the proof of [44, Theorem 3.6]. Its channel  $\Gamma$  obeys  $\Theta = (1 - \delta)\Gamma + \delta\Theta_{\text{rem}}$  for a channel  $\Theta_{\text{rem}}$  (omit it if  $\delta = 0$ ). This proves both bounds. There is no filler channel.  $\square$

A finite tracial factorization with exact target support has a unitary expansion  $\sum_g A_g \otimes B_g$  (compare [43, Theorem 2.2]): each bath-entry Kraus vector must belong to the target Kraus span by Choi positivity. The anchors make all square  $B_g$  unitary, and the triangle columns give  $B_1^* B_y = B_x^* B_z$ . After gauging  $V_g = B_1^* B_g$ , these are exactly the relations of  $P_\Phi$ . The two-anchor reference input gives

$$|\tau(V_g^* V_h)| \leq 2d_\diamond(\Theta, \Phi), \quad g \neq h. \quad (4)$$

If no finite representation separates the whole finite label set, some fixed pair is equal in every finite representation. Otherwise a finite direct sum of pair-separating representations separates all pairs. Thus every exact-support finite tracial mixture is at diamond distance at least  $1/2$  from  $\Phi$ .

*Bounded branch of Theorem 5.1.* Suppose  $B = \sup_n L_\Phi^*(n) < \infty$ . Near-minimizers have bounded  $S$ ,  $a \rightarrow 0$  and  $\ell \rightarrow 0$ . Their channels have a convergent subsequence with limit  $\Theta$  at distance at most  $1/16$  from  $\Phi$ , with exact target support. The compactness proof gives spectral layers with weights  $w_\lambda = k_\lambda \lambda$  and

$$S(p) = H(w) + \sum_\lambda w_\lambda \log k_\lambda \leq B.$$

All layers have exact support. Discard those with  $\log k_\lambda > 16(B + 1)$ ; their mass is below  $1/16$ . The remaining normalized barycenter has bath cap  $2^{16(B+1)}$ . Even if it has infinitely many layers, the preceding barycenter argument leaves at most  $d^4 + 1$  of them, without changing the channel. Lemma C.2 at  $\gamma = 1/32$  gives one exact-support factor of dimension at most  $33(d^4 + 1)2^{16(B+1)}$ , at distance less than  $5/32$  from  $\Phi$ . Equation (4) gives a separating representation with trace gap below  $5/16$ . This proves the displayed bound on  $R_{\text{sep}}$ .

Conversely, tensor a representation with gap at most  $1/2$  to degree  $t = \lceil \log_2(8r(r - 1)) \rceil$ . The exact unitary table has zero production and leakage and half-diamond error at most  $r(r - 1)2^{-t}/2 \leq 1/16$ . It gives  $B \leq tR_{\text{sep}}$ . A representation merely separating the labels acquires a strict trace gap by adding a trivial summand, then a finite tensor power makes the gap at most  $1/2$ . Robust rank streaming (Theorem 7.6(c)) gives logarithmic service memory.

Finally the matrix image of  $P_\Phi$  is finitely generated. Malcev's residual-finiteness theorem [67] gives a finite quotient separating each of the finitely many label pairs; take their finite product. Its regular representation has trace overlaps exactly  $\delta_{gh}$  and obeys all relations. It therefore gives an exact finite tracial factorization of  $\Phi$ . The converse follows from the exact-support expansion above. The same fixed-pair argument proves that an exact finite tracial mixture for this gadget also implies a single exact finite tracial factorization.  $\square$

### C.3 Ultrapower separation and closure

**Lemma C.3** (Ultrapower label criterion). *A canonical untwisted anchored gadget is in  $\overline{\mathcal{F}}$  if and only if some homomorphism  $P_\Phi \rightarrow U(\mathcal{R}^\omega)$  separates its named labels. If there is no such homomorphism, some fixed distinct pair is identified by every such homomorphism.*

*Proof.* For a channel in the closure, Haagerup–Musat’s factorization in  $\mathcal{R}^\omega$  gives a unitary whose coefficient expansion has exact canonical support [44, Theorem 3.6]; the expansion is that of [43, Theorem 2.2] (compare [65, Proposition 3.9]). Faithfulness of the trace makes every orthogonal Kraus coefficient zero. The singleton anchors make the remaining coefficients unitary; triangle column orthogonality gives the relations of  $P_\Phi$  after gauging by the identity coefficient. Equality of the target Choi matrix makes distinct-label trace overlaps zero. This separates the labels.

Conversely, let  $\pi$  separate the labels. Work throughout with finite matrix approximants of the finitely many images  $\pi(v_g)$ , so that every construction below stays inside matrix algebras with normalized trace. Replace  $\pi$  by  $\pi \oplus 1$ , with equal tracial weights on the two summands. For  $g \neq h$  the overlap in  $\mathcal{R}^\omega$  becomes  $(1 + \tau(\pi(v_g)^* \pi(v_h)))/2$ , of modulus strictly below one: there the trace is faithful, so for a unitary  $u$ ,  $|\tau(u)| \leq 1$  with equality only when  $u$  is a scalar unitary  $e^{i\theta}I$ ; then the overlap is  $(1 + e^{i\theta})/2$ , and  $\theta = 0$  is excluded because  $\pi$  separates the labels. So the finitely many overlaps have modulus at most  $1 - \delta$  for some  $\delta > 0$ , and along  $\omega$  those of the matrix approximants have modulus at most  $1 - \delta/2$ . Tensor powers, or tensor-conjugate powers (the tensor-power trick of [78, Lemma 2.2]), drive these finitely many overlaps to zero while preserving every relation up to defects that vanish along the ultrafilter. The resulting finite matrix tuples have vanishing triangle defects. Form the actual physical gadget blocks with lower-right entry  $-V_x V_y$ . Their slot Choi matrices approach the canonical table, because equal-label slot defects and distinct-label overlaps tend to zero. This yields finite tracial factorizations converging to  $\Phi$ ; the same amplification is quantified in Appendix F.8.

Finally, if each pair were separated by some homomorphism, the finite direct sum of those maps would separate every pair and embed in  $\mathcal{R}^\omega$ . Its contrapositive gives the fixed pair assertion. This argument concerns finite label separation, rather than hyperlinearity of all of  $P_\Phi$ .  $\square$

#### C.4 Closure-only and nonclosure branches

At any fixed entropy bound  $B$  and coarse distance  $u < 1/2$ , absence of finite label separation implies

$$\inf\{a + \ell : S \leq B, d_\diamond \leq u\} > 0. \quad (5)$$

Otherwise compactness would give an exact-support finite mixture at distance below  $1/2$ , contradicting the anchor obstruction. In particular  $L_\Phi^*(n) \rightarrow \infty$  on the nonseparating branch. If  $\Phi \in \overline{\mathcal{F}}$ , positive-Gram repair and scalar completion (Lemma 4.3) give exact-target flat rounds with finite  $S_\delta$  and  $a_\delta \rightarrow 0$ . Then  $L_\Phi^*(n) \leq S_\delta + na_\delta$ ; take  $n \rightarrow \infty$  and then  $\delta \rightarrow 0$  to obtain  $o(n)$ . The closure service upper is Theorem 4.1(a). If service memory had a bounded subsequence, the precise extracted production bound of Theorem 7.3 would give  $a = O((1 + \log n)/n)$  at bounded  $S$ , contradicting (5). This proves the middle branch.

For the last branch we need the following qualitative amplification, specific to the gadget.

**Lemma C.4** (Support amplification). *If  $\Theta \in \overline{\mathcal{F}}$  has exact target support and  $d_\diamond(\Theta, \Phi) < 1/2$ , then  $\Phi \in \overline{\mathcal{F}}$ .*

*Proof.* Take channels  $\Psi_j \rightarrow \Theta$  with finite tracial factorizations, with unitaries  $U_j = \sum_g A_g \otimes X_{jg} + L_j$ ,  $\|L_j\|_2^2 = \ell_j \rightarrow 0$  in normalized joint trace. Polar extensions of their singleton anchor blocks give square unitaries  $R_{jg}$  with  $\|X_{jg} - R_{jg}\|_2 \leq 2\sqrt{d\ell_j}$ . The table  $W_j = \sum_g A_g \otimes R_{jg}$  has norm at most  $\sqrt{2}$  and  $\|U_j - W_j\|_2 \leq (1 + 2\sqrt{d})\sqrt{\ell_j}$ . Its column defects therefore tend to zero. Gauging by  $R_{j1}^*$  gives unitaries  $V_{jg}$ ,  $V_{j1} = I$ , with triangle error at most  $\kappa_j = 2\sqrt{d}(1 + \sqrt{2})(1 + 2\sqrt{d})\sqrt{\ell_j}$ . Anchor testing and polar error give trace overlaps at most  $2d_\diamond(\Psi_j, \Phi) + 2\sqrt{d\ell_j}$ , eventually bounded by  $c_0 < 1$ .

Construct actual unitary gadget blocks with lower-right entry  $-V_{jx}V_{jy}$ , not  $-V_{jz}$ . Every slot response  $Q_a$  is within  $\kappa_j$  of its canonical label response. Equal-label slots therefore have  $1 - |\tau(Q_a^*Q_b)|^2 \leq 4\kappa_j^2$ ; different-label slots eventually have overlap at most  $c_1 < 1$ . Replace all unitary responses by  $(V \otimes \bar{V})^{\otimes k}$ . This multiplicative map preserves actual unitary blocks; scalar slot signs remain outside the map. Slot correlations become  $|\tau(Q_a^*Q_b)|^{2k}$ . There are at most  $2d$  slots and their squared scalar coefficients sum to  $d$ . The Choi error is thus at most  $d(4k\kappa_j^2 + c_1^{2k})$ . Choose  $k$  first and  $j$  second to make this arbitrarily small, proving closure membership.  $\square$

For nonclosure  $\Phi$ , the compact set  $\{\Theta \in \bar{\mathcal{F}} : d_\diamond(\Theta, \Phi) \leq 1/8\}$  has a strictly positive minimum leakage  $\eta$  by this lemma; if the set is empty, use  $\eta = 1$ . Put  $\eta' = \min(\eta, 1)$  and  $a_* = \min\{1/(2048d^2 \ln 2), \eta'^2/(32 \ln 2)\}$ . A qualifying round with  $a \leq a_*$  tracializes to a channel in closure at Choi distance  $h \leq \sqrt{2 \ln 2} a_* \leq \min(1/(32d), \eta'/4)$ . Its target diamond distance is at most  $1/16 + dh \leq 3/32 < 1/8$  and its leakage is at most  $2/(15n) + h$ , below  $\eta$  for sufficiently large  $n$ . This is impossible. Hence  $L_\Phi^*(n) \geq na_*$  eventually. The pure Stinespring round gives  $L_\Phi^*(n) \leq n \log r$ . Theorem 7.3 gives the linear rank-rate memory lower at every  $\epsilon \leq 1/16$ ; the exact independent-cell service gives the linear upper. This completes Theorem 5.1.

### C.5 Regime (iii) and non-hyperlinear groups

A countable group is hyperlinear exactly when it is isomorphic to a subgroup of  $U(\mathcal{R}^\omega)$  [69, Remark 8.4 and Theorem 8.5].

*Proof of Proposition 5.2.* (a) $\Rightarrow$ (b). By Theorem 5.1(iii), distinct labels  $g \neq h$  are identified by every homomorphism  $P_\Phi \rightarrow U(\mathcal{R}^\omega)$ , and  $P_\Phi$  is finitely presented. Put  $\zeta = v_g^{-1}v_h$ . Then  $q(\zeta) = g^{-1}h \neq 1$ , so  $\zeta \neq 1$ , and every such homomorphism kills  $\zeta$ .

(b) $\Rightarrow$ (a). Let  $P = \langle S \mid R_0 \rangle$  with  $S$  and  $R_0$  finite. Drop empty relators and replace each relator  $\sigma = s^{\pm 1}$  of length one by its conjugate  $s\sigma s^{-1}$ ; the normal closure is unchanged. Choose a word  $w$  of length at least two representing  $\zeta$ , appending  $ss^{-1}$  if necessary. Let  $\Phi$  be the canonical gadget of  $G = P$ , the generating set  $S$  and the list  $R = R_0 \cup \{ww^{-1}\}$ . Every word in  $R$  has length at least two and equals 1 in  $P$ . The tagged prefix of  $ww^{-1}$  at position  $|w|$  names  $\zeta$ , so  $\zeta$  and 1 are distinct labels.

In  $P_\Phi$  the triangle  $(s, s^{-1}, \emptyset)$  gives  $v_{[s^{-1}]} = v_{[s]}^{-1}$ . For a word  $\sigma_1 \cdots \sigma_m \in R$ , the triangles  $(p_{i,j-1}, \sigma_j, p_{i,j})$  give  $v_{[p_{i,j}]} = v_{[\sigma_1]} \cdots v_{[\sigma_j]}$  by induction on  $j$ ; at  $j = m$  the prefix is empty, so  $v_{[\sigma_1]} \cdots v_{[\sigma_m]} = 1$ . Hence  $s \mapsto v_{[s]}$  defines a homomorphism  $\theta : P \rightarrow P_\Phi$ , and  $\theta(\zeta) = v_{[w]} = v_\zeta$ . For every homomorphism  $\pi : P_\Phi \rightarrow U(\mathcal{R}^\omega)$ , the composite  $\pi\theta$  kills  $\zeta$ , so  $\pi(v_\zeta) = 1 = \pi(v_1)$ . Theorem 5.1 places  $\Phi$  in regime (iii).

(b) $\Rightarrow$ (c). A homomorphism that kills  $\zeta \neq 1$  is not injective, so  $P$  is not hyperlinear.

(c) $\Rightarrow$ (b). Let  $P$  be finitely presented and suppose that every  $\zeta \neq 1$  survives some homomorphism  $\pi_\zeta : P \rightarrow U(\mathcal{R}^\omega)$ . Enumerate  $P \setminus \{1\}$  as  $\zeta_1, \zeta_2, \dots$  and choose orthogonal projections  $p_k \in \mathcal{R}$  with  $\tau(p_k) = 2^{-k}$ . Each corner  $p_k \mathcal{R} p_k$  is a hyperfinite  $\text{II}_1$  factor, so there is a  $*$ -isomorphism  $\theta_k : \mathcal{R} \rightarrow p_k \mathcal{R} p_k$ . It scales  $\|\cdot\|_2$  by  $2^{-k/2}$ , so applying it to representing sequences gives an injective  $*$ -homomorphism  $\theta_k^\omega : \mathcal{R}^\omega \rightarrow p_k \mathcal{R}^\omega p_k$  with  $\theta_k^\omega(1) = p_k$ . The orthogonal sums  $\pi(x) = \sum_k \theta_k^\omega(\pi_{\zeta_k}(x))$  converge in  $\|\cdot\|_2$  on bounded sets and define a homomorphism  $\pi : P \rightarrow U(\mathcal{R}^\omega)$ . Since  $p_k \pi(\zeta_k) p_k = \theta_k^\omega(\pi_{\zeta_k}(\zeta_k)) \neq p_k$ , the map  $\pi$  is injective, so  $P$  is hyperlinear. Contrapositively, a finitely presented group that is not hyperlinear has an element  $\zeta \neq 1$  killed by every homomorphism into  $U(\mathcal{R}^\omega)$ .  $\square$

## D A bounded mixture profile for phased monomial dynamics

This appendix isolates a tractable structured case of the mixture profile used in Section 4. For phased-monomial dynamics, a finite tile description bounds each component bath uniformly in the approximation accuracy. The argument concerns mixtures, not a bounded exact single-factor realization.

Fix product bases on  $\mathbb{C}^d \otimes \mathbb{C}^M$ . A phased-monomial unitary has exactly one nonzero entry of modulus one in each row and column. Let  $\mathcal{C}_{\text{mon}}(d)$  be its reduced channels with bath  $I_M/M$ , for arbitrary finite  $M$ , and let  $\mathcal{F}_B^{\text{mon}}$  be finite convex mixtures of such factors with component bath dimension at most  $B$ .

**Theorem D.1** (Bounded exact mixture). *Put*

$$E(d) = \sum_{k=1}^d \binom{d}{k}^2 k!, \quad B(d) = d \left\lceil (d+1)^{(E(d)+1)/2} \right\rceil.$$

*Then  $\mathcal{C}_{\text{mon}}(d) \subseteq \mathcal{F}_{B(d)}^{\text{mon}}$ , this latter set is compact, and  $\overline{\mathcal{C}_{\text{mon}}(d)} = \mathcal{F}_{B(d)}^{\text{mon}}$ . Every member is a mixture of at most  $d^4 + 1$  components. In particular, for every  $\Phi \in \overline{\mathcal{C}_{\text{mon}}(d)}$ ,*

$$D_{J,\Phi}(\gamma) \leq \left\lceil (d^4 + 1)B(d)/\gamma \right\rceil, \quad 0 < \gamma \leq 1.$$

*No bounded exact single-factor bath is asserted.*

*Proof.* An edge label  $e$  is a nonempty partial bijection  $A_e \rightarrow B_e$  of subsets of  $[d]$ ; there are  $E(d)$  labels. An input tile is a collection of labels whose domains partition  $[d]$ , an output tile one whose ranges partition  $[d]$ . Each contains at most  $d$  edges; repeated labels inside a tile are impossible. For a monomial unitary, group the system slots of an input bath point  $b$  by their output bath point  $y$ . Each group gives one edge, and the resulting input tile is  $T_b$ . Reversing the grouping gives the output tiles  $S_y$ .

Let  $x_T, y_S$  be normalized tile counts. They lie in the rational polytope

$$x, y \geq 0, \quad \sum_T x_T = 1, \quad \sum_{T:e \in T} x_T = \sum_{S:e \in S} y_S \quad (e \text{ any edge}).$$

Multiplying these balances by  $|A_e|$  and summing gives  $\sum_S y_S = 1$ , so the polytope is compact. The constraint matrix has  $E(d) + 1$  rows, entries  $0, \pm 1$ , and each column has Euclidean norm at most  $\sqrt{d+1}$ . A vertex has at most  $E(d) + 1$  positive coordinates with independent columns. Selecting a nonsingular square minor and using Cramer's rule and Hadamard's determinant bound gives a common denominator

$$D \leq \left\lceil (d+1)^{(E(d)+1)/2} \right\rceil.$$

Realize  $Dx_T$  input points and  $Dy_S$  output points. Pair equally labelled edge stubs using the balances. This gives a bipartite multigraph, but parallel edges between a pair of bath points must not be left as separate Kraus terms: they would create coherences absent from the tile formula.

Take a  $d$ -fold cyclic cover. For every parallel-edge family assign distinct shifts in  $\mathbb{Z}_d$ ; there are at most  $d$  edges in a family. An edge with shift  $a$  joins  $(b, t)$  to  $(y, t + a)$ . Thus no covered bath pair carries two edges. The domain partition at each input and range partition at each output make the covered system–bath map a genuine permutation unitary, with bath dimension  $dD \leq B(d)$ .

For a tile  $T$  and phases  $|z_s| = 1$ , define

$$V_{e,z} = \sum_{s \in A_e} z_s |e(s)\rangle \langle s|, \quad \mathcal{E}_{T,z}(X) = \sum_{e \in T} V_{e,z} X V_{e,z}^*.$$

The original channel is  $\sum_T x_T \bar{\mathcal{E}}_T$ , where the bar is the conditional phase average over input bath points with tile  $T$ . This identity includes all coherences. Decompose  $(x, y)$  into vertices of the tile polytope. For each vertex realization, sample independently at its covered input points the original conditional phase law of that point's tile. Because the cover separates bath pairs, its expected channel is exactly the corresponding vertex tile average. Averaging vertices recovers the original channel.

For bath sizes up to  $B(d)$  there are finitely many permutations and compact phase tori; their channels form a compact set. Its convex hull is compact in the finite-dimensional real channel space, and Carathéodory reduces the averages to at most  $d^4 + 1$  actual components. This proves containment and compactness. Conversely every finite mixture of these components is approximated by block copies of finite tracial factorizations with identity filler (Lemma 4.2), which are again phased monomial. This proves the closure equality and single-factor profile bound.  $\square$

**Corollary D.2** (Product-diagonal-preserving finite-radius dynamics). *Consider a reversible finite-radius quantum cellular automaton on a finite-dimensional lattice with product trace. If it preserves the product diagonal maximal abelian algebra, every finite-time reduced channel of a fixed  $d$ -dimensional probe, and every ordinary or Cesàro limit of these channels, belongs to  $\mathcal{F}_{B(d)}^{\text{mon}}$ .*

*Proof.* At any finite time the probe matrix algebra embeds into a finite window  $M_d \otimes M_M$ . The images  $w_{ab}$  of its matrix units are *partial* normalizers of the window diagonal algebra, not generally unitaries. Since  $w_{aa}$  is diagonal and  $w_{ab}$  is a partial isometry normalizing diagonals, it has at most one nonzero entry per row and column, each of modulus one. Trace preservation gives  $\text{rank } w_{aa} = M$ . Choose a basis  $\{v_b\}_{b=1}^M$  of  $\text{ran } w_{11}$  from the preferred window basis. The vectors  $w_{a1}v_b$  form that full basis up to phases, so  $W|a, b\rangle = w_{a1}v_b$  is a phased-monomial unitary implementing the embedding. Theorem D.1 applies, and compactness passes to limits and time averages.  $\square$

This statement concerns maximally mixed baths and a specified product diagonal algebra. It does not prohibit a hard channel from Gaussian baths with arbitrary non-Gaussian controllers, from blocked or dressed diagonal structures, or from general quantum local dynamics. Physically incoherent operations have a related projector–phase-permutation description [16]; the theorem's bath cap and exact-mixture assertion are stated without a priority claim.

## E A coherent entropy–production compiler

This appendix proves Theorem 6.3. The bath source is used to specify a finite bank; it is not an online supply. The two points that permit coherent responses are a full response-label work estimate and a simultaneous history-frame induction. Neither a forced-label scalar estimate nor independence of the observer's histories would suffice.

### E.1 Compact source and full matrix work

Restrict the source to its support. Write  $k = \dim K$ ,  $l = \dim L \leq rk$ ,  $\rho_g = B_g \rho B_g^*$ ,  $S = S(\rho)$ , and

$$\sigma_\mu = \sum_g \mu_g \rho_g, \quad a = S(\sigma_\mu) - S, \quad \sigma = r^{-1} \sum_g \rho_g, \quad a_u = S(\sigma) - S.$$

Here  $\mu_g = \text{Tr}(A_g^* A_g)/d$  is the actual Choi-input weight. Minimizing the relative-entropy center gives

$$a_u \leq a/(r\mu_{\min}), \quad \mu_{\min} \geq 1/d. \quad (6)$$

Indeed, evaluate the uniform center problem at  $\sigma_\mu$  and use  $r^{-1} \leq \mu_g/(r\mu_{\min})$ . The gap-selected version of Lemma F.1 gives a design source  $\xi$  with

$$\begin{aligned} \log_2 k' &\leq 64(S+1) + 16, & \log_2(1/p_{\min}(\xi)) &\leq 192(S+1) + 51, \\ na_\xi &\leq C_r \log_2(e(n+1))(1+na), & c_\xi &\leq 9/16 < 3/4. \end{aligned} \quad (7)$$

The filter changes half trace distance by at most  $1/32$ , so a Gram entry changes by at most  $1/16$ . Selecting its threshold between source square-root eigenvalues supplies the displayed spectral floor with no second, horizon-dependent deletion. Common restrictions preserve every isometry and multiplication triangle. These are offline design choices. In the rest of the proof use this compact source and omit its subscript.

If  $a_u \geq a_0 = (512r \ln 2)^{-1}$ , an exact service with  $n$  fresh pure  $r$ -dimensional Stinespring cells and a charged clock costs  $O_r(n + \log(n+1))$ . Equation (7) bounds this by the claimed quantity. Thus assume  $a_u < a_0$  and  $c \leq 3/4$ . The anchored triangle identities imply that  $\sum_g A_g \otimes B_g^{\otimes t}$  is isometric for every integer  $t$ : the off-diagonal column product in each gadget block is  $(B_1^* B_y - B_x^* B_z)/2$ , and tensor powers preserve that equality. This argument requires the untwisted table; it is not a rule for arbitrary Kraus expansions.

Put  $P = \rho^{\otimes t}$ ,  $T_g = B_g^{\otimes t}$  and

$$\Sigma_t = r^{-1} \sum_g \rho_g^{\otimes t}, \quad R_t = (1-z)\Sigma_t + zI/l^t.$$

In particular  $\Sigma_t$  is not  $\sigma^{\otimes t}$ . Concavity and subadditivity give  $0 \leq S(\Sigma_t) - tS \leq ta_u$ . For eigenvalues  $p_i, q_j$  of  $P, R_t$ , define the label matrices, using natural logarithms in the work variable,

$$M_\beta(g, h) = \sum_{ij} p_i \overline{T_{g,ji}} T_{h,ji} e^{\beta w_{ij}}, \quad w_{ij} = \ln(p_i/q_j), \quad D = M'_0.$$

Each summand is a nonnegative multiple of a positive rank-one label matrix. If  $G_{gh} = \text{Tr}(\rho B_g^* B_h)$  and  $\epsilon_0 = (r-1)c^t$ , then

$$(1 - \epsilon_0)I \leq M_0 \leq (1 + \epsilon_0)I, \quad M_{-1} \leq (1 + \epsilon_1)I, \quad \epsilon_1 = (r-1)\gamma^t + (r-1)z, \quad (8)$$

where  $\gamma = c + 2\sqrt{2r \ln 2 a_u} \leq 7/8$ . For the second assertion, before regularization,

$$M_{-1}(g, h) = r^{-1} \sum_b [\text{Tr}(\rho_b B_h B_g^*)]^t.$$

Its diagonal is at most one. Pinsker applied to  $D_2(\rho_b \|\sigma) \leq r a_u$  bounds every off-diagonal summand by  $\gamma^t$ , since  $\text{Tr}(\rho_g B_h B_g^*) = G_{gh}$ . Gershgorin gives the upper bound. The flat regularizer contributes a positive matrix with trace at most  $r$ , giving the displayed  $z$  term. This is an operator estimate on all labels, not only their diagonal.

Direct summation and operator monotonicity of  $\log$  give

$$\text{Tr } D = r\{[S(\Sigma_t) - tS] \ln 2 + D_{\text{nat}}(\Sigma_t \| R_t)\} \leq T_0 := r t a_u \ln 2 - r \ln(1-z).$$

The scalar function  $\phi(w) = w - 1 + e^{-w}$  is nonnegative; hence  $D - M_0 + M_{-1} \succeq 0$ . With  $e = \epsilon_0 + \epsilon_1$  and  $m = T_0 + (r-1)e$ , the preceding lower bound and trace estimate yield

$$-eI \leq D \leq mI, \quad 0 \leq D - M_0 + M_{-1} \leq (m+e)I, \quad M_0'' \leq VI, \quad V = 2(1+L)(m+e), \quad (9)$$

where  $L = \max\{t \ln(1/p_{\min}), t \ln l + \ln(1/z)\}$ . For  $-L \leq w \leq L$ ,  $w^2 \leq 2(1+L)\phi(w)$ : for  $w \leq 0$  use  $\phi(w) \geq w^2/2$ ; for  $w \geq 0$ , the derivative of  $2(1+w)\phi(w) - w^2$  is  $2w(1 - e^{-w})$ . Sum this scalar inequality against the positive label matrices. The same procedure, using the Taylor bound on  $e^{\theta w}$ , proves

$$M_\theta \leq \exp\left(\epsilon_0 + \theta m + \frac{\theta^2 V}{2(1 - \theta L/3)}\right) I, \quad \sum_g T_g P T_g^* = r \Sigma_t \leq r R_t / (1 - z). \quad (10)$$

The first inequality holds for  $0 \leq \theta < 3/L$ . No operator monotonicity of the exponential is used. The second inequality is the incoming-flux certificate; its factor is  $r$ , not  $r^t$ .

## E.2 Finite integer code and incoming flux

Set

$$\eta = z = \frac{\epsilon^2}{2^{20} r (n+1)^{12}}, \quad x = \ln \frac{2^{16} (n+1)^8}{\epsilon^2}, \quad t = \left\lceil \frac{\ln(8r/\eta)}{\ln(8/7)} \right\rceil.$$

Then  $\epsilon_0 \leq \eta/8$ ,  $\epsilon_1 \leq r\eta$ ,  $m = O_r(ta_u + \eta)$ ,  $V = O_r((1+L)(ta_u + \eta))$ . Write  $h = 1 + L$ ,  $x_+ = x + n\epsilon_0$ , and

$$\theta = \min\{(2h)^{-1}, \sqrt{x_+/(nV)}\},$$

$$B_+ = n(m + \eta) + 2\sqrt{nVx_+} + 2hx_+, \quad B_- = n(\eta + \epsilon_1) + x.$$

Alive energies are the integers  $u = -\lceil B_-/\eta \rceil$  through  $v = \lceil B_+/\eta \rceil$ . Put  $b = e^\eta$ ,  $s_i = \lceil -\ln p_i/\eta \rceil + 8$ ,  $t_j = \lceil -\ln q_j/\eta \rceil + 8$ ,  $E_{\min} = u - s_{\max}$ ,  $E_{\max} = v + t_{\max}$ ,  $D_b = \eta^{-2} e^{-\eta E_{\min}}$ , and  $d_E = \lceil D_b b^E \rceil$ . Stock energies run from  $u - s_{\max}$  through  $v - s_{\min}$ . There are  $N = E_{\max} - E_{\min} + 1$  physical sectors.

Inside sector  $E$ , pack source coordinates  $(i, E - s_i, a)$  of sizes  $d_{E-s_i}$ . The ceiling margins ensure

$$e^{-10\eta} p_i \leq d_{E-s_i}/d_E \leq p_i, \quad \sum_i d_{E-s_i} \leq d_E.$$

Indeed the ceiling ratio differs from  $b^{-s_i}$  by at most  $1 + \eta^2$ , since every stock count is at least  $\eta^{-2}$ . Pack output coordinates analogously with  $q_j, t_j$ . Denote the coordinate isometries by  $C_p, C_q$  and the common input range by  $P_{\text{code}}$ . On that range set

$$\hat{B}_g = C_q(T_g \otimes I_{\text{stock}})C_p^*.$$

All adjoint products are common compressions of  $T_g^* T_h \otimes I$ . The coded round is therefore isometric on  $\mathbb{C}^d \otimes P_{\text{code}}$  and can be completed to a unitary on the full system and bank. The stock and its coordinates specify matrix entries, not an additional retained tensor factor.

In every alive sector its normalized trace mean is

$$\tau_E(\hat{B}_g^* e^{\beta \eta E} \hat{B}_h) = e^{\beta \eta E} M_E^\beta(g, h), \quad M_E^\beta = \sum_{ij} \frac{d_{E-s_i}}{d_E} \overline{T_{g,ji}} T_{h,ji} e^{\beta \eta (t_j - s_i)}.$$

The summands remain positive label matrices. Thus

$$\|M_E^0 - I\| \leq \epsilon_0 + 10\eta,$$

$$M_E^\theta \leq \kappa_+ I, \quad \kappa_+ = e^{\epsilon_0 + \theta(m+\eta) + \theta^2 V}, \quad (11)$$

$$M_E^{-1} \leq \kappa_- I, \quad \kappa_- = e^{\eta + \epsilon_1}.$$

Means vanish outside the alive interval. The padding fraction is at most  $10\eta$ , independently of off-diagonal means.

Amplify all sectors by the same dyadic factor  $M$ , providing a common export port. With  $\Omega = \bigoplus_E (D_b b^E M)^{-1} I_{d_E M}$ , the virtual input Gibbs operator is bounded by  $P$  tensored with the stock Gibbs operator. Equation (10), output coding, and  $q_j b^{t_j} \leq b^9$  imply

$$\sum_g \hat{B}_g \Omega \hat{B}_g^* \leq \frac{r b^9}{1 - \eta} \Omega, \quad \sum_E \tau_E (P_E \Gamma_Q P_E) \leq r e^{12\eta}, \quad \Gamma_Q = \sum_g \hat{B}_g^* P_Q \hat{B}_g. \quad (12)$$

The input restriction commutes with the virtual Gibbs operator and can only decrease it. Also  $\|\Gamma_Q\| \leq r$ . Consequently inflow does not acquire a factor equal to the number of sectors.

### E.3 Concentration, reshaping and export

Here is the concentration estimate used in the chronological argument. For a frame  $F : \mathbb{C}^m \rightarrow (\bigoplus_{p=1}^N \mathbb{C}^{D_p}) \otimes \mathbb{C}^s$ , put  $R = F F^*$ ,  $g = \|F^* F\|$ ,  $\alpha = \max_p \|\text{Tr}_{D_p} R_{pp}\|$ , and  $D_{\min} = \min_p D_p$ . For a Hermitian memory-only operator  $A$  and independent block Haar mixers  $V = \bigoplus_p V_p$ ,

$$\Pr\{\|F^*[(V^* A V - \mathbb{E} V^* A V) \otimes I] F\| \geq s_0\} \leq 2m \exp\left(-\frac{D_{\min} s_0^2}{2(1+N)\alpha g \|A\|^2}\right). \quad (13)$$

For completeness, use Hermitian Hilbert–Schmidt orthonormal bases  $T_{p,a}$  and the product-group Laplacian  $\mathcal{L} = \sum_{p,a} D_p^{-1} \partial_{p,a}^2$ . Writing  $X = V^* A V$ , its centered diagonal and off-diagonal blocks all have Laplacian  $-2$ . Therefore the centered compression  $Z$  satisfies  $\mathcal{L} Z = -2Z$ . The completeness identities  $\sum_a T_{p,a} Y T_{p,a} = \text{Tr}(P_p Y P_p) P_p$  and  $\sum_a T_{p,a}^2 = D_p P_p$  give

$$\sum_{p,a} D_p^{-1} (\partial_{p,a} Z)^2 \leq \frac{2(1+N)\alpha \|A\|^2}{D_{\min}} F^* F.$$

To see the  $1+N$  factor, apply  $(XT - TX)R(XT - TX)^* \leq 2XTRTX + 2TXRX$ . The two unscaled sums are bounded respectively by  $\alpha \|A\|^2 I$  and  $N\alpha \|A\|^2 I$  after partial trace. Haar integration by parts then bounds  $f(t) = \mathbb{E} \text{Tr} e^{tZ}$  by  $m e^{ct^2/4}$  with  $c = 2(1+N)\alpha g \|A\|^2 / D_{\min}$ . The required trace derivative inequality follows by diagonalizing  $Z$  and integrating the scalar convexity inequality for the exponential. Optimize exponential Markov at  $t = 2s_0/c$  and apply it to  $-Z$  as well. This proof explains both unequal-sector validity and the memory-only restriction; it does not permit an operator on future history labels.

Reshape  $F$  by  $\tilde{F}_{(a,h),s} = \overline{F_{(a,s),h}}$ . Its Gram is  $\text{Tr}_{\text{memory}} R$ , whereas its sector spectator marginal is the transpose of  $F_p^* F_p$ . Hence  $\tilde{g} \leq N\alpha$  and  $\tilde{\alpha} \leq g$ . Apply (13) to  $\Gamma_Q^T$  and use (12). When  $g \leq 2$ , the simultaneous inflow bound is

$$\alpha_{\text{call}} \leq r e^{12\eta} \alpha + 2r \sqrt{N(1+N)\alpha \mathcal{H} / D_{\min}}. \quad (14)$$

Each failure probability is at most  $2se^{-\mathcal{H}}$ . Conjugation and transpose here are algebraic reshaping, not physical antiunitaries.

After a sector Haar mixer, export a common  $e$ -dimensional port and replace it by a flat one. Column Grams and energy-weighted Grams are preserved. The new spectator marginal has blocks given by Haar conjugation of  $I \otimes |u\rangle\langle v|$ . Applying the single-sector estimate to their Hermitian parts gives

$$\alpha_{\text{next}} \leq \alpha_{\text{call}} / e^2 + 4\sqrt{2\alpha_{\text{call}} \mathcal{H} / D_{\min}}. \quad (15)$$

There are at most  $2Ne^2$  tests. Explicitly, partial trace of the retained register gives a factor  $1/e$  from the flat import's purifier; the mean contributes another  $1/e$ . The block-to-operator factor  $e$  in the fluctuation cancels the first factor. These are exact partial trace identities, not independence assumptions about frame entries.

#### E.4 One simultaneous history-frame induction

Initialize sector zero flat. Before each round and after each of the first  $n - 1$  rounds choose block mixers offline, then fix them publicly. After round  $v < n$  replace  $j_v = \lfloor v \log_2 r/2 \rfloor - \lfloor (v - 1) \log_2 r/2 \rfloor$  port qubits. There is no final replacement. For analysis alone delete input padding after the premixer and output energies outside the alive interval after the round. Let  $F_v$  be the surviving history frame and

$$G_v = F_v^* F_v, \quad H_v^\beta = F_v^* e^{\beta \eta \mathbf{E}} F_v, \quad \beta = \theta, -1.$$

Its columns are histories; the initial purifier, exports and import purifiers are spectators, all used only in this dilation. Initially  $G_0 = 1$  and  $\alpha_0 \leq D_{\min}^{-1}$ .

Impose concentration on all  $r^2$  unweighted label blocks, both weighted families, the padding projection, and each sector inflow. Use Hermitian real and imaginary parts with threshold  $\eta/(2r)$  so every full label fluctuation is at most  $\eta$ . The Haar mean is a sum of  $M_E^\beta$  tensored with positive parent-sector Grams. Thus the full matrix bounds (11) imply, assuming parent norm at most two,

$$\begin{aligned} \|E_{\text{prep}}\| &\leq 21\eta, & \|G_{\text{call}} - I_r \otimes G_{v-1}\| &\leq \delta_G := 2\epsilon_0 + 41\eta, \\ H_{\text{call}}^\beta &\leq \kappa_\beta I_r \otimes H_{v-1}^\beta + \eta I, & H_v^\beta &\leq (1 + n\eta) \kappa_\beta^v I. \end{aligned}$$

Output cuts decrease positive weighted Grams; energy-preserving export preserves them exactly. Markov in Loewner order at  $B_-, B_+$  gives

$$\|E_{\text{exit}}\| \leq \tau := 2(1 + n\eta)e^{-x}, \quad \|G_v - I\| \leq v(\delta_G + \tau) < 1/16. \quad (16)$$

For the upper tail, if  $\theta = \sqrt{x_+/(nV)}$ , the variance term is  $x_+$  and the square-root buffer costs  $2x_+$ . In the other case put  $y = \sqrt{nV}/(2h\sqrt{x_+}) \leq 1$ ; the variance-minus-buffer part is  $x_+(y^2 - 2y - 1) \leq -x_+$ . In both cases  $n\epsilon_0 - x_+ = -x$ . Integer endpoints only help. The lower tail is  $(1 + n\eta)e^{-x}$ . This also bounds every child before the output cut by two, closing the induction.

Let  $c_v = 2^{2\lfloor v \log_2 r/2 \rfloor} r^{-v} \in (1/4, 1]$ . Applying  $\sqrt{u + c\sqrt{u}} \leq \sqrt{u} + c/2$  to (14) and (15) gives

$$\sqrt{c_v D_{\min} \alpha_v} \leq e^{6\eta} \sqrt{c_{v-1} D_{\min} \alpha_{v-1}} + 7\sqrt{r}(N+1)\sqrt{\mathcal{H}+1}.$$

Consequently, for  $v < n$ ,

$$\alpha_v \leq \mathcal{B}/D_{\min}, \quad \mathcal{B} = 1024r(n+1)^2(N+1)^2(\mathcal{H}+1).$$

The final round requires no spectator bound after it.

Here is an explicit, noncircular amplification. Set

$$\begin{aligned} H_b &= \sum_E d_E, & e_* &= 2^{\lceil \log_2 r/2 \rceil}, \\ T_w &= \eta \max\{|E_{\min}|, |E_{\max}|\}, \\ Z_0 &= 1 + \ln[128(n+1)N(r+1)^2 e_*^2] + \ln H_b + n \ln r, \\ K_0 &= 512r^2(n+1)(N+1)^2/\eta, \\ A_0 &= 4 + T_w + \ln K_0 + Z_0, \\ M &= 2^{\lceil \log_2 [4e^{T_w} K_0 A_0^2] \rceil}, & \mathcal{H} &= Z_0 + \ln M. \end{aligned}$$

All quantities preceding  $M$  are fixed without  $M$ . Then  $M \geq e_*$ ,  $\mathcal{H} \leq 3A_0$ , and

$$D_{\min} \geq M\eta^{-2} \geq 4r\sqrt{(1+N)\mathcal{B}\mathcal{H}} e^{T_w}/\eta.$$

This follows, for example, from  $\sqrt{(1+N)\mathcal{B}\mathcal{H}} \leq 32\sqrt{r}(n+1)(N+1)^{3/2}(\mathcal{H}+1)$  and  $M \geq e^{T_w} K_0(\mathcal{H}+1)$ . Column count is at most  $r^n$  and spectator dimension at most  $H_b M r^n$ . There are at most  $6r^2 + N + 1$  pre-mixer tests and  $2Ne_*^2$  export tests. The tail bounds and a union bound make each successive good choice exist with probability greater than  $1/2$ . Fix choices in chronological order. The tested frames depend only on the responses and earlier fixed matrices, never on an observer.

## E.5 Complete adaptive error and the resource ledger

Purify an arbitrary adaptive, reference-assisted observer, including its measurement records. Its ideal final dilation has history coefficients  $\psi_y$  with  $\sum_y \|\psi_y\|^2 = 1$ . The good physical dilation has the same coefficients  $\sum_y \psi_y \otimes F_n|y\rangle$ : all coded rounds have the canonical coefficients  $A_g$ , and all other bank operations act only on the bath. At each analytical deletion the removed vector has norm at most  $\sqrt{21\eta}$  or  $\sqrt{\tau}$ . The intervening physical circuit, retaining exports in the comparison space, is isometric. Telescoping and the final polar isometry of  $F_n$  therefore give vector error at most

$$n(\sqrt{21\eta} + \sqrt{\tau}) + n(\delta_G + \tau) < \epsilon.$$

This does not suppose discarded branches never return. Half trace distance of pure states is at most their vector distance, and partial trace contracts it. Thus the same public circuit works for every observer in the complete-experiment metric. No projection, history register, runtime random seed or postselection is used by the device.

Encode the body into  $\lceil \log_2 H_b \rceil$  qubits, tensor the dyadic port, and add a pure reversible clock. Outside padding is invariant and initially empty. Precisely,

$$Q = \lceil \log_2 H_b \rceil + \log_2 M + \lceil \log_2(n+1) \rceil, \\ P = Q - \log_2(d_0 M) \leq Q, \quad C = 0, \quad J = \lfloor (n-1) \log_2 r/2 \rfloor.$$

Every retained workspace is counted; in this unrestricted-unitary specification model the fixed controlled matrices need no synthesis workspace. A retained fixed service-wire padding costs only  $O_d(1)$ . The physical initial source is flat on sector zero, not on the entire bank.

The total natural-log energy width is  $O_r(nta_u + (1+L)x + 1)$ , by  $\sqrt{xy} \leq x + y$ . The body contributes this width plus  $2\ln(1/\eta) + \ln N + O(1)$ . The amplification adds  $T_w + \ln K_0 + 2\ln A_0 + O(1)$ ; in particular  $n \ln r$  occurs inside  $\ln A_0$ , not as  $n$  qubits. Hence  $Q \leq C_r[1 + nta_u + (1+L)x]$ . For  $y = \ln[128(n+1)/\epsilon]$ , the compact-source bounds give  $t = O_r(y)$ ,  $L \leq C_r y(S+1)$  and  $nta_u \leq C_{d,r} y^2(1+na)$ . Thus  $Q \leq C_{d,r} y^2(1+S+na)$ , proving the stated squared-log bound. The selected spectral floor enters only the work-range parameter  $L$ ; concentration, chronological induction and the complete adaptive error estimate above are unchanged. All constants are independent of the original rank and spectrum.

## F Static entropic comparisons and their boundaries

This appendix separates source compression, exact support repair and profile optimization. Compression preserves rare support leakage *multiplicatively*. Repair is controlled by a weighted trace norm, not by the same probability. Keeping these currencies distinct is essential to the open comparison (2).

### F.1 Soft compression at source-entropy scale

**Lemma F.1** (Soft source filter). *For an isometric ensemble  $\rho_g = B_g \rho B_g^*$  of  $r$  members, with weights  $\mu_g$  and gain  $a = S(\sum_g \mu_g \rho_g) - S(\rho)$ , put*

$$f_s(p) = (\sqrt{p} - \sqrt{s})_+^2, \quad T = 2^{-64(S(\rho)+1)-14}, \quad T/4 \leq s \leq T, \\ Z_f = \text{Tr } f_s(\rho), \quad \nu = f_s(\rho)/Z_f.$$

*Then  $Z_f \geq 31/32$ ,  $\|\nu - \rho\|_1/2 \leq 1/32$ ,  $\log_2 \text{rank } \nu \leq 64(S(\rho) + 1) + 16$ , and*

$$a_\nu \leq h_2(b) + b \log_2(r-1), \quad b = \min\{(\ln 2)a/Z_f, (r-1)/r\}, \\ na_\nu \leq 2(1 + na) \log_2(er(n+1)).$$

*For a general unitary round on a  $d$ -dimensional system the same rank and distance estimates hold. For  $\beta = (\ln 2)a/Z_f \leq 1/2$ ,*

$$a_\nu \leq h_2(\beta) + \beta \log_2(d^2 - 1) + 4d\beta/\ln 2. \quad (17)$$

*Its channel error changes by at most  $1/32$  and any positive Choi test increases by at most  $Z_f^{-1}$ . An equivalent smaller round has bath dimension at most  $d^2 \text{rank } \nu$ . One can select  $s$  from this interval so that, in addition,*

$$\log_2[1/p_{\min}(\nu)] \leq 192(S(\rho) + 1) + 51. \quad (18)$$

*No gap in the original source spectrum is assumed.*

*Proof.* For states  $X, Y$ , diagonalizing both gives classical probability arrays  $x_i|\langle i|j\rangle|^2, y_j|\langle i|j\rangle|^2$  whose relative entropy equals  $D_{\text{nat}}(X\|Y)$ . Jensen gives

$$\|\sqrt{X} - \sqrt{Y}\|_2^2 = 2(1 - \text{Tr } \sqrt{X}\sqrt{Y}) \leq D_{\text{nat}}(X\|Y).$$

The positive part is the Hilbert–Schmidt metric projection onto the positive cone and is nonexpansive. Indeed  $\text{Tr}[(H - H_+)(C - H_+)] = -\text{Tr}(H_-C) \leq 0$  for  $C \succeq 0$ ; apply this twice and add. With  $A_g = (\sqrt{\rho_g} - \sqrt{s}I)_+$  and  $\bar{A} = \sum_g \mu_g A_g$ ,

$$Z_f - \|\bar{A}\|_2^2 \leq \sum_g \mu_g \|A_g - (\sqrt{\sigma} - \sqrt{s}I)_+\|_2^2 \leq (\ln 2)a.$$

The canonical-purification mixture  $\sum_g \mu_g |\text{vec } A_g/\sqrt{Z_f}\rangle\langle \text{vec } A_g/\sqrt{Z_f}|$  has rank at most  $r$  and largest eigenvalue at least  $1 - (\ln 2)a/Z_f$ . Its entropy bounds the filtered Holevo information by partial-trace data processing, giving the first production estimate. Splitting  $a$  into the ranges  $a \leq 1/n$ ,  $1/n < a \leq 1/2$ , and  $a > 1/2$  gives the uniform estimate.

Eigenvalues below  $2^{-64(S+1)}$  have total mass at most  $1/64$ . Above it,  $p - f_s(p) \leq 2\sqrt{ps} \leq p/64$ . Thus  $1 - Z_f \leq 1/32$ . Since  $f_s(\rho) \leq \rho$ ,  $\rho = Z_f \nu + (1 - Z_f)\rho_{\text{res}}$ , proving distance and positive-test domination. The nonzero filtered eigenvalues have  $p > s \geq T/4$ , which proves the rank bound. To select a spectral floor, collect the distinct square roots of source eigenvalues strictly between  $A = \sqrt{T}/2$  and  $B = \sqrt{T}$ , and include  $A, B$  as endpoints. There are  $N \leq 4/T$  interior points. A longest consecutive gap has length at least  $\sqrt{T}/[2(N+1)]$ . Let  $\sqrt{s}$  be its midpoint. Every square-root eigenvalue above this midpoint is at distance at least  $\sqrt{T}/[4(N+1)]$ , including those outside the interval. Every positive filtered eigenvalue is therefore at least  $T/[16(N+1)^2] \geq T^3/400$ . Normalization increases it because  $Z_f \leq 1$ . Taking logarithms proves (18). The mass, domination and root-contraction proofs hold for every  $s \leq T$ , so this selection preserves their conclusions.

For the general round let  $\omega = U(\tau_d \otimes \rho)U^*$  and  $\sigma = \text{Tr}_A \omega$ . Then  $D_{\text{nat}}(\omega \| \tau_d \otimes \sigma) = (\ln 2)a$ . Apply the preceding root contraction on the joint space with threshold  $\sqrt{s/d}$ . The normalized canonical purification of  $\omega_\nu = U(\tau_d \otimes \nu)U^*$  misses the system Bell subspace on  $AA'$  with probability  $e \leq \beta$ . Consequently  $S(AA') \leq h_2(e) + e \log_2(d^2 - 1)$ . Strong subadditivity and purification duality imply

$$a_\nu \leq \log_2 d - S(A) + S(AA').$$

The system entropy-deficit term cannot be omitted (a pure-source SWAP would violate that assertion). Bell closeness gives  $\|\omega_{\nu,A} - \tau_d\|_1 \leq 2\sqrt{e}$ ; hence

$$\log_2 d - S(A) \leq \log_2[d \text{Tr}(\omega_{\nu,A}^2)] \leq 4de/\ln 2.$$

This proves (17). Always  $0 \leq a_\nu \leq 2 \log_2 d$ : the lower bound because the Choi-input bath map is unital, the upper by the Araki–Lieb inequality. In particular, with  $C_d = 2 + 2 \log_2 d + 5d$ ,  $na_\nu \leq 2C_d(1 + na) \log_2(e(n+1))$ . Finally restrict the input to  $K' = \text{supp } \nu$  and the output to the span of the  $d^2$  system blocks of  $U$  applied to  $K'$ . Its dimension is at most  $d^2 \text{rank } \nu$ . The restricted joint isometry extends to a unitary after separate input/output embeddings; entropy and channel are unchanged. These are static constructions, not free physical resets.  $\square$

## F.2 Profile structure, extraction and the horizon envelope

For the matrix form in Proposition 6.2, factor the positive Gram as  $Y_{gh} = B_g^* B_h$ , where  $B_g = Y^{1/2}(|g\rangle \otimes I)$ . Its diagonal makes each  $B_g$  isometric. The two products of the row map  $[\sqrt{\mu_g} B_g \sqrt{\rho}]_g$  give the output ensemble and  $\omega$ , so their nonzero spectra coincide. Its source marginal is  $\rho$ . Concavity and the entropy-of-mixture bound give  $0 \leq a \leq H(\mu)$ . The infimum of affine nondecreasing costs is concave and nondecreasing. Orthogonal responses on a pure source give  $Z(0) = 0$  and  $Z(t) \leq tH(\mu) \leq t \log r$ . Concavity at zero gives  $Z(ct) \leq cZ(t)$  for  $c \geq 1$  and  $Z(s) + Z(t) \geq Z(s+t)$ .

Tensor all responses and the source to degree  $t_r = \lceil \log_2(8(r-1)) \rceil$ . Exact untwisted triangles survive, the scalar overlaps are powered, and  $\|G - I\| \leq 1/8$ . For every reference input the output difference is  $T(G^\top - I)T^*$  with  $\|T\|_2^2 = 1$ , so its half trace norm is at most  $\|G - I\|/2$ . The bath output is  $\sum_g \mu_g \rho_g^{\otimes t_r}$ , whose marginals are  $\sigma$ . Subadditivity bounds production by  $t_r a$  and entropy by  $t_r S$ . This proves  $L^*(n) \leq t_r Z(n)$  with no compiler dependency.

To prove Theorem 6.4, retain the error in Theorem 7.1: simultaneous selection with  $\lambda = 2$  gives one round with

$$S \leq Q, \quad u \leq \epsilon, \quad \ell \leq \frac{2\epsilon}{n(1-\epsilon)}, \quad na \leq \frac{2[P + C + h_2(\epsilon) + \epsilon \log_2 n]}{1-\epsilon}.$$

Its passing-prefix flag belongs to the observer, not a postselection operation of the device. Theorem H.23 gives, above a fixed threshold, an exact family with gap at most  $1/4$  and

$$S' \leq 2.08S + 0.07, \quad a' \leq K_{d,r}[a + \sqrt{\ell} \log_2(e/\ell)].$$

This estimate follows directly from its commuting source restriction and its positive absorber: their masses are  $O_{d,r}(\sqrt{\ell})$ , and the conditional-entropy mixture cost is  $O_{d,r}(\sqrt{\ell} \log(e/\ell))$ , with no ambient dimension term. For  $\ell = 0$  use the exact frame directly. Insert the selected leakage, use  $P \leq Q$ , and absorb  $\epsilon \log n$  into the square-root term. The pure orthogonal baseline handles finitely many smaller  $n$ . This proves the claimed any-exchange bound. For high precision put  $w = n\epsilon \leq \log_2^{-2}(en)$ . Then

$$\sqrt{n\epsilon} \log_2(en/\epsilon) = \sqrt{w} \log_2(en^2/w) \leq 2 + \frac{2}{e \ln 2} < 3.1.$$

The last maximum follows by differentiating  $\sqrt{w} \ln(1/w)$ .

More generally the *same* selected exact family gives, for every  $T \geq 0$ , with  $B = 1 + Q + C$  and  $b = \sqrt{\epsilon/n} \log_2(en/\epsilon)$ ,

$$Z(T) \leq KB + KT(B/n + b). \quad (19)$$

Thus a genuine all-family lower bound  $Z(T) \geq cT^\beta$ ,  $0 < \beta < 1$ , implies  $B \geq c' \min\{n^\beta, b^{-\beta/(1-\beta)}\}$ : choose  $T \asymp B^{1/\beta}$  so the left side exceeds  $2KB$  and one of  $B/n, b$  pays the difference. This conditional implication does not substitute the restricted Houghton theorem for an all-family bound.

For Theorem 6.6, apply the same repair to a near-minimizer of  $L^*(N)$  and evaluate its linear cost at  $n \leq N$ . The cost is at most  $C[1 + L^*(N) + nN^{-1/2} \log_2(eN)]$ . Infimize over integer  $N \geq \max(n, N_0)$ . Compiler application and the pure baseline give the operational envelope. If  $L^*(N) = N^{\beta+o(1)}$  with  $0 < \beta < 1/2$ , choose  $N \asymp n^{1/(\beta+1/2)}$ ; for  $\beta \geq 1/2$  take  $N \asymp n$ . These give the exponents stated in the main text, not a matched bound below  $1/2$ .

### F.3 Coherent restriction to an exact leakage kernel

The following certificate uses probability mass on a genuinely exact subspace. It does not infer such a subspace from a small mean defect.

**Lemma F.2** (Source projection without commutation). *For a unitary round on  $\mathbb{C}^d \otimes \mathbb{C}^M$ , let  $a(\rho)$  be its Choi-input bath entropy production. If  $P$  is any bath projection,  $\nu = \text{Tr } \rho(I - P) \leq 1/2$ ,  $w = 1 - \nu$  and  $\rho_P = P\rho P/w$ , then*

$$S(\rho_P) \leq S(\rho)/w, \quad \|\rho - \rho_P\|_1/2 \leq \sqrt{\nu}, \quad \|\sqrt{\rho} - \sqrt{\rho_P}\|_F^2 \leq 4\nu.$$

For  $b = (\sqrt{(\ln 2)a(\rho)} + 2\sqrt{\nu})^2 \leq 1/2$ ,

$$a(\rho_P) \leq h_2(b) + b \log_2(d^2 - 1) + 4db/\ln 2. \quad (20)$$

In particular, for  $a(\rho), \nu \leq 1/64$  the new production is  $O_d((a(\rho) + \nu) \log_2(e/(a(\rho) + \nu)))$ . Always  $a(\rho_P) \leq 2 \log_2 d$ .

*Proof.* Decompose  $\rho = \sqrt{\rho}P\sqrt{\rho} + \sqrt{\rho}(I - P)\sqrt{\rho}$ . The normalized first summand has the same nonzero spectrum as  $\rho_P$ . Entropy concavity proves the source-entropy bound. Projecting and normalizing a purification gives overlap  $\sqrt{w}$ , proving the trace-distance estimate by partial trace. On  $\text{ran } P$ , put  $A = P\sqrt{\rho}P$  and  $B = \sqrt{P\rho P}$ . Compression concavity gives  $A \leq B$ , so  $\text{Tr } AB \geq \text{Tr } A^2$ . Also

$$\text{Tr } A^2 = w - \text{Tr}[P\sqrt{\rho}(I - P)\sqrt{\rho}P] \geq w - \nu.$$

The subtracted term is at most  $\text{Tr}[(I - P)\rho] = \nu$ . Consequently  $\text{Tr } \sqrt{\rho}P\sqrt{\rho}P \geq (1 - 2\nu)/\sqrt{1 - \nu} \geq 1 - 2\nu$ , giving the root bound.

For general source states  $\rho, \sigma$  set  $\eta = \|\sqrt{\rho} - \sqrt{\sigma}\|_F$ . Write  $\omega_\rho = U(\tau_d \otimes \rho)U^*$  and  $T\rho = \text{Tr}_A \omega_\rho$ . The root-relative-entropy inequality proved above and  $D_{\text{nat}}(\omega_\rho \|\tau_d \otimes T\rho) = (\ln 2)a(\rho)$  give

$$\|\sqrt{\omega_\sigma} - \sqrt{\tau_d \otimes T\rho}\|_F \leq \sqrt{(\ln 2)a(\rho)} + \eta.$$

Canonically purify both roots on  $ABA'B'$ . The comparison vector is  $|\Omega_d\rangle_{AA'} \otimes \text{vec } \sqrt{T\rho}$ . The actual  $AA'$  marginal therefore misses the Bell line by probability at most  $b = (\sqrt{(\ln 2)a(\rho)} + \eta)^2$ . Bell pinching gives  $S(AA') \leq h_2(b) + b \log_2(d^2 - 1)$  for  $b \leq 1/2$ . The system marginal has half distance at most  $\sqrt{b}$  from  $\tau_d$ , whence  $\log_2 d - S(A) \leq \log_2[d \text{Tr } \omega_{\sigma, A}^2] \leq 4db/\ln 2$ . Strong subadditivity gives  $S(A|B) \geq S(A|BB') = S(A') - S(AA')$ , and the canonical purification makes  $A'$  isospectral with  $A$ . Thus  $a(\sigma) = \log_2 d - S(A|B) \leq \log_2 d - S(A) + S(AA')$ . Take

$\sigma = \rho_P$  and use  $\eta \leq 2\sqrt{\nu}$ . For the stated small-parameter regime  $b \leq 8(a + \nu) \leq 1/4$ ;  $h_2(b) \leq b \log_2(e/b)$  proves the asymptotic bound. Araki–Lieb gives the uniform  $2 \log_2 d$  fallback.  $\square$

The logarithm is necessary for a general projection. For qubits take  $U = I \otimes |0\rangle\langle 0| + Z \otimes |1\rangle\langle 1|$  and source  $|0\rangle\langle 0|$  of zero gain. Projection onto  $\sqrt{1-\nu}|0\rangle + \sqrt{\nu}|1\rangle$  produces a pure source whose bath output is  $\text{diag}(1-\nu, \nu)$ , with gain  $h_2(\nu)$ .

**Proposition F.3** (Exact-kernel concentration). *Project a physical gadget round onto its canonical coefficient span,  $U = \sum_g A_g \otimes X_g + R$ , and set  $\Lambda = \text{Tr}_{\text{system}}(R^* R) \succeq 0$ , so  $\ell = \text{Tr} \rho \Lambda / d$ . Suppose  $u \leq 1/16$  and a bath projection  $P$  satisfies  $\Lambda P = 0$  and  $\nu = \text{Tr} \rho(I - P) \leq 1/64$ . Then  $\rho_P$  and  $B_g = X_g|_{\text{ran } P}$  are feasible for  $Z$ , with entropy and actual production as in Lemma F.2. If  $\nu \leq K/n$ , then*

$$Z_\Phi(n) \leq C_{d,r,K} \log_2(en)(1 + S + na) \quad (21)$$

for sufficiently large  $n$ . If  $\nu \leq K \log_2^q(en)/n$  with fixed  $q \geq 0$ , replace the logarithm by  $\log_2^{q+1}(en)$ .

*Proof.* Positivity and  $\Lambda P = 0$  give  $R(I_d \otimes P) = 0$ . The restricted round is thus exactly  $\sum_g A_g \otimes B_g$ . Singleton anchors make each  $B_g$  isometric, and triangle column orthogonality gives  $B_1^* B_y = B_x^* B_z$ . Source contractivity gives  $u_P \leq u + \sqrt{\nu} \leq 3/16$ , so the anchor tests give gap  $|\text{Tr} \rho_P B_g^* B_h| \leq 2u_P \leq 3/8 < 1/2$ . The ensemble output is exactly  $T(\rho_P)$ . For the cost bound,  $S_P \leq 2S$ . If  $a < 1/n$ , monotonicity of  $x \log_2(e/x)$  bounds the small-parameter gain by  $O_{d,K}(\log(en)/n)$ . If  $1/n \leq a < 1/64$ , its horizon cost is at most  $C_d(na + K) \log_2(en)$ . For larger  $a$  use the uniform fallback. The same arithmetic with  $K \log_2^q(en)$  proves the last statement. Every modification specifies a new offline round; no projection or postselection is performed during the service.  $\square$

Two concrete hypotheses imply the required mass estimate. If  $P = \ker \Lambda$  and  $\Lambda \geq \gamma(I - P)$ , then  $\nu \leq d\ell/\gamma$ . An inverse-polylogarithmic positive defect gap therefore suffices at  $\ell = O(1/n)$ . For a permutation round let  $P_{\text{good}}$  project onto triangle-good points and let  $\mathcal{D}$  be computational-basis pinching. If a coherent source obeys  $\rho \geq c\mathcal{D}(\rho)$ , positivity and Lemma 2.3 give

$$\ell \geq \frac{c}{4d} \text{Tr} \rho(I - P_{\text{good}}), \quad \nu \leq 4d\ell/c.$$

Every good point lies in the exact leak kernel. Constant or inverse-polylogarithmic  $c$  therefore suffices, including coherent entries between good and bad points. Composing with Theorem 6.3 gives the stated rank-rate service,  $C = 0$ ,  $P \leq Q$ , with the squared-log compiler factor in addition to the displayed static logarithms.

Neither hypothesis follows from a small mean leakage. Take the non-canonical anchored gadget with labels  $1, a, b, c$  and the two multiplication triangles  $ab = c$  and  $ba = c$ , realized on  $\mathbb{C}^8 = \mathbb{C}^4 \otimes \mathbb{C}^2$  by  $R_a = S_a \otimes e^{i\delta X}$  and  $R_b = S_b \otimes e^{i\delta Z}$ , where  $S_a, S_b$  are commuting diagonal involutions on  $\mathbb{C}^4$  and the lower-right blocks use the products  $R_a R_b$  and  $R_b R_a$ , with  $c$ 's anchor slot carrying  $R_a R_b$ . Proposition F.3 is stated for general gadget rounds, so it applies. Direct projection of its two triangles gives

$$\Lambda = \frac{3}{8} [R_a, R_b]^* [R_a, R_b] = \frac{3}{2} \sin^4 \delta I_8, \quad \ell = \frac{3}{16} \sin^4 \delta.$$

For  $0 < \delta \leq 1/16$  its exact kernel is empty, although its flat source has zero gain and entropy three. The target has a different exact four-dimensional factor. The universal comparison must permit that different round, rather than require restriction of every supplied round.

#### F.4 Source-weighted affine repair

For canonical projected physical responses  $X_g$  let  $Y = [X_g^* X_h] \succeq 0$ . Let  $\mathcal{P}_{\text{aff}}$  replace diagonal blocks by  $I$  and average each off-diagonal class generated by  $(1, y) \sim (x, z)$  and adjunction. Actual labels keep these classes separate from the diagonal. Define

$$\eta_\rho(U) = \|(I_r \otimes \sqrt{\rho})[\mathcal{P}_{\text{aff}}(Y) - Y](I_r \otimes \sqrt{\rho})\|_1.$$

This is basis-independent under a common source change but refers to the canonical gadget coefficients. If  $D = \text{diag}(\sqrt{\mu_g})$ , orthogonality of the projected and leaking parts gives  $\text{Tr}(D \otimes \sqrt{\rho})Y(D \otimes \sqrt{\rho}) = 1 - \ell$ . The weighted affine correction has  $D$ -weighted trace  $\ell$ , whence  $\ell \leq \eta_\rho(U)$ . The converse bound at order  $\ell$  is false.

The repair can control entropy directly, without filtering or a source rank bound. If  $H = (I \otimes \sqrt{\nu})E(I \otimes \sqrt{\nu})$ ,  $\|E\| \leq b$ ,  $q = \text{Tr } H_-$  and  $\sigma = I_r/r \otimes \nu$ , then  $-H \leq br\sigma$ . Sorted eigenvalue monotonicity gives  $\lambda_j(H_-) \leq br\lambda_j(\sigma)$ . This is not an operator inequality for  $H_-$ . Classical relative entropy of these normalized eigenvalue lists implies

$$qS(H_-/q) \leq -\sum_j \lambda_j(H_-) \log_2 \lambda_j(\sigma) \leq br[S(\nu) + \log_2 r].$$

Zero eigenvalues cause no exception; if  $q = 0$  omit this expression. For  $Z_c = r \text{Tr}_{\text{label}} H_-$ ,  $z = rq$ , Araki–Lieb gives  $zS(Z_c/z) \leq br^2[S(\nu) + \log_2 r] + z \log_2 r$ . The mixture source  $\theta = (\nu + Z_c)/(1 + z)$  therefore obeys

$$S(\theta) \leq \frac{(1 + br^2)S(\nu) + br^2 \log_2 r + z \log_2 r}{1 + z} + h_2\left(\frac{z}{1 + z}\right) \leq C_{b,r}(1 + S(\nu)). \quad (22)$$

The bound is independent of dimension and minimum eigenvalue. For isometric Grams,  $\|E\| \leq 2(r - 1)$  by the block-row bound. For physical projected Grams, canonical coefficients have modulus 1 or  $1/\sqrt{2}$ , so  $\|X_g\| \leq \sqrt{2}$  and  $\|E\| \leq 4r$ .

Use the original source  $\nu = \rho$  restricted to its support, and put  $H = (I \otimes \sqrt{\nu})(\mathcal{P}_{\text{aff}}(Y) - Y)(I \otimes \sqrt{\nu})$ ,  $Z_c = r \text{Tr}_{\text{label}} H_-$ ,  $z = \text{Tr } Z_c \leq r\|H\|_1$ . For every positive  $P$  on  $\mathbb{C}^r \otimes K$ ,  $P \leq rI \otimes \text{Tr}_{\text{label}} P$ ; prove this on rank-one terms by Cauchy–Schwarz and then sum. Therefore  $H + I \otimes Z_c \succeq 0$ . Set

$$J = (I \otimes \sqrt{\nu})Y(I \otimes \sqrt{\nu}) + H + I \otimes Z_c, \quad \theta = (\nu + Z_c)/(1 + z), \quad Y' = \frac{(I \otimes \theta^{-1/2})J(I \otimes \theta^{-1/2})}{1 + z}.$$

Then  $Y' \succeq 0$ ,  $Y'_{gg} = I$  and all affine identities are exact. It realizes a frame on at most  $rk$  output dimensions,  $k = \dim K$ , with  $S(\theta) \leq C_r(1 + S(\rho))$  by (22). No eigenvalue lower bound is used.

For  $\omega_0 = (D \otimes \sqrt{\nu})Y(D \otimes \sqrt{\nu})$  the repaired state is  $(\omega_0 + \Delta)/(1 + z)$ , where  $\Delta = D(H + I \otimes Z_c)D \succeq 0$ ,  $\text{Tr } \Delta = \ell + z$ . It has source marginal  $\theta$  and its conditional entropy equals the new production. With  $p = (\ell + z)/(1 + z)$  and  $H_0 = H(\text{label}|\text{source})_{\omega_0/(1-\ell)}$ , flagging this mixture gives

$$a' \leq H_0 + 2p \log_2 r + h_2(p), \quad (1 - \ell)H_0 \leq a_\nu + h_2(\ell) + \ell \log_2 \max\{1, d^2 - r\}.$$

For the latter, purify the source and use a Bell input. The system output and its reference have conditional entropy  $a_\nu$  given the source purifier. Pinching into the canonical Kraus span and its complement costs at most  $h_2(\ell)$ ; the failed conditional entropy is at least  $-\log_2(d^2 - r)$ . The successful block is the conjugate of  $\omega_0$ . If  $r = d^2$  the complement is absent and  $\ell = 0$ . For isometric old responses  $\ell = 0$ ,  $H_0 = a_\nu$ , and the sharper bound  $a' \leq a_\nu + p \log_2 r + h_2(p)$  holds. In particular small  $\eta = \|H\|_1$  gives  $a' \leq a/(1 - \ell) + C_{d,r}\eta \log_2(e/\eta)$ . No logarithmic

multiplier on  $a$  is introduced, because the original source was not filtered. Define the physical relaxed profile

$$A_\Phi^*(n) = \inf\{S + na : u \leq 1/16, \eta_\rho(U) \leq [64(r+1)n]^{-1}\}.$$

The preceding construction proves

$$Z_\Phi(n) \leq C_{d,r}[1 + A_\Phi^*(n) + \log_2(en)], \quad A_\Phi^*(n) \leq t_r Z_\Phi(n). \quad (23)$$

For the first inequality repair directly and use (22),  $\ell \leq \eta = O_r(1/n)$  and  $n\eta \log_2(e/\eta) = O_r(\log(en))$ . The anchor test bounds the repaired gap by  $1/8 + 2\sqrt{d\eta} + d\eta + \eta/2 < 1/2$  for  $n \geq 4096d$ ; use the exact baseline for smaller  $n$ . For the reverse inequality use fixed tensor amplification of an exact family: its defect is zero and its distance is at most  $1/16$ .

A concrete warning uses a cyclic three-label target on system dimension five. With flat bath  $I_3/3$  and  $X = \text{diag}(1, \omega, \omega^2)$ , take singleton entries  $I, X, e^{i\alpha}X^2$  and triangle  $\begin{bmatrix} I & X \\ X & -X^2 \end{bmatrix} / \sqrt{2}$ . This is a physical zero-production round. Its canonical responses are  $I, X, qX^2$ ,  $q = (2e^{i\alpha} + 1)/3$ . Direct projection gives

$$\ell = \frac{4}{15} \sin^2(\alpha/2), \quad \eta_\rho(U) \geq |q - 1|/2 = \sqrt{5/3} \sqrt{\ell}.$$

Thus raw  $\eta = O(\ell)$  cannot supply (2). The target nevertheless has an exact three-dimensional factor. Also scalar projective phases can cancel: use  $B_g \otimes \overline{B_g}$  and  $\rho \otimes \bar{\rho}$ . Exact scalar phases disappear, entropy doubles and production at most doubles. This extends the flat cancellation observation to biased rectangular responses; it does not cancel arbitrary noncentral errors.

## F.5 Square-unitary ensemble dictionary

Define  $T_\Phi(n)$  by minimizing  $S(\theta) + n[\chi + f_r(\eta)]$  over finite square unitaries  $V_g$ ,  $V_1 = I$ , and sources  $\theta$ , subject to  $|\text{Tr } \theta V_g^* V_h| \leq 1/2$  for  $g \neq h$ . Here

$$\chi = S\left(\sum_g \mu_g V_g \theta V_g^*\right) - S(\theta), \quad \eta = \|(I \otimes \sqrt{\theta})[\mathcal{P}_{\text{aff}}([V_g^* V_h]) - [V_g^* V_h]](I \otimes \sqrt{\theta})\|_1.$$

This is a static ensemble gain, not automatically the gain of a physical round built from an inexact tuple. Here  $p = r\eta/(2 + r\eta)$  and  $f_r(\eta) = h_2(p) + p \log_2 r$ . Then

$$T_\Phi(n) \leq Z_\Phi(n) \leq K_r(1 + T_\Phi(n)). \quad (24)$$

The forward direction is exact unitary completion, not polar repair. Given an exact frame, set  $E = B_1$  and complete  $Ex \mapsto B_g x$  to a unitary  $V_g$  on  $L$ , choosing  $V_1 = I$ . Put  $\theta = E\rho E^*$ . Then entropy is unchanged,  $V_g \theta V_g^* = \rho_g$ , and  $E^* V_g^* V_h E = B_g^* B_h$ . Consequently  $\chi = a$  and  $\eta = 0$ . Global multiplication outside the embedded source is not asserted.

Even requiring faithful square sources gives the same infimum. Replace  $\theta$  by  $\theta_t = (1-t)\theta + tI_L/\dim L$ , and add a cyclic diagonal  $r$ -dimensional block with distinct powers and flat source. Use source  $(1-t)\theta_t \oplus tI_r/r$ . Its gap is at most  $(1-t)[(1-t)/2 + t] = (1-t^2)/2$ ; entropy, gain and defect converge to  $S, a, 0$  as  $t \downarrow 0$  for each fixed horizon. The flag is charged. There is no invariant-source assumption or quantitative minimum eigenvalue promise.

For the reverse direction repair directly. Since the old responses are isometric,  $E_{gg} = H_{gg} = 0$ ,  $\text{Tr } H = 0$ , and  $q = \eta/2$ ,  $z = r\eta/2$  exactly. Testing  $H$  against two conjugate off-diagonal identity blocks of operator norm one gives  $|\text{Tr } H_{gh}| \leq \eta/2$ . Thus the scalar gap obeys

$$c' \leq \frac{c + \eta/2}{1 + r\eta/2} \leq \max\{c, 1/r\} \leq 1/2.$$

No tensor restoration is needed. The corrected spectral state has old-component conditional entropy  $\chi$  and correction weight  $p = z/(1+z)$ , so flagging gives  $a' \leq \chi + f_r(\eta)$ . Equation (22) proves the reverse inequality with a constant independent of the horizon. As a corollary, the linear-defect convention  $W(n) = \inf[S + n(\chi + \eta)]$  still obeys  $W(n) \leq Z(n) \leq C_r \log_2(en)(1 + W(n))$ , by  $nf_r(\eta) \leq C_r \log_2(en)(1 + n\eta)$ . Neither comparison supplies a probability-leakage-to-defect bound or consumes the service compiler.

## F.6 Zero production can be rationalized without a leakage penalty

For any general round with  $a = 0$ , diagonalize input and output sources with spectra  $p_i, q_j$ . The doubly stochastic array  $T_{ji} = \|U_{ji}\|_2^2/d$  satisfies  $D_2(T_{ji}p_i \| T_{ji}q_j) = a$ . Therefore  $T_{ji} > 0$  implies  $p_i = q_j$ . Every positive spectral level has equal input and output multiplicity  $k_\lambda$  and carries a square unitary block. Thus

$$\Psi = \sum_{\lambda} w_{\lambda} \Psi_{\lambda}, \quad S(\rho) = H(w) + \sum_{\lambda} w_{\lambda} \log_2 k_{\lambda}, \quad w_{\lambda} = k_{\lambda} \lambda,$$

where each  $\Psi_{\lambda}$  has a finite tracial factorization. Discard levels with  $\log_2 k_{\lambda} > (S+1)/\theta$ ; their mass is less than  $\theta$ . The normalized good mixture is dominated as a positive Choi operator by  $J(\Psi)/(1-\theta)$ . Reduce it to  $d^4 + 1$  atoms and apply the filler-free copy rounding of Lemma C.2 with loss less than  $\gamma$ . The resulting single finite tracial factorization  $\Gamma$  satisfies

$$d_{\diamond}(\Gamma, \Psi) < \theta + \gamma, \quad \log_2 \dim B_{\Gamma} \leq \frac{S+1}{\theta} + \log_2 \frac{d^4 + 2}{\gamma}, \quad J(\Gamma) \leq \frac{J(\Psi)}{(1-\theta)(1-\gamma)}.$$

Every positive Choi effect obeys the same multiplicative estimate, including a rare support-leakage event. There is no inverse-leakage rank penalty. This argument requires exact zero production; it proves neither positive-production flattening nor general leaky rounding.

## F.7 A flat Choi spectrum is not an anchored profile theorem

**Proposition F.4.** *There is a channel with flat nonzero Choi spectrum and bounded  $L^*(n)$ , but every flat-import service has linear memory at error at most  $1/100$ . Thus bounded-profile classification cannot be extended from anchored gadgets merely by assuming flat Choi spectrum.*

*Proof.* Let  $X|j\rangle = |j+1\rangle$ ,  $Z|j\rangle = e^{2\pi i j/8}|j\rangle$ ,  $F|j\rangle = (-1)^j|j\rangle$  and  $P_{\pm} = (I \pm F)/2$ . Take the 48 Weyl matrices  $X^a Z^b$  with  $a \in \{0, 3, 4, 5, 6, 7\}$ ,  $0 \leq b < 8$ , and add  $A = \sqrt{2}XP_+$ ,  $B = \sqrt{2}X^2P_-$  to form a list  $C_i$  of length fifty. Then  $\text{Tr}(C_i^* C_j) = 8\delta_{ij}$  (the two new matrices occupy the excluded cyclic diagonals),  $A^*A + B^*B = 2I$ , and  $AA^* + BB^* = 4P_-$ . Thus  $\Phi(R) = 50^{-1} \sum_i C_i R C_i^*$  is a channel with  $J(\Phi) = \Pi/50$ , where  $\Pi$  has rank fifty. Its complement on  $\tau_8$  is flat of rank fifty, so it has the required flat probe.

Let  $\Psi$  average only the forty-eight Weyl terms and let  $\Lambda(R) = (ARA^* + BRB^*)/2$ . Then  $\Phi = (24/25)\Psi + (1/25)\Lambda$ . The two Choi supports are orthogonal, so  $d_{\diamond}(\Psi, \Phi) = d_J(\Psi, \Phi) = 1/25 < 1/16$ . The controlled Weyl unitary on a flat forty-eight-dimensional bath realizes  $\Psi$  with zero production and zero leakage relative to  $\Pi$ . Hence  $L^*(n) \leq \log_2 48$  for every  $n$ .

Nevertheless  $\Phi(\tau_8) = (I - F/25)/8$  has odd-coordinate probability  $13/25$ . An observer sending fresh  $\tau_8$  inputs forces every released marginal  $\sigma_t$  to have odd probability at least  $51/100$  when experiment error is at most  $1/100$ . Parity pinching gives  $S(\sigma_t) \leq h_2(p_t) + \log_2 4$ , so its entropy deficit is at least  $b = 1 - h_2(51/100) > 0$ . Unitarity and subadditivity give  $S(\rho'_t) - S(\rho_t) \geq \log_2 8 - S(\sigma_t) \geq b$ . Every flat replacement is entropy-deficit nonincreasing, since exporting  $E$  lowers deficit by  $\log \dim E - S(E|A) \geq 0$  and appending a flat state changes it by zero. Summing yields  $P \geq nb$ , hence  $Q \geq nb \geq n/(5000 \ln 2)$  at any exchange. This is outside the anchored scope: the target is nonunital, while every tracial-closure channel is unital. Its cheap supported competitor cannot be amplified into the target by gadget identities.  $\square$

## F.8 Hyperlinear dimension gives flat leaky rounds

*Proof of Remark 2.5.* Write  $d$  for the system dimension,  $r$  for the label count and  $m$  for the number of triangle blocks, so  $2m \leq d$ . We use normalized Frobenius norm,  $\|A\|_{2,k}^2 = \text{Tr}(A^*A)/k$ ;  $\|\cdot\|_{2,d,K}$  uses normalization  $dK$ . Take a map  $v : E \rightarrow U(k)$  with  $v_1 = I$ , triangle defects at most  $e$  in  $\|\cdot\|_{2,k}$ , and  $\|v_g - v_h\|_{2,k} \geq 1$  for distinct labels. Then  $\text{Re } \tau_k(v_g^* v_h) \leq 1/2$ . Put

$$w_g = v_g \oplus \overline{v_g} \oplus I_{2k}, \quad B_g = w_g^{\otimes t}, \quad K = (4k)^t.$$

This is the construction in the proof of [78, Lemma 2.6]. Every triangle defect of  $w$  is at most  $e$ , and tensor telescoping bounds that of  $B$  by  $te$ . Moreover  $\tau_{4k}(w_g^* w_h) = (1 + \text{Re } \tau_k(v_g^* v_h))/2 \in [0, 3/4]$ . Choose the fixed integer

$$t = \left\lceil \frac{\ln(32d(r-1))}{\ln(4/3)} \right\rceil.$$

All off-diagonal moments of  $B$  are then at most  $z = (3/4)^t \leq [32d(r-1)]^{-1}$ .

Form the physical gadget unitary  $U$  from the symbol unitaries  $B_g$ , with full bath state  $\tau_K$ . Compare it to  $W = \sum_g A_g \otimes B_g$ . These operators differ only in the lower-right triangle slots: each discrepancy is  $-(B_x B_y - B_z)/\sqrt{2}$ . Thus

$$\|U - W\|_{2,d,K}^2 = \frac{1}{2d} \sum_{xy=z} \|B_x B_y - B_z\|_{2,K}^2 \leq t^2 e^2.$$

The normalized vectorizations of both  $U$  and  $W$  have norm one: for  $W$  this follows from disjoint Kraus supports and  $\sum_g \|A_g\|_F^2 = d$ . The  $W$  vector lies in the target Choi support. Hence the physical leakage is at most  $t^2 e^2$ , and the half trace distance of its Choi state from the  $W$  Gram state is at most  $te$ . The latter Gram state's half trace distance from  $\mathcal{J}(\Phi)$  is at most  $z \sum_{g \neq h} \sqrt{\mu_g \mu_h}/2 \leq (r-1)z/2$ . Since  $U$  defines a channel, the usual Choi-to-diamond estimate gives

$$u \leq d(te + (r-1)z/2).$$

For  $e = c_\Phi/\sqrt{n}$  with  $c_\Phi = 1/(32dt)$ , this is at most  $3/64 < 1/16$ , and leakage is at most  $1/(1024d^2n) \leq 2/(15n)$ .

For production, the system input  $\tau_d$  and full bath  $\tau_K$  give a jointly maximally mixed state, invariant under  $U$ . The actual bath map therefore fixes  $\tau_K$ . Consequently  $a = 0$  and  $S + na = \log_2 K = t \log_2(4k)$ , proving the bound. A missing finite approximation makes its right-hand side infinite, with the usual extended-value interpretation.

For a fixed finite presentation of  $P_\Phi$ , choose representatives  $q_x$  of the elements  $x \in E$ , with  $q_1$  the empty word, and the finite list  $T = \{q_x q_y^{-1} : x \neq y\}$ . With hlp the hyperlinear profile of [78, Definition 2.3], [78, Proposition 7.2(b)] gives  $\eta(E; 1, Ce) \leq \text{hlp}(T; 1, e)$  for a presentation-dependent  $C$ : each multiplication in  $E$  has a fixed finite relator-area proof, with normalized norm error at most  $Ce$ . Substitution gives the same upper in terms of  $\text{hlp}(T; 1, c_\Phi/(C\sqrt{n}))$ . The defect here is a norm, not its square. No rounding into permutations, no source flattening and no reverse comparison is used.  $\square$

## G Specialized flat-source constructions

The main coherent compiler includes exact flat-source rounds. This appendix retains specialized path and spatial constructions, and the flat leakage repair. The device of [28, Theorem 12.2] cannot by itself reach Shannon cost: it re-randomizes the whole active memory in every round, and the memory then fills the space that the rank room allows (Theorem I.1). The devices below re-randomize only inside blocks, and they keep a classical record of which

block a Kraus history has reached. Three tools make them work: concentration that does not pay for the number of blocks, a bound on the law of paths through the blocks that holds for adaptive observers, and a rounding that makes a leaky flat round exact on the same bath.

### G.1 Frames and blocks

We use the frames of [27, Appendix A]. A *frame* is a linear map  $F : \mathbb{C}^N \rightarrow H \otimes \mathbb{C}^S$ , where  $H = \bigoplus_p H_p$  is the memory, split into *blocks*  $H_p$  of dimension  $D_p$  with projections  $P_p$ , and  $\mathbb{C}^S$  is an exterior that holds purifiers, exported qubits and the purifiers of imported ones. In a device the columns of  $F$  are the vectors attached to the Kraus histories. Put  $F_p = (P_p \otimes I)F$ ,  $R_p = F_p F_p^*$ ,

$$\alpha_p(F) = \|\text{Tr}_{H_p} R_p\|, \quad g(F) = \|F^* F\|.$$

The number  $\alpha_p$  is the largest weight that the exterior marginal of block  $p$  gives to one exterior vector; Haar unitaries inside a block concentrate well when  $\alpha_p/D_p$  is small.

**Lemma G.1** (Blockwise concentration). (a) Let  $A_p$  be Hermitian on  $H_p$  (possibly 0),  $V = \bigoplus_p V_p$  with independent Haar unitaries  $V_p \in U(D_p)$ , and

$$Z = \sum_p \left[ F_p^* ((V_p^* A_p V_p) \otimes I) F_p - \tau_{D_p}(A_p) F_p^* F_p \right].$$

With  $\kappa = \max_p \alpha_p(F) \|A_p\|^2 / D_p$  and  $g_0 = \|\sum_{p: A_p \neq 0} F_p^* F_p\|$ , for every  $s > 0$ ,  $\Pr\{\|Z\| \geq s\} \leq 2N \exp[-s^2/(4g_0\kappa)]$ .

(b) Let all blocks have the same dimension  $D$ , let  $A = [A_{pp'}]$  be Hermitian on  $H$ ,  $X = V^* A V$ , and  $Z = F^* ((X - \sum_p \tau_D(A_{pp}) P_p) \otimes I) F$ . With  $\alpha_* = \max_p \alpha_p(F)$ ,  $g = g(F)$  and  $\Delta = \max_p \#\{p' : A_{pp'} \neq 0\}$ ,

$$\Pr\{\|Z\| \geq s\} \leq 2N \exp\left[-\frac{s^2 D}{2g(1 + \Delta^2) \|A\|^2 \alpha_*}\right].$$

The frame, the blocks and the operators may depend on anything chosen before  $V$ .

The number of blocks enters nowhere: only the ratios  $\alpha_p/D_p$ , through a maximum, and the Gram norm, once. For one block, (a) is [27, Lemma A.1]; the proof uses the semigroup method of Huang and Tropp [52] on the product group  $U(D_1) \times \cdots \times U(D_m)$ .

**Lemma G.2** (Blockwise export and inflow). (a) Let every block be  $H_p = \tilde{H}_p \otimes \mathbb{C}^e$  with a common port  $\mathbb{C}^e$ ,  $N$  blocks in all. Apply independent Haar unitaries  $V'_p$  on the blocks, export the ports and import a maximally mixed  $\mathbb{C}^e$  purified in the exterior. Every block Gram  $F_p^* F_p$  is unchanged, and with probability at least  $\frac{1}{2}$ , for every block  $p$  and every  $\eta > 0$ ,

$$\alpha_p(F') \leq \frac{\alpha_p}{e^2} + 4\sqrt{\frac{\alpha_p g_p \ln(8Se^2 N)}{D_p}} \leq (1 + \eta) \frac{\alpha_p}{e^2} + \frac{4e^2 g_p \ln(8Se^2 N)}{\eta D_p}, \quad g_p = \|F_p^* F_p\|.$$

(b) Let a round map  $\bigoplus_p H_p$  into  $\bigoplus_Q H'_Q$  through  $W = \sum_i A_i \otimes B_i$ , after a block-diagonal unitary, and put  $\Gamma_Q = \sum_i B_i^* P'_Q B_i$ . If every  $\Gamma_Q$  is block diagonal for the old blocks, then  $\alpha_Q(F') \leq \sum_p \|P_p \Gamma_Q P_p\| \alpha_p(F)$ .

For one block, (a) is [27, Lemma A.4]; the union over blocks costs only the logarithm of their number.

## G.2 Paths and their law

Let  $(B_i)$ ,  $B_i : K \rightarrow \mathbb{C}^{k'}$  with  $K = \mathbb{C}^k$ , be an exact rectangular dilation of  $\Phi$ . Fix isometries  $J_1, \dots, J_c : K \rightarrow \mathbb{C}^{k''} \supseteq \mathbb{C}^{k'}$  with orthogonal ranges whose sum contains  $\mathbb{C}^{k'}$ ; after restricting the codomain to the span of the ranges of the  $B_i$ , of dimension at most  $rk$ ,  $c \leq r+1$  suffices. Put  $C_i^{(b)} = J_b^* B_i$ . The *piece matrices*  $M^{(b)} \in M_r$  have entries  $\langle i|M^{(b)}|j \rangle = \tau_k(C_i^{(b)*} C_j^{(b)})$ ; they are positive and  $\sum_b M^{(b)} = I_r$ . Put

$$w(b) = \left\| \sum_{i,j} \langle i|M^{(b)}|j \rangle A_i^* A_j \right\| \leq \text{Tr } M^{(b)}, \quad u' = \log \sum_b w(b),$$

$\kappa_\theta = \sup\{\sum_b q_b w(b)^{-\theta}\}$  over probability vectors  $q$  with  $q_b \leq w(b)$  ( $0 < \theta \leq 1$ ), and  $W(p) = \prod_s w(p_s)$  for *paths*  $p \in [c]^t$ . The *type* of a path is its vector of letter counts.

**Lemma G.3** (The adaptive path law). *(a) If  $X_b$  are Hermitian with  $\|X_b\| \leq \beta$  and  $M^{(b)} \geq 0$  with  $\sum_b M^{(b)} \leq I$ , then  $\|\sum_b X_b \otimes M^{(b)}\| \leq \beta$ .*

*(b) Let an observer of the ideal experiment, adaptive and with quantum memory, end with the vector  $X$ , the  $n$  environments of the Stinespring dilations holding the Kraus labels. Then  $\Pr(p) = \langle X|I \otimes M^{(p_1)} \otimes \dots \otimes M^{(p_n)}|X \rangle$  is a probability law on paths with  $\Pr(b_1, \dots, b_s) \leq w(b_s) \Pr(b_1, \dots, b_{s-1})$ , and  $\Pr\{W(p) < 2^{-\lambda}\} \leq \kappa_\theta^n 2^{-\theta\lambda}$ . The set  $P_t(\lambda) = \{p \in [c]^t : W(p) \geq 2^{-\lambda}\}$  is closed under prefixes, is a union of types, and has at most  $2^{\lambda+tu'}$  elements.*

The law in (b) is not the product law of a fixed label sequence: an observer that superposes labels can make the conditional probability of the letter  $b$  equal to  $w(b)$ , which can exceed  $\max_i \langle i|M^{(b)}|i \rangle$  when  $M^{(b)}$  is not diagonal. This is why the device below counts paths by  $W$ .

**Theorem G.4** (Path-register streaming). *In this setting, for  $n \geq 2$ ,  $0 < \epsilon < 1$  and  $0 < \theta \leq 1$  there is a causal service of  $\Phi$  for  $n$  rounds with error at most  $\epsilon$ ,  $C = 0$ ,  $J = \lfloor \frac{n-1}{2} \log r \rfloor$  and*

$$Q \leq \log k + n \left( u' + \frac{\log \kappa_\theta}{\theta} \right) + \frac{\log(32n/\epsilon^2)}{\theta} + \log c + 4 + O_r(\log(n/\epsilon) + \log(2 + \log k)).$$

The device keeps the memory  $\bigoplus_{p \in P_t} K \otimes \mathbb{C}^m$ , with the path  $p$  stored as its rank in  $P_t = P_t(\lambda)$ ,  $\lambda = (n \log \kappa_\theta + \log(32n/\epsilon^2))/\theta$ . In each round it applies independent Haar unitaries inside the blocks, applies  $W \otimes I_m$ , which appends the piece  $b$  to the path, applies a fixed permutation of register values that keeps the paths in  $P_t$ , applies Haar unitaries inside the new blocks, and exports on the schedule of [28, Theorem 12.2]. Paths that leave  $P_t$  are discarded, and they can come back to good register values through the unitary completion of the permutation; this costs  $(1 + \sqrt{n})\sqrt{\omega}$  in the error, where  $\omega \leq \epsilon^2/(16n)$  is the discarded weight.

To apply the theorem to a round, the blocks are cut from the spectrum of its memory map.

**Lemma G.5** (Top eigenspace). *Let  $B_g : \mathbb{C}^k \rightarrow \mathbb{C}^{k'}$  be isometries,  $\mu$  a probability vector,  $\bar{\tau} = \sum_g \mu_g B_g B_g^*/k$  and  $a = S(\bar{\tau}) - \log k$ . If  $w$  is the mass of  $\bar{\tau}$  outside its top- $k$  eigenspace, then  $w \leq a \ln 2$ .*

**Theorem G.6** (Flat rounds with zero leakage). *Let  $\Phi$  be a gadget channel.*

*(a) Let  $(U, \rho)$  be an exact flat round with bath rank  $k$  and entropy production  $a$ . For  $n \geq 2$  and  $0 < \epsilon < 1$  there is a rank-rate service with error at most  $\epsilon$ ,  $C = 0$  and, when  $a \leq 1/(4d \ln 2)$ ,*

$$Q \leq \log k + 1.443 d \ln 2 n a + 2.89 \sqrt{\bar{\tau}} n \bar{\beta} \log(1/\bar{\beta}) \\ + 2 \log \frac{n}{\epsilon} \log \frac{32n}{\epsilon^2} + O_r(\log \frac{n}{\epsilon} + \log(2 + \log k)),$$

*with  $\bar{\beta} = \max(d a \ln 2, \epsilon^2/n^2)$ . For every  $a$ ,  $Q \leq K_{d,r} \log^2(n/\epsilon)(1 + S + n a)$ .*

- (b) Theorem G.4 holds for every isometry  $W = \sum_i A_i \otimes B_i$  (for a channel with a flat probe, with no condition on the traces) whose flat Gram  $G$  has  $\gamma = \|G - I\|$  with  $n\gamma \leq \frac{1}{2}$ , with the same memory and error at most  $\epsilon + \frac{n}{2}\gamma$ .
- (c) Let  $(U, \rho)$  be a flat round with zero leakage, bath rank  $k$  and entropy production  $a$ , whose gadget family has flat Gram off-diagonals at most  $c \leq \frac{1}{2}$ ; distance  $u \leq \frac{1}{4}$  suffices, since then  $c \leq 2u$ . With  $t = \lceil \log(4rn/\epsilon) / \log(1/c) \rceil$  there is a rank-rate service with error at most  $2\epsilon$ ,  $C = 0$  and  $Q \leq K_{d,r} \log^3(n/\epsilon)(1 + S + na)$ . When  $dt a \ln 2 \leq \frac{1}{4}$  its memory also satisfies

$$Q \leq t \log k + C_{d,r} n t a \log \frac{e}{ta} + O_{d,r}(\log^2(n/\epsilon) + \log(2 + t \log k)).$$

So flat rounds with zero leakage stream. The factor  $\log(1/a)$  that the path count adds to the entropy production is absorbed by the polylogarithmic allowance.

### G.3 Leakage

Leakage is an error, and a device must not carry it. The simplest situation is a round that is exact on most of its bath.

**Lemma G.7** (Restriction to an exact region). *Let  $(U, \rho)$  be a flat round with  $\rho = P_K/k$ , distance  $u$  and entropy production  $a$ , let  $K_0 \subseteq K$  have dimension  $(1 - \nu)k$  with  $0 < \nu \leq \frac{1}{2}$ , and  $\rho_0 = P_{K_0}/((1 - \nu)k)$ . Then  $S(\rho_0) = S + \log(1 - \nu)$ , the round  $(U, \rho_0)$  has distance at most  $(u + \nu)/(1 - \nu)$  and entropy production*

$$a_0 \leq \frac{a + \nu \log(1/\nu)}{1 - \nu} + \log \frac{1}{1 - \nu},$$

and its leakage is 0 if  $U(\xi \otimes \kappa) = \sum_i A_i \xi \otimes X_i \kappa$  for  $\kappa \in K_0$  (an exact region).

**Theorem G.8** (Exact regions and flat permutation rounds). *Let  $\Phi$  be a gadget channel.*

- (a) *A flat round with distance  $u \leq \frac{1}{16}$  and an exact region of codimension  $\nu k$ ,  $\nu \leq \frac{1}{8}$ , gives a rank-rate service with error at most  $2\epsilon$ ,  $C = 0$  and  $Q \leq K_{d,r} \log^3(n/\epsilon)(1 + S + na + n\nu \log(e/\nu))$ .*
- (b) *In a flat permutation round the triangle-good points span an exact region with  $\nu \leq 4d\ell$ . Hence for  $n \geq 32dC_0$  every flat permutation round with distance at most  $\frac{1}{16}$  and leakage  $\ell \leq C_0/n$  gives a rank-rate service with error at most  $2\epsilon$  and  $Q \leq K_{d,r} \log^3(n/\epsilon)(1 + S + na + 4dC_0 \log(en/(4dC_0)))$ .*

Part (b) covers the leakage scale of the lower half,  $\frac{32}{15}\epsilon/n$ , with no rescaling. Leakage of order  $1/n$  per round would add up to a constant error if a device simply used a leaky round; a sequential-measurement bound [35, Theorem 1] shows that the experiment projected onto an exact region in every round stays within  $\sqrt{\sum_t \epsilon_t}$  of the actual experiment, which tolerates only leakage of order  $\epsilon^2/n$ . Restricting the bath once, as above, is better. Without an exact region the round has to be rounded.

**Theorem G.9** (First-order rounding on the same bath). *Let  $\Phi$  be a gadget channel, and let  $(U, \rho)$  with  $\rho = P_K/k$  be a flat round with distance  $u$ , leakage  $\ell \leq \frac{1}{2}$  and entropy production  $a$ . There are isometries  $X'_g : K \rightarrow \mathbb{C}^r \otimes K$  forming a gadget family, whose flat round has the same bath entropy  $S' = S$ , leakage 0, flat Gram off-diagonals at most  $2u + C_1\sqrt{\ell}$ , and*

$$a' \leq a + \ell S + \ell / \ln 2 + 2\bar{m} \log d + h_2(2\bar{m}) + \bar{m} \log(r/\bar{m}), \quad \bar{m} = \ell + (3d^3r + 9dr^5)\sqrt{\ell} \leq \frac{1}{4}.$$

In particular  $a' \leq a + \ell S + C_2 \sqrt{\ell} \log(e/\ell)$  when  $C_m \sqrt{\ell} \leq \frac{1}{4}$ , where

$$C_1 = 2d^{3/2} + d + 3d^3 r + 18dr^2 + 9d^3 r^5, \quad C_m = 1 + 3d^3 r + 9dr^5, \\ C_2 = 1 + C_m(1.39 \log d + 0.70 \log r + 2.2).$$

The same holds for the twisted gadget  $\Phi_\omega$  of Section G.4.

The family is built from the Hilbert–Schmidt projection of the round onto the Kraus operators: shrink it by  $(I + P)^{-1/2}$ , where  $P$  collects the excess of the diagonal blocks, project the off-diagonal blocks onto the identities, absorb the negative part of the correction into the diagonal, and renormalize. No  $\log k$  multiplies the terms of order  $\sqrt{\ell}$ ; the bath entropy enters only through  $\ell S$ .

**Corollary G.10** (The flat upper half up to an additive  $\sqrt{n}$ ). *For a gadget channel,  $n \geq n_F := \lceil \frac{32}{15} C_1^2 \rceil$  and  $0 < \epsilon < 1$ , every flat round with distance at most  $\frac{1}{16}$  and leakage at most  $\frac{2}{15n}$  gives a rank-rate service with error at most  $2\epsilon$ ,  $C = 0$  and*

$$Q \leq K \log^3(n/\epsilon)(1 + S + n a'), \quad n a' \leq n a + \frac{2}{15} S + 0.37 C_2 \sqrt{n} \log(20n);$$

so  $Q \leq K' \log^4(n/\epsilon)(1 + S + n a + \sqrt{n})$ . At leakage at most  $(\log n)^m/n^2$  the additive term is  $O((\log n)^{m/2+1})$ .

So flat rounds stream up to an additive  $\sqrt{n} \log^4(n/\epsilon)$ , and the law holds for every gadget channel whose two-scale profile is attained by flat rounds and is at least  $\sqrt{n}/\text{polylog}(n)$ . The additive term is the price of a rounding at first order in  $\sqrt{\ell}$ . The next subsection shows that a rounding that keeps the bath cannot do better.

#### G.4 Optimality: the twisted torus

Let  $\Phi_\omega$  be the gadget of the relator  $\mathbf{aba}^{-1}\mathbf{b}^{-1}$  of  $\mathbb{Z}^2$  evaluated in the twisted regular representation,  $\lambda(\mathbf{b})\lambda(\mathbf{a}) = \omega\lambda(\mathbf{a})\lambda(\mathbf{b})$  with  $\omega = e^{2\pi i\theta}$  and  $\theta$  irrational. Each word  $w$  evaluates to  $\omega^{f(w)}$  times the section  $\hat{\lambda}(u, v) = \lambda(\mathbf{a})^u \lambda(\mathbf{b})^v$  at its label, for an integer  $f(w)$ ; put  $F = \max |f(w)|$  over the slot words. So the slots of the gadget carry phases, the channel has  $d = 17$  and  $r = 6$  labels, and it has a flat probe. A family  $(B_g)$  is a twisted gadget family if  $B_1^* B_{[y]} = \zeta_t B_{[x]}^* B_{[z]}$  on every triangle  $t$ , with the phases  $\zeta_t$  of the twisted regular representation, which gives one exactly. Lemmas 2.9 and 2.10 hold with these phases.

**Lemma G.11** (Exact flat rounds are trace-norm models). *Let  $(U, \rho = P_K/k)$  be a flat round of a gadget channel, or of  $\Phi_\omega$ , with leakage 0 and entropy production  $a$ , with gadget family  $B_g$ . There are unitaries  $V_g$  on  $\mathbb{C}^k$  with  $V_1 = I$  and, on every triangle, in the normalized trace norm,*

$$\|V_1^* V_{[y]} - \zeta_t V_{[x]}^* V_{[z]}\|_1 \leq 6d \ln 2 \cdot a,$$

and  $|\tau(V_g^* V_h)| \leq |\tau(B_g^* B_h)| + 3d a \ln 2$  for  $g \neq h$ .

So exactness puts the relation defect at second order: a round with entropy production  $a$  gives unitaries whose relations fail by  $O(a)$  in trace norm, not by  $O(\sqrt{a})$ .

**Theorem G.12** (The twisted obstruction). *Let  $\theta$  be irrational, and let  $\|x\|$  denote the distance from  $x$  to the nearest integer.*

(a) *Every zero-leakage round of  $\Phi_\omega$  with a flat bath of rank  $k$  has*

$$a \geq \frac{2\|k\theta\|}{9d \ln 2 \cdot k},$$

*whatever its distance. For  $\theta = (\sqrt{5} - 1)/2$  this gives  $a \geq 2/(27d \ln 2 \cdot k^2) \geq 0.0062/k^2$ .*

- (b) Let  $\theta = (\sqrt{5} - 1)/2$ . For Fibonacci numbers  $N = F_m$ , the clock-and-shift model on  $\mathbb{C}^N$  with commutator  $e^{2\pi i F_{m-1}/N}$  is a flat round of  $\Phi_\omega$  with  $a = 0$ ,  $S = \log N$ , leakage  $\ell < 4\pi^2 F^2/N^4$  and distance at most  $2\pi F\sqrt{d}/N^2$ . A rounding that replaces these rounds by zero-leakage flat rounds with  $S' \leq \lambda S + C_0$  and  $a' \leq a + C_1 \ell^\gamma (\log 1/\ell)^m$ , for constants  $C_0, C_1, m$ , must have  $\lambda \geq 2\gamma$ .
- (c) Theorem G.9 turns the rounds of (b) into zero-leakage flat rounds with  $S' = \log N$  and  $a' = O((\log N)/N^2)$ ; so (a) is sharp up to the logarithm.

At a fixed bath ( $\lambda = 1$ ) the loss must therefore be at least of order  $\sqrt{\ell}$ , which Theorem G.9 attains, and a loss of order  $\ell$  needs at least twice the bath entropy. Rounding at first order on the same bath is optimal. The upper half of the law at the full leakage scale needs a rounding that balances the two costs instead (the flat subcase in Section 9). The proofs of this section are in Appendices K and L.

## H Conserved bias and spatial constructions

A round whose bath is not flat is *biased*. Flattening a biased bath is not free: gluing the level sets of the bath into a flat bath pays the square root of the entropy production, because level sets are  $\ell^1$  objects while entropy production is an  $\ell^2$  quantity (Theorem H.12). Nor can a device simply re-randomize a biased memory, since that flattens it and fills the rank-grown space (Theorem I.1). Theorem 6.3 supplies general exact-support streaming. This appendix gives specialized conserved-bias and spatial constructions, followed by the biased leakage repair used in the main profile bounds. Throughout,  $(B_g)$  denotes the gadget family of a zero-leakage round on  $K = \text{supp } \rho$  (Lemma 2.10), and  $G_{gh} = \text{Tr}(\rho B_g^* B_h)$  its weighted Gram.

### H.1 Tensor powers and conserved bias

**Lemma H.1** (Tensor powers). *For a zero-leakage round  $(U, \rho)$  of a gadget channel and  $t \geq 1$ , the family  $(B_g^{\otimes t})$  extends to a round with bath  $\rho^{\otimes t}$ , leakage 0, weighted Gram  $(G_{gh}^t)$ , distance at most  $\frac{1}{2}(r-1) \max_{g \neq h} |G_{gh}|^t$ , bath entropy  $tS$  and entropy production at most  $t a$ . In particular, if  $|G_{gh}| \leq c < 1$  for  $g \neq h$  and  $t = \lceil \log(4rn/\epsilon) / \log(1/c) \rceil$ , every off-diagonal is at most  $\epsilon/(4rn)$ .*

So a weighted Gram bounded away from one costs only a factor  $O(\log(n/\epsilon))$ , for any bath. The first class that streams is the one in which the bias is a conserved quantity of the round.

**Theorem H.2** (Conserved bias). *Let  $(U, \rho)$  be a zero-leakage round of a canonical gadget with a finitely supported level law  $(w_s)$ , such that  $K = \bigoplus_s E_s$ ,  $e_s = \dim E_s$ ,  $\rho = \bigoplus_s (w_s/e_s) I_{E_s}$ , there are mutually orthogonal subspaces  $E'_s$  with  $B_g E_s \subseteq E'_s$  for all  $g$  and  $s$ , and the weighted Gram has off-diagonals at most  $\frac{1}{8}$  (distance  $u \leq \frac{1}{16}$  suffices). The per-level Grams may be arbitrary. Then  $a = \sum_s w_s a_s$  exactly, with  $a_s \geq 0$  the entropy production of level  $s$  at flat input, and for  $n \geq 2$  and  $0 < \epsilon < 1$  there is a rank-rate service with error at most  $2\epsilon$ ,  $C = 0$  and  $Q \leq K_{d,r} \log^3(n/\epsilon)(1 + S + n a)$ , with  $t = \lceil \log(4rn/\epsilon)/2 \rceil$ . When  $1.07 t a \leq 1$  its memory also satisfies*

$$Q \leq t(32S + 5.1) + C_{d,r} n (1.07 t a) \log \frac{e}{1.07 t a} + O_{d,r}(\log^2(n/\epsilon) + \log(2 + tS)).$$

*Without any device: such a round gives a zero-leakage flat round with flat Gram off-diagonals at most  $\frac{1}{4}$ ,  $S' \leq 32S + 5.1$  and  $a' \leq 1.07 a$ .*

The device truncates the levels at surprisal  $32S$ , which costs no binary entropy because  $a$  is a conserved average, flattens the level index into a flat bath with multiplicities proportional

to  $w_s/e_s$ , and streams by Theorem G.6(c). So conserved bias never beats flat rounds by more than constants. Leaky rounds of this kind stream too, up to the additive term of Corollary G.10; the tool is a restriction that commutes with the bath.

**Lemma H.3** (Restriction for biased baths). *Let  $(U, \rho)$  be any round and  $P_0$  a projection commuting with  $\rho$ , with  $\nu = \text{Tr}((I - P_0)\rho) < 1$  and  $\rho = (1 - \nu)\rho_0 \oplus \nu\rho_\perp$ . Then  $S(\rho_0) \leq S(\rho)/(1 - \nu)$ , the round  $(U, \rho_0)$  has distance at most  $(u + \nu)/(1 - \nu)$ , leakage at most  $\ell/(1 - \nu)$  and*

$$a(\rho_0) \leq \frac{a + h_2(\nu)}{1 - \nu},$$

*and its leakage is 0 if  $U$  has the form  $\sum_i A_i \otimes X_i$  on  $\mathbb{C}^d \otimes \text{ran } P_0$ .*

**Corollary H.4** (Leaky conserved bias). *Let  $(U, \rho)$  be a round of a canonical gadget with a finitely supported level law  $(w_s)$  and  $K = \bigoplus_s E_s$ ,  $\rho = \bigoplus_s (w_s/e_s)I_{E_s}$ , and, with  $W = U(I \otimes \iota_K)$  for the inclusion  $\iota_K$  of  $K$ ,  $W(\mathbb{C}^d \otimes E_s) \subseteq \mathbb{C}^d \otimes E'_s$  with the  $E'_s$  mutually orthogonal, distance  $u \leq \frac{1}{16}$  and leakage  $\ell \leq c_1/n$ . For  $n \geq n_{CB}(d, r, c_1)$  and  $0 < \epsilon < 1$  there is a rank-rate service with error at most  $2\epsilon$ ,  $C = 0$  and  $Q \leq K' \log^4(n/\epsilon)(1 + S + na + \sqrt{n})$ .*

## H.2 Rounds that are vacuous for the law

A zero-leakage round of a channel whose labels a finite-dimensional representation separates needs no device: the channel has logarithmic memory (Corollary 7.8). The following lemma shows that a zero-leakage round can only be “finite-dimensional” in that sense.

**Lemma H.5** (Compressions of finite-dimensional representations). *Let  $(U, \rho)$  be a zero-leakage round of a gadget channel with distance  $u < \frac{1}{2}$ . If  $B_g^* B_h = J^* \pi(g)^* \pi(h) J$  for a unitary representation  $\pi$  of  $P_\Phi$  on a finite-dimensional space and an isometry  $J$ , then  $\pi$  separates the labels, and  $Q^*(n, \epsilon) = O_\Phi(\log(n/\epsilon))$ . This applies to every square round ( $B_g K \subseteq K$  for every label), abelian or not.*

So for a channel that needs more than logarithmic memory, every zero-leakage round at distance below  $\frac{1}{2}$  is non-square, and its Gram operators are not jointly a compression of a finite-dimensional representation.

## H.3 Permutation rounds and the balance condition

A family is *block-monomial* if  $K = \bigoplus_{\omega \in \Omega_0} K_\omega$ , there are mutually orthogonal subspaces  $K'_\eta$  ( $\eta \in \Omega \supseteq \Omega_0$ ) with  $K'_\omega = K_\omega$  for  $\omega \in \Omega_0$ , and each  $B_g$  maps  $K_\omega$  isometrically onto  $K'_{f(g)\omega}$  for maps  $f(g)$  with  $f(1) = \text{id}$ ; it is *free* if  $f(g)\omega \neq f(h)\omega$  for  $g \neq h$  and  $\omega \in \Omega_0$ . A bath is block diagonal if  $\rho = \bigoplus_\omega \rho_\omega$ , with base law  $p(\omega) = \text{Tr } \rho_\omega$ .

**Lemma H.6** (Free block-monomial rounds). *In a free block-monomial zero-leakage round with a block-diagonal bath, the base family  $\lambda_g|\omega\rangle = |f(g)\omega\rangle$  is a gadget family, and with the bath  $p$  it defines an exact round (Lemma 2.9(c)) with  $S_{\text{base}} = H(p) \leq S(\rho)$  and  $a_{\text{base}} \leq a$ . Here each  $f(g)$  is an injection  $\Omega_0 \rightarrow \Omega$ ; the base round is a free exact permutation round when these injections extend to permutations of  $\Omega$  that are triangle-good on the support of  $p$ .*

So it suffices to stream free exact permutation rounds. The devices for them keep a set of points, twirl a few slots attached to each point, and move the points around inside fixed windows.

**Theorem H.7** (Point-block devices). *Let a free permutation round of a gadget channel be given on  $\Omega$ , with windows  $\Omega_0 \subseteq \dots \subseteq \Omega_n \subseteq \Omega$  fixed in advance on which it is triangle-good*

and collision-free. Consider the device whose memory is  $\mathbb{C}^\Omega \otimes \mathbb{C}^m$  with point blocks  $|\omega\rangle \otimes \mathbb{C}^m$ , which starts in  $p_0 \otimes I/m$  for a law  $p_0$  on  $\Omega_0$  and in round  $t$  applies a fixed permutation  $\beta_t$  of the points of  $\Omega_{t-1}$ , independent Haar unitaries on the point blocks, the round, an optional fixed partial injection of points, sends the points outside  $\Omega_t$  to a garbage register by a fixed unitary completion, applies Haar unitaries on the new point blocks and exports on the schedule of [28, Theorem 12.2]. Let  $\kappa(y)$  be the probability that the ideal killed chain of the label sequence  $y$  (start from  $p_0$ , permute by  $\beta_t$ , move by  $f(y_t)$  and the injection, kill outside  $\Omega_t$ ) is killed, and  $\omega_{\text{ideal}} = \sum_y \Pr(y) \kappa(y)$  for the observer's ideal law of label sequences. For  $n \geq 2$ ,  $0 < \epsilon < 1$  and  $0 < \zeta \leq \epsilon^2/(64n)$  there is  $m$  with  $\log m = O_r(\log(n/\zeta) + \log \log |\Omega|)$  such that for every observer the discarded weight is at most  $\omega_{\text{ideal}} + \zeta$ , the error is at most  $(1 + \sqrt{n})\sqrt{\omega_{\text{ideal}} + \epsilon^2/(16n)}$ , and at the start of every round the memory has mass at most  $n(\omega_{\text{ideal}} + \zeta)$  outside the window. The memory is  $\log \max_t |\Omega_t| + \log m + O(1)$  qubits at the rank rate.

The quantum side of the device is thus unconditional; what remains is the containment of a classical chain. Discarded branches can come back to the window through the unitary completion, which is why the error is  $(1 + \sqrt{n})$  times the square root of the discarded weight and not the weight itself. Random permutations inside classes of points make the chain mix.

**Lemma H.8** (Random class permutations). *Fix windows and partitions of each  $\Omega_{t-1}$  into classes in advance, and let  $\beta_t$  be independent uniformly random permutations of each class; a fixed partial injection of points after each round is allowed. Let  $\kappa_{\text{class}}(y)$  be the kill probability of the class-uniformized chain of  $y$  (start uniform on  $\Omega_0$ , resample uniformly inside the current class before each step, move, kill outside  $\Omega_t$ ), and  $\kappa_\beta(y)$  the fraction of  $\Omega_0$  whose trajectory under the  $\beta$ 's is killed. For every  $y$  and  $z > 0$ ,*

$$\Pr_\beta\{\kappa_\beta(y) \geq \kappa_{\text{class}}(y) + z\} \leq \exp(-2z^2|\Omega_0|/n),$$

so with probability at least  $\frac{1}{2}$ ,  $\max_y \kappa_\beta(y) \leq \max_y \kappa_{\text{class}}(y) + \sqrt{n(n \ln r + \ln 2)/(2|\Omega_0|)}$ .

**Definition H.9** (The balance condition). A free exact permutation round with bath  $p$ , of entropy  $S$  and entropy production  $a$ , satisfies the *balance condition* at  $(n, \epsilon)$  with exponent  $\kappa$  if there are windows  $\Omega_0 \subseteq \dots \subseteq \Omega_n$  on which the round is triangle-good and collision-free and partitions of each  $\Omega_{t-1}$  into classes, all fixed in advance, such that

- (B1)  $|\Omega_0| \geq 512n^3(n \ln r + \ln 2)/\epsilon^4$ ;
- (B2)  $\log \max_t |\Omega_t| \leq (\log(n/\epsilon))^\kappa(1 + S + na)$ ;
- (B3) for every label sequence the class-uniformized chain is killed with probability at most  $\epsilon^2/(32n)$ .

The windows and classes may contain slots: a point of  $\Omega \times [m_t]$  is a point of  $\Omega$  together with a slot that the labels do not move.

**Corollary H.10** (Balance streams). *A free exact permutation round that satisfies the balance condition at  $(n, \epsilon)$  has a rank-rate service with error at most  $0.61\epsilon$  and  $Q \leq (\log(n/\epsilon))^\kappa(1 + S + na) + O_r(\log(n/\epsilon) + \log \log |\Omega_W|)$ , where  $\Omega_W$  is the union of the windows of the balance condition and their images under the slot words. The same holds for a free block-monomial round whose base round, with each  $f(g)$  extended to a permutation of  $\Omega$  that is triangle-good on the windows, satisfies the balance condition.*

The given bath enters only through the budget (B2): on the windows every round is the gadget unitary of the round, which is exact for every memory state there, so the device may start the chain wherever it likes. It must, because the chain cannot start from the given bath.

**Proposition H.11** (The start belongs to the device). *For the gadget of  $\mathbb{Z}$  with labels  $1, a, a^{-1}$ , and every  $n \geq 2$ , there is a bath  $p$  on a cycle  $\mathbb{Z}_{N'}$ ,  $N' \geq 2^n + 3$ , all of whose points are triangle-good and collision-free, with  $S + na = O(\log n)$ , such that a class-uniformized chain started from  $p$  is killed at its first step with probability at least  $\epsilon^2/(32n)$  unless  $\log |\Omega_0| \geq n - 1$ .*

#### H.4 The Faber–Krahn form

For a permutation round the cost  $S + na$  of a bath has a geometric meaning. For  $\varphi : \Omega \rightarrow \mathbb{R}$  put  $\mathcal{E}(\varphi) = \sum_g \sum_x (\varphi(f(g)x) - \varphi(x))^2$ , and let

$$\Lambda(v) = \inf \left\{ \frac{\mathcal{E}(\varphi)}{\|\varphi\|^2} : \varphi \geq 0, \varphi \neq 0, |\text{supp } \varphi| \leq v \right\}$$

be the  $\ell^2$ -isoperimetric profile of the Schreier graph, the Faber–Krahn function of Grigor’yan [39];  $\Lambda_{\text{good}}$  is the same infimum over  $\varphi$  supported on triangle-good, collision-free points. Put  $C_\mu = 2 \ln 2 \sum_{g \neq 1} (1/\mu_g + 1/\mu_1) \leq 4(r-1)d \ln 2$ .

**Theorem H.12** (The entropic Faber–Krahn bound). *For every permutation round and every bath  $p$  supported on triangle-good points:*

- (i)  $\mathcal{E}(\sqrt{p}) \leq C_\mu a$ ;
- (ii)  $a \geq \Lambda(2^{2S+4})/(8C_\mu)$ , and  $a \geq \Lambda_{\text{good}}(2^{2S+4})/(8C_\mu)$  when  $p$  also lives on collision-free points;
- (iii) conversely, every  $\varphi \geq 0$  on the good points with  $\|\varphi\| = 1$  and  $|\text{supp } \varphi| \leq v$  gives the bath  $p = \varphi^2$  of an exact round with  $S \leq \log v$  and  $a \leq (16r/\ln 2) \mathcal{E}(\varphi)$ .

Hence, writing  $L^{\text{perm}}(n)$  for the least  $S + na$  over zero-leakage permutation rounds on  $(\Omega, f)$  whose baths live on triangle-good, collision-free points,

$$\min_S \left[ S + \frac{n \Lambda_{\text{good}}(2^{2S+4})}{8C_\mu} \right] \leq L^{\text{perm}}(n) \leq \min_v \left[ \log v + \frac{16r}{\ln 2} n \Lambda_{\text{good}}(v) \right].$$

The two-scale cost of permutation rounds is therefore the Faber–Krahn profile of the Schreier graph, up to constants and the change  $v \mapsto 16v^2$  of the argument (compare [21, Proposition 3.14], which relates sofic and isoperimetric profiles); the balance condition asks for a device whose windows cost  $\text{polylog} \cdot \min_v [\log v + n\Lambda(v)]$ . The flat bath of a level set pays the  $\ell^1$  (Cheeger) profile instead, which is the square root of the  $\ell^2$  one in the worst case. On groups of exponential growth the Faber–Krahn profile is large.

**Proposition H.13** (Exponential growth costs  $n^{1/3}$ ). *Let the labels generate a group whose balls in  $\mathcal{L} \cup \mathcal{L}^{-1}$  satisfy  $|B(\varrho)| \geq \omega_0^{\varrho}$  for some  $\omega_0 > 1$ , and put  $\phi(v) = \min\{\varrho : |B(\varrho)| \geq v\}$ . Consider a permutation round in which, for some choice of geodesic words  $w_x$ , of length  $|x|$ , for the elements  $x$  of  $B(\varrho_*)$ ,  $\varrho_* = \phi(2^{2S+5})$ , the points  $f(w_x)y$  are distinct for every  $y \in \text{supp } p$ , and  $p$  is supported on triangle-good points (both hold for the regular action). Then*

$$a \geq \frac{1}{128r C_\mu \phi(2^{2S+5})^2} \quad \text{and} \quad S + na \geq \frac{3}{2^{2/3}} \left( \frac{c}{\alpha^2} \right)^{1/3} n^{1/3} - \frac{\beta_0}{\alpha},$$

with  $c = 1/(128r C_\mu)$ ,  $\alpha = 2/\log \omega_0$  and  $\beta_0 = 5/\log \omega_0 + 1$ .

The first step of the proof is the isoperimetric inequality of Coulhon and Saloff-Coste [22, Théorème 1], for the Schreier graph near the bath; a co-area argument then passes from  $\ell^1$  to  $\ell^2$ .

## H.5 Devices that satisfy the balance condition

**Re-folding.** The basic device sorts its units by distance from a centre and refills the inner positions first, while the number of slots per point grows slowly. Points  $c + (\mathbb{Z} + \frac{1}{2})$  around a half-integer centre  $c$  carry  $m_t$  slots at time  $t$ , with  $m_t = \lceil (1 + \gamma)m_{t-1} \rceil$ , so that  $\gamma_t := m_t/m_{t-1} - 1 \geq \gamma$ ; the classes are the pairs  $\{c \pm D\}$  times the slots. After the round the device shifts by a fixed integer and *re-folds by distance*: the units are ranked by distance from  $c$ , the two points of a pair interleaved, and the unit of rank  $N$  goes to the pair  $\lfloor N/(2m_t) \rfloor$ ; units beyond the window are discarded.

**Lemma H.14** (The re-folding chain). *(a) Under independent class-uniform permutations, the distance  $D_t$  of the branch of a fixed label sequence with integer steps  $\delta_t$  is the Markov chain  $X = |D + \sigma\delta_t|$  with a uniform sign  $\sigma$ , followed by  $D' = X - B$  with  $B \sim \text{Bernoulli}((X - \frac{1}{2})\gamma_t)$  if  $(X - \frac{1}{2})\gamma_t \leq 1$ , and  $D' \leq X - 1$  otherwise.*

*(b) Let  $|\delta_t| \leq \delta_{\max}$  ( $\delta_{\max} \geq 1$ ),  $D_0 \leq R_0$ ,  $0 < \eta < 1$ ,  $L = \ln(n(1 + 8/\gamma)/\eta)$  and  $\gamma L \leq 1/(2\delta_{\max}^2)$ . Then*

$$\Pr(\exists t \leq n : D_t \geq R_0 + 2\delta_{\max}\sqrt{2L/\gamma} + 2\delta_{\max} + \tfrac{1}{2}) \leq \eta.$$

The spread is set by the capacity growth  $\gamma$ , not by the shape of the bath: the chain has stationary distance about  $|\delta|/\sqrt{2\gamma}$ , and no automorphism of the group is used.

**Theorem H.15** (Fibred translation devices). *Let a gadget channel have its relations in a group  $G$  in which its labels are distinct, acting on itself by the regular action. Let  $\Omega_B \subseteq G$  carry a coordinate map  $\pi : \Omega_B \rightarrow \mathbb{Z}^k$  whose fibres have a common size  $\Phi_f$ , let the slot words have integer translations  $\pi(w)$ , and let  $B \subset \mathbb{Z}^k$  be a box such that for every  $v \in B$  and every slot word  $w$  the action of  $w$  maps the fibre over  $v$  bijectively onto the fibre over  $v - \pi(w)$ , which lies in  $\Omega_B$ . Let  $r_0 \in \mathbb{Z}^k$ ,  $\delta_i = \max_g |\pi_i(g) - r_{0,i}|$ ,  $\delta_{\max} = \max_i \delta_i \geq 1$ , and let the window  $W_x = \prod_i [c_i - x_i, c_i + x_i] \subseteq B$  have half-integer centres and  $x_i = 1 + 2\delta_i + 2\delta_i\sqrt{2L/\gamma}$ , where  $n \geq 2$ ,  $0 < \epsilon \leq \frac{1}{16}$ ,  $0 < \gamma < 1$ ,  $0 < \eta \leq \min(\epsilon^2/(64n), 1/(4n^3))$ ,  $L = \ln(kn(1 + 8/\gamma)/\eta)$  and  $\gamma L \leq 1/(2\delta_{\max}^2)$ . Let  $\Omega_W$  be  $\pi^{-1}(W_x)$  together with its images under the slot words. Then the channel has a rank-rate service with  $C = 0$  and error at most  $\epsilon$  in which every round acts on the window as the gadget unitary of the regular action, the memory has mass at most  $3/(4n^2)$  outside the window at the start of every round, for every observer, and*

$$Q \leq \log |\Omega_W| + nk \log(1 + \gamma) + O_{r,k}(\log(n/\epsilon) + \log \log |\Omega_W|).$$

For  $\mathbb{Z}^k$  (trivial fibres,  $\gamma = 1/n$ ) this is  $O_{k,r}(\log(n/\epsilon))$ .

**Collars.** The same mechanism works on any Schreier graph that has nested sets with balanced shells. Let  $F_0 \subset F_1 \subset \dots \subset F_{K+1} \subseteq \Omega$  satisfy  $f(g)F_j \subseteq F_{j+1}$  and  $f(g)^{-1}F_j \subseteq F_{j+1}$  for every label and  $j \leq K$ ; put  $L_j = F_j \setminus F_{j-1}$  (the *shells*,  $L_0 = F_0$ ),  $s_j = |L_j|$ ,  $V_j = |F_j|$  and the cut fluxes  $\Phi_g(j) = |f(g)F_j \setminus F_j|$ . This is a *collar* of depth  $K$ .

**Proposition H.16** (Collar devices). *(a) For  $x$  uniform on  $L_j$  ( $j \geq 1$ ),  $f(g)x \in L_{j-1} \cup L_j \cup L_{j+1}$ , and  $s_j p_+(j) = \Phi_g(j)$ ,  $s_j p_-(j) = \Phi_g(j-1)$ : for every label the shell index is a birth-death chain, reversible for the shell sizes, with drift  $(\Phi_g(j) - \Phi_g(j-1))/s_j$ .*

*(b) Use the classes  $\{F_0, L_1, \dots, L_K\}$  times the slots, random class permutations, point-block twirls, the round, and the volume re-fold (units ranked shell by shell and slot by slot inside a shell, ranks kept, units of rank at least  $V_K m_t$  discarded). Suppose  $s_j \in [s, 2s]$  for  $1 \leq j \leq K$ ,  $(\Phi_g(j) - \Phi_g(j-1))_+ \leq \zeta s$ ,  $|F_K \setminus F_0| \leq |F_0|$ ,  $\zeta \leq L_c/K$ ,  $K \geq 8L_c$  and  $m_0 \geq \max(n, 4)$ , where  $L_c = \ln(n(1 + 8K^2)/\eta)$ . Then  $\gamma = 2s(2\zeta + 4L_c/K + 1/m_0)/|F_0|$*

gives kill probability at most  $\eta$  for every label sequence of the class-uniformized chain started in the core. With  $\eta = \min(\epsilon^2/(64n), 1/(4n^3))$ , random class permutations and Theorem H.7, this is a rank-rate service with error at most  $\epsilon$  and memory

$$Q \leq \log |F_{K+1}| + 17.4n L_c/K^2 + O_r(\log(n/\epsilon) + \log \log |F_{K+1}|).$$

The drift of each label is the discrete derivative of its boundary flux, so balanced shells make every label diffusive. For the regular action of  $\mathbb{Z}_m \wr \mathbb{Z}$ , with labels that move the position by at most one and toggle only at the current position, the sets of configurations with lamps in  $[-2K, 2K]$  and position in  $[-K-j, K+j]$  form a collar with constant shells and fluxes, and the cost is  $\min_K[(4K+1)\log m + 17.4nL_c/K^2] = O(n^{1/3}(\log m)^{2/3}L_c^{1/3})$ .

**Theorem H.17** (The collar condition). *Say that  $(\Omega, f)$  satisfies the collar condition with constant  $\kappa$  if there are collars inside the good, collision-free points with shell sizes  $s_j \in [s, 2s]$ , flux increments  $(\Phi_g(j) - \Phi_g(j-1))_+ \leq \zeta s$ ,  $|F_K \setminus F_0| \leq |F_0|$  and  $\zeta \leq 1/K$  such that (a) for every  $v \geq 2$  there is one of depth  $K$  with  $\log |F_{K+1}| \leq \kappa(1 + \log v)(\log(2 + \log v))^\kappa$  and  $K^{-2} \leq \kappa(\log(2 + \log v))^\kappa \Lambda(v)$ , and (b) for every  $K_0$  there is one of depth at least  $K_0$  with  $\log |F_{K+1}| \leq \kappa K_0^\kappa$ . Then, with the devices of Proposition H.16(b) run at  $m_0 \geq \max(n, 4)$ , every bath on  $(\Omega, f)$  satisfies the balance condition with windows of size*

$$\log \max_t |\Omega_t| \leq (\log(n/\epsilon) + \log(2 + S))^{\kappa'} (1 + S + na), \quad \kappa' = \kappa'(\kappa), \quad n \geq n_0.$$

For the regular action of a group of exponential growth, collars of size  $e^{O(K \log K)}$  at every depth  $K$  suffice.

**Theorem H.18** (Houghton groups). *Let  $q \geq 3$  and let a gadget channel have its relations in Houghton's group  $H_q$  and its labels distinct there, given by slot words of length at most  $\ell$  whose translation parts are also bounded by  $\ell$ . The implied constants below depend on  $q$ , on  $d$  and  $r$ , on a bound  $\varrho_0$  for the defect radii  $\varrho(w^{-1})$  of the inverse slot words, defined in the proof, and on the growth rate of the label subgroup. There is a rank-rate service with error at most  $\epsilon$  in which every round acts on a window of the regular action of  $H_q$ , with memory mass at most  $3/(4n^2)$  outside it, and*

$$Q = O(q^{5/3}(\ell^2 L)^{1/3} n^{1/3} (\log n)^{2/3} + q^2 \ell^2 L \log(q\ell L)) + O(\log(n/\epsilon)), \quad L = O(\log(n/\epsilon)).$$

If the labels generate a subgroup of exponential growth, every bath on the regular action has  $S + na \geq c_q n^{1/3} - C_q$ , so  $Q \leq K \log^2(n/\epsilon)(1 + S + na)$  for  $n \geq n_0$ : the regular-action rounds of the channel stream. Otherwise the channel has  $Q^*(n, \epsilon) = O(\log(n/\epsilon))$  by Corollary 7.8.

The window comes from a coarse fibration of  $H_q$  by its translation part, whose cocycles stay in a finite symmetric group whose support grows linearly with the window (Lemma M.1); this is the mechanism of the corrected extension bound [26, Proposition 7.1]. The comparison with the bath is Proposition H.13. The dichotomy in the statement is Chou's theorem: a finitely generated elementary amenable group has polynomial or exponential growth [17, Theorem 3.2], and in the first case it is virtually nilpotent, hence residually finite.

**Proposition H.19** (The ball device, and the configuration-lamp group of [28]). *On a regular action, the device that starts at a point  $x_0$ , uses the word balls  $\Omega_t = \{f(w)x_0 : |w| \leq t\}$  as windows and point classes, and never re-folds, never kills a branch; Theorem H.7 gives error at most  $0.53\epsilon$  and  $Q \leq n \log(2r + 1) + O_r(\log(n/\epsilon))$ . For the configuration-lamp group  $\Gamma$  of [28], every bath on the regular action has  $S + na \geq c'n/(\log n)^2$ , so the ball device streams every regular-action round of the configuration-lamp channel.*

The cheap rounds of the configuration-lamp channel are not regular-action rounds: its sofic models give  $L^*(n) \leq n^{1/2+o(1)}$ , and the law holds for it by Corollary 7.9 with no device of this section. The configuration-lamp group also shows that the regular action can be far from the cheapest one,  $n/(\log n)^2$  against  $n^{1/2+o(1)}$ .

## H.6 Leaky biased rounds

**Theorem H.20** (Leaky permutation rounds). *Let a permutation round have any diagonal bath  $p$ , distance  $u \leq \frac{1}{16}$ , leakage  $\ell \leq C_0/n$  and  $n \geq 64dC_0$ . There is a free exact permutation round, on the product model  $f^{\times t}$  with  $t = \lceil \log_{15/4}(r^2 n^2/2) \rceil$  and a bath on triangle-good, collision-free tuples, with distance 0,*

$$S' \leq \frac{tS}{(1-\nu_1)(1-\nu_2)}, \quad n a' \leq (1 + O(\nu_1)) t n a + O(t d C_0 \log n) + O(1),$$

where  $\nu_1 \leq 4d\ell \leq \frac{1}{16}$  and  $\nu_2 \leq 1/n^2$ . So  $S' + n a' \leq O(\log(rn))(1 + S + n a + dC_0 \log n)$ , and the round streams if the output satisfies the balance condition with windows inside the good, collision-free tuples.

For general biased rounds we need a rounding that works for every bath. Write  $W = U(I \otimes \iota_K) = G + L$  for the restriction of the round to its support, with  $G = \sum_g A_g \otimes X_g$  the Hilbert–Schmidt projection onto the Kraus operators and  $L$  the rest, and call  $\Lambda = \text{Tr}_d(L^* L)$  on  $K$  the leak operator; then  $\ell = \text{Tr}(\rho \Lambda)/d$ .

**Lemma H.21** (Entropy production is a conditional entropy). *For every round and every state  $\sigma$  on  $K$ ,  $a(\sigma) = H(C|R)$  for the state of  $C$ , the output together with the reference of a maximally entangled input, and  $R$ , a purifying reference of  $\sigma$ . If the round is exact on  $K$  with gadget family  $B_g$ , then  $a(\sigma) = H(L|R)$  for the state  $\sum_{g,h} \sqrt{\mu_g \mu_h} |g\rangle \langle h| \otimes \sqrt{\sigma} B_g^* B_h \sqrt{\sigma}$  on  $\mathbb{C}^r \otimes K$ , whose  $K$ -marginal is  $\sigma$ .*

**Lemma H.22** (Operator domination of the defects). *If  $W = G + L$  is an isometry and  $0 < t < 1$ , then  $\pm(G^* G - I) \leq (tI + (1 + 1/t)L^* L)/(1 - t)$ . For a gadget round, with  $\beta_t = (tI + (1 + 1/t)\Lambda)/(1 - t)$  on  $K$ , every diagonal defect satisfies  $-\beta_t \leq X_g^* X_g - I \leq \beta_t$ , and every triangle defect  $R_t = X_1^* X_{[y]} - X_{[x]}^* X_{[z]}$  satisfies  $\begin{pmatrix} 2\beta_t & R_t \\ R_t^* & 2\beta_t \end{pmatrix} \geq 0$ .*

Every defect of a leaky round is thus dominated in operator order by one operator built from the leak. The rounding puts the correction that this domination requires into the bath rather than into the family.

**Theorem H.23** (Biased first-order rounding). *Let  $\Phi$  be a gadget channel and  $(U, \rho)$  a round with any finite bath, distance  $u$ , leakage  $\ell$ , entropy production  $a$  and bath entropy  $S$ , with  $s := \sqrt{d\ell} \leq 1/(46c_r)$ , where  $c_r = 1 + 2(r - 1)(r^2 - 1)$ . There is a zero-leakage round of  $\Phi$  with*

- (1) *weighted Gram off-diagonals at most  $2u(1 + 1.05c_r s) + 4.2c_r s$ , which is at most  $\frac{1}{4}$  when  $u \leq \frac{1}{16}$ ;*
- (2) *bath entropy  $S' \leq (2 + 3.4c_r s)S + 3.1c_r s$ ;*
- (3) *entropy production  $a' \leq \frac{a + h_2(\nu)}{(1 - \nu)(1 - \ell_0)} + \frac{h_2(\ell_0) + \ell_0 \log(d^2 - r)}{1 - \ell_0} + 2p \log r + h_2(p)$ , with  $\nu \leq c_r s$ ,  $\ell_0 \leq \ell/(1 - \nu)$  and  $p \leq 2.2c_r s$ .*

*At the scale of the law: if  $\ell \leq c_1/n$  and  $n \geq n_{BF} := \lceil 2116c_r^2 d c_1 \rceil$ , then  $S' \leq 2.08S + 0.07$  and  $n a' \leq (1 + O(n^{-1/2})) n a + O_{d,r,c_1}(\sqrt{n} \log n)$ .*

The construction restricts the bath to the eigenvectors of  $\rho$  on which the leak is small (a restriction that commutes with  $\rho$ ), dominates every defect by  $\beta_s$ , projects the off-diagonal blocks onto the identities, fills the diagonal, and absorbs the correction into the bath,  $\rho' = (\rho_0 + \sqrt{\rho_0} Z \sqrt{\rho_0})/(1 + z)$ . The new label–bath state of Lemma H.21 is a mixture of the old Kraus part and a positive correction of weight  $p$ , so the entropy production moves linearly in  $\sqrt{\ell}$ , and the bath entropy at most doubles. No bath dimension, spread or truncation enters.

**Corollary H.24** (The upper half up to an additive  $\sqrt{n} \log n$ ). *For a canonical untwisted anchored gadget, for  $n \geq \max(n_0, n_{BF})$  (with  $c_1 = \frac{2}{15}$ ) and  $0 < \epsilon \leq \frac{1}{16}$ ,*

$$\frac{L^*(n) - 13.6 - 0.14 \log n}{7.4} \leq Q^*(n, \epsilon) \leq C_{d,r} \log_2^2 \left( \frac{128(n+1)}{\epsilon} \right) (1 + L^*(n) + \sqrt{n} \log n).$$

The upper has  $C = 0$ ,  $P \leq Q$  and the stated rank-rate exchange. Thus the two-scale law holds for every such gadget with  $L^*(n) \geq \sqrt{n}/\text{polylog}(n)$ , whatever its optimal rounds look like. The proofs of this section are in Appendix M.

## I Limits of specified constructions

This appendix gives method-scoped obstructions: the device of [28, Theorem 12.2] cannot reach Shannon cost, a twisted rounding cannot keep the bath at second order, a rounding cannot keep square families square, and the one-round method of [27] cannot certify the Houghton exponent.

### I.1 Global scrambling

Let  $\Phi_{\mathbb{Z}}$  be the gadget of  $\mathbb{Z}$  with labels  $1, \mathbf{a}, \mathbf{a}^{-1}$ :  $d = 5$  and label weights  $\mu = (\frac{2}{5}, \frac{3}{10}, \frac{3}{10})$ .

**Theorem I.1** (Global scrambling pays rank room). *(a) Let  $k \geq 3$ ,  $U$  the cyclic shift on  $\mathbb{C}^k$  and  $0 < \vartheta < \pi/2$ . The maps*

$$B_1 = \begin{pmatrix} I \\ 0 \end{pmatrix}, \quad B_{\mathbf{a}} = \begin{pmatrix} \cos \vartheta U \\ \sin \vartheta U \end{pmatrix}, \quad B_{\mathbf{a}^{-1}} = \begin{pmatrix} \cos \vartheta U^* \\ -\sin \vartheta U^* \end{pmatrix}$$

*form an exact rectangular dilation of  $\Phi_{\mathbb{Z}}$  with  $k' = 2k$ , so its room is exactly one bit, while its flat round has leakage 0, distance 0 and entropy production  $a = h_2(0.6 \sin^2 \vartheta) = \Theta(\vartheta^2 \log(1/\vartheta))$ .*

- (b) Let  $\mathcal{A}$  be a  $D$ -dimensional subspace of  $\mathbb{C}^k \otimes \mathbb{C}^m$  and  $Y_g = (B_g \otimes I)\mathcal{A}$ . Every projection  $\Pi$  with  $\text{Tr}(\Pi P_{Y_1}) \geq (1 - \eta)D$  and  $\text{Tr}(\Pi P_{Y_{\mathbf{a}}}) \geq (1 - \eta)D$  has  $\text{rank } \Pi \geq D(2 - 2\eta/(1 - \cos \vartheta))$ .*
- (c) Run the device of [28, Theorem 12.2] for a gadget channel and an exact rectangular dilation with  $q = k'/k$ , with independent Haar unitaries  $V_t$  on the active space of dimension  $N_{t-1}$ , and let  $\sigma_t$  be the memory state for the observer that inputs  $\tau_d$  in every round. Then  $\mathbb{E}[\text{Tr} \sigma_t^2 \mid \sigma_{t-1}] \leq s_2 \text{Tr} \sigma_{t-1}^2 + 2(1 - s_2)/N_{t-1}$  with  $s_2 = \sum_g \mu_g^2$ . If  $qs_2 < 1$ , then for each  $t$ , with probability at least  $\frac{1}{2}$  over the Haar unitaries, every projection  $\Pi$  with  $\text{Tr}(\Pi \sigma_t) \geq 1 - \eta$  has  $\text{rank } \Pi \geq (1 - \eta)^2(1 - qs_2)D_t/(12q)$ , where  $D_t$  is the dimension of the active space after round  $t$ .*

In that device the round-one memory branches forced by anchor inputs are the flat states  $P_{Y_g}/D_0$ , so by (b) a truncation that keeps a fraction  $1 - \eta$  of every forced branch keeps  $\log(2 - 2\eta/(1 - \cos \vartheta))$  bits of growth: 0.9997 bit at  $\vartheta = 0.1$  and  $\eta = 10^{-6}$ , where  $a = 0.053$ . By (c) the smoothed rank of the accumulated memory is within  $O(1)$  bits of the whole active space. The device reads only the flat Gram (Theorem 7.6(a)), re-randomization flattens the bias, and the memory fills the rank-grown space. This blocks the route, not the law: the round of (a) is an exact flat round, which path registers stream at its Shannon cost (Theorem G.6(a)).

## I.2 Rounding at second order

The following statement was a natural target for the upper half at the full leakage scale: a leaky flat round can be made exact on the same bath, up to a constant, at a cost of the order of its leakage.

**Corollary I.2** (Second-order rounding is false). *There are no constants  $C_0, C_1$  such that every flat round of  $\Phi_\omega$ ,  $\theta = (\sqrt{5} - 1)/2$ , with distance at most  $\frac{1}{16}$  and leakage  $\ell$  has a zero-leakage flat round with  $S' \leq S + C_0$ , distance at most  $\frac{1}{4}$  and  $a' \leq a + C_1 \ell \log(1/\ell)$ .*

This is Theorem G.12(b) with  $\lambda = \gamma = 1$ . The witness is a twisted gadget, a gadget evaluated in the twisted group von Neumann algebra of  $\mathbb{Z}^2$ ; it has a flat probe, lies in the closure of finite tracial channels, and has the structure that the arguments of Sections G and H use.

## I.3 Square fibres

A round is *square* if its slot operators are unitaries of one space. For leaky rounds of block-monomial form a rounding that keeps the monomial pattern exists (it is part of the construction of Theorem H.23), but it cannot keep the fibres square.

**Proposition I.3** (Square families cannot be rounded to square families).

- (a)  $\Phi_\omega$  has square flat rounds with  $a = 0$  and leakage tending to 0 (Theorem G.12(b)), but no zero-leakage square round.
- (b)  $\Phi_H$  has square flat rounds with  $a = 0$ , distance at most  $\frac{1}{16}$  and leakage tending to 0, but no zero-leakage square round at distance below  $\frac{1}{2}$ . No gadget channel whose labels no finite-dimensional representation separates has a zero-leakage square round at distance below  $\frac{1}{2}$ .

So a rounding of leaky rounds must in general leave the square class: the target for leaky block-monomial rounds is a family with rectangular fibres, whose room is of order  $\sqrt{\ell}$ .

## I.4 The one-round method

The lower bound  $cn^{1/(2 \log 107+2)}$  for Houghton's channel comes from the one-round theorem [27, Theorem 5.5], whose proof certifies  $N$  approximately commuting copies of  $S_3$  in the memory of one round and converts each into a sixteenth of a bit ([27, Appendix B, proof of Theorem 5.5]). The certification needs

$$10^8 \sqrt{N} \max\{N, A^*(3N)\} 2^{17} \sqrt{\nu} \leq 1, \quad \nu = \ell + a, \quad (25)$$

where  $A^*(M)$  is the largest area of a commutator  $[\sigma_p^{\pm 1}, \sigma_q^{\pm 1}]$ ,  $\sigma_j = \mathbf{a}^j \alpha \mathbf{a}^{-j}$ ,  $0 \leq p, q \leq M$ ,  $|p - q| \geq 2$ , in Johnson's presentation [27, Section 4.1].

**Proposition I.4** (The one-round method is capped at  $n^{1/6}$ ). *Every  $N \geq 8$  that satisfies (25) has  $N \leq 7.13 \cdot 10^{-6} \nu^{-1/5}$ , whatever upper bound on  $A^*$  is used. Consequently, combined with Theorem 7.1 and Proposition 7.2 at the rank rate, the method certifies  $Q_{\Phi_H}^*(n, \epsilon) \geq Q_0$  only for  $Q_0 = O(n^{1/6})$ .*

The cap comes from the quadratic lower bound on far-commutator areas [27, Proposition 4.4] together with the row sums of the sector bound, which make  $N$  copies cost  $N^{5/2}$  in (25). Lemma 8.3 removes the row sums when the copies commute exactly, and Theorem 8.5 then reaches  $n^{1/3}$ ; for general devices a proof of the exponent  $\frac{1}{3}$  needs a sector bound that is local in the same sense for copies that commute only approximately, or an argument that reads more than one round. The proofs of this section are in Appendix S.

## J Proofs for Sections 2 and 7

Two facts are used throughout the appendices.

- (F1) *Anchors.* For a gadget channel the anchor  $z_g$  of a label  $g$  is a diagonal slot with coefficient 1 whose row and column contain no other slot. Hence  $A_h|z_g\rangle = \delta_{gh}|z_g\rangle$  and  $\langle z_g|A_h = \delta_{gh}\langle z_g|$ .
- (F2) *Reading an entry.* If  $d_\diamond(\Psi, \Phi) \leq u$ , then  $|\langle e|\Psi(|e\rangle\langle f|)|f\rangle - \langle e|\Phi(|e\rangle\langle f|)|f\rangle| \leq 2u$  for orthonormal  $e, f$ . Indeed, feed  $(|ee\rangle + |ff\rangle)/\sqrt{2}$  with a reference; the difference  $\Delta$  of the two outputs is Hermitian and traceless with  $\|\Delta\|_1 \leq 2u$ , so every entry has modulus at most  $\|\Delta\| \leq \frac{1}{2}\|\Delta\|_1 \leq u$ , and the entry at  $(ee, ff)$  is half the difference above.

### J.1 Proofs for Section 2

*Proof of Lemma 2.3.* Write  $U_f = \sum_a c_a E_a \otimes P_{\pi_a}$ , where  $P_\pi$  is the permutation matrix of  $\pi$ ,  $\pi_a = f([w_a])$  for every slot except the lower right slot of a triangle  $(x, y, z)$ , where  $\pi_a = f([x])f([y])$ . Since  $\rho = \text{diag}(p)$ , the channel is  $\Psi_p = \sum_\omega p(\omega)\Phi_\omega$  with

$$\Phi_\omega(X) = \sum_{a,b} c_a \bar{c}_b \mathbf{1}[\pi_a \omega = \pi_b \omega] E_a X E_b^*,$$

whose Kraus operators are  $K_q = \sum_{a:\pi_a \omega = q} c_a E_a$ ,  $q \in \Omega$ . So  $\mathcal{J}(\Phi_\omega) = \frac{1}{d} \sum_q |K_q\rangle\rangle\langle\langle K_q|$  and  $\ell(\Phi_\omega) = \frac{1}{d} \sum_q \|P_\perp K_q\|_F^2$ , with  $P_\perp$  the Hilbert–Schmidt projection off  $\text{span}\{A_g\}$ . The  $A_g$  have disjoint supports. Let  $S_g$  be the set of slots of label  $g$ ,  $\Sigma_g = \|A_g\|_F^2$ , and  $s_g = \sum_{a \in S_g, \pi_a \omega = q} |c_a|^2$  the squared norm of the part of  $K_q$  on  $S_g$ ; that part loses exactly  $s_g^2/\Sigma_g$  to its projection onto  $A_g$ , so  $\|P_\perp K_q\|_F^2 = \sum_g s_g(\Sigma_g - s_g)/\Sigma_g$ .

At a triangle-good  $\omega$  every slot of label  $g$  sends  $\omega$  to  $f(g)\omega$ : for the lower right slot of  $(x, y, z)$ ,  $f([x])f([y])\omega = f([z])\omega$  and its label is  $[z]$ . So all of  $S_g$  lies in one  $K_q$ ,  $s_g \in \{0, \Sigma_g\}$ , and  $\ell(\Phi_\omega) = 0$ , whatever the collisions. If a triangle  $(x, y, z)$  fails at  $\omega$ , its lower right slot (label  $[z]$ ,  $|c|^2 = \frac{1}{2}$ ) and the anchor of  $[z]$  ( $|c|^2 = 1$ ) send  $\omega$  to different points, so some  $K_q$  has  $\frac{1}{2} \leq s \leq \Sigma - \frac{1}{2}$  for  $g = [z]$ , and  $s(\Sigma - s)/\Sigma = 1/(1/s + 1/(\Sigma - s)) \geq \min(s, \Sigma - s)/2 \geq \frac{1}{4}$ . So  $\ell(\Phi_\omega) \geq 1/(4d)$ . Leakage is linear in the Choi state, so  $\ell = \sum_\omega p(\omega)\ell(\Phi_\omega) \geq p(\text{bad})/(4d)$ .

For the collisions, by (F1) the only slot in the column  $z_g$  is the anchor of  $g$ , so  $\langle z_g|\Psi_p(|z_g\rangle\langle z_h|)|z_h\rangle = p(\{\omega : f(g)\omega = f(h)\omega\})$ , while for  $\Phi$  the entry is  $\tau(\lambda(h)^* \lambda(g)) = 0$ . By (F2) the mass is at most  $2u$ .  $\square$

*Proof of Lemma 2.6.* Enlarging  $u$  or  $\ell_0$  enlarges the set of rounds over which the infimum is taken. For  $\epsilon \leq \frac{1}{16}$ ,  $\epsilon \leq \frac{1}{16}$  and  $\frac{32}{15}\epsilon/n \leq \frac{2}{15n}$ , so  $L_\Phi(n, \epsilon) \geq L^*(n)$ . For  $K \geq c/c_1$  we have  $c/(Kn) \leq c_1/n$ , so every round admissible in  $L_\Phi(Kn; u, c/(Kn))$  is admissible in  $L_\Phi(n; u, c_1/n)$ , and  $S + na \leq S + Kna$  since  $K \geq 1$  and  $a \geq 0$ .  $\square$

*Proof of Proposition 2.7.* Hold  $n$  registers of  $\lceil \log r \rceil$  qubits, all in  $|0\rangle$ . In round  $t$  apply a unitary that extends the Stinespring isometry  $\xi \mapsto \sum_i A_i \xi \otimes |i\rangle$  from the input and  $|0\rangle$  of register  $t$  to the input and register  $t$ , release the output, and keep the register. Register  $t$  is a fresh environment that is never touched again, so the service implements  $\Phi^{\otimes n}$  exactly for every observer, with  $J = 0$  and  $Q = n \lceil \log r \rceil$ .  $\square$

*Proof of Lemma 2.9.* (a) The gadget unitary is block diagonal:  $1 \times 1$  blocks at the empty word and at the symbols, and one  $2 \times 2$  block  $\frac{1}{\sqrt{2}} \begin{pmatrix} I & \lambda(y) \\ \lambda(x) & -\lambda(x)\lambda(y) \end{pmatrix}$  for each triangle. Since  $A_g = \sum_{[w_a]=g} c_a E_a$ , the operator  $W$  has the same pattern with each slot entry replaced by  $c_a B_{[w_a]}$ . Distinct blocks occupy disjoint rows and columns of  $\mathbb{C}^d$ , so  $W^*W = I$  if and only if

each block of  $W$  has orthonormal block columns. A  $1 \times 1$  block is  $B_g$  for the label  $g$  of its position, and every label occurs there because every label has an anchor; so the  $1 \times 1$  blocks are isometric if and only if every  $B_g$  is. The block of a triangle is  $\frac{1}{\sqrt{2}} \begin{pmatrix} B_1 & B_{[y]} \\ B_{[x]} & -B_{[z]} \end{pmatrix}$ , since  $[x][y] = [z]$ . Its column Gram matrix has diagonal blocks  $\frac{1}{2}(B_1^*B_1 + B_{[x]}^*B_{[x]})$  and  $\frac{1}{2}(B_{[y]}^*B_{[y]} + B_{[z]}^*B_{[z]})$ , which are  $I$  when the  $B_g$  are isometries, and off-diagonal block  $\frac{1}{2}(B_1^*B_{[y]} - B_{[x]}^*B_{[z]})$ . Then  $\sum_g B_g^*B_g = rI$ . Replacing  $B_g$  by  $VB_g$  replaces  $W$  by  $(I \otimes V)W$ .

(b) Let  $\rho_*$  be the input marginal of the flat probe. The output-and-reference state and the environment state of the pure state  $(V_\Phi \otimes I)\psi_*$ ,  $V_\Phi = \sum_i A_i \otimes |i\rangle$ , have the same nonzero spectrum, and the environment state is the matrix  $[\text{Tr}(A_i \rho_* A_j^*)]$ . The output is flat of rank  $r$ , so this matrix is  $I/r$ :  $\text{Tr}(\rho_* A_j^* A_i) = \delta_{ij}/r$ . Applying  $\text{Tr}(\rho_* \cdot) \otimes \text{id}$  to  $W^*W = \sum_{i,j} A_i^* A_j \otimes B_i^* B_j = I$  gives  $\frac{1}{r} \sum_i B_i^* B_i = I$ .

(c) Let  $\iota$  embed  $\mathbb{C}^k$  in  $\mathbb{C}^{k'}$ . The orthogonal complements of the ranges of  $I_d \otimes \iota$  and of  $W$  have the same dimension  $d(k' - k)$ , so  $W$  extends to a unitary  $U$  on  $\mathbb{C}^d \otimes \mathbb{C}^{k'}$  with  $U(I \otimes \iota) = W$ . With  $\rho = \iota^*/k$ ,  $\Psi(X) = \text{Tr}_{k'} W(X \otimes I_k/k)W^* = \sum_{i,j} A_i X A_j^* \text{Tr}(B_j^* B_i)/k = \Phi(X)$ , so the distance and the leakage are 0. The state  $T(\rho)$  lives on  $\mathbb{C}^{k'}$ , so  $a = S(T(\rho)) - \log k \leq \log(k'/k)$ , and  $S(\rho) = \log k$ . Hence every dilation is admissible in  $L_\Phi(n; 0, 0)$  with cost at most  $\log k + n \log(k'/k)$ .  $\square$

*Proof of Lemma 2.10.* Write  $U = \sum_{o,o'} |o\rangle\langle o'| \otimes A_{oo'}$  with  $A_{oo'} \in M_M$ , and  $\xi = \sqrt{\rho}$ . Let  $\mathcal{H}$  be  $M_M$  with the Hilbert–Schmidt inner product. The Choi state  $\mathcal{J}(\Psi)$  is the marginal on  $\mathbb{C}^d \otimes \mathbb{C}^d$  of the unit vector  $v = d^{-1/2} \sum_{o,o'} |o\rangle\langle o'| \otimes A_{oo'} \xi$  in  $\mathbb{C}^d \otimes \mathbb{C}^d \otimes \mathcal{H}$  [28, proof of Lemma C.1], and  $\ell = \|((I - \Pi_\Phi) \otimes I)v\|^2$ . The range of  $\Pi_\Phi$  is  $\text{span}\{|A_i\rangle\rangle\}$ , so  $\ell = 0$  means  $v = \sum_i |A_i\rangle\rangle \otimes Y_i$  for some  $Y_i \in \mathcal{H}$ , that is,  $d^{-1/2} A_{oo'} \xi = \sum_i (A_i)_{oo'} Y_i$  for all  $o, o'$ . By linear independence of the  $A_i$  there are coefficients  $\beta_{oo'}^i$  with  $Y_i = d^{-1/2} \sum_{o,o'} \beta_{oo'}^i A_{oo'} \xi = d^{-1/2} X_i \xi$ , where  $X_i = \sum \beta_{oo'}^i A_{oo'}$ . Hence  $A_{oo'} \xi = \sum_i (A_i)_{oo'} X_i \xi$ , and since  $\xi$  maps onto  $K$ ,  $A_{oo'}|_K = \sum_i (A_i)_{oo'} X_i|_K$ . Then  $U(|o'\rangle \otimes \kappa) = \sum_o |o\rangle \otimes A_{oo'} \kappa = \sum_i A_i |o'\rangle \otimes X_i \kappa$  for  $\kappa \in K$ , so  $U$  restricted to  $\mathbb{C}^d \otimes K$  is  $W = \sum_i A_i \otimes X_i|_K$ , an isometry, and  $\Psi(X) = \sum_{i,j} A_i X A_j^* \text{Tr}(X_i \rho X_j^*)$ . With a flat probe, Lemma 2.9(b) applied to  $W$  gives  $\sum_i X_i^* X_i = rI$  on  $K$ . For a gadget channel Lemma 2.9(a) applies to  $W$ . By (F1),  $\langle z_g | \Psi(|z_g\rangle\langle z_h|) | z_h \rangle = \text{Tr}(\rho X_h^* X_g)$ , while  $\Phi$  gives 0 for  $g \neq h$ , so (F2) gives  $|\text{Tr}(\rho X_h^* X_g)| \leq 2u$ .  $\square$

## J.2 The lower half

*Proof of Theorem 7.1.* Fix the *test observer*: in every round  $t$  it inputs half of a fresh maximally entangled pair on the input and a reference  $R_t$ , measures  $\{\Pi_\Phi, I - \Pi_\Phi\}$  on the output and  $R_t$ , keeps the outcome bit and discards the output and  $R_t$ . Let  $p_t$  be the probability that the tests of rounds  $1, \dots, t-1$  pass,  $q_t$  the probability that test  $t$  fails given that, and  $F$  the first failure ( $F = \infty$  if there is none).

*First budget.*  $p_{t+1} = p_t(1 - q_t)$ , so  $\sum_{t=1}^n p_t q_t = 1 - p_{n+1} = \Pr(F \leq n) =: f$ . The ideal experiment never fails, since  $\mathcal{J}(\Phi)$  is supported on  $\Pi_\Phi$ , and the outcome record is part of the observer’s final state, so  $f \leq \epsilon$ . Hence  $p_t \geq 1 - \epsilon$  for  $t \leq n+1$ ,  $\sum_t p_t \geq n(1 - \epsilon)$  and  $\sum_t p_t q_t \leq \epsilon$ .

*The rounds.* Let  $\rho_t$  be the memory state given that tests  $1, \dots, t-1$  passed. Earlier pairs are discarded, so  $\rho_t$  does not depend on the input of round  $t$ . Round  $t$  applies a unitary  $U_t$  on the input and the  $Q$  memory qubits (imports happen between rounds), so  $(U_t, \rho_t)$  is a round with bath dimension  $2^Q$ ; let  $\Psi_t$  be its channel. With a maximally entangled input the failure probability of the test is  $\text{Tr}[(I - \Pi_\Phi)\mathcal{J}(\Psi_t)]$ , so the leakage of  $\Psi_t$  is  $\ell_t = q_t$ .

*Second budget.* Let  $\mathbf{L}$  be the average, over the branches of the test record, of the entropy of the memory, a branch being frozen when its test fails. Initially  $\mathbf{L} = S(\beta_0) = Q - P$ , and always  $\mathbf{L} \leq Q$ . In round  $t$  the passing branch, of weight  $p_t$ , has memory  $\rho_t$  and input marginal  $\tau_d$ , so

after the round its memory is  $T_t(\rho_t)$ , of entropy  $S(\rho_t) + a_t$ . The test acts on other systems, so it splits  $T_t(\rho_t) = (1 - q_t)\sigma_{\text{pass}} + q_t\sigma_{\text{fail}}$ , and  $S(T_t(\rho_t)) \leq (1 - q_t)S(\sigma_{\text{pass}}) + q_tS(\sigma_{\text{fail}}) + h_2(q_t)$ . An export of  $j$  qubits lowers the entropy by at most  $j$  (subadditivity), and an import of  $j$  qubits in a fixed independent state of entropy  $j - c$  raises it by  $j - c$ . Summing,

$$Q \geq L_{\text{final}} \geq Q - P - C + \sum_t p_t a_t - \sum_t p_t h_2(q_t).$$

By the chain rule for the stopping time,  $\sum_t p_t h_2(q_t) = H(F) \leq h_2(f) + f \log n \leq h_2(\epsilon) + \epsilon \log n$ , since  $F$  takes the value  $\infty$  with probability  $1 - f$  and  $h_2(x) + x \log n$  increases on  $[0, \frac{1}{2}]$ . So  $\sum_t p_t a_t \leq B'$ .

*Markov.* Put  $w_t = p_t / \sum_s p_s$  and  $\varsigma = 1/(n(1 - \epsilon))$ , so that  $\sum_t w_t q_t \leq \epsilon \varsigma$  and  $\sum_t w_t a_t \leq B' \varsigma$ . Let  $A_1 = \{t : q_t > \lambda \epsilon \varsigma\}$  and  $A_2 = \{t : a_t > \frac{\lambda}{\lambda - 1} B' \varsigma\}$ . If  $w(A_1) > 0$ , then  $\epsilon \varsigma \geq \sum_{A_1} w_t q_t > \lambda \epsilon \varsigma w(A_1)$ , so  $w(A_1) < 1/\lambda$ ; likewise  $w(A_2) < (\lambda - 1)/\lambda$ , and the sets have weight 0 if  $\epsilon = 0$  or  $B' = 0$ . Since the weights sum to 1, some round  $t$  with  $w_t > 0$  lies in neither set:  $\ell_t \leq \lambda \epsilon / (n(1 - \epsilon))$  and  $a_t \leq \frac{\lambda}{\lambda - 1} B' / (n(1 - \epsilon))$ .

*Distance.* Let the observer perform the tests in rounds  $1, \dots, t - 1$  and then feed an arbitrary input with a reference in round  $t$ . Its state after round  $t$  is  $p_t \cdot (\text{pass}, \sigma) \oplus (1 - p_t) \cdot (\text{fail}, \cdot)$  in the service and  $(\text{pass}, \omega)$  in the ideal experiment, where  $\sigma = (\Psi_t \otimes \text{id})(\text{input})$  and  $\omega = (\Phi \otimes \text{id})(\text{input})$ . By data processing  $\frac{1}{2} \|p_t \sigma - \omega\|_1 + \frac{1}{2} (1 - p_t) \leq \epsilon$ , and  $\frac{1}{2} \|\sigma - \omega\|_1 \leq \frac{1}{2} \|p_t \sigma - \omega\|_1 + \frac{1}{2} (1 - p_t) \leq \epsilon$ . Taking the supremum over inputs,  $d_\diamond(\Psi_t, \Phi) \leq \epsilon$ .  $\square$

**Theorem J.1** (Explicit extraction constants). *Let  $\Phi$  have a flat probe of rank  $r$ . Every causal service of  $\Phi$  for  $n \geq 1$  rounds with error  $\epsilon \leq \frac{1}{16}$  at the rank rate has a round with bath dimension  $2^Q$ , distance at most  $\epsilon$ , leakage at most  $\frac{32}{15} \epsilon / n$  and  $n a \leq 6.4 Q + 13.52 + 0.134 \log n$ . Consequently*

$$Q \geq \frac{L_\Phi(n, \epsilon) - 13.6 - 0.14 \log n}{7.4} \geq \frac{L^*(n) - 13.6 - 0.14 \log n}{7.4}.$$

At any exchange, a service with purity  $P + C \leq B$  and error  $\epsilon \leq \frac{1}{16}$  has  $Q \geq \inf\{S(\rho) : d_\diamond \leq \epsilon, \ell \leq \frac{32}{15} \epsilon / n, a \leq \frac{32}{15} (B + 0.34 + \epsilon \log n) / n\}$ .

*Proof of Theorem 7.3.* Proposition 7.2 gives  $P + C < 3Q + 6 =: B$ . Apply Theorem 7.1 with  $\lambda = 2$ . Since  $\epsilon \leq \frac{1}{16}$ ,  $1/(1 - \epsilon) \leq \frac{16}{15}$ , so the round has distance at most  $\epsilon$ , leakage at most  $\frac{32}{15} \epsilon / n$  and  $n a \leq \frac{32}{15} (3Q + 6 + h_2(\epsilon) + \epsilon \log n)$ . As  $h_2$  increases on  $[0, \frac{1}{2}]$ ,  $h_2(\epsilon) \leq h_2(\frac{1}{16}) < 0.3373$ , and  $\epsilon \log n \leq (\log n)/16$ , so  $n a \leq 6.4 Q + 13.52 + 0.134 \log n$ . The bath has  $Q$  qubits, so  $S(\rho_t) \leq Q$ . Hence  $L_\Phi(n, \epsilon) \leq S(\rho_t) + n a \leq 7.4 Q + 13.52 + 0.134 \log n$ , which is the first inequality, and Lemma 2.6 gives the second. At any exchange, apply Theorem 7.1 with  $\lambda = 2$  and the given  $B$ , and use  $h_2(\epsilon) \leq 0.34$ .  $\square$

### J.3 The rank form

*Proof of Theorem 7.6.* (a) The proof of [28, Theorem 12.2] builds frames  $F_t : \mathbb{C}^{r^t} \rightarrow \mathbb{C}^{D_t} \otimes \mathbb{C}^S$ , one column per Kraus history, chooses the unitary before each round by the concentration lemma [27, Lemma A.1] and the unitary after it by the export lemma [27, Lemma A.4], and keeps  $\|F_t^* F_t - I\| \leq t\epsilon/n$  and  $\alpha(F_t) \leq A_*/D_0$ . The identity  $\text{Tr}(B_i^* B_j) = k\delta_{ij}$  enters that proof in one place, the Haar mean of the Gram blocks; the identity  $\sum_i B_i^* B_i = rI$  enters the step on exterior marginals, and it holds here by Lemma 2.9(b); and the identity matrix enters the final error step. We change these steps only.

*Gram step.* The Haar mean of the block  $(i, j)$  of the new Gram matrix,  $F^*((V^*(B_i^* B_j \otimes I)V) \otimes I)F$ , is  $\tau_k(B_i^* B_j) F^* F = G_{ij} F^* F$ . Replace the induction hypothesis by  $e_t := \|F_t^* F_t - G^{\otimes t}\| \leq e t \epsilon / n$ , the histories ordered as tensor products. Then  $g(F_t) \leq \|G\|^t + e t \epsilon / n \leq$

$e(1 + \epsilon) < 2e$ , because  $\|G\|^t \leq (1 + \|G - I\|)^n \leq e$ . In the concentration step  $g < 2e$  replaces  $g \leq 2$ , which multiplies the required  $D_0^2$  by at most  $e$ . As in [28] every block then deviates from its mean by at most  $\epsilon/(rn)$ , so the new Gram matrix is within  $\epsilon/n$  of  $(F^*F) \otimes G$ , and  $\|(F^*F) \otimes G - G^{\otimes t}\| \leq \|G\| e_{t-1}$ . Hence  $e_t \leq \|G\| e_{t-1} + \epsilon/n$  and  $e_t \leq \sum_{s < t} \|G\|^s \epsilon/n \leq e t \epsilon/n$ . Exports and imports preserve the Gram matrix.

*Error step.* Put  $\Gamma = G^{\otimes n}$ . For  $n \geq 2$ ,  $\|G - I\| \leq \frac{1}{2}$ , and  $(1 - x)^n \geq e^{-2nx}$  for  $x \leq \frac{1}{2}$ , so  $\Gamma \geq e^{-2}I$ . For positive  $X, Y$  and  $Z = X^{1/2} - Y^{1/2}$  one has  $X^{1/2}Z + ZY^{1/2} = X - Y$ ; evaluating at a unit eigenvector of the Hermitian  $Z$  gives  $\|Z\| \leq \|X - Y\|/(\lambda_{\min}(X^{1/2}) + \lambda_{\min}(Y^{1/2}))$ . So  $\|(F_n^*F_n)^{1/2} - \Gamma^{1/2}\| \leq e\epsilon/e^{-1} = e^2\epsilon$ . With the polar decomposition  $F_n = T(F_n^*F_n)^{1/2}$ ,  $T$  an isometry, the purified final vector  $\sum_y \psi_y \otimes F_n|y\rangle$  of the service is within  $e^2\epsilon$  of  $(I \otimes T) \sum_y \psi_y \otimes \Gamma^{1/2}|y\rangle$ , since  $\sum_y \psi_y \otimes |y\rangle$  is a unit vector.

*Comparison.* Let  $e_i = G^{1/2}|i\rangle$ , so that  $\langle e_j, e_i \rangle = G_{ji}$ . Applying  $\text{id} \otimes \tau_k$  to  $W^*W = I$  gives  $\sum_{i,j} G_{ij} A_i^* A_j = I$ , so  $V_G : \xi \mapsto \sum_i A_i \xi \otimes e_i$  is an isometry and  $\text{Tr}_E V_G X V_G^* = \sum_{i,j} G_{ji} A_i X A_j^* =: \Phi_W(X)$ . The observer's vectors  $\psi_y$  depend only on the Kraus histories, so  $\sum_y \psi_y \otimes \Gamma^{1/2}|y\rangle$  is the purified final state of the ideal experiment with  $\Phi_W$  in every round, and  $T$  acts on systems the observer never holds. A hybrid over the  $n$  rounds changes the final state by at most  $n d_\diamond(\Phi_W, \Phi)$ . Finally  $\Phi_W - \Phi = \text{Tr}_E[(I \otimes H)V_I(\cdot)V_I^*]$  with  $V_I = \sum_i A_i \otimes |i\rangle$  and  $\langle j|H|i\rangle = G_{ji} - \delta_{ij}$ , and  $|\text{Tr}[(Z \otimes H)Y]| \leq \|Z\| \|H\| \text{Tr} Y$  for  $Y \geq 0$ , so  $d_\diamond(\Phi_W, \Phi) \leq \frac{1}{2} \|G - I\|$ . The error is at most  $e^2\epsilon + \frac{n}{2} \|G - I\|$ , and the memory and purity are those of [28, Theorem 12.2].

(b)  $(B_1^{\otimes t})^* B_{[y]}^{\otimes t} = (B_1^* B_{[y]})^{\otimes t} = (B_{[x]}^* B_{[z]})^{\otimes t}$ , tensor powers of isometries are isometries, and  $\tau_{k^t}((B_g^{\otimes t})^* B_h^{\otimes t}) = \tau_k(B_g^* B_h)^t$ .

(c) With  $t$  as stated,  $c^t \leq \epsilon/(2rn)$ . The flat Gram  $G^{(t)}$  of  $(B_g^{\otimes t})$  has unit diagonal and off-diagonal entries of modulus at most  $c^t$ , so  $\|G^{(t)} - I\| \leq (r - 1)c^t < \epsilon/(2n)$ ; then  $n\|G^{(t)} - I\| \leq 1$  and  $\frac{n}{2}\|G^{(t)} - I\| < \epsilon/4$ . Apply (a), which uses the flat probe of every gadget channel, to  $(B_g^{\otimes t})$  with dimensions  $k^t$  and  $k'^t$ .  $\square$

*Proof of Corollary 7.7.* The lower bound is Theorem 7.3. If  $U_\Phi(n) < \infty$  the infimum defining it is attained, since  $\log k \leq U_\Phi(n) + 1$  leaves finitely many pairs  $(k, k')$ . Theorem 7.6(a) with  $G = I$ , which is [28, Theorem 12.2], applied to a minimizing dilation gives a service with error  $\epsilon$ ,  $J = \lfloor \frac{n-1}{2} \log r \rfloor \leq \frac{n}{2} \log r$  and  $Q \leq \log k + n \log(k'/k) + O_r(\log(n/\epsilon) + \log(2 + \log k))$ . By Lemma 2.9(c) and Lemma 2.6,  $L_\Phi(n, \epsilon) \leq L_\Phi(n; 0, 0) \leq U_\Phi(n)$ .  $\square$

*Proof of Corollary 7.8.* Let  $\pi$  be a finite-dimensional unitary representation of  $P_\Phi$  that separates the labels, and  $\pi' = \pi \oplus 1$ , of dimension  $D$ . The family  $B_g = \pi'(g)$  is square and unitary, and  $B_1^* B_{[y]} = \pi'([y]) = \pi'([x])^* \pi'([z])$  on every triangle, because  $[x][y] = [z]$  in  $P_\Phi$  and  $\pi'(1) = I$ ; so it is a gadget family with room 0. For labels  $g \neq h$ ,  $\pi'(g)^* \pi'(h) = \pi(g^{-1}h) \oplus 1$  with  $\pi(g^{-1}h) \neq I$ . A unitary  $V \neq I$  on  $\mathbb{C}^{D-1}$  has  $|\text{Tr} V + 1| < D$ : if  $V$  is not scalar then  $|\text{Tr} V| < D - 1$ , and if  $V = \lambda I$  with  $\lambda \neq 1$  then  $|\lambda(D - 1) + 1| < D$ . So  $z := \max_{g \neq h} |\tau_D(B_g^* B_h)| < 1$ . Theorem 7.6(c) with  $c = z$ ,  $\delta = 0$  and  $\epsilon/(e^2 + 1)$  in place of  $\epsilon$  gives a rank-rate service with error at most  $\epsilon$  and  $Q \leq t \log D + O_r(\log(n/\epsilon) + \log(2 + t \log D))$ ,  $t = \lceil \log(2(e^2 + 1)rn/\epsilon) / \log(1/z) \rceil$ . So  $Q^*(n, \epsilon) = O_\Phi(\log(n/\epsilon))$ .  $\square$

*Proof of Corollary 7.9. Lower.* Let  $(U, \rho)$  have distance at most  $\frac{1}{16}$ , and put  $\nu = \ell + a$ . By [28, Corollary 10.2(a)], applied to the round, which has distance at most  $\frac{1}{8}$ , there are  $C_R, y_0$  with  $S(\rho) \geq y/(32(\log y)^{2\kappa})$  for every  $y \geq y_0$  with  $C_R \nu y \leq 1$ . For  $\nu \leq v \leq v_0 := 1/(C_R y_0)$  take  $y = 1/(C_R v)$ :  $S \geq f(v) := c v^{-1} (\log 1/v)^{-2\kappa}$ , and  $f$  decreases on  $(0, v_0]$  if  $v_0$  is small. Now let the round also have leakage at most  $\frac{2}{15n}$ . If  $a \leq \frac{2}{15n}$ , then  $\nu \leq \frac{4}{15n} \leq v_0$  and  $S \geq f(\frac{4}{15n}) \geq c n / (\log n)^{2\kappa}$  for large  $n$ . If  $a > \frac{2}{15n}$  and  $2a \leq v_0$ , then  $\nu \leq 2a$  and  $S \geq f(2a) \geq (c/(2a)) (\log(15n/4))^{-2\kappa}$ , since  $2a > \frac{4}{15n}$ ; by the arithmetic-geometric mean

inequality  $S + na \geq \sqrt{2cn}(\log(15n/4))^{-\kappa}$ . If  $2a > v_0$ , then  $na \geq nv_0/2$ . In every case  $S + na \geq c'\sqrt{n}/(\log n)^\kappa$  for  $n \geq n_0$ .

*Upper.* In the proof of [28, Corollary 12.3], a  $(1 - s^{-1})$ -expansive  $s^{-1}$ -morphism [21, Definition 3.4] of the chunk of labels, of degree  $\sigma_E(s)$  with  $s \geq 2K_G$ , gives on its good points an exact rectangular dilation: distinct labels never collide at good points, so the label traces are exactly 0, with room  $\log(k'/k) \leq 2K_G/(s \ln 2)$  and  $\log k \leq \log \sigma_E(s) \leq C_E s(\log s)^\lambda$ . With  $s = \max(2K_G, \lceil \sqrt{n} \rceil)$  this gives  $U_\Phi(n) \leq C\sqrt{n}(\log n)^\lambda$ . Corollary 7.7 then gives  $Q^*(n, \epsilon) \leq U_\Phi(n) + K_r(\log(n/\epsilon) + \log(2 + U_\Phi(n))) \leq K(\log(n/\epsilon))^{\kappa+\lambda+1}(1 + L^*(n))$  by the lower bound. For the configuration-lamp channel  $\lambda = 1$  [28, Theorem 3.3(b)] and  $\kappa = 6 + \log 107$  [28, Section 11], so  $L^*(n)$ ,  $U_\Phi(n)$  and, by Theorem 7.3,  $Q^*(n, \epsilon)$  are all  $n^{1/2+o(1)}$ .  $\square$

## K Proofs for Section G: blocks and paths

Two further facts are used.

(F3) *Partial traces.* For a frame  $F$  with  $R = FF^*$  and operators  $Y$  on the memory,  $\text{Tr}_{\text{mem}}[(Y \otimes I)R(Y \otimes I)^*] = \text{Tr}_{\text{mem}}[(Y^*Y \otimes I)R]$ ; the map  $Y \mapsto \text{Tr}_{\text{mem}}[(Y \otimes I)R]$  is positive; and  $\text{Tr}_{\text{mem}}[(P_p \otimes I)R] = \text{Tr}_{H_p} R_p$ .

(F4) *Reduced states.* If  $\rho(u)$  denotes the observer's reduced state of a vector  $u$ , then  $\frac{1}{2}\|\rho(u) - \rho(v)\|_1 \leq \frac{1}{2}\|u\|_1\langle u| - |v\rangle\langle v| \|_1 \leq \frac{1}{2}\|u - v\|(\|u\| + \|v\|)$ , and isometries on systems the observer does not hold do not change  $\rho(\cdot)$ . Errors are always compared in this form, so that vectors of different spaces, such as a physical device and a proof object, are never subtracted.

### K.1 Concentration and export

*Proof of Lemma G.1.* (a) Write  $Z = \sum_p Z_p$  with  $Z_p$  depending only on  $V_p$ . For each  $p$  fix a Hermitian orthonormal basis  $(T_{p,a})_a$  of  $M_{D_p}$  for the Hilbert–Schmidt inner product, so that  $\sum_a T_{p,a} Y T_{p,a} = \text{Tr}(Y)I$  and  $\sum_a T_{p,a}^2 = D_p I$ , and let  $\partial_{p,a}$  be the derivative along  $V_p \mapsto V_p e^{iuT_{p,a}}$  at  $u = 0$ . Haar measure is invariant under these flows, so  $\mathbb{E}[(\partial f)h] = -\mathbb{E}[f \partial h]$  for smooth matrix-valued  $f, h$ . Put  $\mathbb{L}_p = \sum_a \partial_{p,a}^2$ .

*Generator.* With  $X = V_p^* A_p V_p$ ,  $\partial_{p,a} X = i[X, T_{p,a}]$  and  $\partial_{p,a}^2 X = -[T_{p,a}, [T_{p,a}, X]]$ ; summing,  $\mathbb{L}_p X = -2D_p X + 2\text{Tr}(X)I$ , and  $\text{Tr} X = \text{Tr} A_p$ , so  $\mathbb{L}_p Z_p = -2D_p Z_p$ . Also  $\partial_{p,a} Z = \partial_{p,a} Z_p$ .

*Carré du champ.*  $\partial_{p,a} Z = F_p^*(Y_{p,a} \otimes I)F_p$  with  $Y_{p,a} = i(XT_{p,a} - T_{p,a}X)$  Hermitian, so  $\sum_a (\partial_{p,a} Z)^2 = F_p^*[\sum_a (Y_{p,a} \otimes I)R_p(Y_{p,a} \otimes I)]F_p$ . By  $(P - P')R(P - P')^* \leq 2PRP^* + 2P'RP'^*$ ,  $\sum_a (XT_a \otimes I)R_p(T_a X \otimes I) = (X \otimes I)(I \otimes \text{Tr}_{H_p} R_p)(X \otimes I) \leq \alpha_p X^2 \otimes I$ , and  $\sum_a (T_a X \otimes I)R_p(XT_a \otimes I) = I \otimes \text{Tr}_{H_p}[(X \otimes I)R_p(X \otimes I)] \leq \|X\|^2 \alpha_p I$ ; the last inequality is tested on an exterior vector  $v$ , for which  $R_v = (I \otimes \langle v|)R_p(I \otimes |v\rangle) \geq 0$  satisfies  $\text{Tr}(X R_v X) \leq \|X\|^2 \text{Tr} R_v$ . Hence  $\sum_a (\partial_{p,a} Z)^2 \leq 4\alpha_p \|A_p\|^2 F_p^* F_p$ , and this is 0 when  $A_p = 0$ .

*Trace exponential.* Let  $f(\theta) = \mathbb{E} \text{Tr} e^{\theta Z}$ . Integration by parts with  $\mathbb{L}_p Z_p = -2D_p Z_p$  gives  $f'(\theta) = \sum_p \mathbb{E} \text{Tr}(Z_p e^{\theta Z}) = \sum_p (2D_p)^{-1} \sum_a \mathbb{E} \text{Tr}[(\partial_{p,a} Z)(\partial_{p,a} e^{\theta Z})]$ . With  $\partial e^{\theta Z} = \theta \int_0^1 e^{u\theta Z} (\partial Z) e^{(1-u)\theta Z} du$  and  $\int_0^1 e^{u\lambda + (1-u)\mu} du \leq (e^\lambda + e^\mu)/2$  in an eigenbasis of  $Z$ ,  $\text{Tr}[(\partial Z)(\partial e^{\theta Z})] \leq \theta \text{Tr}[(\partial Z)^2 e^{\theta Z}]$ . So  $f'(\theta) \leq \theta \mathbb{E} \text{Tr}[\sum_p (2\alpha_p \|A_p\|^2 / D_p) F_p^* F_p e^{\theta Z}] \leq 2\theta \kappa g_0 f(\theta)$ , using  $\sum_{p:A_p \neq 0} F_p^* F_p \leq g_0 I$  and  $\text{Tr}(Y e^{\theta Z}) \leq \|Y\| \text{Tr} e^{\theta Z}$  for  $Y \geq 0$ . Hence  $f(\theta) \leq N e^{\theta^2 \kappa g_0}$ , and  $\Pr\{\lambda_{\max}(Z) \geq s\} \leq e^{-\theta s} f(\theta) = N e^{-s^2/(4\kappa g_0)}$  at  $\theta = s/(2\kappa g_0)$ ; the same holds for  $-Z$ .

(b) Let  $\tilde{T}_{p,a}$  be  $T_a$  on  $H_p$  and 0 elsewhere, and  $\partial_{p,a}$  the derivative along  $V_p \mapsto V_p e^{iuT_a}$ ; then  $\partial_{p,a} X = i[X, \tilde{T}_{p,a}]$ . Since  $\sum_a \tilde{T}_{p,a} Y \tilde{T}_{p,a} = \text{Tr}(P_p Y P_p) P_p$  and  $\sum_a \tilde{T}_{p,a}^2 = D P_p$ ,

$\sum_{p,a}[\tilde{T}_{p,a}, [\tilde{T}_{p,a}, X]] = 2D(X - \sum_p \tau(X_{pp})P_p)$ , and  $\tau(X_{pp}) = \tau(A_{pp})$  does not depend on  $V$ ; so the Laplacian of the product group satisfies  $LZ = -2DZ$ . Here the equal dimensions are needed. With  $Y_{p,a} = i(X\tilde{T}_{p,a} - \tilde{T}_{p,a}X)$ ,

$$\sum_{p,a}(\partial_{p,a}Z)^2 \leq 2F^* \left[ \sum_{p,a} (X\tilde{T}_{p,a} \otimes I)R(\tilde{T}_{p,a}X \otimes I) + \sum_{p,a} (\tilde{T}_{p,a}X \otimes I)R(X\tilde{T}_{p,a} \otimes I) \right] F.$$

The first sum is  $(X \otimes I)(\sum_p P_p \otimes \text{Tr}_{H_p} R_p)(X \otimes I) \leq \alpha_* \|A\|^2 I$ . The second is  $\sum_p P_p \otimes \text{Tr}_{\text{mem}}[(XP_p X \otimes I)R]$  by (F3), and  $XP_p X = V^*AP_p AV$  because  $V$  commutes with  $P_p$ . The positive operator  $AP_p A$  lives on the blocks of  $N(p) = \{p' : A_{p'p} \neq 0\}$ , so by Cauchy–Schwarz  $AP_p A \leq |N(p)| \sum_{p' \in N(p)} P_{p'} AP_p AP_{p'} \leq |N(p)| \sum_{p' \in N(p)} \|A_{p'p}\|^2 P_{p'} =: \Lambda_p$ , which commutes with  $V$ . Hence  $V^*AP_p AV \leq \Lambda_p$  and, by (F3),  $\text{Tr}_{\text{mem}}[(XP_p X \otimes I)R] \leq |N(p)| \sum_{p'} \|A_{p'p}\|^2 \text{Tr}_{H_{p'}} R_{p'} \leq \Delta^2 \|A\|^2 \alpha_* I$ . So  $\sum(\partial Z)^2 \leq 2(1 + \Delta^2) \|A\|^2 \alpha_* F^* F$  for every  $V$ . As in (a),  $f'(\theta) \leq \theta(1 + \Delta^2) \|A\|^2 \alpha_* g f(\theta)/D$ ,  $f(\theta) \leq N \exp(\theta^2(1 + \Delta^2) \|A\|^2 \alpha_* g/(2D))$ , and the tail follows at  $\theta = sD/((1 + \Delta^2) \|A\|^2 \alpha_* g)$ .  $\square$

For a free permutation round with point blocks, the Hermitian and anti-Hermitian parts of  $\lambda_g^* \lambda_h \otimes I$  ( $g \neq h$ ) couple each block to at most two blocks, so  $\Delta = 2$  there.

*Proof of Lemma G.2.* (a) Export and import rearrange the columns of each block isometrically: the port moves to the exterior, and a maximally entangled pair joins the new port and a new exterior register. So the block Grams do not change. The new exterior marginal of block  $p$  is  $Y_p \otimes I_e/e$  with  $Y_p = \text{Tr}_{\tilde{H}_p}[(V_p' \otimes I)R_p(V_p' \otimes I)^*]$ , a function of  $V_p'$  alone, so  $\alpha_p(F') = \|Y_p\|/e$ , and  $\mathbb{E}Y_p = (I_e/e) \otimes \text{Tr}_{H_p} R_p$  has norm  $\alpha_p/e$ . As in [27, Lemma A.4], the frame  $\hat{F}_p : \mathbb{C}^S \rightarrow H_p \otimes \mathbb{C}^N$  with entries  $\hat{F}_{(q,y),\sigma} = \overline{F_{(q,\sigma),y}}$  satisfies  $\text{Tr}_{H_p}[(X \otimes I)R_p] = \hat{F}_p^*(X^T \otimes I)\hat{F}_p$ ,  $\hat{F}_p^* \hat{F}_p = \text{Tr}_{H_p} R_p$  and  $\alpha(\hat{F}_p) = g_p$ , and the block  $(k, l)$  of  $Y_p$  is  $\text{Tr}_{H_p}[(X_{kl} \otimes I)R_p]$  with  $X_{kl}^T = \overline{V_p'}^*(I \otimes |k\rangle\langle l|)\overline{V_p'}$ , which is Haar. Part (a) of Lemma G.1 for one block, applied to  $\hat{F}_p$  ( $S$  columns) and to the Hermitian parts of  $I \otimes |k\rangle\langle l|$  at level  $z_p = 2\sqrt{\alpha_p g_p \ln(8Se^2 N)/D_p}$ , fails for each of the  $2e^2$  events with probability at most  $2Se^{-\ln(8Se^2 N)} = 1/(4e^2 N)$ ; so block  $p$  fails with probability at most  $1/(2N)$ , and some block with probability at most  $\frac{1}{2}$ . Otherwise each block of  $Y_p$  is within  $2z_p$  of its mean, the  $e \times e$  block matrix within  $2ez_p$ , so  $\|Y_p\| \leq \alpha_p/e + 2ez_p$  and  $\alpha_p(F') \leq \alpha_p/e^2 + 2z_p$ . The second form is  $2\sqrt{uv} \leq \eta u + v/\eta$  with  $u = \alpha_p/e^2$  and  $v = 4e^2 g_p \ln(8Se^2 N)/D_p$ .

(b) The block- $Q$  part of the frame after the round has  $R'_Q = \sum_i (P'_Q B_i V \otimes I)R(V^* B_i^* P'_Q \otimes I)$ , so by (F3)  $\text{Tr}_{H'_Q} R'_Q = \text{Tr}_{\text{mem}}[(V^* \Gamma_Q V \otimes I)R]$ . If  $\Gamma_Q$  is block diagonal,  $V^* \Gamma_Q V = \bigoplus_p V_p^* (P_p \Gamma_Q P_p) V_p \leq \sum_p \|P_p \Gamma_Q P_p\| P_p$ , and (F3) gives  $\alpha_Q(F') \leq \sum_p \|P_p \Gamma_Q P_p\| \alpha_p(F)$ .  $\square$

For a free permutation round with point blocks, a basis permutation before the round (absorbed in  $F$ ) and a fixed partial injection  $\iota$  of points after it,  $\Gamma_Q = \sum_g P_{(\iota\lambda_g)^{-1}(Q)} \otimes I_m$  is diagonal,  $(\iota\lambda_g)^{-1}(Q)$  is at most one point for each  $g$ , and no two labels send a point to the same point. So  $\|P_p \Gamma_Q P_p\| \leq 1$  for every point  $p$ , at most  $r$  points have  $\|P_p \Gamma_Q P_p\| = 1$ , and  $\alpha_Q(F') \leq r \max_p \alpha_p(F)$ : this is the inflow factor of [28, Theorem 12.2].

## K.2 Paths

*Proof of Lemma G.3.* The matrix  $M^{(b)}$  is the Gram matrix of the vectors  $C_j^{(b)}$  in the normalized Hilbert–Schmidt space, so  $M^{(b)} \geq 0$ , and  $\sum_b C_i^{(b)*} C_j^{(b)} = B_i^* (\sum_b J_b J_b^*) B_j = B_i^* B_j$  gives  $\sum_b M^{(b)} = [\tau_k(B_i^* B_j)] = I_r$ . With the Stinespring isometry  $V : \xi \mapsto \sum_i A_i \xi \otimes |i\rangle$ ,  $\tilde{A}^{(b)} := \sum_{i,j} \langle i|M^{(b)}|j\rangle A_i^* A_j = V^*(I \otimes M^{(b)})V$ , so  $\tilde{A}^{(b)} \geq 0$ ,  $\sum_b \tilde{A}^{(b)} = I$ , and  $w(b) = \|\tilde{A}^{(b)}\| \leq \|M^{(b)}\| \leq \text{Tr } M^{(b)}$ .

(a)  $(\beta I - X_b) \otimes M^{(b)}$  and  $(\beta I + X_b) \otimes M^{(b)}$  are positive; summing over  $b$  and using  $\sum_b I \otimes M^{(b)} \leq I$  gives  $-\beta I \leq \sum_b X_b \otimes M^{(b)} \leq \beta I$ .

(b) In the ideal experiment round  $s$  applies  $V_s : \xi \mapsto \sum_i A_i \xi \otimes |i\rangle_{E_s}$ , and  $E_s$  is never touched again. Let  $N_{\leq s} = M^{(b_1)} \otimes \dots \otimes M^{(b_s)}$  on  $E_1, \dots, E_s$ . Summing over the later letters (the matrices  $M^{(b)}$  sum to  $I$ ),  $\Pr(b_1, \dots, b_s) = \langle X | N_{\leq s} | X \rangle$ , and since nothing acts on  $E_{\leq s}$  after round  $s$  this equals  $\langle X_s | N_{\leq s} | X_s \rangle$  for the global vector  $X_s$  right after round  $s$ . Write  $X_s = (V_s \otimes I)Y_s$ , with  $Y_s$  the vector just before the use of round  $s$ . Then  $\Pr(b_{\leq s}) = \langle Y_s | \tilde{A}^{(b_s)} \otimes N_{< s} | Y_s \rangle \leq w(b_s) \langle Y_s | I \otimes N_{< s} | Y_s \rangle = w(b_s) \Pr(b_{< s})$ , because  $(w(b_s)I - \tilde{A}^{(b_s)}) \otimes N_{< s} \geq 0$  and  $E_{< s}$  is untouched between round  $s-1$  and  $Y_s$ . The same computation with  $\sum_b \tilde{A}^{(b)} = I$  shows that  $\Pr$  is a probability law. So the conditional law  $q$  of  $b_s$  given  $b_{< s}$  satisfies  $q_b \leq w(b)$ , and  $\mathbb{E}[w(b_s)^{-\theta} | b_{< s}] \leq \kappa_\theta$ ; by the tower property  $\mathbb{E}[W(p)^{-\theta}] \leq \kappa_\theta^n$ , and Markov's inequality gives  $\Pr\{W(p) < 2^{-\lambda}\} = \Pr\{W(p)^{-\theta} > 2^{\theta\lambda}\} \leq \kappa_\theta^n 2^{-\theta\lambda}$ . Since  $w \leq 1$ ,  $W$  does not increase along extensions, so  $P_t(\lambda)$  is prefix-closed;  $W(p)$  depends only on the type of  $p$ ; and  $\sum_{p \in [c]^t} W(p) = 2^{tu'}$  with  $W(p) \geq 2^{-\lambda}$  on  $P_t(\lambda)$  gives  $|P_t(\lambda)| \leq 2^{\lambda+tu'}$ .  $\square$

*Proof of Theorem G.4. The device.* Let  $h = \log r$ ,  $E_p = 2^{\lceil h/2 \rceil}$ ,  $J_t = \lfloor th/2 \rfloor$ ,  $j_t = J_t - J_{t-1}$  and  $e_t = 2^{j_t}$  (the export schedule of [28, Theorem 12.2]),  $x = \log(32n/\epsilon^2)$ ,  $\lambda = (n \log \kappa_\theta + x)/\theta$  and  $P_t = P_t(\lambda)$ . The memory consists of a register with  $2^{\lceil \lambda + nu' \rceil} \geq \max_t |P_t|$  values, holding  $\text{rank}_t(p)$ , the rank of  $p$  in a fixed enumeration of  $P_t$ ; a register  $\mathbb{C}^c$ ;  $K$ ; and  $\mathbb{C}^m = \mathbb{C}^{m'} \otimes \mathbb{C}^{E_p}$ , whose last factor is the port. So  $Q = \lceil \lambda + nu' \rceil + \lceil \log c \rceil + \lceil \log k \rceil + \lceil \log m \rceil$ . Initially the register holds  $\text{rank}_0(\emptyset)$ ,  $\mathbb{C}^c$  holds  $|1\rangle$  and  $K \otimes \mathbb{C}^m$  is maximally mixed. Round  $t$  applies: (1) controlled on  $\text{rank}_{t-1}(p)$ , a unitary  $V_{t,p}$  on  $K \otimes \mathbb{C}^m$ ; (2) a unitary extension of  $W \otimes I_m$  from  $\mathbb{C}^d \otimes |1\rangle \otimes K \otimes \mathbb{C}^m$  to  $\mathbb{C}^d \otimes \mathbb{C}^c \otimes K \otimes \mathbb{C}^m$ , identifying  $J_b K$  with  $|b\rangle \otimes K$ , and the release of the output; (3) a permutation of the register values and  $[c]$  that maps  $(\text{rank}_{t-1}(p), b)$  to  $(\text{rank}_t(pb), 1)$  whenever  $pb \in P_t$  (an injective partial map, completed to a permutation); (4) controlled on  $\text{rank}_t(p')$ , a unitary  $V'_{t,p'}$ ; (5) if  $t < n$ , the export of  $j_t$  port qubits and the import of  $j_t$  maximally mixed qubits. All unitaries are fixed in advance, so  $C = 0$  and  $J = J_{n-1}$ .

*Proof object.* The untruncated device has memory  $\bigoplus_{p \in [c]^t} H_p$ ,  $H_p \cong K \otimes \mathbb{C}^m$  of dimension  $D_b = km$ , replaces step (3) by the relabelling of block  $p \otimes |b\rangle$  as block  $pb$ , and uses unitaries  $V_{t,p}, V'_{t,p'}$  for all paths. It is an isometric evolution, and on the blocks of  $P_{t-1}$  it coincides with the device up to the relabelling of the register. Its frame  $F_t : \mathbb{C}^{r^t} \rightarrow (\bigoplus_p H_p) \otimes \mathbb{C}^{S_t}$  has one column per Kraus history;  $F_0$  is one unit vector with  $\alpha(F_0) = 1/D_b$ ; and the column  $(y, i)$  of block  $pb$  after the round is  $(C_i^{(b)} \otimes I_m \otimes I)(V_{t,p} \otimes I)F_{t-1,p}|y\rangle$ . Let  $G_{t,\tau} = \sum_{p \text{ of type } \tau} F_{t,p}^* F_{t,p}$  and  $\Gamma_{t,\tau} = \sum_{p \text{ of type } \tau} M^{(p_1)} \otimes \dots \otimes M^{(p_t)}$ .

*Induction.* Put  $\delta_1 = \epsilon/2$ ,  $\eta = \epsilon^2/(32n^2(n+1)^{c-1})$ ,  $L_* = \ln(8D_b r^n E_p^2 c^n)$  and  $A = 4e(1 + 8E_p^2 n^2 L_*)$ . For  $0 \leq t \leq n$  we show (I1)  $\|F_t^* F_t - I\| \leq t\delta_1/n$ ; (I2)  $\alpha_p(F_t) \leq A/D_b$  for every  $p$  (for  $t \leq n-1$ ); (I3)  $\|G_{t,\tau} - \Gamma_{t,\tau}\| \leq t\eta$  for every type  $\tau$ . All hold at  $t = 0$ . Assume them at  $t-1$ , so  $g(F_{t-1}) \leq 2$ .

*Choice of  $V_t$ .* Apply Lemma G.1(a) to  $F_{t-1}$  with the following operators; always  $\|A_p\| \leq r$ , since  $\|B_i\|^2$  and  $\|C_i^{(b)}\|^2$  are at most  $\|\sum_i B_i^* B_i\| = r$ , so  $\kappa \leq (A/D_b)r^2/D_b$  and  $g_0 \leq 2$ . (E1) For each  $(i, j)$ , the Hermitian and anti-Hermitian parts of  $B_i^* B_j \otimes I_m$  in every block, at level  $s_1 = \delta_1/(2rn)$ : the block  $(i, j)$  of the full Gram matrix after the round is  $\sum_p F_p^* [(V_p^* (B_i^* B_j \otimes I_m) V_p) \otimes I] F_p$ , because children of different letters or different parents are orthogonal, and its mean is  $\tau(B_i^* B_j) F^* F = \delta_{ij} F^* F$ ; on success the  $r \times r$  block matrix deviates by at most  $2rs_1 = \delta_1/n$ , which gives (I1). (E2) For each type  $\tau'$  of length  $t$ , each letter  $b$ , each  $(i, j)$  and the Hermitian or anti-Hermitian part of  $C_i^{(b)*} C_j^{(b)} \otimes I_m$  on the blocks of type  $\tau' - e_b$  ( $A_p = 0$  elsewhere), at level  $s_3 = \eta/(2cr)$ . The new type Gram is  $G'_{\tau'} = \sum_b \sum_{p \in \tau' - e_b} F_p^* [(V_p^* X^{(b)} V_p) \otimes I] F_p$  with  $X^{(b)} = [C_i^{(b)*} C_j^{(b)} \otimes I_m]_{ij}$ , whose mean is  $\sum_b G_{t-1, \tau' - e_b} \otimes M^{(b)}$  by Schur's lemma; on success it deviates from that mean by at most

$2crs_3 = \eta$ , and Lemma G.3(a) with  $X_b = G_{t-1, \tau' - e_b} - \Gamma_{t-1, \tau' - e_b}$  gives  $\|G_{t, \tau'} - \Gamma_{t, \tau'}\| \leq (t-1)\eta + \eta$ , which is (I3). Each event fails with probability at most  $2r^n \exp[-s^2 D_b^2 / (8Ar^2)]$ ; there are  $N_E = 2r^2(1 + c(n+1)^{c-1})$  events and  $s_3 \leq s_1$ , so all succeed with probability at least  $\frac{1}{2}$  if

$$D_b^2 \geq 8Ar^2 \ln(4r^n N_E) / s_3^2, \quad s_3 = \epsilon^2 / (64crn^2(n+1)^{c-1}). \quad (26)$$

Fix such a  $V_t$ .

*Inflow.* By (F3), for the child block  $pb$ ,  $\text{Tr}_{H_{pb}}(F'_{pb} F_{pb}^*) = \text{Tr}_{H_p}[(V_p^* \Gamma_b V_p) \otimes I] R_p$  with  $\Gamma_b = \sum_i C_i^{(b)*} C_i^{(b)} \otimes I_m \leq \sum_i B_i^* B_i \otimes I_m = rI$ , so  $\alpha_{pb} \leq r a_{t-1}$  with  $a_{t-1} := \max_p \alpha_p(F_{t-1})$ . Each child has a single parent; this is where the tree structure of paths is used.

*Choice of  $V'_t$ .* Lemma G.2(a) with  $e = e_t$ ,  $\eta = 1/n$ ,  $g_p \leq 2$ , at most  $c^t$  blocks and  $S_t e_t^2 \leq D_b r^n E_p^2$  gives  $a_t \leq (1 + 1/n)r a_{t-1} / e_t^2 + 8E_p^2 n L_* / D_b$ ; in rounds with  $j_t = 0$ ,  $a_t \leq r a_{t-1}$  directly. With  $c_t = 2^{2j_t} r^{-t} \in (\frac{1}{4}, 1]$  and  $c_t r / e_t^2 = c_{t-1}$ ,  $c_t a_t \leq (1 + 1/n)c_{t-1} a_{t-1} + 8E_p^2 n L_* / D_b$ , so  $c_t a_t \leq (1 + 1/n)^t (1 + 8E_p^2 n^2 L_*) / D_b$  and  $a_t \leq A / D_b$ , which is (I2).

*Error.* Purify the observer. The device acts on the input and the memory only through  $W$ , so the untruncated final vector is  $\tilde{x}_n = \sum_y \psi_y \otimes F_n |y\rangle = (I \otimes F_n)X$ , with the observer's vectors  $\psi_y$  of the ideal experiment and  $X = \sum_y \psi_y \otimes |y\rangle$  a unit vector, and  $\|\tilde{x}_n\| = 1$ . Let  $\Pi$  project onto the blocks of  $P_n$ ,  $\omega = \|(I - \Pi)\tilde{x}_n\|^2$ ,  $x_n$  the physical final vector and  $T$  the polar isometry of  $F_n$ .

1. *Polar step.*  $\|\tilde{x}_n - (I \otimes T)X\| \leq \|(F_n^* F_n)^{1/2} - I\| \leq \|F_n^* F_n - I\| \leq \delta_1$ , since  $|\sqrt{u} - 1| \leq |u - 1|$  for  $u \geq 0$ .
2. *Truncation.* The untruncated dynamics never changes the prefix of a path, the observer does not act on the memory, and  $P_t$  is prefix-closed; so the projection onto “first exit from  $P$  at time  $t$ ” commutes with all later rounds, and with  $\Pi_{\text{fe}=t}$  that projection,  $\sum_t \|\Pi_{\text{fe}=t} \tilde{x}_n\|^2 = \omega$ . Let  $g_t$  be the good part of  $\tilde{x}_t$  (blocks of  $P_t$ ), embedded in the physical memory, and  $\mathcal{U}_t$  the physical round together with the observer's operation. On the good part the physical round is the untruncated one, and the permutation of step (3) maps good inputs bijectively onto good outputs, so  $\mathcal{U}_t g_{t-1} = g_t + d_t$  with  $\|d_t\| = \|\Pi_{\text{fe}=t} \tilde{x}_t\| = \|\Pi_{\text{fe}=t} \tilde{x}_n\|$ . From  $x_t - g_t = \mathcal{U}_t(x_{t-1} - g_{t-1}) + d_t$  and  $x_0 = g_0$ ,  $\|x_n - g_n\| \leq \sum_t \|d_t\| \leq \sqrt{n\omega}$ . Discarded branches may re-enter good register values; the bound does not care.
3. *Discarded weight.*  $P_n$  is a union of types, so  $\omega = \sum_{\tau \notin P_n} \langle X | I \otimes G_{n, \tau} | X \rangle \leq \sum_{\tau \notin P_n} \langle X | I \otimes \Gamma_{n, \tau} | X \rangle + |\mathcal{T}_n| n\eta$ , using  $|\langle X | I \otimes Y | X \rangle| \leq \|Y\|$  and (I3); here  $|\mathcal{T}_n| \leq (n+1)^{c-1}$  is the number of types, so  $|\mathcal{T}_n| n\eta \leq \epsilon^2 / (32n)$ . The first term is the probability under the path law that  $W(p) < 2^{-\lambda}$ , at most  $\kappa_\theta^n 2^{-\theta\lambda} = 2^{-x} = \epsilon^2 / (32n)$  by Lemma G.3(b). So  $\omega \leq \epsilon^2 / (16n)$ .
4. By (F4), with  $\|x_n\| = \|\tilde{x}_n\| = 1$  and  $\|g_n\| \leq 1$ , the observer's state from  $x_n$  is within  $\sqrt{n\omega}$  of that from  $g_n$ , which equals that from  $\Pi \tilde{x}_n$ ; that is within  $\sqrt{\omega}$  of the state from  $\tilde{x}_n$ , which is within  $\delta_1$  of the ideal state from  $(I \otimes T)X$ . The error is at most  $\delta_1 + (1 + \sqrt{n})\sqrt{\omega} \leq \epsilon/2 + (1 + \sqrt{n})\epsilon/(4\sqrt{n}) < \epsilon$ .

*Memory.*  $Q \leq \lambda + nu' + \log c + \log k + \log m + 4$ . Condition (26) reads  $D_b \geq K_{r,c} n^{c+3} \epsilon^{-2} (\ln D_b + n)^{1/2}$  up to constants, since  $A = O(n^2 L_*)$  and  $L_* = O_r(n + \ln D_b)$ ; take  $m'$  least with  $D_b = km' E_p$  satisfying it. Then  $\log m = \log(D_b/k) = O_r(\log(n/\epsilon) + \log(2 + \log k))$ , as in the proof of [28, Theorem 12.2].  $\square$

### K.3 Flat rounds

*Proof of Lemma G.5.* Each  $B_g B_g^* / k \leq I/k$ , so the eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots$  of  $\bar{\tau}$  are at most  $1/k$ , and  $a = \sum_i \lambda_i \log(1/(k\lambda_i))$  with nonnegative terms. There are at least  $k$  positive

eigenvalues, so  $x := \lambda_k > 0$ ; put  $y = kx \in (0, 1]$ . For  $i > k$ ,  $\lambda_i \leq x$ , so the tail contributes at least  $w \log(1/y)$ . For  $i \leq k$ ,  $\log(1/(k\lambda_i)) \geq (1 - k\lambda_i)/\ln 2$  and  $\lambda_i \geq x$ , so the top contributes at least  $(x/\ln 2) \sum_{i \leq k} (1 - k\lambda_i) = (x/\ln 2)(k - k(1 - w)) = yw/\ln 2$ . Hence  $a \geq (w/\ln 2)(y - \ln y) \geq w/\ln 2$ .  $\square$

*Proof of Theorem G.6.* (a) By Lemma 2.10 with  $u = 0$ , the round restricted to  $K = \text{supp } \rho$  is an exact rectangular dilation  $(B_g)$ , and since the  $A_g$  have disjoint supports,  $T(\rho) = \sum_g \mu_g B_g B_g^*/k =: \bar{\tau}$ , so  $a = S(\bar{\tau}) - \log k$ . Restrict  $\mathbb{C}^{k'}$  to the span of the ranges  $(k' \leq rk)$  and choose the pieces:  $J_1 K$  the top- $k$  eigenspace of  $\bar{\tau}$ , and the complement, padded, cut into  $k$ -dimensional pieces,  $c \leq r + 1$  in all. Then  $\langle g | M^{(1)} | g \rangle = \text{Tr}(P_{J_1 K} \omega_g)$  with  $\omega_g = B_g B_g^*/k$ , and

$$\beta := \sum_{b \geq 2} w(b) \leq \sum_{b \geq 2} \text{Tr } M^{(b)} = \sum_g (1 - \langle g | M^{(1)} | g \rangle) \leq \mu_{\min}^{-1} \sum_g \mu_g (1 - \langle g | M^{(1)} | g \rangle) \leq d a \ln 2$$

by Lemma G.5, since  $\mu_{\min} \geq 1/d$ . As  $w(1) \leq 1$ ,  $u' \leq \log(1 + \beta) \leq 1.443\beta$ . For  $q \leq w$ ,  $\sum_b q_b w(b)^{-\theta} \leq \sum_b w(b)^{1-\theta} \leq 1 + (c-1)\bar{\beta}^{1-\theta}$  by Hölder's inequality over  $b \geq 2$ , so  $\kappa_\theta \leq 1 + (c-1)\bar{\beta}^{1-\theta}$ . Take  $\theta = 1/\log(1/\bar{\beta})$  with  $\bar{\beta} = \max(\beta, \epsilon^2/n^2) \leq \frac{1}{4}$ ; this needs  $a \leq 1/(4d \ln 2)$ . Then  $\theta \leq \frac{1}{2}$  and  $\bar{\beta}^{-\theta} = 2$ , so  $(c-1)\bar{\beta}^{1-\theta} \leq 2\sqrt{c-1}\bar{\beta}$  and  $\log \kappa_\theta / \theta \leq 2.885\sqrt{c-1}\bar{\beta} \log(1/\bar{\beta})$ . Also  $\log(32n/\epsilon^2)/\theta = \log(1/\bar{\beta}) \log(32n/\epsilon^2) \leq 2 \log(n/\epsilon) \log(32n/\epsilon^2)$ , and if  $\bar{\beta} = \epsilon^2/n^2$  then  $n\bar{\beta} \log(1/\bar{\beta}) = 2(\epsilon^2/n) \log(n/\epsilon) \leq 2$ . Theorem G.4 gives the first bound, since  $x \log(1/x)$  increases on  $(0, 1/e)$  and  $\beta \leq d a \ln 2$ . For the second: if  $a \geq \epsilon^2/n^2$  then  $\log(e/a) \leq 1.443 + 2 \log(n/\epsilon)$ , and if  $a < \epsilon^2/n^2$  then  $n a \log(e/a) \leq (\epsilon^2/n)(1.443 + 2 \log(n/\epsilon)) \leq 2.45$ ; so  $n a \log(e/a) \leq 2.45 + n a (1.443 + 2 \log(n/\epsilon))$ , and every term is at most  $K \log^2(n/\epsilon)(1 + \log k + n a)$ . If  $a > 1/(4d \ln 2)$ , Theorem 7.6(a) with  $G = I$ , which is [28, Theorem 12.2], gives  $Q \leq \log k + n \log(k'/k) + O_r(\log(n/\epsilon) + \log(2 + \log k))$  with  $\log(k'/k) \leq \log r \leq 4d \ln 2 \log r \cdot a$ .

(b) The changes to the proof of Theorem G.4 are those of Theorem 7.6(a). (I1) becomes  $\|F_t^* F_t - G^{\otimes t}\| \leq e^{1/2} t \delta_1 / n$ : the mean of the block  $(i, j)$  after a round is  $\tau(B_i^* B_j) F^* F$ , so the deviation  $e_t$  from  $G^{\otimes t}$  satisfies  $e_t \leq \|G\| e_{t-1} + \delta_1 / n$ , and  $\|G\|^n \leq e^{n\gamma} \leq e^{1/2}$ ; then  $g(F_t) \leq 4$ , and 2 is replaced by 4 in (26) and in Lemma G.2(a). (I2) is unchanged, since  $\sum_i B_i^* B_i = rI$  by Lemma 2.9(b). (I3) becomes  $\|G_{t,\tau} - \Gamma_{t,\tau}\| \leq e^{1/2} t \eta$ ; we run the construction with  $\eta/\sqrt{e}$  in place of  $\eta$ , which restores the discarded-weight bound  $\omega \leq \epsilon^2/(16n)$  and changes only constants. This uses Lemma G.3(a) with  $\sum_b M^{(b)} = G$ , which gives  $\|\sum_b X_b \otimes M^{(b)}\| \leq \beta \|G\|$ . Applying  $\text{id} \otimes \tau_K$  to  $W^* W = I$  gives  $\sum_b A^{(b)} = \sum_{i,j} G_{ij} A_i^* A_j = I$ , so the path law is still a probability law and Lemma G.3(b) holds verbatim. In the polar step,  $\Gamma = G^{\otimes n} \geq (1 - \gamma)^n I \geq e^{-1} I$ , and as in Theorem 7.6(a),  $\|(F_n^* F_n)^{1/2} - \Gamma^{1/2}\| \leq e^{1/2} \delta_1 / e^{-1/2} = e \delta_1$ ; so  $\tilde{x}_n$  is within  $e \delta_1$  of  $(I \otimes T)(I \otimes \Gamma^{1/2})X$ , the final vector of the experiment that applies  $\Phi_W(X) = \sum G_{ji} A_i X A_j^*$  in every round, and a hybrid over the rounds costs  $n d_\infty(\Phi_W, \Phi) \leq \frac{n}{2} \gamma$ . With  $\delta_1 = \epsilon/(2e)$  the error is at most  $\epsilon + \frac{n}{2} \gamma$ .

(c) By Lemma 2.10 the round gives isometries  $B_g = X_g|_K$  forming a gadget family with flat Gram  $G$ ,  $G_{gg} = 1$  and  $|G_{gh}| \leq c$ . By Theorem 7.6(b) the family  $(B_g^{\otimes t})$  is a gadget family with Gram entries  $G_{gh}^t$ , so  $\gamma_t := \|G^{(t)} - I\| \leq (r-1)c^t \leq \epsilon/(4n)$ ,  $n\gamma_t \leq \frac{1}{2}$  and  $\frac{n}{2}\gamma_t \leq \epsilon/8$ . Its flat entropy production  $a_t = S(\sum_g \mu_g \omega_g^{\otimes t}) - t \log k$  is at most  $t a$  by subadditivity, since every one-factor marginal of  $\sum_g \mu_g \omega_g^{\otimes t}$  is  $\bar{\tau}$ . Restrict the codomain to the span of the ranges of the  $B_g^{\otimes t}$ , of dimension at most  $r k^t$ , and take the pieces of (a) for the  $t$ -fold family; Lemma G.5 applies to it, so  $\beta_t \leq d a_t \ln 2 \leq d t a \ln 2$ . Part (b) with the choice of  $\theta$  of (a) gives the second bound when  $d t a \ln 2 \leq \frac{1}{4}$ ; otherwise Theorem 7.6(c), run at error  $2\epsilon/(e^2 + 1)$  so that its error is at most  $2\epsilon$ , gives  $Q \leq t' \log k + n t' \log(k'/k) + O_r(\cdot)$  with its own  $t' \leq t + 2$  and  $\log(k'/k) \leq \log r \leq 4d t \ln 2 \log r \cdot a$ . For the polylogarithmic form,  $t \leq \log(4rn/\epsilon) + 1 \leq 3 \log(n/\epsilon) + \log(8r)$  for  $n \geq 2$  and  $\epsilon < 1$ , and  $t a \log(e/(ta)) \leq t a \log(e/a)$ .  $\square$

## L Proofs for Section G: leakage and the twisted torus

We write  $S_u(X) = -\text{Tr } X \log X$  for  $X \geq 0$ , the entropy of an unnormalized operator, and use two facts.

(F5) *Subadditivity.* For  $M, N \geq 0$ ,  $S_u(M + N) \leq S_u(M) + S_u(N)$ , because  $\log$  is operator monotone, so  $\text{Tr } M \log(M + N) \geq \text{Tr } M \log M$ . A positive operator of trace  $m$  and rank at most  $R$  has  $S_u \leq m \log(R/m)$ .

(F6) *Concavity, unnormalized.* If  $A, B \geq 0$  have traces  $m_A, m_B$  with  $m_A + m_B = m$ , then  $S_u(A + B) \geq S_u(A) + S_u(B) - m h_2(m_A/m)$ .

### L.1 Exact regions

*Proof of Lemma G.7.* Write  $\rho = (1-\nu)\rho_0 + \nu\rho_\perp$  with  $\rho_\perp$  flat on  $K \ominus K_0$ . The channel and  $T$  are linear in the bath, so  $\Psi_{\rho_0} - \Phi = (\Psi_\rho - \Phi - \nu(\Psi_{\rho_\perp} - \Phi))/(1-\nu)$  with  $d_\diamond(\Psi_{\rho_\perp}, \Phi) \leq 1$ , which gives the distance. Since  $\rho_\perp \leq I/(\nu k)$ ,  $U(\tau_d \otimes \rho_\perp)U^* \leq I/(d\nu k)$ , and the partial trace gives  $T(\rho_\perp) \leq I/(\nu k)$ , so  $S(T(\rho_\perp)) \geq \log(\nu k)$ . Concavity gives  $S(T(\rho)) \geq (1-\nu)S(T(\rho_0)) + \nu \log(\nu k)$ ; with  $S(T(\rho)) = a + \log k$  this reads  $S(T(\rho_0)) \leq \log k + (a + \nu \log(1/\nu))/(1-\nu)$ , and subtracting  $S(\rho_0) = \log((1-\nu)k)$  gives the bound on  $a_0$ . If  $U(\xi \otimes \kappa) = \sum_i A_i \xi \otimes X_i \kappa$  on  $\mathbb{C}^d \otimes K_0$ , then  $\Psi_{\rho_0}(X) = \sum_{i,j} A_i X A_j^* \text{Tr}(X_i \rho_0 X_j^*)$ , whose Choi state is supported in the Kraus span.  $\square$

*Proof of Theorem G.8.* (a) By Lemma G.7,  $(U, \rho_0)$  has leakage 0,  $S(\rho_0) \leq S$  and distance at most  $(\frac{1}{16} + \frac{1}{8})/\frac{7}{8} = \frac{3}{14} < \frac{1}{4}$ , and since  $\log(1/(1-\nu)) \leq \frac{8}{7}\nu/\ln 2 \leq 2\nu$  for  $\nu \leq \frac{1}{8}$ ,  $na_0 \leq \frac{8}{7}(na + n\nu \log(1/\nu)) + 2n\nu \leq 2(na + n\nu \log(e/\nu))$ . By Lemma 2.10 its gadget family has flat Gram off-diagonals at most  $2 \cdot \frac{3}{14} < \frac{1}{2}$ , and Theorem G.6(c) applies.

(b) At a triangle-good point every slot of label  $g$  acts by  $f(g)$ , so  $U(\xi \otimes |\omega\rangle) = \sum_g A_g \xi \otimes |f(g)\omega\rangle$ , and by linearity the span of the good points of  $K$  is an exact region. By Lemma 2.3,  $\nu = p(\text{bad}) \leq 4d\ell$ . For  $\ell \leq C_0/n$  and  $n \geq 32dC_0$ ,  $\nu \leq \frac{1}{8}$  and (a) applies;  $x \log(e/x)$  increases on  $(0, 1]$ , so  $n\nu \log(e/\nu) \leq 4dC_0 \log(en/(4dC_0))$ .  $\square$

### L.2 First-order rounding

*Proof of Theorem G.9. Decomposition.* Let  $W = U(I \otimes \iota_K)$ , and expand it in the orthogonal basis  $\{A_g/\|A_g\|_F\} \cup \{E_\alpha\}$  of  $M_d$ , where  $(E_\alpha)$  is a Hilbert–Schmidt orthonormal basis of  $\text{span}\{A_g\}^\perp$ :  $W = G + L$  with  $G = \sum_g A_g \otimes X_g$  and  $L = \sum_\alpha E_\alpha \otimes L_\alpha$ . Then:

- (f1)  $\ell = \|L\|_F^2/(dk)$ . Indeed  $\mathcal{J}(\Psi)$  is the marginal of  $v = d^{-1/2}(\sum_g |A_g\rangle\rangle \otimes X_g \xi + \sum_\alpha |E_\alpha\rangle\rangle \otimes L_\alpha \xi)$ ,  $\xi = P_K/\sqrt{k}$  (proof of Lemma 2.10), and  $I - \Pi_\Phi$  keeps exactly the second sum.
- (f2)  $\|W\|_F^2 = dk = \|G\|_F^2 + \|L\|_F^2$  and  $\|G\|_F^2 = \sum_g d\mu_g k \tau(X_g^* X_g)$ , so  $\sum_g \mu_g \tau(X_g^* X_g) = 1 - \ell$ .
- (f3)  $T(\rho) = \sigma_X + \sigma_L$  with  $\sigma_X = \sum_g \mu_g X_g X_g^*/k$  and  $\sigma_L = \text{Tr}_d L(\tau_d \otimes \rho)L^* \geq 0$ ,  $\text{Tr } \sigma_L = \ell$ : the cross terms vanish because  $\text{Tr}(A_g^* E_\alpha) = 0$ , and  $\text{Tr}(A_h^* A_g) = d\mu_g \delta_{gh}$ .
- (f4)  $X_g = \sum_{a \in S_g} \bar{c}_a W_a / \sum_{a \in S_g} |c_a|^2$  over the slots  $a$  of label  $g$ , where  $W_a$  is the block of  $W$  at the position of  $a$  and  $\|W_a\| \leq 1$ . Every slot coefficient has modulus 1 or  $1/\sqrt{2}$ , so  $\|X_g\| \leq \sqrt{2} \leq d$ .

*Defects.* From  $W^*W = I$ ,  $G^*G - I = -(G^*L + L^*G + L^*L)$ , whose unnormalized trace norm is at most  $2\|G\|_F\|L\|_F + \|L\|_F^2 \leq dk(2\sqrt{\ell} + \ell)$ . By the block structure of Lemma 2.9(a) and (F1), the anchor block of  $G^*G$  at  $z_g$  is  $X_g^* X_g$ , and the off-diagonal block of a triangle  $t = (x, y, z)$  is  $\frac{1}{2}R_t$  with  $R_t := X_1^* X_{[y]} - X_{[x]}^* X_{[z]}$ . A block has trace norm at most that of the operator, so in the normalized trace norm on  $K$ ,  $\|X_g^* X_g - I\|_1 \leq 3d\sqrt{\ell}$  and  $\|R_t\|_1 \leq 6d\sqrt{\ell}$ .

*Construction.* Let  $Y^0 = [X_g^* X_h]_{g,h} \geq 0$  on  $\mathbb{C}^r \otimes K$ .

1.  $P := \sum_g (X_g^* X_g - I)_+$ , so  $\tau(P) \leq 3dr\sqrt{\ell}$ ;  $T := (I + P)^{-1/2}$  and  $Y_A := (I_r \otimes T)Y^0(I_r \otimes T)$ . Since  $X_g^* X_g \leq I + P$ ,  $(Y_A)_{gg} \leq I$ .
2. Link two ordered pairs of distinct labels if an identity  $(1, [y]) \sim ([x], [z])$  of a triangle or the adjoint relation  $(g, h) \sim (h, g)$  joins them. On a class, of at most  $r^2$  pairs, the constraints say that each block is a fixed common block or its adjoint. Let  $\hat{Y}_{\text{off}}$  be the Frobenius-orthogonal projection of the off-diagonal blocks of  $Y_A$  onto the constraints: on each class the average of the transported blocks, followed by  $R \mapsto (R + R^*)/2$  if the class closes on itself through an adjoint. Each identity edge has defect  $\|TR_t T\|_1 \leq \|R_t\|_1 \leq 6d\sqrt{\ell}$  and adjoint edges have none, so every block moves by at most  $3r^2 \cdot 6d\sqrt{\ell}$  in  $\|\cdot\|_1$ .
3.  $E := \hat{Y}_{\text{off}} - (Y_A)_{\text{off}}$  is Hermitian with zero diagonal blocks, hence traceless, and  $\text{Tr } |E|/k \leq \sum_{g \neq h} \|E_{gh}\|_1 \leq 18dr^4\sqrt{\ell}$ .
4. For  $N \geq 0$  on  $\mathbb{C}^r \otimes K$ ,  $N \leq r I_r \otimes \text{Tr}_{\mathbb{C}^r} N$ : the average of  $N$  over conjugation by the  $r^2$  Weyl unitaries of  $\mathbb{C}^r$  is  $\frac{I}{r} \otimes \text{Tr}_{\mathbb{C}^r} N$ , and  $N$  is one of  $r^2$  positive terms. With  $E = E_+ - E_-$  and  $Z := r \text{Tr}_{\mathbb{C}^r}(E_-)$ ,  $E + I_r \otimes Z \geq E_+ \geq 0$  and  $\tau(Z) = r \text{Tr } E_-/k = r \text{Tr } |E|/(2k) \leq 9dr^5\sqrt{\ell}$ .
5.  $Y_1 := Y_A + \Delta + E + I_r \otimes Z$  with  $\Delta := \bigoplus_g (I - (Y_A)_{gg}) \geq 0$ . Then  $Y_1 \geq 0$ ,  $(Y_1)_{gg} = I + Z$  and  $(Y_1)_{gh} = \hat{Y}_{gh}$  for  $g \neq h$ .
6.  $Y := S_c Y_1 S_c$  with  $S_c := I_r \otimes (I + Z)^{-1/2}$ . Then  $Y \geq 0$  and  $Y_{gg} = I$ , and the off-diagonal blocks satisfy every identity and the adjoint relations, since a common congruence acts on both sides; the identities with  $[y] = 1$  read  $I = I$ , because in a triangle  $[y] = 1$  if and only if  $[x] = [z]$ .
7.  $X'_g := Y^{1/2}(|g\rangle \otimes I_K)$  has  $X_g'^* X'_h = Y_{gh}$ : these are isometries with the exact identities. By Lemma 2.9(a) and (c) they give a round with bath flat on a copy of  $K$  inside  $\mathbb{C}^r \otimes K$ , with leakage 0 and  $S' = \log k$ .

*Gram.* For  $g \neq h$  the anchor entry of  $\Psi$  is  $\langle z_g | \Psi(|z_g\rangle \langle z_h|) | z_h \rangle = \text{Tr}(\rho W_{z_h z_g}^* W_{z_g z_g})$ , with  $W_{z_g z_g} = X_g + \Lambda_g$  and  $\Lambda_g = \sum_\alpha (E_\alpha)_{z_g z_g} L_\alpha$ . By Cauchy–Schwarz and  $\sum_\alpha |(E_\alpha)_{z_g z_g}|^2 \leq 1$ ,  $\Lambda_g^* \Lambda_g \leq \sum_\alpha L_\alpha^* L_\alpha$ , so  $\|\Lambda_g \xi\|_F^2 \leq \text{Tr}(L^* L)/k = d\ell$ . By (F2) the entry has modulus at most  $2u$ , and with  $\|X_g\| \leq d$ ,  $|\tau(X_g^* X_h)| \leq 2u + 2d\sqrt{d\ell} + d\ell$ . Then  $|\tau((Y_A)_{gh}) - \tau(X_g^* X_h)| = |\tau((I - T^2)X_g^* X_h)| \leq d^2\tau(P)$ , since  $0 \leq I - T^2 \leq P$ ; the projection moves  $\tau$  by at most  $18dr^2\sqrt{\ell}$ ; and  $|\tau(Y_{gh}) - \tau(\hat{Y}_{gh})| = |\tau((I - (I + Z)^{-1})\hat{Y}_{gh})| \leq d^2\tau(Z)$ , since  $\|\hat{Y}_{gh}\| \leq d^2$ . With  $\ell \leq \sqrt{\ell}$ ,  $|\tau(Y_{gh})| \leq 2u + (2d^{3/2} + d + 3d^3r + 18dr^2 + 9d^3r^5)\sqrt{\ell} = 2u + C_1\sqrt{\ell}$ .

*Entropy.* The new round's bath map applied to its flat bath is  $\sum_g \mu_g X'_g X'_g^*/k$ , whose nonzero spectrum is that of  $DYD/k$  with  $D := \text{diag}(\sqrt{\mu_g}) \otimes I_K$ , and  $\text{Tr}(DYD)/k = \sum_g \mu_g = 1$ ; so  $a' = S(DYD/k) - \log k$ .

1.  $Y = S_c Y_A S_c + S_c Y_B S_c$  with  $Y_B := \Delta + E + I_r \otimes Z \geq 0$ .  $D$  commutes with  $S_c$ , so  $DYD = M_A + M_B$  with  $M_A := DS_c Y_A S_c D \geq 0$  and  $M_B := DS_c Y_B S_c D \geq 0$ , and by (F5)  $S(DYD/k) \leq S_u(M_A/k) + S_u(M_B/k)$ .
2.  $m_B := \text{Tr } M_B/k \leq \text{Tr}(DY_B D)/k$ , since  $S_c^2 \leq I$ , and  $\text{Tr}(DY_B D)/k = \sum_g \mu_g \tau(I - TX_g^* X_g T) + \tau(Z) = \ell + \sum_g \mu_g \tau(X_g(I - T^2)X_g^*) + \tau(Z) \leq \ell + d^2\tau(P) + \tau(Z) \leq \bar{m}$ , by (f2).  $M_B$  has rank at most  $rk$ , so by (F5)  $S_u(M_B/k) \leq m_B \log k + m_B \log(r/m_B)$ .
3. With  $\mathcal{X} := [X_1 \cdots X_r] : \mathbb{C}^r \otimes K \rightarrow \mathbb{C}^M$  and  $R := T(I + Z)^{-1/2}$ , a contraction,  $S_c Y_A S_c = (I \otimes R)^* \mathcal{X}^* \mathcal{X} (I \otimes R)$ , so  $M_A/k$  has the nonzero spectrum of  $A' := \sum_g \mu_g X_g R R^* X_g^*/k \leq \sigma_X$ . Put  $B' := \sigma_X - A' \geq 0$ ,  $m_X := \text{Tr } \sigma_X = 1 - \ell$ ,  $m_A := \text{Tr } A' = 1 - m_B$  and  $m_{B'} := \text{Tr } B' = m_B - \ell$ . By (F6),  $S_u(A') \leq S_u(\sigma_X) - S_u(B') + h_2(m_{B'}/m_X)$ . Since

$B' \leq \sigma_X \leq d^2 I/k$ ,  $S_u(B') \geq m_{B'} \log(k/d^2)$ . By (f3) and concavity,  $\log k + a = S(\sigma_X + \sigma_L) \geq m_X S(\sigma_X/m_X)$ , so  $S_u(\sigma_X) = m_X S(\sigma_X/m_X) - m_X \log m_X \leq \log k + a + \ell/\ln 2$ .

4. Adding, the terms in  $\log k$  combine to  $(m_B - m_{B'}) \log k = \ell \log k$ :

$$a' \leq a + \ell \log k + \ell/\ln 2 + 2m_{B'} \log d + h_2(m_{B'}/m_X) + m_B \log(r/m_B).$$

With  $m_{B'} \leq m_B \leq \bar{m} \leq \frac{1}{4}$ ,  $m_X \geq \frac{1}{2}$ ,  $h_2$  increasing on  $[0, \frac{1}{2}]$  and  $x \log(r/x)$  increasing on  $(0, r/e]$ , this is  $a' \leq a + \ell S + \ell/\ln 2 + 2\bar{m} \log d + h_2(2\bar{m}) + \bar{m} \log(r/\bar{m})$ .

*Constants.*  $\bar{m} \leq C_m \sqrt{\ell}$  with  $C_m = 1 + 3d^3 r + 9dr^5$ . Use  $\log(e/\ell) \geq \max(1.443, \log(1/\ell))$ . Then  $\ell/\ln 2 \leq 1.443\sqrt{\ell} \leq \sqrt{\ell} \log(e/\ell)$ ;  $2\bar{m} \log d \leq 1.39C_m \log d \cdot \sqrt{\ell} \log(e/\ell)$ ;  $\bar{m} \log r \leq 0.70C_m \log r \cdot \sqrt{\ell} \log(e/\ell)$ ; and  $h_2(2\bar{m}) + \bar{m} \log(1/\bar{m}) \leq 2\bar{m} \log(e/(2\bar{m})) + \bar{m} \log(1/\bar{m}) = 3\bar{m} \log(1/\bar{m}) + 0.886\bar{m}$ . As  $x \log(1/x)$  increases on  $(0, 1/e)$  and  $C_m \geq 1$ ,  $3\bar{m} \log(1/\bar{m}) \leq 3C_m \sqrt{\ell} \log(1/\sqrt{\ell}) = 1.5C_m \sqrt{\ell} \log(1/\ell)$ , and  $0.886\bar{m} \leq 0.614C_m \sqrt{\ell} \log(e/\ell)$ . The total is at most  $C_2 \sqrt{\ell} \log(e/\ell)$  with  $C_2 = 1 + C_m(1.39 \log d + 0.70 \log r + 2.2)$ , since  $1.5 + 0.614 < 2.2$ .

*The twisted gadget.* For  $\Phi_\omega$  the slots carry phases, the identities read  $B_1^* B_{[y]} = \zeta_t B_{[x]}^* B_{[z]}$ , and the triangle defect becomes  $R_t = \varphi_y X_1^* X_{[y]} - \bar{\varphi}_x \varphi_{xy} X_{[x]}^* X_{[z]}$  with the phases  $\varphi_w$  of the slot words. The constraints of step 2 are then that each block of a class is a phase times a common block or its adjoint. The section  $\hat{\lambda}$  of the twisted regular representation satisfies them all exactly, with every block  $\hat{\lambda}(g)^* \hat{\lambda}(h)$  a nonzero unitary, so no class forces its blocks to vanish, the projection of step 2 is the phase-transported average, and the identities with  $[y] = 1$  carry the phase 1. Every other step is unchanged.  $\square$

*Proof of Corollary G.10.* For  $n \geq n_F$  and  $\ell \leq \frac{2}{15n}$ ,  $C_1 \sqrt{\ell} \leq C_1 \sqrt{2/(15n_F)} \leq \frac{1}{4}$ , so Theorem G.9 gives a zero-leakage flat round with  $S' = S$ , flat Gram off-diagonals at most  $\frac{2}{16} + \frac{1}{4} < \frac{1}{2}$ , and, since  $C_m \leq C_1$ ,  $a' \leq a + \ell S + C_2 \sqrt{\ell} \log(e/\ell)$ . Theorem G.6(c) uses only the identities, the flat Gram bound and the flat entropy production, so it applies with  $c = \frac{1}{2}$  and gives  $Q \leq K \log^3(n/\epsilon)(1 + S + n a')$ . Here  $n \ell S \leq \frac{2}{15} S$  and  $n C_2 \sqrt{\ell} \log(e/\ell) \leq C_2 \sqrt{2n/15} \log(15en/2) \leq 0.37 C_2 \sqrt{n} \log(20n)$ , since  $\sqrt{2/15} < 0.3652$  and  $15e/2 < 20.4$ . At  $\ell \leq (\log n)^m/n^2$ ,  $n C_2 \sqrt{\ell} \log(e/\ell) \leq C_2 (\log n)^{m/2} \log(en^2) = O((\log n)^{m/2+1})$ .  $\square$

### L.3 The twisted torus

Write  $\hat{\lambda}(u, v) = \lambda(a)^u \lambda(b)^v$  for the section of the twisted regular representation; every slot word  $w$  with label  $(u, v)$  has  $\lambda(w) = \omega^{f(w)} \hat{\lambda}(u, v)$  for an integer  $f(w)$  that depends only on the commutation rule, and every unitary pair  $(\mathcal{A}, \mathcal{B})$  with  $\mathcal{B}\mathcal{A} = \zeta \mathcal{A}\mathcal{B}$  satisfies the same identities with  $\zeta^{f(w)}$  in place of  $\omega^{f(w)}$ . So  $\Phi_\omega$  has the Kraus operators  $A_g = \sum_{a \in S_g} c_a \varphi_a E_a$  with  $\varphi_a = \omega^{f(w_a)}$ . Put  $F = \max_a |f(w_a)|$ .

*Proof of Lemma G.11.* Let  $\bar{\tau} = \sum_g \mu_g B_g B_g^*/k$ ,  $E$  a top- $k$  eigenspace of  $\bar{\tau}$  and  $w = 1 - \text{Tr}(P_E \bar{\tau})$ .

1. *Mass outside E.* By Lemma G.5,  $w \leq a \ln 2$ ; phases do not enter  $\bar{\tau}$ . Put  $C_g = P_E B_g$  and  $D_g = (I - P_E) B_g$ . Then  $\sum_g \mu_g \tau(D_g^* D_g) = \text{Tr}((I - P_E) \bar{\tau}) = w$ , so  $\|D_g\|_2^2 \leq w/\mu_g$ .
2. *The identities split.* The ranges of  $P_E$  and  $I - P_E$  are orthogonal, so  $B_g^* B_h = C_g^* C_h + D_g^* D_h$  and  $C_1^* C_{[y]} - \zeta_t C_{[x]}^* C_{[z]} = \zeta_t D_{[x]}^* D_{[z]} - D_1^* D_{[y]}$ . By Hölder's inequality the right side has  $\|\cdot\|_1 \leq w(1/\sqrt{\mu_x \mu_z} + 1/\sqrt{\mu_1 \mu_y})$ .
3. *Polar parts.* Identify  $E$  with  $\mathbb{C}^k$  and write  $C_g = V_g |C_g|$  with  $V_g$  unitary. Since  $C_g^* C_g = I - D_g^* D_g \leq I$  and  $1 - s \leq 1 - s^2$  on  $[0, 1]$ ,  $\|C_g - V_g\|_1 = \tau(I - |C_g|) \leq \tau(D_g^* D_g) \leq w/\mu_g$ . All operators involved are contractions and  $\|AB\|_1 \leq \|A\|_1 \|B\|$ , so replacing  $C_g$  by  $V_g$  for  $g \in \{1, [x], [y], [z]\}$  changes the defect by at most  $\sum_g w/\mu_g$ .

4. *Normalization.* Replacing every  $V_g$  by  $V_1^* V_g$  changes neither the defects nor any  $\tau(V_g^* V_h)$ , and makes  $V_1 = I$ .
5. *Gram.*  $|\tau(D_g^* D_h)| \leq \|D_g\|_2 \|D_h\|_2 \leq w/\mu_{\min}$  and  $|\tau(C_g^* C_h) - \tau(V_g^* V_h)| \leq \|C_g - V_g\|_1 + \|C_h - V_h\|_1 \leq 2w/\mu_{\min}$ .

Finally the defect is at most  $6w/\mu_{\min} \leq 6d a \ln 2$ , since  $\mu_{\min} \geq 1/d$ .  $\square$

**Lemma L.1** (Thin rounds). *Let  $\mathcal{S}$  be the cyclic shift and  $\mathcal{D} = \text{diag}(e^{2\pi i p j/N})$  on  $\mathbb{C}^N$ ,  $\gcd(p, N) = 1$ ,  $N > 4$ , so that  $\mathcal{D}\mathcal{S} = e^{2\pi i p/N} \mathcal{S}\mathcal{D}$ , and let  $U$  be the gadget unitary of  $\Phi_\omega$  with  $\mathbf{a} \mapsto \mathcal{S}$ ,  $\mathbf{b} \mapsto \mathcal{D}$ . With the flat bath  $I/N$  this is a square round with  $a = 0$ ,  $S = \log N$ , leakage  $\ell \leq F^2 e^2$  and distance at most  $F\sqrt{d}e$ , where  $e = |e^{2\pi i p/N} - \omega|$ .*

*Proof.* The slot operators are  $M_a = \theta_a \mathcal{S}^{u_a} \mathcal{D}^{v_a}$  with  $\theta_a = \zeta^{pf(w_a)}$ ,  $\zeta = e^{2\pi i/N}$ . Labels have  $|u|, |v| \leq 1$ , so for  $N > 4$ ,  $\tau_N(M_b^* M_a) = \theta_a \bar{\theta}_b$  if  $a$  and  $b$  carry the same label and 0 otherwise; the channel is therefore  $\sum_g A'_g X A_g^*$  with  $A'_g = \sum_{a \in S_g} c_a \theta_a E_a$ . Since  $|\theta_a - \varphi_a| \leq |f(w_a)|e \leq Fe$  and  $A_g$  lies in the Kraus span of  $\Phi_\omega$ ,  $\ell = \frac{1}{d} \sum_g \|P_\perp A'_g\|_F^2 \leq \frac{1}{d} \sum_g \|A'_g - A_g\|_F^2 = \frac{1}{d} \sum_a |c_a|^2 |\theta_a - \varphi_a|^2 \leq F^2 e^2$ . The Stinespring isometries  $V' = \sum_g A'_g \otimes |g\rangle$  and  $V = \sum_g A_g \otimes |g\rangle$  have  $\|V' \rho V'^* - V \rho V^*\|_1 \leq 2\|V' - V\| \leq 2\|V' - V\|_F \leq 2F\sqrt{d}e$ , so the distance is at most  $F\sqrt{d}e$ . The bath is flat and  $T$  is unital, so  $a = 0$ .  $\square$

*Proof of Theorem G.12.* (a) By Lemma 2.10, with the phases of  $\Phi_\omega$ , the round gives isometries  $B_g$  with  $B_1^* B_{[y]} = \zeta_t B_{[x]}^* B_{[z]}$ , and Lemma G.11 gives unitaries  $V_g$  on  $\mathbb{C}^k$ ,  $V_1 = I$ , with every identity holding to  $\delta := 6d \ln 2 \cdot a$  in  $\|\cdot\|_1$ . Among the triangles are  $(\mathbf{a}, \mathbf{a}^{-1}, \emptyset)$ ,  $(\mathbf{a}, \mathbf{b}, \mathbf{ab})$  and  $(\mathbf{ab}, \mathbf{a}^{-1}, \mathbf{aba}^{-1})$ , with  $[\mathbf{aba}^{-1}] = [\mathbf{b}]$ . With phases  $\zeta_1, \zeta_3, \zeta_4$  of modulus one they give  $\|V_{\mathbf{a}^{-1}} - \zeta_1 V_{\mathbf{a}}^*\|_1 \leq \delta$ ,  $\|V_{\mathbf{ab}} - \zeta_3 V_{\mathbf{a}} V_{\mathbf{b}}\|_1 \leq \delta$  and  $\|V_{\mathbf{a}^{-1}} - \zeta_4 V_{\mathbf{ab}}^* V_{\mathbf{b}}\|_1 \leq \delta$ . Multiplication by unitaries and adjoints preserve  $\|\cdot\|_1$ , so  $\|\zeta_1 V_{\mathbf{a}}^* - \zeta_3 \zeta_4 V_{\mathbf{b}}^* V_{\mathbf{a}}^* V_{\mathbf{b}}\|_1 \leq 3\delta$ ; multiplying by  $V_{\mathbf{b}}$  on the left and by  $V_{\mathbf{b}}^* V_{\mathbf{a}}$  on the right,  $\|C - \varphi I\|_1 \leq 3\delta$  with  $C := V_{\mathbf{b}} V_{\mathbf{a}}^* V_{\mathbf{b}}^* V_{\mathbf{a}}$  and  $\varphi := \zeta_1 \zeta_3 \zeta_4$ . The section  $\hat{\lambda}$  satisfies the same identities exactly with the same phases, so the same steps give  $\hat{\lambda}(\mathbf{b}) \hat{\lambda}(\mathbf{a})^* \hat{\lambda}(\mathbf{b})^* \hat{\lambda}(\mathbf{a}) = \varphi I$ . The left side does not change if  $\hat{\lambda}(\mathbf{a}), \hat{\lambda}(\mathbf{b})$  are rephased, so it equals  $\lambda(\mathbf{b}) \lambda(\mathbf{a})^{-1} \lambda(\mathbf{b})^{-1} \lambda(\mathbf{a})$ , and inverting  $\lambda(\mathbf{b}) \lambda(\mathbf{a}) = \omega \lambda(\mathbf{a}) \lambda(\mathbf{b})$  shows that this is  $\bar{\omega} I$ . Hence  $\varphi = \bar{\omega}$ . Now  $\det C = 1$ ; write the eigenvalues of  $C$  as  $\varphi e^{i\psi_j}$  with  $\psi_j \in (-\pi, \pi]$ . Then  $\varphi^k e^{i \sum_j \psi_j} = 1$ , so  $\sum_j \psi_j \in 2\pi k\theta + 2\pi\mathbb{Z}$  and  $|\sum_j \psi_j| \geq 2\pi \|k\theta\|$ .  $C$  is normal and  $|e^{i\psi} - 1| = 2|\sin(\psi/2)| \geq 2|\psi|/\pi$  on  $[-\pi, \pi]$ , so  $\|C - \varphi I\|_1 = \frac{1}{k} \sum_j |e^{i\psi_j} - 1| \geq \frac{2}{\pi k} \sum_j |\psi_j| \geq 4\|k\theta\|/k$ . Hence  $4\|k\theta\|/k \leq 3\delta = 18d \ln 2 \cdot a$ . For  $\theta = (\sqrt{5} - 1)/2$  every partial quotient is 1. For a convergent denominator  $q_j$ ,  $\|q_j \theta\| = 1/(q_j(\alpha_{j+1} + q_{j-1}/q_j))$  with  $\alpha_{j+1} < 2$ , so  $\|q_j \theta\| > 1/(3q_j)$ ; for  $q_j \leq q < q_{j+1}$ ,  $\|q\theta\| \geq \|q_j \theta\|$  by the best approximation property; so  $\|q\theta\| \geq 1/(3q)$  for every  $q \geq 1$ , and  $a \geq 2/(27d \ln 2 \cdot k^2)$ .

(b) For  $N = F_m$  and  $p = F_{m-1}$ ,  $|\theta - p/N| < 1/N^2$ , so  $e = |e^{2\pi i p/N} - \omega| \leq 2\pi|p/N - \theta| < 2\pi/N^2$ , and Lemma L.1 gives the stated round. An exact flat round with  $S' \leq \lambda S + C_0$  has bath rank  $k' \leq 2^{C_0} N^\lambda$ , so by (a)  $a' \geq 2/(27d \ln 2 \cdot k'^2) \geq 0.0062 \cdot 4^{-C_0} N^{-2\lambda}$ . If also  $a' \leq C_1 \ell^\gamma (\log 1/\ell)^m$ , then, since  $x^\gamma (\log 1/x)^m$  increases for small  $x$ ,  $a' \leq C_1 (4\pi^2 F^2)^\gamma N^{-4\gamma} (\log(N^4/(4\pi^2 F^2)))^m$ . As  $N \rightarrow \infty$  this forces  $2\lambda \geq 4\gamma$ .

(c) For  $N$  large,  $C_m \sqrt{\ell} \leq \frac{1}{4}$ , and Theorem G.9 for  $\Phi_\omega$  gives a zero-leakage flat round with  $S' = \log N$  and  $a' \leq \ell \log N + C_2 \sqrt{\ell} \log(e/\ell) = O((\log N)/N^2)$ .  $\square$

## M Proofs for Section H

### M.1 Tensor powers and conserved bias

*Proof of Lemma H.1.* By Theorem 7.6(b),  $(B_g^{\otimes t})$  is a gadget family, so  $W_t = \sum_g A_g \otimes B_g^{\otimes t}$  is an isometry and extends to a unitary (Lemma 2.9(c)). Its channel with bath  $\rho^{\otimes t}$  is

$X \mapsto \sum_{g,h} \text{Tr}(\rho^{\otimes t} (B_h^* B_g)^{\otimes t}) A_g X A_h^* = \sum_{g,h} G_{gh}^t A_g X A_h^*$ , whose Choi state lies in the Kraus span: leakage 0. Its distance is at most  $\frac{1}{2} \|G^{\otimes t} - I\|$ , as in the comparison step of the proof of Theorem 7.6(a), and  $\|G^{\otimes t} - I\| \leq (r-1) \max_{g \neq h} |G_{gh}|^t$  since the diagonal is 1.  $S(\rho^{\otimes t}) = tS$ . Since the  $A_g$  have disjoint supports,  $T_t(\rho^{\otimes t}) = \sum_g \mu_g (B_g \rho B_g^*)^{\otimes t}$ , every one-factor marginal of which is  $T(\rho)$ ; subadditivity gives  $S(T_t(\rho^{\otimes t})) \leq tS(T(\rho))$ , so  $a_t \leq t a$ .  $\square$

*Proof of Theorem H.2.* 1. *Levels are sub-rounds.* For  $\kappa \in E_{s'}$ ,  $s' \neq s$ , and  $v \in E'_s$ ,  $\langle \kappa, B_g^* v \rangle = \langle B_g \kappa, v \rangle = 0$ , so every  $B_g^*$  maps  $E'_s$  into  $E_s$ . Hence  $B_1^* B_{[y]}$  and  $B_{[x]}^* B_{[z]}$  are block diagonal over  $\{E_s\}$ , and each  $B_g^{(s)} := B_g|_{E_s} : E_s \rightarrow E'_s$  is a gadget family. The weighted Gram is  $G = \sum_s w_s M^{(s)}$  with  $M_{gh}^{(s)} = \tau_{E_s}(B_g^{(s)*} B_h^{(s)})$ ,  $|M_{gh}^{(s)}| \leq 1$ .

2. *The key identity.*  $T(\rho) = \bigoplus_s w_s T_s(I/e_s)$  on the orthogonal  $E'_s$ , with  $T_s(I/e_s) = \sum_g \mu_g B_g^{(s)} B_g^{(s)*} / e_s$ . So  $S(T(\rho)) = H(w) + \sum_s w_s S(T_s(I/e_s))$  and  $S(\rho) = H(w) + \sum_s w_s \log e_s$ , hence  $a = \sum_s w_s a_s$  with  $a_s = S(T_s(I/e_s)) - \log e_s \geq 0$ , since  $T_s(I/e_s)$  has eigenvalues at most  $1/e_s$ .

3. *Truncation.* The surprisals  $\log(e_s/w_s)$  are nonnegative with  $w$ -mean  $S$ . Keep  $\Sigma' = \{s : \log(e_s/w_s) \leq S/\delta\}$  with  $\delta = \frac{1}{32}$ ; Markov's inequality gives  $w(\Sigma \setminus \Sigma') \leq \delta$ . Because  $a$  is a conserved average, dropping levels costs no binary entropy.

4. *Flattening.* Put  $\eta = \frac{1}{32}$ ,  $\nu := \max_{\Sigma'} (e_s/w_s)/\eta$ ,  $m_s := \lceil \nu w_s / e_s \rceil$  ( $s \in \Sigma'$ ),  $\widetilde{K} := \bigoplus_{\Sigma'} E_s \otimes \mathbb{C}^{m_s}$ ,  $N := \dim \widetilde{K}$  and  $\widetilde{B}_g := \bigoplus_{\Sigma'} B_g^{(s)} \otimes I_{m_s}$  into  $\bigoplus_{\Sigma'} E'_s \otimes \mathbb{C}^{m_s}$ ; a direct sum with orthogonal codomains, it is a gadget family. On  $\Sigma'$ ,  $\nu w_s / e_s \geq 1/\eta$ , so  $\nu w_s \leq e_s m_s \leq (1+\eta)\nu w_s$  and  $\nu w(\Sigma') \leq N \leq (1+\eta)\nu w(\Sigma')$ ; the weights  $\hat{w}_s := e_s m_s / N$  lie in  $[w_s / ((1+\eta)w(\Sigma')), (1+\eta)w_s / w(\Sigma')]$ .

5. *Gram.*  $\tau_N(\widetilde{B}_g^* \widetilde{B}_h) = \sum_{\Sigma'} \hat{w}_s M_{gh}^{(s)}$ , which differs from  $G_{gh}$  by at most  $\sum_{\Sigma'} |\hat{w}_s - w_s| + w(\Sigma \setminus \Sigma') \leq (\eta + \delta)/(1 - \delta) + \delta < 0.096$ . Its off-diagonals are below  $\frac{1}{8} + 0.096 < \frac{1}{4}$ , and its diagonal is 1.

6. *Costs.*  $\log N \leq \log(1+\eta) + \log \nu \leq 0.045 + S/\delta + \log(1/\eta) = 32S + 5.045$ . The flat bath on  $\widetilde{K}$  has  $\widetilde{T}(I/N) = \bigoplus_{\Sigma'} \hat{w}_s T_s(I/e_s) \otimes I/m_s$  and  $\log N = H(\hat{w}) + \sum \hat{w}_s \log(e_s m_s)$ , so its entropy production is  $a' = \sum_{\Sigma'} \hat{w}_s a_s \leq \frac{1+\eta}{1-\delta} \sum_s w_s a_s = \frac{33}{31} a < 1.07 a$ .

7. *Streaming.* Apply Theorem G.6(c) with  $c = \frac{1}{4}$ , so  $t = \lceil \log(4rn/\epsilon)/2 \rceil$ , and  $\log k = \log N$ . Its memory term  $C n (ta') \log(e/(ta'))$  increases in  $a'$  while  $ta' \leq 1$ , so  $a' \leq 1.07a$  may be inserted. If  $d t a' \ln 2 > \frac{1}{4}$ , Theorem G.6(c) gives only its polylogarithmic form, and we use instead the exact service of Proposition 2.7, with  $Q = n \lceil \log r \rceil$ , error 0 and  $J = C = 0$ . There  $x := 1.07 t a \geq ta' > x_0 := 1/(4d \ln 2)$ , and  $x \log(e/x)$  increases on  $(0, 1]$ , so the middle term of the display is at least  $C_{d,r} n x_0 \log(e/x_0) \geq n \lceil \log r \rceil$  once  $C_{d,r} \geq \lceil \log r \rceil / (x_0 \log(e/x_0))$ . Since  $t \leq 3 \log(n/\epsilon) + \log(8r)$ , also  $n a \geq n x_0 / (1.07 t)$ , so  $n \lceil \log r \rceil \leq K_{d,r} \log^3(n/\epsilon) (1 + S + n a)$  for  $K_{d,r}$  large enough. The statement without a device is steps 3–6 alone.  $\square$

*Proof of Lemma H.3.* Since  $P_0$  commutes with  $\rho$ ,  $S(\rho) = h_2(\nu) + (1-\nu)S(\rho_0) + \nu S(\rho_\perp) \geq (1-\nu)S(\rho_0)$ . The channel,  $T$  and the leakage are linear in the bath:  $\Psi_{\rho_0} - \Phi = [(\Psi_\rho - \Phi) - \nu(\Psi_{\rho_\perp} - \Phi)]/(1-\nu)$  with  $d_\diamond(\Psi_{\rho_\perp}, \Phi) \leq 1$ , and  $\ell(\rho) \geq (1-\nu)\ell(\rho_0)$ .  $T$  is unital, so  $S(T(\rho_\perp)) \geq S(\rho_\perp)$ , and concavity gives  $S(T(\rho)) \geq (1-\nu)S(T(\rho_0)) + \nu S(\rho_\perp)$ . Subtracting  $S(\rho)$  gives  $(1-\nu)a(\rho_0) \leq a + h_2(\nu)$ . If  $U$  has Kraus form on  $\mathbb{C}^d \otimes \text{ran } P_0$ , then  $\Psi_{\rho_0}(X) = \sum_{i,j} \text{Tr}(\rho_0 X_j^* X_i) A_i X A_j^*$ , whose Choi state lies in the Kraus span.  $\square$

*Proof of Corollary H.4.* 1. *Levels are sub-rounds.*  $W_s := W|_{\mathbb{C}^d \otimes E_s}$  is an isometry into  $\mathbb{C}^d \otimes E'_s$  and  $\rho$  is block diagonal, so  $T(\rho) = \bigoplus_s w_s T_s$  on the orthogonal  $E'_s$  with

$T_s := T(I_{E_s}/e_s)$ . Hence  $a = \sum_s w_s a_s$  with  $a_s := S(T_s) - \log e_s \geq 0$ , as in the proof of Theorem H.2. Leakage is linear in the bath:  $\ell = \sum_s w_s \ell_s$ .

2. *Dropping levels.* The surprisals  $\log(e_s/w_s) \geq 0$  have  $w$ -mean  $S$ , so  $\Sigma_1 := \{s : \log(e_s/w_s) \leq 32S\}$  has  $w(\Sigma_1^c) \leq \frac{1}{32}$ . With  $\theta_0 := 1/(16C_m^2)$ ,  $\Sigma_2 := \{s \in \Sigma_1 : \ell_s \leq \theta_0\}$  has  $w(\Sigma_1 \setminus \Sigma_2) \leq \ell/\theta_0 \leq 16C_m^2 c_1/n$ . Restrict to  $\bigoplus_{\Sigma_2} E_s$ , a projection commuting with  $\rho$ , with  $\nu := 1 - w(\Sigma_2) \leq \frac{1}{32} + 16C_m^2 c_1/n$  and weights  $w'_s := w_s/(1 - \nu)$ . By step 1 and  $a_s \geq 0$ ,  $a_0 = \sum_{\Sigma_2} w'_s a_s \leq a/(1 - \nu)$  with no binary entropy;  $\ell_0 \leq \ell/(1 - \nu)$ ,  $u_0 \leq (u + \nu)/(1 - \nu)$  and  $S_0 \leq S/(1 - \nu)$  by Lemma H.3.
3. *First-order rounding per level.* For  $s \in \Sigma_2$ ,  $(U, I_{E_s}/e_s)$  is a flat round with  $\ell_s \leq \theta_0 \leq \frac{1}{2}$  and  $C_m \sqrt{\ell_s} \leq \frac{1}{4}$ . Theorem G.9 gives isometries  $X_g^{(s)} : E_s \rightarrow \mathbb{C}^r \otimes E_s$  with the exact identities and  $a'_s \leq a_s + \ell_s \log e_s + C_2 \sqrt{\ell_s} \log(e/\ell_s)$ ; its Gram steps (the shrink, the projection and the renormalization) move each flat Gram entry by at most  $C'_1 \sqrt{\ell_s}$ , where  $C'_1 := 3d^3 r + 18dr^2 + 9d^3 r^5$ .
4. *Direct sum.*  $X'_g := \bigoplus_{\Sigma_2} X_g^{(s)}$  has mutually orthogonal codomains  $\mathbb{C}^r \otimes E_s$ , so  $X_g'^* X_h' = \bigoplus_s X_g^{(s)*} X_h^{(s)}$  and the identities hold; with bath  $\rho_0$  the new round again has conserved bias with flat levels, and step 1 gives  $a' = \sum_s w'_s a'_s$ . The function  $x \mapsto \sqrt{x} \log(e/x)$  is concave on  $(0, 1]$ , so by Jensen's inequality and  $\log e_s \leq \log(e_s/w_s) \leq 32S$  on  $\Sigma_1$ ,  $a' \leq a/(1 - \nu) + 32\ell_0 S + C_2 \sqrt{\ell_0} \log(e/\ell_0)$ . For the Gram,  $\sum_s w'_s \tau_s(X_g^{(s)*} X_h^{(s)}) = \text{Tr}(\rho_0 X_g'^* X_h')$ , which the anchor bound (28) bounds by  $2u_0 + 2\sqrt{d\ell_0} + d\ell_0$ ; the rounding adds at most  $C'_1 \sum_s w'_s \sqrt{\ell_s} \leq C'_1 \sqrt{\ell_0}$ . For  $n \geq n_{CB}$  the weighted Gram off-diagonals are at most  $\frac{6}{31} + O(n^{-1/2}) \leq 0.2$ .
5. *Streaming.* The tensor square (Lemma H.1) has conserved bias, with flat levels  $E_s \otimes E_{s'}$ , whose images lie in the mutually orthogonal  $(\mathbb{C}^r \otimes E_s) \otimes (\mathbb{C}^r \otimes E_{s'})$ , weighted Gram off-diagonals at most  $0.04 < \frac{1}{8}$ , bath entropy  $2S_0$  and entropy production at most  $2a'$ . Theorem H.2 gives error at most  $2\epsilon$  and  $Q \leq K \log^3(n/\epsilon)(1 + 2S_0 + 2na')$ . Finally  $\ell_0 \leq 1.04c_1/n$  for  $n \geq n_{CB}$ , so  $32n\ell_0 S \leq 34c_1 S$ , and  $\sqrt{x} \log(e/x)$  increases on  $(0, 1/e)$ , so  $nC_2 \sqrt{\ell_0} \log(e/\ell_0) \leq C_2 \sqrt{1.04c_1 n} \log(en/c_1)$ . Hence  $Q \leq K' \log^4(n/\epsilon)(1 + S + na + \sqrt{n})$ .  $\square$

## M.2 Rounds that are vacuous

*Proof of Lemma H.5.*  $G_{gh} = \text{Tr}(\rho B_g^* B_h) = \text{Tr}(\rho J^* \pi(g)^* \pi(h) J)$ . If  $\pi(g) = \pi(h)$  for labels  $g \neq h$ , then  $G_{gh} = \text{Tr}(\rho J^* J) = 1$ , while  $|G_{gh}| \leq 2u < 1$  by Lemma 2.10. So  $\pi$  separates the labels, and Corollary 7.8 applies. If  $B_g K \subseteq K$  for every label, each  $B_g$  is an isometry of the finite-dimensional  $K$  into itself, hence unitary; replacing every  $B_g$  by  $B_1^* B_g$ , as in Lemma 2.9(a), leaves every  $B_g^* B_h$  unchanged and makes  $B_1 = I_K$ ; the identity  $B_1^* B_{[y]} = B_{[x]}^* B_{[z]}$  reads  $B_{[z]} = B_{[x]} B_{[y]}$ , and these are the defining relations of  $P_\Phi$ . So  $g \mapsto B_g$  extends to a unitary representation  $\pi$  of  $P_\Phi$  on  $K$ , and  $B_g^* B_h = \pi(g)^* \pi(h)$  with  $J = I$ .  $\square$

## M.3 Permutation rounds and devices

*Proof of Lemma H.6.* 1. *Injectivity.*  $B_g$  is an isometry, so it maps the orthogonal  $K_\omega$  to orthogonal subspaces; hence  $f(g)$  is injective on  $\Omega_0$ .

2. *Identities.* For  $\omega \in \Omega_0$ ,  $B_1^* B_{[y]}$  restricted to  $K_\omega$  is a unitary onto  $K_{f([y])\omega}$  if  $f([y])\omega \in \Omega_0$  and 0 otherwise,  $B_1^*$  being the projection onto  $K$ .  $B_{[x]}^*$  kills the complement of  $\text{ran } B_{[x]} = \bigoplus_{\omega' \in \Omega_0} K'_{f([x])\omega'}$ , so  $B_{[x]}^* B_{[z]}$  restricted to  $K_\omega$  is a unitary onto  $K_{\omega''}$  if  $f([z])\omega = f([x])\omega''$  for some  $\omega'' \in \Omega_0$ , and 0 otherwise. The two operators are equal, so they vanish together, and when they are nonzero their targets coincide:  $f([y])\omega = \omega''$ .

This is  $\lambda_1^* \lambda_{[y]} |\omega\rangle = \lambda_{[x]}^* \lambda_{[z]} |\omega\rangle$ , and by step 1 the  $\lambda_g$  are isometries, so the base family is a gadget family.

3. *Gram.* For  $g \neq h$ ,  $B_h K_\omega = K'_{f(h)\omega} \perp K'_{f(g)\omega} = B_g K_\omega$  by freeness, so  $P_{K_\omega} B_g^* B_h P_{K_\omega} = 0$ ,  $\text{Tr}(\rho B_g^* B_h) = 0$  for every block-diagonal  $\rho$ , and likewise  $\text{Tr}(\text{diag}(p) \lambda_g^* \lambda_h) = 0$ ; by Lemma 2.10 the distance is 0.
4. *Entropy.*  $S(\rho) = H(p) + \sum_\omega p(\omega) S(\hat{\rho}_\omega) \geq H(p)$ , with  $\hat{\rho}_\omega = \rho_\omega / p(\omega)$ .
5. *Entropy production.*  $T(\rho)$  is block diagonal over the  $K'_\eta$ ; the block  $\eta$  has mass  $T_{\text{base}} p(\eta) = \sum_g \mu_g p(f(g)^{-1} \eta)$  and normalized state  $\sum_g [\mu_g p(f(g)^{-1} \eta) / T_{\text{base}} p(\eta)] u_g \hat{\rho}_{f(g)^{-1} \eta} u_g^*$  with  $u_g$  unitary, there being at most one  $\omega$  for each  $g$  by step 1. Concavity gives  $S(T(\rho)) \geq H(T_{\text{base}} p) + \sum_\eta \sum_g \mu_g p(f(g)^{-1} \eta) S(\hat{\rho}_{f(g)^{-1} \eta}) = H(T_{\text{base}} p) + \sum_\omega p(\omega) S(\hat{\rho}_\omega)$ . Subtracting  $S(\rho)$  gives  $a \geq H(T_{\text{base}} p) - H(p) = a_{\text{base}}$ .  $\square$

*Proof of Theorem H.7.* Absorb the fixed permutations  $\beta_t$  and injections into the frames. Let  $F_t^{\text{tr}}$  be the frame of the evolution projected onto the windows after round  $t$ , with columns indexed by label histories  $y \in [r]^t$ ,  $G_{t,Q} = F_{t,Q}^{\text{tr}*} F_{t,Q}^{\text{tr}}$  for the points  $Q$  of  $\Omega_t$ , and  $\Gamma_{t,Q} = \sum_y \text{Pr}_y(\text{alive at } t \text{ at } Q) |y\rangle \langle y|$  for the ideal killed chain. For weights  $c \in [0, 1]^{\Omega_t}$  put  $e_t(c) = \|\sum_Q c_Q (G_{t,Q} - \Gamma_{t,Q})\|$ . The blocks are the points, of dimension  $D = m$ .

(1) *Concentration, inflow, export.* Every Gram quantity used below has the form  $F^*((V^* A V - \mathbb{E} V^* A V) \otimes I) F$  with  $A = \sum_Q c_Q (\lambda_g^* P'_Q \lambda_h \otimes I_m)$ ,  $c \in [0, 1]^{\Omega_t}$  and  $\|A\| \leq 1$ .  $A$  maps each point to at most one point, so its Hermitian and anti-Hermitian parts couple each block to at most two blocks, and Lemma G.1(b) applies with  $\Delta \leq 2$ . Lemma G.2(b) gives the inflow factor  $r$  exactly, and with Lemma G.2(a) the unrolling of the proof of Theorem G.4 gives  $\alpha_* \leq A/D$  with  $A = 4e(1 + 8E_p^2 n^2 L')$  and  $L' = \ln(8SE_p^2 |\Omega|)$ ; initially  $\alpha_x(F_0) = p_0(x)/m \leq 1/D$ . The variance proxy of Lemma G.1(b) is  $O(n^2 L'/D^2)$ , and a union over the events below costs  $O(n \log r + \log n)$  in the exponent; so each event holds at level  $s$  for  $m = \text{poly}(n/s) \cdot \text{polylog} |\Omega|$ .

(2) *Means.* For  $g \neq h$ ,  $\lambda_g^* P'_Q \lambda_h$  has zero diagonal, since there are no collisions, so the mean Gram is diagonal in the labels. For  $g = h$ ,  $\mathbb{E}[V_x^*(P_x \lambda_g^* P'_Q \lambda_g P_x) V_x] = f_{x \rightarrow Q}^{(g)} P_x$  with  $f_{x \rightarrow Q}^{(g)} = \mathbf{1}[f(g)x = Q]$ .

(3) *Set-resolved Gram.* By (2),  $\sum_Q c'_Q G_{t,Q} = \sum_g [\sum_x (T_g c')_x G_{t-1,x}] \otimes |g\rangle \langle g| + \text{Dev}(c')$  with  $(T_g c')_x = \sum_Q c'_Q f_{x \rightarrow Q}^{(g)} \in [0, 1]$ , and the ideal  $\Gamma$  satisfies the same recursion without  $\text{Dev}$ . Block diagonality in  $g$  gives  $e_t(c') \leq \max_g e_{t-1}(T_g c') + \|\text{Dev}(c')\|$ , and  $e_0 = 0$ . The weight vectors needed are the images under  $T_{g_{s+1}} \cdots T_{g_t}$  of the indicators of the exit sets and of all points: at most  $2n(n+1)r^n$  vectors, all fixed before the twirls they enter. On the events of (1) at level  $s$ ,  $e_t \leq ts$  for all of them.

(4) *Error.* Let  $x_t$  be the physical vector,  $g_t$  the vector of the projected evolution embedded in the physical memory,  $\mathcal{U}_t$  the physical round and  $d_t = \mathcal{U}_t g_{t-1} - g_t$ , the part of the projected vector sent to garbage at round  $t$ . Then  $x_t - g_t = \mathcal{U}_t(x_{t-1} - g_{t-1}) + d_t$  and  $x_0 = g_0$ , so  $\|x_t - g_t\| \leq \sum_{s \leq t} \|d_s\|$ . Since  $g_t$  lies in the window, the memory's mass outside the window at time  $t$  is at most  $(\sum_{s \leq t} \|d_s\|)^2 \leq n\omega$  with  $\omega := \sum_t \|d_t\|^2$ . Next,  $\|d_t\|^2 = \langle X | I \otimes \text{Exit}_t | X \rangle$  with  $\text{Exit}_t$  the Gram of the exit set at  $t$ , so by (3)  $\|d_t\|^2 \leq \sum_y \text{Pr}(y) \kappa_t(y) + ts$  and  $\omega \leq \omega_{\text{ideal}} + n^2 s$ . The full Gram after a round is  $G_{t-1}^{\text{tr}} \otimes I + \text{Dev}_t$  with  $\|\text{Dev}_t\| \leq s'$  by (1), and truncation removes  $\text{Exit}_t$ , so  $G_n^{\text{tr}} = I - \sum_t (\text{Exit}_t - \text{Dev}_t) \otimes I \leq (1 + ns')I$ . With  $F_n^{\text{tr}} = T(G_n^{\text{tr}})^{1/2}$ ,  $T$  an isometry, and  $(\sqrt{u} - 1)^2 \leq |1 - u|$  for  $u \geq 0$ ,  $\|g_n - (I \otimes T)X\|^2 \leq \langle X | I \otimes |I - G_n^{\text{tr}}| | X \rangle \leq \omega + 3ns'$ , since  $|I - G| = (I - G) + 2(G - I)_+ \leq (I - G) + 2ns'I$ . By (F4) the error is at most  $\sqrt{n\omega} + \sqrt{\omega + 3ns'}$ . With  $s = s' = \zeta/n^2$ ,  $\omega \leq \omega_{\text{ideal}} + \zeta$ , and since  $\zeta \leq \epsilon^2/(64n)$  the error is at most  $(1 + \sqrt{n})\sqrt{\omega_{\text{ideal}} + \epsilon^2/(16n)}$ . The memory is  $\log \max_t |\Omega_t| + \log m + O(1)$  at the rank rate, with the export schedule of [28, Theorem 12.2].  $\square$

*Proof of Lemma H.8.*  $f(g_t)\beta_t$  followed by the injection is injective, so the alive points form a set; given the past,  $\beta_t$  sends the  $a_C$  alive points of a class  $C$  to a uniformly random  $a_C$ -subset of  $C$ , independently over classes. One point under independent class-uniform permutations follows the class-uniformized chain, so  $\mathbb{E}\kappa_\beta(y) = \kappa_{\text{class}}(y)$ . Let  $h_t(x)$  be the class-uniformized kill probability from a point  $x$  after step  $t$ , a function of the class of  $x$ , and  $M_t = \text{killed}_t/|\Omega_0| + \sum_{x \text{ alive at } t} h_t(x)/|\Omega_0|$ . Then  $(M_t)$  is the Doob martingale of  $\kappa_\beta(y)$ , with  $M_0 = \kappa_{\text{class}}(y)$  and  $M_n = \kappa_\beta(y)$ . Its increment at step  $t$  is  $\sum_C$  (a sum over an  $a_C$ -sample without replacement of values in  $[0, 1]$ , minus its mean)/ $|\Omega_0|$ . By Hoeffding's comparison of sampling with and without replacement [48, Theorem 4] each class sum is sub-Gaussian with variance proxy  $a_C/4$ , and independence over classes gives a proxy at most  $(\sum_C a_C)/(4|\Omega_0|^2) \leq 1/(4|\Omega_0|)$  per step; the Azuma–Hoeffding inequality [5] gives the tail, and a union over the  $r^n$  label sequences gives the second claim.  $\square$

*Proof of Corollary H.10.* Use the windows and classes of the balance condition, with a start uniform on  $\Omega_0$ . By (B1) and Lemma H.8, some fixed choice of class permutations has  $\max_y \kappa_\beta(y) \leq \max_y \kappa_{\text{class}}(y) + \epsilon^2/(32n) \leq \epsilon^2/(16n)$  by (B3). With these permutations fixed, Theorem H.7 with  $\zeta = \epsilon^2/(64n)$  and  $p_0$  uniform on  $\Omega_0$  has  $\omega_{\text{ideal}} \leq \epsilon^2/(16n)$ , hence error at most  $(1 + \sqrt{n})\sqrt{\epsilon^2/(8n)} \leq 0.61\epsilon$  for  $n \geq 2$ , and memory  $\log \max_t |\Omega_t| + O_r(\log(n/\epsilon) + \log \log |\Omega_W|)$ , which (B2) bounds. Here  $\Omega$  may be replaced by  $\Omega_W$ : the ideal chain is killed outside the windows and one round moves a window point by a slot word, so the device only touches  $\Omega_W$ , and the injections are completed to permutations of  $\Omega_W$  by a fixed unitary completion, as in Theorem H.15. On the windows every round is the gadget unitary of the round, which is exact for every memory state there. For a free block-monomial round, Lemma H.6 gives a free exact permutation round of the same channel, its base round, with  $S_{\text{base}} \leq S$  and  $a_{\text{base}} \leq a$ ; streaming the base round streams the given one.  $\square$

*Proof of Proposition H.11.* Let  $N = 2^n$ , let  $b_w$  be the law  $\propto \sin^2$  on an interval of  $w = \lceil \sqrt{n} \rceil$  points,  $u$  the uniform law on  $[0, N)$ , and  $p = (1 - \frac{1}{n})b_w + \frac{1}{n}u$  on  $\mathbb{Z}_N$ . Every point of the cycle is triangle-good and collision-free, so the round has distance and leakage 0 for every bath. The entropy of a mixture is at most the average entropy plus the mixing entropy and at least the average entropy, so  $a(p) \leq h_2(\frac{1}{n}) + (1 - \frac{1}{n})a(b_w) + \frac{1}{n}a(u)$  and  $S(p) \leq h_2(\frac{1}{n}) + \log w + 1$ , using  $H(u) = n$ . Theorem H.12(iii) with  $\varphi = \sqrt{b_w}$  gives  $a(b_w) \leq (16r/\ln 2)\mathcal{E}(\sqrt{b_w}) = O(1/w^2)$ , and  $a(u) \leq \log(1 + 2/N)$ . So  $S + na \leq \log w + O(1) + n h_2(\frac{1}{n}) = O(\log n)$ . A chain started from  $p$  is killed at its first step with probability  $p(\Omega_0^c) \geq \frac{1}{n}(1 - |\Omega_0 \cap [0, N)|/N)$ , which exceeds  $\frac{1}{2n} \geq \epsilon^2/(32n)$  unless  $|\Omega_0| \geq N/2$ .  $\square$

## M.4 The Faber–Krahn form

*Proof of Theorem H.12.* Write  $P_g := f(g)_*p$ , so that  $P_1 = p$ ,  $H(P_g) = S$  and  $\mathcal{T}p = \sum_g \mu_g P_g =: Q$ .

(i) The Holevo identity gives  $a = H(Q) - \sum_g \mu_g H(P_g) = \sum_g \mu_g D(P_g \| Q)$ , so  $D(P_g \| Q) \leq a/\mu_g$ . For laws  $P, Q$ ,  $D(P \| Q) \geq -2 \log \sum \sqrt{PQ} \geq 2(1 - \sum \sqrt{PQ})/\ln 2 = \|\sqrt{P} - \sqrt{Q}\|^2/\ln 2$ , by monotonicity of the Rényi divergences in their order and  $-\log y \geq (1 - y)/\ln 2$ . So  $\|\sqrt{P_g} - \sqrt{p}\|^2 \leq 2\|\sqrt{P_g} - \sqrt{Q}\|^2 + 2\|\sqrt{p} - \sqrt{Q}\|^2 \leq 2 \ln 2 \cdot a(1/\mu_g + 1/\mu_1)$ . Reindexing by the bijection  $f(g)$ ,  $\|\sqrt{P_g} - \sqrt{p}\|^2 = \sum_x (\sqrt{p(f(g)x)} - \sqrt{p(x)})^2$ , the  $g$ -term of  $\mathcal{E}(\sqrt{p})$ ; sum over  $g \neq 1$ .

(ii) For  $s > 0$  put  $\varphi_s := (\sqrt{p} - s)_+$ . As  $t \mapsto (t - s)_+$  is 1-Lipschitz,  $\mathcal{E}(\varphi_s) \leq \mathcal{E}(\sqrt{p})$ ;  $\text{supp } \varphi_s = \{p > s^2\}$  has fewer than  $1/s^2$  points; and on  $\{p \geq 4s^2\}$ ,  $\sqrt{p} - s \geq \sqrt{p}/2$ , so  $\|\varphi_s\|^2 \geq \frac{1}{4}p(\{p \geq 4s^2\})$ . Markov's inequality for the surprisal  $-\log p(X) \geq 0$ , of mean  $S$ , gives  $p(\{p < 4s^2\}) \leq S/\log(1/(4s^2))$ . At  $4s^2 = 2^{-2S-2}$  this is at most  $S/(2S+2) < \frac{1}{2}$ , so

$\|\varphi_s\|^2 \geq \frac{1}{8}$  and  $|\text{supp } \varphi_s| < 2^{2S+4}$ . Hence  $\mathcal{E}(\sqrt{p}) \geq \mathcal{E}(\varphi_s) \geq \Lambda(2^{2S+4})/8$ , and (i) gives the claim.  $\varphi_s$  is supported in  $\text{supp } p$ , which gives the form with  $\Lambda_{\text{good}}$ .

(iii)  $S \leq \log |\text{supp } p| \leq \log v$ .  $D(P\|Q) \leq \log(1 + \chi^2(P\|Q)) \leq \chi^2(P\|Q)/\ln 2$ . With  $P := P_g$  and  $P_h := f(h)_*p$ :  $(P - Q)^2 = (\sum_h \mu_h(P - P_h))^2 \leq \sum_h \mu_h(P - P_h)^2$ ,  $(P - P_h)^2 \leq 2(\sqrt{P} - \sqrt{P_h})^2(P + P_h)$ , and  $Q \geq \mu_g P$ ,  $Q \geq \mu_h P_h$ , so  $\chi^2(P\|Q) \leq \sum_h 2(\mu_h/\mu_g + 1)\|\sqrt{P} - \sqrt{P_h}\|^2$ . Since  $\sqrt{f(g)_*\varphi^2} = f(g)_*\varphi$  for  $\varphi \geq 0$ ,  $\|\sqrt{P} - \sqrt{P_h}\|^2 \leq 2\|f(g)_*\varphi - \varphi\|^2 + 2\|\varphi - f(h)_*\varphi\|^2 \leq 4\mathcal{E}(\varphi)$ , so  $\chi^2(P_g\|Q) \leq 8\mathcal{E}(\varphi)(1/\mu_g + r)$  and  $a = \sum_g \mu_g D(P_g\|Q) \leq (8\mathcal{E}(\varphi)/\ln 2)(r + r)$ . The round is exact because  $p$  lives on good, collision-free points.

The left inequality of the dictionary is (ii) for baths on the good points; the right one is (iii) at a minimizer of  $\mathcal{E}(\varphi)/\|\varphi\|^2$  with  $|\text{supp } \varphi| \leq v$ .  $\square$

*Proof of Proposition H.13.* 1. *Isoperimetry.* Let  $A \subseteq \text{supp } p$  be finite with  $|A| < 2^{2S+4}$ , and  $\varrho := \phi(2|A|) \leq \varrho_*$ . For  $x \in A$  the points  $f(w_g)x$ ,  $g \in B(\varrho)$ , are distinct, and  $|B(\varrho)| \geq 2|A|$ , so at least  $|B(\varrho)|/2$  of them lie outside  $A$ ; hence  $\sum_{g \in B(\varrho)} |f(w_g)A \setminus A| \geq |A||B(\varrho)|/2$ . Since  $|f(uv)A \setminus A| \leq |f(v)A \setminus A| + |f(u)A \setminus A|$ , a word of length at most  $\varrho$  has  $|f(w)A \setminus A| \leq \varrho|\partial A|$  with  $|\partial A| := \sum_{s \in S_G} |f(s)A \setminus A|$ ,  $S_G := \mathcal{L} \cup \mathcal{L}^{-1}$ . So  $|\partial A| \geq |A|/(2\phi(2|A|))$ . This is the argument of [22, Théorème 1], for the Schreier graph near the bath.

2. *From  $\ell^1$  to  $\ell^2$ .* For  $\varphi \geq 0$  supported in  $\text{supp } p$  with  $|\text{supp } \varphi| \leq v < 2^{2S+4}$ , put  $F := \varphi^2$ . Each level set  $A_t := \{F > t\}$  satisfies step 1, and  $\sum_x |\mathbf{1}_A(f(s)x) - \mathbf{1}_A(x)| = 2|f(s)^{-1}A \setminus A|$ , so the co-area formula gives  $\sum_{s \in S_G} \sum_x |F(f(s)x) - F(x)| = 2 \int_0^\infty |\partial A_t| dt \geq \int_0^\infty |A_t| dt / \phi(2v) = \|\varphi\|^2 / \phi(2v)$ , as  $\phi$  is nondecreasing. By Cauchy-Schwarz the left side is at most  $\sqrt{\mathcal{E}_{S_G}(\varphi)} \cdot 2\sqrt{|S_G|}\|\varphi\|$ . So  $\mathcal{E}_{S_G}(\varphi) \geq \|\varphi\|^2 / (4|S_G|\phi(2v)^2)$ , and with  $|S_G| \leq 2r$  and  $\mathcal{E} = \mathcal{E}_{\mathcal{L}} \geq \mathcal{E}_{S_G}/2$  (the term of an inverse label equals that of the label after reindexing),  $\Lambda_{\text{supp } p}(v) \geq 1/(16r\phi(2v)^2)$ .

3. By Theorem H.12(ii), through  $\varphi_s$ , which is supported in  $\text{supp } p$ ,

$$a \geq \Lambda_{\text{supp } p}(2^{2S+4})/(8C_\mu) \geq 1/(128rC_\mu\phi(2^{2S+5})^2).$$

4. By submultiplicativity and Fekete's lemma  $|B(\varrho)| \geq \omega_0^\varrho$ , so  $\phi(v) \leq \log v / \log \omega_0 + 1$  and  $\phi(2^{2S+5}) \leq \alpha S + \beta_0$ . With  $u := \alpha S + \beta_0 \geq \beta_0$ ,  $S + na \geq (u - \beta_0)/\alpha + nc/u^2$ , whose minimum over  $u > 0$ , at  $u^3 = 2nc\alpha$ , is  $\frac{3}{2^{2/3}}(c/\alpha^2)^{1/3}n^{1/3} - \beta_0/\alpha$ .  $\square$

## M.5 Re-folding devices and collars

*Proof of Lemma H.14.* (a) After the class-uniform permutation the branch is uniform on its class: on both points  $c \pm D$  of its pair and on all  $m_{t-1}$  slots. The round moves the point by the label's translation and the device shifts it by a fixed integer, so the new distance is  $X = |D + \sigma\delta_t|$  with  $\sigma$  uniform, and the slot is still uniform. The source pair at distance  $X$  has rank  $j = X - \frac{1}{2}$ ; its  $2m_{t-1}$  units have ranks  $2jm_{t-1} + 2k + s$  (slot  $k$ , point  $s \in \{0, 1\}$ ), and target pair  $i$  holds the ranks  $[2m_t i, 2m_t(i + 1))$ . A unit of slot  $k$  goes below target pair  $j$  if and only if  $2k + s < 2j(m_t - m_{t-1})$ , that is, if and only if  $k < j(m_t - m_{t-1})$ , for both points. So if  $j\gamma_t \leq 1$  a fraction exactly  $j\gamma_t$  of the slots goes to pair  $j - 1$  and the rest to pair  $j$ , since pair  $j$  ends at  $2m_t(j + 1) \geq 2jm_{t-1} + 2m_{t-1}$ ; and if  $j\gamma_t > 1$  all units go below pair  $j$ .

(b) Put  $\lambda = \sqrt{\gamma L/2}/\delta_{\max} \leq 1/(2\delta_{\max}^2) \leq \frac{1}{2}$  and  $D_1 = \delta_{\max} + \frac{1}{2} + 2\lambda\delta_{\max}^2/\gamma$ . For  $D \geq D_1$ ,  $X = D + \sigma\delta$  and  $q := \min(1, (X - \frac{1}{2})\gamma_t) \geq \min(1, (D - \delta_{\max} - \frac{1}{2})\gamma) \geq 2\lambda\delta_{\max}^2$ , and in both cases of (a)  $\mathbb{E}[e^{\lambda D'} | X] \leq e^{\lambda X}(1 - q(1 - e^{-\lambda}))$ . With  $1 - e^{-\lambda} \geq \lambda/2$  and  $\cosh(\lambda\delta) \leq e^{\lambda^2\delta^2/2}$ ,  $\mathbb{E}[e^{\lambda D'} | D] \leq e^{\lambda D} \exp(\lambda^2\delta_{\max}^2/2 - q\lambda/2) \leq e^{\lambda D} e^{-\lambda^2\delta_{\max}^2/2} \leq (1 - \theta)e^{\lambda D}$  with  $\theta = \lambda^2\delta_{\max}^2/4$ , as  $\lambda^2\delta_{\max}^2/2 \leq 1$ . For  $D < D_1$ ,  $D' \leq X < D_1 + \delta_{\max}$ , so  $e^{\lambda D'} \leq K := e^{\lambda(D_1 + \delta_{\max})}$ . Hence  $V_t = \mathbb{E}e^{\lambda D_t}$  satisfies  $V_{t+1} \leq (1 - \theta)V_t + K$  and  $V_t \leq e^{\lambda R_0} + K/\theta$ . By Markov's inequality and

a union over  $t \leq n$ ,  $\Pr(\exists t : D_t \geq y) \leq n(e^{\lambda R_0} + K/\theta)e^{-\lambda y}$ . At  $y = R_0 + 2\delta_{\max} + \frac{1}{2} + 2L/\lambda$ :  $L/\lambda = \delta_{\max}\sqrt{2L/\gamma} = 2\lambda\delta_{\max}^2/\gamma$ , so  $y - D_1 - \delta_{\max} = R_0 + L/\lambda$ , and the bound is at most  $ne^{-2L} + n(8/(\gamma L))e^{-L} \leq ne^{-L}(1 + 8/(\gamma L)) \leq \eta$ , because  $ne^{-L} = \eta/(1 + 8/\gamma)$  and  $L \geq 1$ . Finally  $2L/\lambda = 2\delta_{\max}\sqrt{2L/\gamma}$ .  $\square$

*Proof of Theorem H.15. The device.* The memory holds the points of  $\Omega_W$ ,  $k$  slot registers, one per coordinate, with  $m_t^{(i)}$  slots in use at time  $t$ ,  $m_0 \geq \max(n, 1/\gamma)$  and  $m_t = \lceil (1 + \gamma)m_{t-1} \rceil$ , an ancilla  $\mathbb{C}^{m'}$  with a port, and a garbage flag. The windows  $\Omega_t$  are the units (a point of  $\pi^{-1}(W_x)$  and slots) at time  $t$ , and  $\Omega_0$  is one pair in each coordinate ( $R_0 = \frac{1}{2}$ ) times the fibre times the slots. The classes are the products of the coordinate pairs  $\{c_i \pm D_i\}$ , the fibre and the slots. Round  $t$  applies an independent uniformly random permutation of each class, fixed once and for all before any observer; Haar unitaries on the point blocks  $\{\text{unit}\} \otimes \mathbb{C}^{m'}$ ; the round, the gadget unitary of the regular action on  $\Omega_W$  completed to a unitary of the memory; the shift by  $-r_0$  and, in each coordinate, the re-folding injection of Lemma H.14 by distance, with a fixed bijection between the fibres over the old and the new coordinates; sends units without a target to garbage; and applies Haar unitaries on the new point blocks and exports. Every operation other than the round acts on the memory alone, and on the window the round is the gadget unitary of the regular action tensored with the identity on slots and ancillas.

*Exactness and the class chain.* Every point of the window is triangle-good and collision-free, since the action is the regular action of a group in which the relations hold and the labels are distinct; so the round is a free permutation round on the window. By the fibre hypothesis a uniform point of the fibre over  $v$  is mapped to a uniform point of the fibre over  $v - \pi(g)$ ; the class permutation makes the signs and slots of the coordinates independent and uniform; so the class-uniformized chain is the product of  $k$  independent chains of Lemma H.14(a), with steps  $\pi_i(g_t) - r_{0,i}$ . Lemma H.14(b) with  $\eta/k$  per coordinate gives  $\max_y \kappa_{\text{class}}(y) \leq \eta$ ; a coordinate with  $\delta_i = 0$  never moves out.

*Discarded weight and error.* Put  $z = \min(\epsilon^2/(64n), 1/(4n^3))$  and choose  $|\Omega_0|$ , through  $m_0$ , with  $|\Omega_0| \geq n(n \ln r + \ln 2)/(2z^2)$ , so that  $\log m_0 = O(\log(n/\epsilon))$ . By Lemma H.8 there are class permutations, fixed from then on, with  $\max_y \kappa_\beta(y) \leq \eta + z$ . Theorem H.7 with  $\zeta = z$ , its Haar choices made after the permutations, gives for every observer  $\omega_{\text{ideal}} \leq \eta + z \leq \epsilon^2/(32n)$  and  $\omega \leq \eta + 2z \leq 3/(4n^3)$ . So the error is at most  $(1 + \sqrt{n})\sqrt{\epsilon^2/(32n) + \epsilon^2/(16n)} \leq 0.53\epsilon$  for  $n \geq 2$ , and the memory mass outside the window at the start of every round is at most  $n\omega \leq 3/(4n^2)$ .

*Memory.*  $Q \leq \log |\Omega_W| + \sum_i \log m_n^{(i)} + \log m' + O(1)$ , with  $\log m_n^{(i)} \leq \log m_0 + n \log(1 + \gamma + 1/m_0) \leq \log m_0 + n \log(1 + \gamma) + 1/\ln 2$  since  $m_0 \geq n$ ,  $\log m_0 = O(\log(n/\epsilon))$ , and  $\log m' = O_r(\log(n/\epsilon) + \log \log |\Omega_W|)$  by Theorem H.7.

For  $\mathbb{Z}^k$  the fibres are trivial and the slot words move each coordinate by a bounded amount, so  $|\Omega_W| \leq \prod_i (2x_i + O(1))$ ; with  $\gamma = 1/n$  the condition  $\gamma L \leq 1/(2\delta_{\max}^2)$  holds for  $n \geq n_0$ ,  $x_i = O(\delta_i \sqrt{nL})$ , and  $Q = O_{k,r}(\log(n/\epsilon))$ .  $\square$

*Proof of Proposition H.16.* (a)  $x \in L_j \subseteq F_j$  gives  $f(g)x \in F_{j+1}$ , and  $x \notin F_{j-1}$  with  $f(g)^{-1}F_{j-2} \subseteq F_{j-1}$  gives  $f(g)x \notin F_{j-2}$ . The points of  $F_j$  sent outside  $F_j$  all lie in  $L_j$ , since  $f(g)F_{j-1} \subseteq F_j$ , and there are  $|f(g)F_j \setminus F_j| = \Phi_g(j)$  of them. For the down moves,  $F_{j-1} \cap f(g)L_j = F_{j-1} \setminus f(g)F_{j-1}$ : a point  $y$  of  $F_{j-1}$  outside  $f(g)F_{j-1}$  has  $f(g)^{-1}y \in F_j \setminus F_{j-1} = L_j$ , and conversely; and  $|F_{j-1} \setminus f(g)F_{j-1}| = |f(g)F_{j-1} \setminus F_{j-1}| = \Phi_g(j-1)$  since  $|f(g)F_{j-1}| = |F_{j-1}|$ . So  $s_j p_+(j) = \Phi_g(j) = s_{j+1} p_-(j+1)$ .

(b) *Re-fold.* Ranks are kept and capacities grow ( $V_j m_t \geq V_j m_{t-1}$ ), so no unit moves outward. A unit of shell  $X \geq 1$  with slot  $k$  and index  $\text{idx} < s_X$  goes to a shell at most  $X - 1$  if and only if  $ks_X + \text{idx} < V_{X-1}(m_t - m_{t-1})$ ; the fraction of slots doing so is at least  $\min(1, V_{X-1}\gamma_t/s_X) - 1/m_{t-1} \geq \Gamma' := \min(1, \gamma|F_0|/(2s)) - 1/m_0$ , since  $V_{X-1} \geq |F_0|$ ,  $s_X \leq 2s$ ,

$\gamma_t \geq \gamma$  and  $m_{t-1} \geq m_0$ . Slots are uniform and independent of the point, so, given the past, the unit drops by at least one shell with probability at least  $\Gamma'$ . Reaching the garbage needs shell  $K+1$  after a round, hence shell  $K$  before it.

*Drift.* Let  $\lambda := \min(1, \Gamma'/2 - \zeta)$ . For  $D \geq 2$  the move  $X - D \in \{-1, 0, 1\}$  has mean  $(\Phi_g(D) - \Phi_g(D-1))/s_D \leq \zeta$  by (a), so  $\mathbb{E}e^{\lambda(X-D)} \leq \exp(\lambda\zeta + \lambda^2/2)$  (Hoeffding's lemma), and  $\mathbb{E}[e^{\lambda D'} | X] \leq e^{\lambda X}(1 - \Gamma'(1 - e^{-\lambda})) \leq e^{\lambda X} \exp(-\Gamma'(\lambda - \lambda^2/2))$ . Since  $\lambda \leq 1$  and  $\lambda \leq \Gamma'/2 - \zeta$ ,  $\Gamma'(1 - \lambda/2) \geq \Gamma'/2 \geq \zeta + \lambda$ , so the exponent  $\lambda\zeta + \lambda^2/2 - \Gamma'\lambda(1 - \lambda/2)$  is at most  $-\lambda^2/2$  and  $\mathbb{E}[e^{\lambda D'} | \text{past}] \leq (1 - \lambda^2/4)e^{\lambda D}$ . For  $D \leq 1$ ,  $e^{\lambda D'} \leq e^{2\lambda}$ .

*Kill.* With  $U_t := \mathbb{E}[e^{\lambda D_t} \mathbf{1}\{\tau \geq t\}]$ ,  $\tau$  the first visit to shell  $K$ :  $U_t \leq (1 - \lambda^2/4)U_{t-1} + e^{2\lambda}$ , so  $U_t \leq 1 + 4e^{2\lambda}/\lambda^2 \leq 1 + 30/\lambda^2$ , and  $\Pr(\tau \leq n) \leq \sum_{t \leq n} \Pr(\tau \geq t, D_t \geq K) \leq n(1 + 30/\lambda^2)e^{-\lambda K}$ .

*Scales.*  $|F_K \setminus F_0| = \sum_{1 \leq j \leq K} s_j \geq Ks$ , so  $|F_0| \geq Ks$ . With  $\gamma$  as stated,  $\gamma|F_0|/(2s) = 2\zeta + 4L_c/K + 1/m_0 \leq 6L_c/K + \frac{1}{4} \leq 1$ , so  $\Gamma' = 2\zeta + 4L_c/K$  and  $\lambda = 2L_c/K \leq \frac{1}{4}$ . The kill is at most  $n(1 + 30K^2/(4L_c^2))e^{-2L_c} \leq n(1 + 8K^2)e^{-2L_c} = \eta e^{-L_c} \leq \eta$ , since  $L_c \geq 1$ . And  $n\gamma \leq (2n/K)(2L_c/K + 4L_c/K + 1/m_0) \leq 12nL_c/K^2 + 2/K$ , as  $m_0 \geq n$ , so the slot register costs  $\log m_0 + n\gamma/\ln 2 + O(1) \leq \log m_0 + 17.4nL_c/K^2 + O(1)$ . The rest is the proof of Theorem H.15, with the core  $F_0$  times the initial slots as  $\Omega_0$  and  $\eta = \min(\epsilon^2/(64n), 1/(4n^3))$ .  $\square$

*Proof of Theorem H.17.* Put  $v := 2^{2S+4}$  and take the collar of (a), of depth  $K$ . If  $K \geq 8L_c$ , with  $L_c$  at this depth, Proposition H.16(b) applies, since  $\zeta \leq 1/K \leq L_c/K$ , and gives  $\log \max_t |\Omega_t| \leq \kappa(2S+5)(\log(2S+6))^\kappa + 17.4nL_c\kappa(\log(2S+6))^\kappa \Lambda(2^{2S+4}) + O_r(\log(n/\epsilon) + \log \log |F_{K+1}|)$ , and Theorem H.12(ii) gives  $n\Lambda(2^{2S+4}) \leq 8C_\mu n a$ . Otherwise let  $K_0 \geq 16$  be the least integer with  $K_0 \geq 8 \ln(n(1 + 8K_0^2)/\eta)$ , and take the collar of (b) for this  $K_0$ , of depth  $K' \geq K_0$ . The function  $K \mapsto K - 8 \ln(n(1 + 8K^2)/\eta)$  increases for  $K \geq 16$ , so  $K' \geq 8L_c(K')$  and Proposition H.16(b) applies: its size term is at most  $\kappa K_0^\kappa = \text{polylog}(n/\epsilon)$ , and its  $\gamma$ -term is  $17.4nL_c/K'^2 \leq 17.4nL_c/K^2 \leq 17.4nL_c\kappa(\log(2S+6))^\kappa \Lambda(2^{2S+4})$ , since  $K' \geq K_0 > K$ , which is bounded as before. In both cases  $L_c = O(\log(n/\epsilon) + \log K')$  and  $\log K' \leq \log |F_{K'+1}|$ , so all factors are polylogarithmic in  $n/\epsilon$  and  $2 + S$ ; (B1) and (B3) hold by the construction of Proposition H.16(b). For the regular action of a group of exponential growth, step 2 of the proof of Proposition H.13 gives  $\Lambda(v) \geq c/(\log v)^2$ ; at  $K \asymp \log v / \log \log v$  a collar of size  $e^{O(K \log K)}$  has  $\log |F_{K+1}| = O(\log v)$  and  $K^{-2} \leq C(\log \log v)^2 \Lambda(v)$ , which is (a), and (b) is immediate.  $\square$

**Houghton groups.** For  $q \geq 3$ ,  $H_q$  acts on  $X_q = \{1, \dots, q\} \times \{1, 2, \dots\}$  by permutations that are eventually translations on each ray;  $t(g) \in \mathbb{Z}^q$ , with  $\sum_i t_i(g) = 0$ , is the eventual translation of  $g$  on the rays, a homomorphism with kernel the finitary symmetric group  $\text{FSym}(X_q)$ . The generators  $g_j$  ( $2 \leq j \leq q$ ) have  $t(g_j) = e_j - e_1$ :  $g_j$  moves ray 1 one step towards the origin, sends  $(1, 1)$  to  $(j, 1)$ , and moves ray  $j$  one step outwards. Put  $s(\vec{m}) = g_2^{m_2} \circ \dots \circ g_q^{m_q}$  for  $\vec{m} \in \mathbb{Z}^{q-1}$ ; every element is uniquely  $\sigma \circ s(\vec{m})$  with  $\sigma \in \text{FSym}(X_q)$ , since  $x \circ s(\vec{m})^{-1}$  has zero translation for the  $\vec{m}$  read off  $t(x)$ . The *defect radius* of  $g$  is  $\varrho(g) = \min\{R \geq 1 : g(i, k) = (i, k + t_i(g)) \text{ for all } i \text{ and all } k \geq R\}$ .

**Lemma M.1** (Cocycles of Houghton groups). (a)  $\varrho(g \circ h) \leq \max(\varrho(h), \varrho(g) + \|t(h)\|_\infty)$  and  $\varrho(g^{-1}) \leq \varrho(g) + \|t(g)\|_\infty$ , and  $t(c) = 0$  implies  $c \in \text{Sym}(X_{<\varrho(c)})$ , where  $X_{<R} = \{(i, k) : k < R\}$  has  $q(R-1)$  points.

(b) Let  $w$  be an element with  $\|t(w)\|_\infty \leq \ell$  and  $\varrho(w^{-1}) \leq \varrho_0$ , and let  $\pi(w)$  be the  $\mathbb{Z}^{q-1}$ -part of  $t(w)$ . For  $x = \sigma \circ s(\vec{m})$  with  $|m_j - \frac{1}{2}| \leq x_0$  for all  $j$ ,  $x \circ w^{-1} = (\sigma \circ c) \circ s(\vec{m} - \pi(w))$  with  $c \in \text{Sym}(X_{<R_*})$ ,  $R_* := \varrho_0 + \ell + 4 + 7(q-1)(x_0 + 1 + \ell)$ , and  $c$  depends only on  $\vec{m}$  and  $w$ .

*Proof.* (a) For  $k \geq \max(\varrho(h), \varrho(g) + \|t(h)\|_\infty)$ ,  $h(i, k) = (i, k + t_i(h))$  with  $k + t_i(h) \geq \varrho(g)$ , so  $g(h(i, k)) = (i, k + t_i(h) + t_i(g))$ . For  $k \geq \varrho(g) + \|t(g)\|_\infty$  put  $k' := k - t_i(g) \geq \varrho(g)$ : then

$g(i, k') = (i, k)$ , so  $g^{-1}(i, k) = (i, k - t_i(g))$ . If  $t(c) = 0$ ,  $c$  fixes every  $(i, k)$  with  $k \geq \varrho(c)$ , hence permutes the finite complement  $X_{<\varrho(c)}$ .

(b)  $x \circ w^{-1} = \sigma \circ s(\vec{m}) \circ w^{-1} = \sigma \circ c \circ s(\vec{m}')$  with  $\vec{m}' := \vec{m} - \pi(w)$  and  $c := s(\vec{m}) \circ w^{-1} \circ s(\vec{m}')^{-1}$ , which has zero translation and depends only on  $\vec{m}$  and  $w$ . Since  $\varrho(g_j^{\pm 1}) = 2$  and each factor changes each coordinate of  $t$  by at most one, (a) gives  $\varrho(s(\vec{m})) \leq 2 + |\vec{m}|_1$  and  $\varrho(s(\vec{m})^{-1}) \leq 2 + 2|\vec{m}|_1$ . With  $h_1 := s(\vec{m}')^{-1}$  and  $h_2 := w^{-1} \circ h_1$ :  $\varrho(h_1) \leq 2 + 2|\vec{m}'|_1$ ,  $\|t(h_1)\|_\infty \leq |\vec{m}'|_1$ ,  $\varrho(h_2) \leq \max(\varrho(h_1), \varrho_0 + |\vec{m}'|_1)$ ,  $\|t(h_2)\|_\infty \leq \ell + |\vec{m}'|_1$ , and  $\varrho(c) = \varrho(s(\vec{m}) \circ h_2) \leq \max(\varrho(h_2), 2 + |\vec{m}|_1 + \ell + |\vec{m}'|_1)$ . In the box,  $|\vec{m}|_1 \leq (q-1)(x_0 + 1)$  and  $|\vec{m}'|_1 \leq (q-1)(x_0 + 1 + \ell)$ , so  $\varrho(c) \leq R_\star$  and  $c \in \text{Sym}(X_{<R_\star})$  by (a).  $\square$

*Proof of Theorem H.18.* Use the right regular action  $f(g)x = x \circ g^{-1}$ , a homomorphism, with every point triangle-good and collision-free. Let  $\ell$  bound  $\|t(w)\|_\infty$  and  $\varrho_0$  bound  $\varrho(w^{-1})$  over the slot words. Given the window half-width  $x_0$ , put  $\Omega_B := \{\sigma \circ s(\vec{m}) : \sigma \in \text{Sym}(X_{<R_\star}), \vec{m} \in B_{\text{out}}\}$  with  $B_{\text{out}}$  the box of half-width  $x_0 + \ell$  around  $\frac{1}{2} \cdot \mathbf{1}$ , and  $\pi(\sigma \circ s(\vec{m})) := \vec{m}$ . The fibres have the common size  $(q(R_\star - 1))!$ , and by Lemma M.1(b) every slot word maps the fibre over  $\vec{m}$ , for  $\vec{m}$  in the box  $B$  of half-width  $x_0$ , bijectively onto the fibre over  $\vec{m} - \pi(w)$ , by  $\sigma \mapsto \sigma \circ c$ . So Theorem H.15 applies with  $k = q - 1$ ,  $r_0 = 0$  and  $\delta_i \leq \ell$ , at the window half-width  $x_0 = 1 + 2\ell + 2\ell\sqrt{2L/\gamma}$ , that is  $\gamma = 8\ell^2 L / (x_0 - 1 - 2\ell)^2$ ; its condition  $\gamma L \leq 1/(2\ell^2)$  holds as soon as  $x_0 \geq 1 + 2\ell + 4\ell^2 L$ . Its memory is  $\log((q(R_\star - 1))!) + (q-1)O(\log x_0) + (q-1)n \log(1 + \gamma) + O(\log(n/\epsilon)) = O(q^2 x_0 \log(qx_0) + qn\ell^2 L/x_0^2) + O(\log(n/\epsilon))$ . Take  $x_0 = \max((n\ell^2 L/(q \log n))^{1/3}, 1 + 2\ell + 4\ell^2 L)$ : the memory is  $O(q^{5/3}(\ell^2 L)^{1/3} n^{1/3} (\log n)^{2/3} + q^2 \ell^2 L \log(q\ell L)) + O(\log(n/\epsilon))$ .

$H_q$  is elementary amenable, being an extension of  $\text{FSym}(X_q)$  by  $\mathbb{Z}^{q-1}$ , and so is the subgroup generated by the labels. By Chou's theorem [17, Theorem 3.2] that subgroup has polynomial or exponential growth. In the first case it is virtually nilpotent, hence residually finite, and its finite quotients give a finite-dimensional representation of  $P_\Phi$  that separates the labels, so Corollary 7.8 applies. In the second case Proposition H.13, on the regular action, gives  $S + na \geq c_q n^{1/3} - C_q$  for every bath; since  $L = O(\log(n/\epsilon))$ , both terms of the memory are at most  $K \log^2(n/\epsilon)(1 + S + na)$  for  $n \geq n_0$ .  $\square$

*Proof of Proposition H.19.* (a) With point classes no permutation is needed, and the branch of a label sequence at time  $t$  is a point  $f(w)x_0$  with  $|w| \leq t$ , inside  $\Omega_t$ ; so the ideal killed chain is never killed, and Theorem H.7 gives error at most  $(1 + \sqrt{n})\sqrt{\epsilon^2/(16n)} \leq 0.43\epsilon$  and memory  $\log|\Omega_n| + O_r(\log(n/\epsilon))$ . The ball of radius  $n$  in the labels and their inverses has at most  $(2r + 1)^n$  points.

(b) Let  $A$  be finite and  $r_A := 12(\lfloor \log \log |A| \rfloor + 2)$ , so that  $|A| < 2^{2^{\lfloor r_A/12 \rfloor - 1}}$ . By [28, Proposition 7.3], some generator  $s \in \{p, q, t, c, a\}$  of  $\Gamma$  has  $|As \setminus A| > |A|/r_A$ ; the generators are labels of the lamp channel, and for the right regular action  $f(s)A = As^{-1}$  with  $|As^{-1} \setminus A| = |A \setminus As| = |As \setminus A|$ . So the label boundary  $|\partial A| = \sum_{s \in S_G} |f(s)A \setminus A|$  exceeds  $|A|/\Psi(|A|)$  with  $\Psi(v) := 12(\log \log v + 2) \geq r_A$ . The co-area and Cauchy–Schwarz steps of the proof of Proposition H.13, with  $|\partial A_t| \geq |A_t|/\Psi(v)$  for the level sets of  $\varphi^2$ , give  $\mathcal{E}_{S_G}(\varphi) \geq \|\varphi\|^2/(|S_G|\Psi(v)^2)$  and  $\Lambda(v) \geq 1/(4r\Psi(v)^2)$ . By Theorem H.12(ii),  $a \geq c/(\log(2S + 4))^2$  for every bath on the regular action, so  $S + na \geq c'n/(\log n)^2$ : if  $S \geq n/(\log n)^2$  this is immediate, and otherwise  $\log(2S + 4) = O(\log n)$ . So the window of (a), of at most  $n \log(2r + 1)$  bits, is at most  $(\log(n/\epsilon))^3(1 + S + na)$  for  $n \geq n_0$ .  $\square$

## M.6 Leaky biased rounds

*Proof of Theorem H.20.* 1. By Lemma 2.3,  $\nu_1 := p(\text{bad}) \leq 4d\ell \leq 4dC_0/n \leq \frac{1}{16}$ .

2. At a triangle-good  $\omega$ ,  $U_f(\xi \otimes |\omega\rangle) = \sum_g A_g \xi \otimes |f(g)\omega\rangle$ . Lemma H.3 on the good points, a diagonal projection that commutes with  $p$ , gives a bath  $p_1$  with leakage 0, distance at most  $(\frac{1}{16} + \frac{1}{16})/\frac{15}{16} = \frac{2}{15}$ ,  $S_1 \leq S/(1 - \nu_1)$  and  $a_1 \leq (a + h_2(\nu_1))/(1 - \nu_1)$ .
3. By Lemma 2.3,  $p_1\{\omega : f(g)\omega = f(h)\omega\} \leq 2 \cdot \frac{2}{15} = \frac{4}{15}$  for  $g \neq h$ .
4. Lemma H.1 applied to the round of step 2: on tuples of good points  $U_{f \times t}$  has Kraus form with the product model, the bath is  $p_1^{\otimes t}$ ,  $S = tS_1$ ,  $a \leq ta_1$ , and two labels collide at a tuple if and only if they collide in every coordinate, so the collision mass is at most  $(r^2/2)(4/15)^t \leq 1/n^2$  for  $t = \lceil \log_{15/4}(r^2 n^2/2) \rceil$ .
5. Lemma H.3 again removes the colliding tuples, a diagonal projection with  $\nu_2 \leq 1/n^2$ . The result is a free exact permutation round, whose weighted Gram is the identity, so its distance is 0 (Lemma 2.10);  $S' \leq tS_1/(1 - \nu_2)$  and  $a' \leq (ta_1 + h_2(\nu_2))/(1 - \nu_2)$ . Then  $na' \leq (1 + O(\nu_1))tna + tn h_2(4dC_0/n)(1 + O(n^{-2})) + O(1)$ , and  $n h_2(4dC_0/n) \leq 4dC_0 \log(en/(4dC_0)) = O(dC_0 \log n)$ .

Theorem H.7 and Lemma H.8 need freeness and exactness only at the points that the windows contain, so for the output the windows of the balance condition must lie inside the good, collision-free tuples, an exit into a bad or colliding tuple counting as a kill.  $\square$

*Proof of Lemma H.21.* Let  $\Phi_d$  be maximally entangled on the input and a reference  $R_{\text{in}}$ , and  $\Omega := (U \otimes I)(|\Phi_d\rangle \otimes |\psi_\sigma\rangle)$ , a pure state of the output,  $\mathbb{C}^M$ ,  $R_{\text{in}}$  and  $R$ . Its  $\mathbb{C}^M$ -marginal is  $T(\sigma)$  and its  $R$ -marginal has the spectrum of  $\sigma$ . As  $\Omega$  is pure,  $S(CR) = S(\mathbb{C}^M) = S(T(\sigma))$ , so  $H(C|R) = S(CR) - S(R) = a(\sigma)$ . If the round is exact on  $K$ , then  $W = \sum_g A_g \otimes B_g$  there and  $\text{Tr}(A_h^* A_g) = d\mu_g \delta_{gh}$ , so  $T(\sigma) = \sum_g \mu_g B_g \sigma B_g^* = \tilde{X} \tilde{X}^*$  with  $\tilde{X} := \sum_g \sqrt{\mu_g} B_g \sqrt{\sigma} \langle g| : \mathbb{C}^r \otimes K \rightarrow \mathbb{C}^M$ . Then  $\tilde{X}^* \tilde{X} = \sum_{g,h} \sqrt{\mu_g \mu_h} |g\rangle \langle h| \otimes \sqrt{\sigma} B_g^* B_h \sqrt{\sigma}$  has the nonzero spectrum of  $T(\sigma)$  and  $K$ -marginal  $\sum_g \mu_g \sqrt{\sigma} B_g^* B_g \sqrt{\sigma} = \sigma$ , so its conditional entropy given  $K$  is  $a(\sigma)$ .  $\square$

We also use the leaky form of this identity. For a state  $\sigma$  on  $K$  with leakage  $\ell_\sigma = \text{Tr}(\sigma\Lambda)/d$  put  $\omega^0 := \sum_{g,h} \sqrt{\mu_g \mu_h} |g\rangle \langle h| \otimes \sqrt{\sigma} X_g^* X_h \sqrt{\sigma}$ , of trace  $1 - \ell_\sigma$ . Then

$$(1 - \ell_\sigma) H(L|R)_{\omega^0/(1-\ell_\sigma)} \leq a(\sigma) + h_2(\ell_\sigma) + \ell_\sigma \log(d^2 - r). \quad (27)$$

Indeed, write  $|\psi_\sigma\rangle = (\sqrt{\sigma} \otimes I)|\Gamma\rangle$  with  $|\Gamma\rangle = \sum_\kappa |\kappa\kappa\rangle$ , and  $W = G + L$ . The part of  $\Omega$  coming from  $G$  is  $\sum_g d^{-1/2} |A_g\rangle \rangle \otimes (X_g \sqrt{\sigma} \otimes I)|\Gamma\rangle$ , and the part coming from  $L$  lies in the range of  $\Pi^\perp \otimes I$ ,  $\Pi := \Pi_\Phi \otimes I_R$ , by Hilbert–Schmidt orthogonality. Let  $\sigma_{CR}$  be the  $CR$ -marginal of  $\Omega$ . Since  $\text{Tr}_M[(A \otimes I)|\Gamma\rangle \langle \Gamma|(B \otimes I)^*] = (B^* A)^T$ , the block  $(g, h)$  of  $\Pi \sigma_{CR} \Pi$  in the orthonormal basis  $|A_g\rangle \rangle / \sqrt{d\mu_g}$  is the entrywise conjugate of the block  $(g, h)$  of  $\omega^0$ ; conjugation preserves spectra and commutes with partial traces, and  $\text{Tr}(\Pi \sigma_{CR} \Pi) = 1 - \ell_\sigma$ , so  $H(C|R)_{\Pi \sigma_{CR} \Pi/(1-\ell_\sigma)} = H(L|R)_{\omega^0/(1-\ell_\sigma)} =: H_0$ . The pinched state  $\tilde{\sigma} := \Pi \sigma_{CR} \Pi + \Pi^\perp \sigma_{CR} \Pi^\perp$  has the  $R$ -marginal of  $\sigma_{CR}$ . With  $A := \Pi \sqrt{\sigma_{CR}}$ ,  $\Pi \sigma_{CR} \Pi = A A^*$  has the nonzero spectrum of  $\sqrt{\sigma_{CR}} \Pi \sqrt{\sigma_{CR}}$ , and  $\sigma_{CR} = \sqrt{\sigma_{CR}} \Pi \sqrt{\sigma_{CR}} + \sqrt{\sigma_{CR}} \Pi^\perp \sqrt{\sigma_{CR}}$ , so concavity gives the inequality of Groenewold and Lindblad [61],  $S(\sigma_{CR}) \geq (1 - \ell_\sigma) S(\Pi \sigma_{CR} \Pi/(1 - \ell_\sigma)) + \ell_\sigma S(\Pi^\perp \sigma_{CR} \Pi^\perp / \ell_\sigma)$ , while  $S(\tilde{\sigma})$  is  $h_2(\ell_\sigma)$  plus the same average; so  $H(C|R)_{\tilde{\sigma}} \leq a(\sigma) + h_2(\ell_\sigma)$ . Finally  $\tilde{\sigma}$  is block diagonal for  $\Pi \oplus \Pi^\perp$  on  $C$ , so with the flag  $F$  of the block,  $H(C|R)_{\tilde{\sigma}} = H(F|R) + H(C|FR) \geq (1 - \ell_\sigma) H_0 - \ell_\sigma \log(d^2 - r)$ , since the  $\Pi^\perp$  block lives on a space of dimension  $d^2 - r$ .

*Proof of Lemma H.22.* From  $W^* W = I$ ,  $H := G^* G - I = -(G^* L + L^* G + L^* L)$ . From  $(\sqrt{t}G \pm L/\sqrt{t})^*(\sqrt{t}G \pm L/\sqrt{t}) \geq 0$ ,  $\mp(G^* L + L^* G) \leq tG^* G + L^* L/t$ . Hence  $H \leq tG^* G + L^* L/t$  and  $-H \leq tG^* G + (1 + 1/t)L^* L$ . The first gives  $(1 - t)G^* G \leq I + L^* L/t$ , so  $tG^* G \leq (tI + L^* L)/(1 - t)$ , and inserting it,  $\pm H \leq [tI + (1 + 1/t)L^* L]/(1 - t) \leq \bar{B} := (tI + (1 + 1/t)L^* L)/(1 - t)$ . For  $|p\rangle \in \mathbb{C}^d$  write  $Y_{pq} = (\langle p| \otimes I)Y(|q\rangle \otimes I)$ . Then  $\bar{B}_{pp} = (tI + (1 + 1/t)(L^* L)_{pp})/(1 - t) \leq \beta_t$ , because  $(L^* L)_{pp} \geq 0$  and  $\sum_p (L^* L)_{pp} = \Lambda$ . By

(F1),  $G(|z_g\rangle \otimes \kappa) = |z_g\rangle \otimes X_g \kappa$ , so  $H_{z_g z_g} = X_g^* X_g - I$ , and  $-\beta_t \leq H_{z_g z_g} \leq \beta_t$ . Compress  $\bar{B} + H \geq 0$  and  $\bar{B} - H \geq 0$  to two slots  $p, q$ , conjugate the second by  $\text{diag}(I, -I)$  and add:  $\begin{pmatrix} \bar{B}_{pp} & H_{pq} \\ H_{qp} & \bar{B}_{qq} \end{pmatrix} \geq 0$ . For the two columns  $p, q$  of a triangle block,  $(G^* G)_{pq} = \frac{1}{2} R_t$  and  $I_{pq} = 0$ ; raising the diagonal blocks to  $\beta_t$  keeps positivity, and multiplying by 2 gives the claim.  $\square$

Conjugating the last matrix by  $\text{diag}(I, e^{i\phi})$  or by  $\text{diag}(I, -I)$ , or swapping its blocks, and scaling, shows  $\begin{pmatrix} 2|c|\beta_t & cD \\ \bar{c}D^* & 2|c|\beta_t \end{pmatrix} \geq 0$  for every triangle defect or adjoint  $D$  and  $c \in \mathbb{C}$ . Hence, if  $E$  is Hermitian on  $\mathbb{C}^r \otimes K$  with zero diagonal blocks and  $E_{pq} = \sum_e c_{pq,e} D_e$  with every  $D_e$  a triangle defect or its adjoint up to a phase, then  $\pm E + I_r \otimes 2\bar{c}\beta_t \geq 0$  with  $\bar{c} := \max_p \sum_{q \neq p} \sum_e |c_{pq,e}|$ : embed one such  $2 \times 2$  block matrix at the positions  $(p, q)$  and  $(q, p)$  for every  $p < q$  and every term, sum, and add  $\bigoplus_p (2\bar{c} - \sum_{q \neq p} \sum_e 2|c_{pq,e}|) \beta_t \geq 0$ . For the correction  $E$  that projects the off-diagonal blocks of  $Y^0 = [X_g^* X_h]$  onto the identities (the average over each class of linked label pairs, as in the proof of Theorem G.9), each block difference telescopes along at most  $r(r-1) - 1$  edges, so  $\bar{c} \leq (r-1)(r(r-1) - 1) \leq (r-1)(r^2 - 1)$ .

*Proof of Theorem H.23. Facts.* For every state  $\sigma$  on  $K$ ,  $\ell_\sigma = \text{Tr}(\sigma \Lambda)/d$ , as in (f1) of the proof of Theorem G.9. Applying  $\text{Tr}_d$  to  $W^* W = I$ , the cross terms vanish and  $\text{Tr}_d(G^* G) = \sum_g d\mu_g X_g^* X_g$ , so  $\sum_g \mu_g X_g^* X_g = I - \Lambda/d$  and  $\sum_g \mu_g \text{Tr}(\sigma X_g^* X_g) = 1 - \ell_\sigma$ . For  $g \neq h$  and any state  $\sigma$  on  $K$  with distance  $u_\sigma$ ,

$$|\text{Tr}(\sigma X_g^* X_h)| \leq 2u_\sigma + 2\sqrt{\text{Tr} \sigma \Lambda} + \text{Tr} \sigma \Lambda : \quad (28)$$

the anchor block of  $W$  is  $W_{z_g z_g} = X_g + \lambda_g$  with  $\lambda_g = \sum_\alpha (E_\alpha)_{z_g z_g} L_\alpha$  and  $\lambda_g^* \lambda_g \leq \Lambda$  by Cauchy–Schwarz;  $|\text{Tr}(\sigma W_{z_h z_h}^* W_{z_g z_g})| \leq 2u_\sigma$  by (F1) and (F2);  $\|W_{z_g z_g} \sqrt{\sigma}\|_F \leq 1$  since  $W_{zz}^* W_{zz} \leq (W^* W)_{zz} = I$ ;  $\|\lambda_g \sqrt{\sigma}\|_F^2 \leq \text{Tr} \sigma \Lambda$ ; expand  $X = W_{zz} - \lambda$  and use Cauchy–Schwarz.

1. *Commuting restriction.* Write  $\rho = \sum_i \rho_i |e_i\rangle\langle e_i|$  ( $\rho_i > 0$ ) and  $\lambda_i := \langle e_i | \Lambda | e_i \rangle$ , so  $\sum_i \rho_i \lambda_i = d\ell = s^2$ . Put  $\theta := s/c_r$ ,  $J := \{i : \lambda_i \leq \theta\}$ ,  $P_0 := \sum_{i \in J} |e_i\rangle\langle e_i|$  and  $K_0 := \text{ran } P_0$ . Markov's inequality gives  $\nu := \sum_{i \notin J} \rho_i \leq s^2/\theta = c_r s \leq \frac{1}{46}$ .  $P_0$  commutes with  $\rho$ , so Lemma H.3 gives  $\rho_0 := P_0 \rho P_0 / (1 - \nu)$  with  $S_0 \leq S/(1 - \nu)$ ,  $a_0 \leq (a + h_2(\nu))/(1 - \nu)$ ,  $u_0 \leq (u + \nu)/(1 - \nu)$  and  $\ell_0 \leq \ell/(1 - \nu)$ . On  $K_0$ ,  $W_0 = G_0 + L_0$  with the  $X_g$  restricted and  $\Lambda_0 = P_0 \Lambda P_0$ . In the basis  $\{e_i\}_{i \in J}$ ,  $\rho_0 \log(1/\rho_0)$  is diagonal with nonnegative entries, so  $J_0 := \text{Tr}(\Lambda_0 \rho_0 \log(1/\rho_0)) = \sum_{i \in J} \lambda_i \rho_{0,i} \log(1/\rho_{0,i}) \leq \theta S_0$ . From now on everything is on  $K_0$ , where  $\rho_0$  is invertible;  $\xi := \sqrt{\rho_0}$ .
2. *Domination.*  $Y^0 := [X_g^* X_h]_{g,h} \geq 0$  on  $\mathbb{C}^r \otimes K_0$ , and  $\beta := (sI + (1 + 1/s)\Lambda_0)/(1 - s)$ ; Lemma H.22 applies to  $W_0$  with  $t = s$ .
3. *Projection.* Let  $E$  be the class-average correction, so that the off-diagonal blocks of  $Y^0 + E$  satisfy every identity, and  $Z_E := 2(r-1)(r^2-1)\beta \geq 2\bar{c}\beta$ . Then  $\pm E + I_r \otimes Z_E \geq 0$ .
4. *Fill.*  $Z_D := \beta$ ,  $Z := Z_D + Z_E = c_r \beta$ ,  $F := \bigoplus_g (I + Z_D - X_g^* X_g) \geq 0$  by Lemma H.22, and  $Y_1 := Y^0 + F + (E + I_r \otimes Z_E) \geq 0$ . Then  $(Y_1)_{gg} = I + Z$  for every  $g$  and  $(Y_1)_{gh} = Y_{gh}^0 + E_{gh}$  for  $g \neq h$ , and every identity holds: the off-diagonal ones by step 3, the diagonal ones because all diagonal blocks are equal.
5. *Absorber in the bath.*  $z := \text{Tr}(\rho_0 Z)$  and  $\rho' := (\rho_0 + \xi Z \xi)/(1 + z)$ , a state on  $K_0$  with  $\rho' \geq \rho_0/(1 + z)$ . Put  $Y := (1 + z)^{-1} (I \otimes \rho'^{-1/2} \xi) Y_1 (I \otimes \xi \rho'^{-1/2})$ . Then  $Y \geq 0$ ,  $Y_{gg} = \rho'^{-1/2} (\rho_0 + \xi Z \xi) \rho'^{-1/2} / (1 + z) = I$ , and every identity survives the common congruence.  $X'_g := Y^{1/2}(|g\rangle \otimes I) : K_0 \rightarrow \mathbb{C}^r \otimes K_0$  are isometries with  $X'_g{}^* X'_h = Y_{gh}$ . With the embedding  $\iota := X'_1$ , the bath  $X'_1 \rho' X'_1{}^*$  and the family  $X'_g X'_1{}^*$  on  $\text{ran } X'_1$ , Lemma 2.9(c) extends  $\sum_g A_g \otimes X'_g X'_1{}^*$  to a unitary. The new round has channel  $\sum_{g,h} \text{Tr}(\rho' X_h^* X_g') A_g(\cdot) A_h^*$ , hence leakage 0, bath entropy  $S' := S(\rho')$ , and bath map with the spectrum of  $\sum_g \mu_g X'_g \rho' X'_g{}^*$ .

6. *Entropy production.*  $(I \otimes \sqrt{\rho'})Y(I \otimes \sqrt{\rho'}) = Y_1^\xi/(1+z)$  with  $Y_1^\xi := (I \otimes \xi)Y_1(I \otimes \xi)$ . By Lemma H.21,  $a' = H(L|R)_{\omega'}$  with  $\omega' := DY_1^\xi D/(1+z)$ ,  $D := \text{diag}(\sqrt{\mu_g}) \otimes I$ . Let  $\omega^0 := D(I \otimes \xi)Y^0(I \otimes \xi)D$ , of trace  $1 - \ell_0$ ,  $\hat{\omega}^0 := \omega^0/(1 - \ell_0)$ , and  $\Delta := D(I \otimes \xi)(F + E + I \otimes Z_E)(I \otimes \xi)D \geq 0$ . As  $E$  has zero diagonal blocks,  $\text{Tr} \Delta = \sum_g \mu_g \text{Tr}(\rho_0(I + Z_D - X_g^* X_g)) + \text{Tr}(\rho_0 Z_E) = \ell_0 + z$ . Hence

$$\omega' = (1-p)\hat{\omega}^0 + p\tau, \quad p := (\ell_0 + z)/(1+z), \quad \tau := \Delta/(\ell_0 + z) :$$

a mixture with a positive correction. The mixture bound  $(H(L|R) \leq H(LX|R) \leq h_2(p) + (1-p)H(L|R)_{\hat{\omega}^0} + pH(L|R)_\tau$  for an adjoined index  $X$  and  $H(L|R)_\tau \leq \log r$  give  $a' \leq (1-p)H_0 + p \log r + h_2(p) \leq H_0 + 2p \log r + h_2(p)$  with  $H_0 := H(L|R)_{\hat{\omega}^0} \geq -\log r$ . By (27) for  $(U, \rho_0)$ ,  $(1 - \ell_0)H_0 \leq a_0 + h_2(\ell_0) + \ell_0 \log(d^2 - r)$ ; with step 1 this is item 3.

7. *Bath entropy.*  $\log$  is operator monotone and  $\rho' \geq \rho_0/(1+z)$ , so  $S(\rho') \leq -\text{Tr} \rho' \log \rho_0 + \log(1+z) = (S_0 + \text{Tr}(\xi Z \xi \log(1/\rho_0)))/(1+z) + \log(1+z)$ . Since  $\xi$  commutes with  $\log \rho_0$ ,  $\text{Tr}(\xi Z \xi \log(1/\rho_0)) = \text{Tr}(Z \rho_0 \log(1/\rho_0)) = \frac{c_r}{1-s}(sS_0 + (1+1/s)J_0) \leq \frac{(c_r s + s + 1)S_0}{1-s}$  by step 1, as  $\theta = s/c_r$ . So  $S' \leq S_0(1 + (1 + (c_r + 1)s)/(1-s)) + z/\ln 2$ .
8. *Gram.* For  $g \neq h$ ,  $\text{Tr}(\rho' X_g'^* X_h') = \text{Tr}((Y_1^\xi)_{gh})/(1+z) = [\text{Tr}(\rho_0 X_g^* X_h) + \text{Tr}(\xi E_{gh} \xi)]/(1+z)$ . Compressing  $\pm E + I \otimes Z_E \geq 0$  to the blocks  $g, h$  and conjugating by  $\xi$  gives  $\begin{pmatrix} \xi Z_E \xi & \pm \xi E_{gh} \xi \\ \pm \xi E_{hg} \xi & \xi Z_E \xi \end{pmatrix} \geq 0$ ; a matrix  $\begin{pmatrix} A & X \\ X^* & A \end{pmatrix} \geq 0$  has  $X = A^{1/2} C A^{1/2}$  with  $\|C\| \leq 1$ , so  $\|\xi E_{gh} \xi\|_1 \leq \text{Tr}(\rho_0 Z_E)$ . With (28) for  $\rho_0$ ,  $|\text{Tr}(\rho' X_g'^* X_h')| \leq 2u_0 + 2\sqrt{d\ell_0} + d\ell_0 + \text{Tr}(\rho_0 Z_E)$ .
9. *Constants.*  $c_r \geq 7$ , so  $s \leq 1/322$  and  $\nu \leq c_r s \leq \frac{1}{46}$ ;  $d\ell_0 \leq s^2/(1-\nu)$ . Then  $z = c_r(s + (1+1/s)d\ell_0)/(1-s) \leq c_r s(1 + (1+s)/(1-\nu))/(1-s) \leq 2.05c_r s$ ,  $\text{Tr}(\rho_0 Z_E) = (c_r - 1)z/c_r \leq 2.05(c_r - 1)s$  and  $p \leq \ell_0 + z \leq 2.2c_r s$ . For the Gram,  $2(u + \nu)/(1-\nu) \leq 2u(1 + 1.05c_r s) + 2.05c_r s$  and  $2\sqrt{d\ell_0} + d\ell_0 \leq 2.1s$ , so the bound is at most  $2u(1 + 1.05c_r s) + 4.2c_r s$ , which at  $u \leq \frac{1}{16}$  and  $c_r s \leq \frac{1}{46}$  is at most  $0.128 + 0.092 < \frac{1}{4}$ . For the bath,  $(1 + (c_r + 1)s)/(1-s) \leq 1 + 1.3c_r s$  and  $S_0 \leq S(1 + 1.03c_r s)$ , so  $S' \leq (2 + 1.3c_r s)(1 + 1.03c_r s)S + 2.05c_r s/\ln 2 \leq (2 + 3.4c_r s)S + 3.1c_r s$ . At the scale of the law,  $s = \sqrt{d\ell} \leq \sqrt{dc_1/n} \leq 1/(46c_r)$  when  $n \geq 2116c_r^2 dc_1$ , and then  $c_r s \leq \frac{1}{46}$  gives  $S' \leq 2.074S + 0.068 \leq 2.08S + 0.07$ . In item 3,  $1/((1-\nu)(1-\ell_0)) \leq 1 + 1.1c_r s$ ,  $h_2$  is increasing on  $[0, \frac{1}{2}]$ ,  $n \cdot 2p \log r \leq 4.4c_r \sqrt{dc_1/n} \log r$ ,  $n h_2(\ell_0)/(1-\ell_0) \leq 1.1c_1 \log(en/c_1)$  and  $n\ell_0 \log(d^2 - r)/(1-\ell_0) \leq 2.2c_1 \log d$ ; and  $n h_2(x) \leq nx \log(e/x) = O(\sqrt{n} \log n)$  for  $x = O(n^{-1/2})$ . This gives  $na' \leq (1 + O(n^{-1/2}))na + O_{d,r,c_1}(\sqrt{n} \log n)$ .  $\square$

*Proof of Corollary H.24.* The lower inequality, with these constants, is Theorem J.1. For the upper one take a round with  $u \leq \frac{1}{16}$ ,  $\ell \leq \frac{2}{15n}$  and  $S + na \leq L^*(n) + 1$ . Theorem H.23 with  $c_1 = \frac{2}{15}$  gives a zero-leakage round with weighted Gram off-diagonals at most  $\frac{1}{4}$ ,  $S' \leq 2.08S + 0.07$  and  $na' \leq (1 + O(n^{-1/2}))na + O(\sqrt{n} \log n)$ , so  $S' + na' \leq 2.1(S + na) + O(\sqrt{n} \log n)$  for large  $n$ . Apply Theorem 6.3, which gives the displayed squared-log upper with flat imports and charged initial purity.  $\square$

## N Proofs for Section 8

We use the rules of [28, Appendix C.1], numbered as there, for unitaries  $u, v$  and  $\|W\| \leq 1$ : (W2)  $\chi(uv) \leq \chi(u) + \chi(v)$  and  $\chi(u^*) = \chi(u)$ ; (W4)  $|E(uWu^*) - E(W)| \leq 2\chi(u)$ . Work in the Hilbert–Schmidt space of operators on the Hilbert space  $\mathcal{H}$ , where left multiplication acts on the system and right multiplication on the reference;  $\rho$  has finite rank, so  $\xi = \sqrt{\rho}$  is Hilbert–Schmidt and purifies  $\rho$ , the reference has entropy  $S(\rho)$ , and (W2), (W4) and every step below hold verbatim.

*Proof of Lemma 8.3.* Write  $c_i = \sum_{k \in \mathbb{Z}/3} \omega_3^k P_i^k$  with  $\omega_3 = e^{2\pi i/3}$ , and  $Q_{\mathbf{a}} = P_{i-1}^{a_{i-1}} \dots P_0^{a_0}$ . The  $c_k$  with  $k < i$  commute, so  $\sum_{\mathbf{a}} Q_{\mathbf{a}}^* Q_{\mathbf{a}} = I$ , and  $c_i, x_i$  commute with  $Q_{\mathbf{a}}$ . Measure  $c_0, \dots, c_{N-1}$  one after another on the system, with outcomes  $A_i$ :  $\Pr(A_{<i} = \mathbf{a}, A_i = k) = \|P_i^k Q_{\mathbf{a}} \xi\|_F^2 =: p(\mathbf{a}, k)$ .

1. The reference is untouched, so  $S(\rho) \geq I(A_1 \dots A_N : \text{Ref}) = \sum_i I(A_i : \text{Ref} \mid A_{<i})$ .
2. By  $\|(c-I)Y\|_F^2 = 3 \sum_{k \neq 0} \|P^k Y\|_F^2$  and  $[c_i, Q_{\mathbf{a}}] = 0$ ,  $\Pr(A_i \neq 0) = \frac{1}{3} \sum_{\mathbf{a}} \|Q_{\mathbf{a}}(c_i - I)\xi\|_F^2 = \Delta_i^2/3$ , an equality.
3.  $x_i P_i^{-k} x_i = P_i^k$ ,  $x_i$  commutes with  $Q_{\mathbf{a}}$ , and right multiplication by  $x_i$  is unitary, so  $\sqrt{p(\mathbf{a}, -k)} = \|P_i^k Q_{\mathbf{a}} x_i \xi\|_F$  and  $\sqrt{p(\mathbf{a}, k)} = \|P_i^k Q_{\mathbf{a}} \xi x_i\|_F$ ; hence  $\sum_{\mathbf{a}, k} (\sqrt{p(\mathbf{a}, k)} - \sqrt{p(\mathbf{a}, -k)})^2 \leq \sum_{\mathbf{a}} \|Q_{\mathbf{a}}[x_i, \xi]\|_F^2 = \eta_i^2$ , and the terms  $k = 1$  and  $k = 2$  are equal. The inequality  $H(r_0, r_1, r_2) \geq s - (\sqrt{r_1} - \sqrt{r_2})^2 / \ln 2$ ,  $s = r_1 + r_2$ , of [28, equation (16)] is homogeneous, so it applies to the unnormalized weights given each  $\mathbf{a}$ , and  $H(A_i \mid A_{<i}) \geq \Delta_i^2/3 - \eta_i^2/(2 \ln 2)$ .
4. Let the reference measure  $c_i$  from the right, with outcome  $B_i$ . Left and right multiplications commute, and by  $\|[c, Y]\|_F^2 = 3 \sum_{k \neq l} \|P^k Y P^l\|_F^2$  and  $[c_i, Q_{\mathbf{a}} \xi] = Q_{\mathbf{a}}[c_i, \xi]$ ,  $\Pr(A_i \neq B_i) = \theta_i^2/3$ , an equality.
5. By data processing and Fano's inequality for three values, with the concavity of  $h_2$  over  $\mathbf{a}$ ,  $I(A_i : \text{Ref} \mid A_{<i}) \geq I(A_i : B_i \mid A_{<i}) \geq H(A_i \mid A_{<i}) - h_2(\theta_i^2/3) - \theta_i^2/3$ .

Summing over  $i$  proves the bound. On the flat state of a tensor product of standard representations,  $\Delta_i^2 = 3$  and  $\theta_i = \eta_i = 0$ , so the bound is  $N = S$ .  $\square$

*Proof of Lemma 8.4.*  $U$  agrees with  $U_\pi$  on  $\mathbb{C}^d \otimes K$  and distinct slots occupy distinct matrix positions, so  $T(\rho) = \sum_a (|c_a|^2/d) \rho_a$  with  $\rho_a = \pi(w_a) \rho \pi(w_a)^*$ , the weights summing to 1. Every  $\rho_a$  has the spectrum of  $\rho$ , so  $a = S(\bar{\rho}) - \sum_a (|c_a|^2/d) S(\rho_a) = \sum_a (|c_a|^2/d) D(\rho_a \parallel \bar{\rho})$  with  $\bar{\rho} = T(\rho)$  (Holevo's identity). The anchor slot of a symbol  $w$ , and that of the empty word, have coefficient 1 and hence weight  $1/d$ , so  $D(\rho_w \parallel \bar{\rho}) \leq d a$  and  $D(\rho \parallel \bar{\rho}) \leq d a$ . For states,  $\|\sqrt{\sigma} - \sqrt{\tau}\|_F^2 = 2 - 2 \text{Tr} \sqrt{\sigma} \sqrt{\tau} \leq -2 \ln \text{Tr}(\sqrt{\sigma} \sqrt{\tau}) \leq D(\sigma \parallel \tau)$  in nats, since the Petz–Rényi divergences increase with the order. So  $\|\sqrt{\rho_w} - \sqrt{\bar{\rho}}\|_F$  and  $\|\sqrt{\rho} - \sqrt{\bar{\rho}}\|_F$  are at most  $\sqrt{d a \ln 2}$ , and since  $\pi(w) \xi \pi(w)^* = \sqrt{\rho_w}$  and right multiplication by a unitary preserves the norm,  $\chi(\pi(w)) = \|\sqrt{\rho_w} - \sqrt{\bar{\rho}}\|_F \leq 2\sqrt{d a \ln 2}$ .  $\square$

**Theorem N.1** (Explicit genuine-round constants). *Let  $(U, \rho)$  be a representation round of  $\Phi_H$  with distance at most  $\frac{1}{8}$  and entropy production  $a$ , with any leakage, and let  $N_{\text{ex}}$  be the largest  $N$  for which  $(c_i, x_i)_{i < N}$  satisfy the hypotheses of Lemma 8.3. Then*

$$S(\rho) \geq 0.375 \min\left(N_{\text{ex}}, \frac{1}{60\delta_{\text{cent}}} - \frac{2}{3}\right), \quad \delta_{\text{cent}} = \max(\chi(\pi(\mathbf{a})), \chi(\pi(\alpha))) \leq 16.981\sqrt{a}.$$

For a genuine round  $N_{\text{ex}} = \infty$ , so

$$S(\rho) \geq \frac{a^{-1/2}}{2717} - \frac{1}{4} \quad \text{and} \quad S + n a \geq 0.0097 n^{1/3} - \frac{1}{4} \quad \text{for every } n \geq 1.$$

*Proof of Theorems 8.5 and N.1. Displacement.*  $\gamma$  is a symbol, a prefix of  $(\alpha \alpha \alpha^{-1})^3$ , with an anchor  $z_\gamma$ , and the empty word has an anchor 0. By (F1),  $\langle z_\gamma | \Psi(|z_\gamma\rangle\langle 0|) | 0 \rangle = \text{Tr}(\rho \pi(\gamma))$ , while for  $\Phi_H$  it is  $\tau(\lambda(\gamma)) = 0$ , since  $\gamma \neq 1$  in  $H_3$ . By (F2),  $|\text{Tr} \rho \pi(\gamma)| \leq 2u \leq \frac{1}{4}$ , so  $E(\pi(\gamma))^2 = 2 - 2 \text{Re} \text{Tr}(\rho \pi(\gamma)) \geq \frac{3}{2}$ .

*Transport.*  $c_i = \pi(\mathbf{a})^{3i} \pi(\gamma) \pi(\mathbf{a})^{-3i}$ , so by (W4) and (W2),  $\Delta_i \geq \sqrt{1.5} - 2\chi(\pi(\mathbf{a})^{3i}) \geq \sqrt{1.5} - 6i\delta_{\text{cent}}$ . The words of  $c_i$  and  $x_i$  have lengths  $6i + 4$  and  $6i + 1$  in  $\mathbf{a}^{\pm 1}$  and  $\alpha$ , so by (W2)  $\theta_i \leq (6i + 4)\delta_{\text{cent}}$  and  $\eta_i \leq (6i + 1)\delta_{\text{cent}}$ .

*Copies.* Let  $N = \min(N_{\text{ex}}, \lfloor (0.1/\delta_{\text{cent}} + 2)/6 \rfloor)$ . For  $i < N$ ,  $\theta_i, \eta_i \leq (6N - 2)\delta_{\text{cent}} \leq 0.1$  and  $6i\delta_{\text{cent}} \leq 0.1$ . Lemma 8.3, which holds on any Hilbert space for states of finite rank, applies directly with these  $\Delta_i, \theta_i, \eta_i$ . Each term is at least  $(\sqrt{1.5} - 0.1)^2/3 - 0.01/(2 \ln 2) - h_2(\frac{1}{300}) - \frac{1}{300} = 0.37891 > 0.375$ , so  $S(\rho) \geq 0.375N \geq 0.375 \min(N_{\text{ex}}, 1/(60\delta_{\text{cent}}) - \frac{2}{3})$ .

*Centrality.*  $\mathbf{a}$  and  $\alpha$  are symbols, so Lemma 8.4 gives  $\delta_{\text{cent}} \leq 2\sqrt{104 \ln 2} \sqrt{a} = 16.981\sqrt{a}$ .

*Genuine rounds.* In  $H_3$ ,  $\mathbf{a}^{3i}\gamma\mathbf{a}^{-3i}$  is a three-cycle and  $\mathbf{a}^{3i}\alpha\mathbf{a}^{-3i}$  a transposition on a triple of consecutive points of the line through the first two rays, shifted by  $3i$ , with  $\alpha\gamma\alpha = \gamma^{-1}$  [27, Section 4]; triples for different  $i$  are disjoint. So the relations of Lemma 8.3 hold in every representation of  $H_3$ ,  $N_{\text{ex}} = \infty$ , and  $S \geq 0.375(a^{-1/2}/(60 \cdot 16.981) - \frac{2}{3}) \geq a^{-1/2}/2717 - \frac{1}{4}$ . Finally  $\min_{x>0}(Ax^{-1/2} + nx) = (2^{1/3} + 2^{-2/3})A^{2/3}n^{1/3}$ , which is at least  $0.0097n^{1/3}$  for  $A = 1/2717$ .  $\square$

**Theorem N.2** (Explicit genuine-service constants). *For  $n \geq n_0$ , every genuine service of  $\Phi_H$  with error  $\epsilon \leq 1/16$  and  $J \leq n \log 26/2$  has  $Q \geq n^{1/3}/385$  if its discard is zero, and  $Q \geq n^{1/3}/420$  if its discard is at most  $1/n^2$ . Consequently the lower constant in Theorem 8.7 may be taken as  $1/420$ . The nongenuine-round threshold in Theorem 8.12(c) may be taken as  $S + na < 0.0097n^{1/3} - 1/4$ .*

*Proof of Theorems 8.6 and N.2.* If  $U$  is genuine on  $V$  and  $\rho$  is supported in  $V$ , the round  $(U, \rho)$  has the channel of the representation round  $(U_\pi, J\rho J^*)$ , since the partial trace over the memory ignores the isometry  $R$ ;  $T_U(\rho) = RT_\pi(J\rho J^*)R^*$  has the spectrum of  $T_\pi(J\rho J^*)$ , and  $S(J\rho J^*) = S(\rho)$ . So Theorem N.1 applies to  $(U, \rho)$ .

*Discard 0.* By Theorem J.1 some round  $t$  of the service has distance at most  $\frac{1}{16}$ ,  $S(\rho_t) \leq Q$  and  $na_t \leq 6.4Q + 13.52 + 0.134 \log n$ , and the conditioned memory state  $\rho_t$  is supported in  $V_t$ , since conditioning only shrinks supports. Theorem N.1 gives  $Q + \frac{1}{4} \geq a_t^{-1/2}/2717$ . If  $Q \geq 14 + 0.14 \log n$ , then  $na_t \leq 7.4Q$ , so  $(Q + \frac{1}{4})Q^{1/2} \geq n^{1/2}/(2717\sqrt{7.4}) \geq n^{1/2}/7392$ , and with  $Q + \frac{1}{4} \leq (1 + \frac{1}{56})Q$ ,  $Q \geq n^{1/3}/384$ . If  $Q < 14 + 0.14 \log n$ , then  $a_t = O(\log n/n)$  and Theorem N.1 gives  $Q + \frac{1}{4} \geq c(n/\log n)^{1/2}$ , which is impossible for  $n \geq n_0$ .

*Discard at most  $1/n^2$ .* Let  $t$  be the round of Theorem J.1 and  $\rho = \rho_t$ . The test observer's memory state at the start of round  $t$ , conditioned on passing, an event of probability at least  $1 - \epsilon$ , puts mass  $\delta' \leq \delta$  outside  $V := V_t$ . Let  $P$  project onto  $V$  and  $\hat{\rho} = P\rho P / \text{Tr}(P\rho)$ . If  $\psi$  purifies  $\rho$ , then  $(P \otimes I)\psi/\sqrt{1 - \delta'}$  purifies  $\hat{\rho}$  and has overlap  $\sqrt{1 - \delta'}$  with  $\psi$ , so  $\|\rho - \hat{\rho}\|_1 \leq 2\sqrt{\delta'} \leq 2\sqrt{\delta}$ , by the pure-state trace-distance formula and contraction under partial trace. The round  $(U, \hat{\rho})$  is genuine, and: its distance is at most  $\epsilon + \frac{1}{2}\|\rho - \hat{\rho}\|_1 \leq \frac{1}{16} + \frac{1}{n} \leq \frac{1}{8}$  for  $n \geq 16$ , since the channel is linear in the bath;  $T$  is a channel on the  $Q$ -qubit memory, so  $\frac{1}{2}\|T(\rho) - T(\hat{\rho})\|_1 \leq \sqrt{\delta} \leq \frac{1}{2}$ , and Audenaert's sharp form of Fannes' inequality [4, Theorem 1] gives  $|S(T(\rho)) - S(T(\hat{\rho}))| \leq \sqrt{\delta}Q + h_2(\sqrt{\delta})$ , and likewise for  $S(\rho) - S(\hat{\rho})$ , so  $a(\hat{\rho}) \leq a(\rho) + 2\sqrt{\delta}Q + 2h_2(\sqrt{\delta})$ ; and  $S(\hat{\rho}) \leq Q$ . With  $\delta \leq 1/n^2$  and  $nh_2(1/n) \leq \log n + \log e$ ,  $na(\hat{\rho}) \leq 8.4Q + 16.41 + 2.134 \log n$ . If  $Q \geq 16.41 + 2.134 \log n$  then  $na(\hat{\rho}) \leq 9.4Q$ , and Theorem N.1 gives  $(Q + \frac{1}{4})Q^{1/2} \geq n^{1/2}/(2717\sqrt{9.4}) \geq n^{1/2}/8331$ , so  $Q \geq n^{1/3}/415.2 \geq n^{1/3}/420$ . Otherwise  $a(\hat{\rho}) = O(\log n/n)$  and Theorem N.1 gives a contradiction for  $n \geq n_0$ .  $\square$

With discard of order  $\epsilon^2/n$  the term  $2\sqrt{\delta}Q$  is of order  $\epsilon Q/\sqrt{n}$ , which dominates the budget  $Q/n$ , and the same argument gives only  $Q \gtrsim n^{1/6}$ .

*Proof of Theorem 8.7.* The lower bound is Theorem 8.6. For the upper bound take the service of Theorem H.18 with  $q = 3$ ; for  $n \geq n_0(\epsilon)$  its memory is  $O(n^{1/3}(\log n)^{2/3}(\log(n/\epsilon))^{1/3})$ . Its round unitary is  $U_t = R''W_cR'$ , where  $R'$  (class permutation and twirls) and  $R''$  (re-fold with its completion, twirls) act on the memory alone and  $W_c$  is the round.  $R'$  maps the window subspace  $V$  (window units, slots and ancilla) onto itself, and on  $\mathbb{C}^d \otimes V$  the round is the gadget unitary of the right regular representation  $\pi$  of  $H_3$ , tensored with the identity on

slots and ancilla. So  $U_t$  is genuine on  $V$ , with  $J$  the identification of  $V$  with a subspace of  $\ell^2(H_3) \otimes \text{slots} \otimes \text{ancilla}$  composed with  $R'$ , and  $R = R''$  on the images of window points under slot words, which lie in  $\Omega_W$ . By Theorem H.15, for every observer the memory has mass at most  $3/(4n^2) \leq (1 - \epsilon)/n^2$  outside  $V$  at the start of every round. So the service is genuine with discard at most  $1/n^2$ .  $\square$

*Proof of Corollary 8.8.* The labels of  $\Phi_H$  generate  $H_3$ , which is elementary amenable. It is not residually finite, since every finite-dimensional representation kills its three-cycles (Remark 8.1), so it is not virtually nilpotent, and by Chou's theorem [17, Theorem 3.2] it has exponential growth. The regular action is free, so the words  $w_x$  of Proposition H.13 give distinct points, and the proposition applies to every finitely supported bath.  $\square$

*Proof of Proposition 8.9.* By Theorem N.1, a genuine round at distance at most  $\frac{1}{8}$  has  $1 + S + na \geq 0.0097n^{1/3} + \frac{3}{4}$ . For  $n \geq n_1$  the service of Theorem H.18 with  $q = 3$  has memory  $O(n^{1/3}(\log n)^{2/3}L^{1/3} + L \log L) + O(\log(n/\epsilon))$  with  $L = O(\log(n/\epsilon))$ , which is at most  $K \log^2(n/\epsilon) n^{1/3}$ , hence at most  $K' \log^2(n/\epsilon)(1 + S + na)$ . For  $2 \leq n < n_1$ , Proposition 2.7 gives  $Q^*(n, \epsilon) \leq n_1 \lceil \log 26 \rceil$ .  $\square$

*Proof of Proposition 8.10.* By [27, Proposition 4.14], for every  $u_0 > 0$  there is  $C(u_0)$  with  $\log D_{\Phi_H}^{\text{ent}}(u_0, v) \leq C(u_0)v^{-1/3}$  for  $0 < v \leq 1$ , where  $D^{\text{ent}}$  is the least bath dimension of a round with distance at most  $u_0$  and  $\ell + a \leq v$ ; the rounds of its proof have flat baths, so  $a = 0$  and  $\ell \leq v$ , and  $S$  is at most the logarithm of the bath dimension. With  $u_0 = \epsilon$  and  $v = \frac{32}{15}\epsilon/n$  they are admissible in  $L_{\Phi_H}(n, \epsilon)$ , so  $L_{\Phi_H}(n, \epsilon) \leq C(\epsilon)(15n/(32\epsilon))^{1/3}$ , and with  $u_0 = \frac{1}{16}$ ,  $v = \frac{2}{15n}$  the same gives  $L_{\Phi_H}^*(n) \leq Cn^{1/3}$ .  $\square$

**The single budget (Remark 8.11).** At the rank rate, [27, Theorem 5.2] with Proposition 7.2 reads a round with  $\ell + a \leq t_n \leq \frac{16}{15}(3Q + 8 + \log n)/n$ . A lower bound  $Q^* \geq Q_0$  by this method requires that the least bath entropy of a round with distance  $\epsilon/(1 - \epsilon)$  and  $\ell + a \leq t_n(Q)$  exceed  $Q$  for all  $Q < Q_0$ . By the proof of Proposition 8.10 that bath entropy is at most  $C t_n(Q)^{-1/3}$ , so the requirement fails as soon as  $C(\frac{16}{15}(3Q + 8 + \log n)/n)^{-1/3} \leq Q$ , which holds for  $Q \geq C'n^{1/4}$ .

*Proof of Theorem 8.12.* (a) Let  $L_{\Phi_H}^*(n) \geq n^{1/3}/p(n)$ . As in the proof of Proposition 8.9,  $Q_{\Phi_H}^*(n, \epsilon) \leq K \log^2(n/\epsilon) n^{1/3}$  for all  $n \geq 2$  (for small  $n$  by Proposition 2.7), so  $Q_{\Phi_H}^*(n, \epsilon) \leq Kp(n) \log^2(n/\epsilon)(1 + L_{\Phi_H}^*(n))$ , the upper half of the law; the lower half is Theorem J.1, and it gives  $Q_{\Phi_H}^*(n, \epsilon) \geq (n^{1/3}/p(n) - 13.6 - 0.14 \log n)/7.4$ . So  $Q_{\Phi_H}^*(n, \epsilon) = n^{1/3} \cdot \text{polylog}(n/\epsilon)$ .

(b) If the law holds,  $L_{\Phi_H}^*(n) \geq Q_{\Phi_H}^*(n, \epsilon)/(K(\log(n/\epsilon))^\kappa) - 1$ , and under Conjecture 8.13 the right side is  $n^{1/3-o(1)}$ .

(c) This is Theorem N.1.  $\square$

## N.1 Constant-field geometry and sector-isotropic sources

We give a direct proof of Corollary O.2. The geometric input is the existing construction of [26, Section 6.2]:  $\gamma$  is exactly a three-cycle at every clock; the  $r_4$  holonomy is a product of disjoint rank-one phase rotations with angles  $F_R = s_R \pi / |R|$ . There are  $M$  regions, each of area between  $m_M = 1 + (M - 1)(M - 2)/2$  and  $M^2$ . These are model identities, not conclusions for an arbitrary round.

**Sector characters force entropy.** Initially suppose the source is clock diagonal. Set  $v_j = j(L-j)/[L(L-1)]$  and  $V_j = \sum_l v_{jl}$ . The normalized three-cycle character in degree  $j$  is

$$\chi_j = 1 - 3v_j.$$

Indeed its eigenvalues are 1 repeated  $L-2$  times and  $\omega, \omega^2$ ; the generating polynomial is  $(1+t)^{L-2}(1-t+t^2) = (1+t)^L - 3t(1+t)^{L-2}$ . These characters are nonnegative for  $L = 4M+4$ . Conditional sector flatness removes conjugations of the cycle and gives  $\chi_\gamma = \mathbb{E}_p \prod_l (1 - 3v_{jl})$ . The identity and  $\gamma$  singleton anchors imply  $0 \leq \chi_\gamma \leq 2u \leq 1/8$ . In the rest of the lower proof it suffices to assume  $\chi_\gamma \leq 1/4$ . The product is at least  $1 - 3V$ , so  $\mathbb{E}V \geq 1/4$ ; on  $V < 1/12$  it is greater than  $3/4$ , giving  $\Pr(V \geq 1/12) \geq 2/3$ . For  $m = \min(j, L-j)$ ,  $\binom{L}{j} \geq 2^m$  and  $Lv_j \leq m$ . The first inequality follows termwise from  $(L-i)/(m-i) \geq 2$  for  $0 \leq i < m$ . Consequently, with  $D_j = \prod_l \binom{L}{j_l}$ ,

$$S = H(p) + \mathbb{E} \log_2 D_j \geq L\mathbb{E}V \geq L/4 = M+1. \quad (29)$$

Source entropy, rather than logarithmic source rank, pays this bound.

**Clock bias costs actual production.** Every literal slot translates the clock and applies a degree-preserving one-particle unitary. It preserves each  $\bigotimes_l \tau_{jl}$ . The actual Choi bath map is therefore classical clock convolution,  $Kp = \sum_s w_s p_s$ , leaving degrees fixed. Identity and both coordinate steps have weights at least  $1/d$  from their singleton slots, with  $d = 104$ . Thus

$$a = H(Kp) - H(p) = \sum_s w_s D_2(p_s \| Kp).$$

The unchanged average  $\log D_j$  cancels exactly. Root relative entropy and the triangle inequality bound each coordinate root displacement by  $4d(\ln 2)a$ . The  $M$ -torus Fourier Poincaré constant is  $4\sin^2(\pi/M) \geq 16/M^2$ . Applied separately to each degree block and then summed, it gives

$$\sum_{c, \mathbf{j}} (\sqrt{p(c, \mathbf{j})} - \text{mean}_c \sqrt{p(c, \mathbf{j})})^2 \leq (d \ln 2) M^2 a / 2.$$

Put  $w(\mathbf{j}) = \sum_c p(c, \mathbf{j})$  and  $q(c, \mathbf{j}) = w(\mathbf{j})/M^2$ . In a nonzero degree block let  $x = M \text{mean}_c \sqrt{p}/\sqrt{w} \in [0, 1]$ . Its variance is  $w(1-x^2)$ , whereas its squared root distance from the uniform clock law is  $2w(1-x) \leq 2w(1-x^2)$ . Cauchy–Schwarz therefore proves

$$\text{TV}(p, q) \leq \|\sqrt{p} - \sqrt{q}\|_F \leq M \sqrt{d(\ln 2)a}. \quad (30)$$

This argument retains all clock–degree and flavour correlations.

**A protected region forces leakage.** Let  $B_0 = \{0, \dots, \lfloor M/4 \rfloor\}^2$ . It has at least  $M^2/16$  clocks and is contained in all  $M$  regions. To check containment, the  $M-1$  bulk regions indexed by  $5 \leq t \leq M+3$  have source  $(M-1, M-1)$ , the column  $(M-1, j)$  for  $1 \leq j \leq M-2$ , and bulk points  $0 \leq i \leq M-3$ ,  $0 \leq j \leq M-2$ ,  $i+j \leq t+M-8$ . Their areas are  $(M-1)^2 - h(h+1)/2$ ,  $h = M+3-t$ . The last region contains  $i, j \geq 0$ ,  $i+j \leq M-3$ , as well as  $(0, M-2)$ . The inequality  $2\lfloor M/4 \rfloor \leq M-3$  proves the claim. At a clock in  $B_0$ ,  $\sum_R F_R^2 \geq \pi^2/M^3$ . There are at most  $M < L/4$  nonzero phase modes and  $\sum_R |F_R| \leq M\pi/m_M \leq \pi$ .

For a uniformly sampled degree- $j$  subset of the  $L$  modes set  $X = \sum_R F_R I_R$ . Sampling without replacement gives

$$\text{Var } X = v_j \left[ \sum_R F_R^2 - (\sum_R F_R)^2 / L \right] \geq \frac{3}{4} v_j \sum_R F_R^2.$$

For independent copies,  $|X - X'| \leq \pi$  and  $1 - |\mathbb{E}e^{iX}|^2 = \mathbb{E}[1 - \cos(X - X')] \geq 4 \text{Var } X / \pi^2$ . Hence  $|\mathbb{E}e^{iX}| \leq \exp[-3v_j \sum_R F_R^2 / (2\pi^2)]$ . Multiplication across conditionally flat flavours shows that on  $B_0 \cap \{V \geq 1/12\}$  the sector relator squared defect obeys

$$e_{c,j}^2 \geq 2(1 - e^{-1/(8M^3)}) \geq 1/(8M^3).$$

No bound on the total phase across flavours is needed.

Project the physical round onto the canonical coefficients and let  $\Lambda$  be its bath leakage operator as in Proposition F.3. Its identity singleton and closing  $r_4$  triangle have the same target label and squared weights  $1, 1/2$ . A common projected endpoint  $z$  must therefore pay

$$\inf_z \{\|\psi - z\|^2 + \frac{1}{2}\|\pi(r_4)\psi - z\|^2\} = \frac{1}{3}\|(\pi(r_4) - I)\psi\|^2.$$

Every other slot contributes nonnegative error, so  $\Lambda \geq (\pi(r_4) - I)^*(\pi(r_4) - I)/3$ . Leakage on the protected event is at least  $1/(24dM^3)$ . Its  $q$ -mass is at least  $(1/16)(2/3) = 1/24$ . Set  $A = 1/(2304d \ln 2)$ ,  $B = 1/(1152d)$ . Equation (30) gives

$$a \leq A/M^2 \implies \ell \geq B/M^3, \quad (31)$$

because its total-variation error is at most  $1/48$ . At leakage  $\ell \leq c/n$ , either  $M \geq (Bn/c)^{1/3}$  or  $S + na \geq M + nA/M^2 \geq 3A^{1/3}n^{1/3}/2^{2/3}$ . This proves the diagonal-source lower with constant  $\min\{(B/c)^{1/3}, 3A^{1/3}/2^{2/3}\}$ .

**Coherent clocks and flags.** Clock pinching commutes with the physical bath map: each slot has one fixed clock translation. Data processing of  $D_2(\rho \|\Delta_c \rho) = S(\Delta_c \rho) - S(\rho)$  gives  $a(\Delta_c \rho) \leq a(\rho)$ . Every literal slot assigned a target label has that label's clock translation. Its projected coefficient and residual have the same translation; each residual square preserves the clock. Thus  $\Lambda$  is clock block diagonal, and pinching preserves leakage. The  $\gamma$  character has zero translation and is preserved as well. To retain the entropy floor directly, coherently copy the clock to an analytical register. Its marginals have entropies  $S(\Delta_c \rho)$  and  $H(p_c)$ , so Araki–Lieb gives  $S(\rho) \geq \sum_c p_c S(\rho_c)$ , where  $\rho_c$  is the normalized clock-diagonal block. Each  $\rho_c$  is an orthogonal degree-sector mixture, hence

$$S(\rho) \geq \sum_{c,j} p(c, j) \log_2 D_j \geq L\mathbb{E}V \geq L/4 = M + 1.$$

There is no clock-entropy subtraction or free pinching assumption. If  $a(\rho) \leq A/M^2$ , the pinched source satisfies (31); otherwise optimize  $M + nA/M^2$ . This proves the coherent-source lower with the same constant as the clock-diagonal case.

For a finite block-diagonal round with coherent flag source, let  $b_i = \text{Tr } P_i \rho$  and  $\rho_i = P_i \rho P_i / b_i$ . Channel and leakage depend only on diagonal flag blocks. All  $\chi_i \geq 0$ , and the overall anchor test gives  $\sum_i b_i \chi_i \leq 1/8$ . Components with  $\chi_i \leq 1/4$  have mass at least  $1/2$ ; those with  $\ell_i \leq 8/(15n)$  have mass at least  $3/4$  by Markov. Their intersection has mass at least  $1/4$ . Flag pinching commutes with the bath map, so data processing gives  $a(\rho) \geq \sum_i b_i a(\rho_i)$ . Coherently copying the flag is an analytical isometry. Its marginals have entropies  $H(b) + \sum_i b_i S(\rho_i)$  and  $H(b)$ ; Araki–Lieb gives  $S(\rho) \geq \sum_i b_i S(\rho_i)$ . Apply the coherent clock bound to the good intersection. These comparisons charge the original source; no physical pinching, purifier or flag supply is introduced.

**Replacement and upper bound.** The triangle-good corner families of [27, Proposition 4.10] have source entropy  $O(m)$ , room  $O(m^{-2})$  and scalar gap at most  $1/2$ . Their actual gain is bounded by that room. Fixed untwisted tensor amplification gives target error at

most  $1/16$ , zero leakage and cost  $O(m + nm^{-2})$ . Choose  $m \asymp n^{1/3}$ ; small  $n$  use the pure exact baseline. The source lies entirely in the new round's exact leak kernel. This different round need not stay close to the smoothed source or belong to its geometry. Finally the fully flat smoothed models themselves, with fixed flavour count selected for coarse error  $1/16$ , have  $a = 0$ , leakage  $O(M^{-3})$  and entropy  $O(M)$  [27, Proposition 4.14]. Taking  $M \asymp n^{1/3}$  proves the class's matching static upper. Neither lower is an all-family operational memory statement.

## N.2 Collective orbital invariance

We prove Theorem O.1, retaining precisely the geometry above but dropping product-sector flatness. For an invariant internal state  $\eta$ , let  $Q$  project onto the nontrivial part of the restriction to  $SU(2)$  acting on a two-dimensional one-particle plane, and put  $p_\eta = \text{Tr } \eta Q$ . All planes are conjugate in  $U(L)$ , so this number does not depend on the plane.

**Entropy without flattening multiplicities.** The three-cycle is a product of two  $SU(2)$  quarter turns on planes  $(1, 2)$  and  $(2, 3)$ : each sends its first basis vector to the second and the second to minus the first. The product sends  $e_1$  to  $e_2$ ,  $e_2$  to  $e_3$  and  $e_3$  to  $e_1$ . For either turn,  $(\Pi(A) - I)^*(\Pi(A) - I) \leq 4Q$ . The Hilbert–Schmidt triangle inequality and invariance give

$$2(1 - \text{Re Tr } \eta \Pi(\gamma)) = \|(\Pi(\gamma) - I)\sqrt{\eta}\|_F^2 \leq 16p_\eta.$$

Choose  $t = \lfloor L/2 \rfloor$  disjoint planes. The restriction to their commuting  $SU(2)$  groups decomposes into tensor products of irreducibles, with arbitrary multiplicity spaces. Schur's lemma makes an invariant state flat only on the irreducible factors:

$$\eta = \bigoplus_{\mathbf{m}} w_{\mathbf{m}} \left[ \bigotimes_{i=1}^t \frac{I_{m_i+1}}{m_i+1} \otimes \sigma_{\mathbf{m}} \right].$$

Here the multiplicity states  $\sigma_{\mathbf{m}}$  need not be flat or separable. Every nontrivial factor costs at least one bit, hence  $S(\eta) \geq \sum_i \Pr(m_i > 0) = tp_\eta$ . For completeness, the finite-dimensional unitary  $SU(2)$  irreducibles have weights  $m, m-2, \dots, -m$  and dimension  $m+1$ . With Lie algebra generators  $E, F, H$ , a maximal-weight vector obeys  $EF^j v = j(m-j+1)F^{j-1}v$ . Positivity of the unitary norm and termination force an integer  $m \geq 0$  and a chain  $j = 0, \dots, m$ ; invariant orthogonal complements complete the decomposition.

Set  $\bar{\eta} = \sum_c q_c \rho_c$  and  $\chi = \text{Tr } \bar{\eta} \Pi(\gamma)$ . The clock blocks of  $\gamma$  are conjugate three-cycles, so invariance makes this their common trace test. It is real, and the anchors give  $|\chi| \leq 2u \leq 1/8$ . The direct conditional-entropy argument above gives

$$S(\rho) \geq \sum_c q_c S(\rho_c) \geq tp_{\bar{\eta}}.$$

Even the weaker  $\chi \leq 1/4$  therefore implies  $p_{\bar{\eta}} \geq 3/32$  and  $S(\rho) \geq bM$ ,  $b = 3/16$ .

**An operator-valued clock law.** Put  $\bar{\rho} = \Delta_c \rho = \bigoplus_c R_c$ ,  $R_c = q_c \rho_c$ . Invariance makes every internal literal unitary act trivially on these blocks; the bath map is clock convolution  $K\bar{\rho} = \sum_s w_s \bar{\rho}_s$ . The translated states have the same spectrum, so the Holevo identity gives

$$\bar{a} = S(K\bar{\rho}) - S(\bar{\rho}) = \sum_s w_s D_2(\bar{\rho}_s \| K\bar{\rho}).$$

The inequality  $\|\sqrt{X} - \sqrt{Y}\|_F^2 \leq D_{\text{nat}}(X \| Y)$  follows by diagonalizing both states: their overlap-weighted eigenvalue arrays give exactly the quantum relative entropy and affinity, and

Jensen gives  $D_{\text{nat}} \geq -2 \ln \text{Tr} \sqrt{X} \sqrt{Y} \geq 2(1 - \text{Tr} \sqrt{X} \sqrt{Y})$ . Identity and coordinate weights at least  $1/d$  bound each root shift energy by  $4d(\ln 2)\bar{a}$ . Apply the torus Poincaré inequality to the Hilbert-space field  $X_c = \sqrt{R_c}$ , not to simultaneously diagonalized densities. With  $A_0 = M^{-1} \sum_c X_c$ ,

$$V = \sum_c \|X_c - M^{-2} \sum_b X_b\|_F^2 = 1 - \text{Tr} A_0^2 \leq \frac{1}{2}(d \ln 2) M^2 \bar{a}.$$

Operator Jensen gives  $A_0^2 \leq \sum_c R_c = \bar{\eta}$ ; square-root monotonicity gives  $A_0 \leq \sqrt{\bar{\eta}}$ . Thus  $\text{Tr} A_0 \sqrt{\bar{\eta}} \geq \text{Tr} A_0^2$ , without commutation. For  $q = (I/M^2) \otimes \bar{\eta}$  the root distance is at most  $2V$ . The square-root trace-distance inequality yields

$$\frac{1}{2} \|\bar{\rho} - q\|_1 \leq M \sqrt{d(\ln 2)\bar{a}} \leq M \sqrt{d(\ln 2)a}. \quad (32)$$

The final inequality, and preservation of leakage and  $\chi$ , follow from clock-pinching covariance and data processing exactly as above. No internal twirl or commuting-density assumption is used.

**A spin-uniform flux estimate.** For a nontrivial  $SU(2)$  irrep,  $m \geq 1$ , and  $|\theta| \leq \pi/4$ , its normalized character satisfies

$$\left| \frac{\sin((m+1)\theta)}{(m+1)\sin\theta} \right| \leq e^{-\theta^2/32}. \quad (33)$$

To verify this uniformly, set  $N = m + 1$ . If  $N|\theta| \leq 1$ , the character is the mean of  $e^{iX}$  for weights  $X = (N-1-2j)\theta$ . Its variance is  $(N^2-1)\theta^2/3 \geq \theta^2$ , and  $|X - X'| \leq 2 < \pi$ . The cosine inequality gives the stronger modulus bound  $e^{-2\theta^2/\pi^2}$ . If  $N|\theta| > 1$ , then  $|\sin x|/x \leq \sin 1$  for  $x \geq 1$ , by monotonicity up to  $\pi$  and  $1/x$  thereafter. Also  $\theta/\sin\theta < 9/8$  and  $\sin 1 \leq 101/120$ . Hence the modulus is at most  $909/960$ , smaller than  $e^{-\theta^2/32} \geq 1 - \pi^2/512$ . The value at zero is understood by continuity.

Let a one-particle holonomy  $W$  have angles  $F_j$  on at most  $M$  disjoint phase modes. Pair each with an unused mode, and let  $C$  swap the partners. Then  $h = WCW^*C^*$  is a product of disjoint  $SU(2)$  rotations of angles  $F_j$ . Restriction of  $\eta$  to their product gives a weighted sum of products of characters. Using (33), the triangle inequality and  $2(1 - e^{-x}) \geq x$  for  $0 \leq x \leq 1$ , when  $\sum_j F_j^2 \leq 1$ ,

$$\|(\Pi(h) - I)\sqrt{\eta}\|_F^2 \geq p_\eta \sum_j F_j^2/32.$$

Invariance under  $C$  and  $W$  bounds the left root norm by  $2\|(\Pi(W) - I)\sqrt{\eta}\|_F$ . Consequently

$$\|(\Pi(W) - I)\sqrt{\eta}\|_F^2 \geq p_\eta \sum_j F_j^2/128. \quad (34)$$

This estimate is independent of all spin, flavour and multiplicity dimensions; it does not assume nonnegative three-cycle characters.

**Optimization and coherent flags.** The existing fields have  $\sum_j F_j^2 < 1$  and  $|F_j| < \pi/4$  for  $M \geq 12$ . On  $B_0$ ,  $\sum_j F_j^2 \geq \pi^2/M^3$ . The same two-slot leakage lower and (34) give, for any fixed plane projection  $Q$ ,

$$\ell \geq \frac{\pi^2}{384dM^3} \text{Tr} \bar{\rho}(P_{B_0} \otimes Q).$$

Invariance makes  $Q$ 's conditional expectation plane-independent. For  $q$  in (32), this probability is at least  $p_{\bar{q}}/16 \geq 3/512$ . Set  $A = 9/(1048576d \ln 2)$  and  $B = \pi^2/(131072d)$ . If  $a \leq A/M^2$ , the comparison changes it by at most  $3/1024$ , so  $\ell \geq B/M^3$ . With  $\ell \leq c/n$  and  $S \geq bM$ , optimization gives

$$S + na \geq c_H(c)n^{1/3}, \quad c_H(c) = \min \left\{ b(B/c)^{1/3}, \frac{3(b^2 A)^{1/3}}{2^{2/3}} \right\}.$$

For coherent model flags the conditional entropy/gain inequalities above still hold. Here each  $\chi_i \in [-1/2, 1]$ , because the three-cycle has order three and is conjugate to its inverse. The overall anchor bound  $\sum_i b_i \chi_i \leq 1/8$  gives mass at least  $1/6$ , not  $1/2$ , on  $\chi_i \leq 1/4$ . Markov gives mass at least  $11/12$  on  $\ell_i \leq 8/(5n)$ ; the intersection has mass at least  $1/12$ . The lower is therefore  $c_H(8/5)n^{1/3}/12$ . The fully flat diffuse upper and different exact corner replacement already proved above finish the theorem.

**The lower bound of Remark 8.14.** By Theorem 7.1 with  $\lambda = 2$  and Proposition 7.2, some round of a rank-rate service with  $\epsilon \leq \frac{1}{16}$  has distance at most  $\frac{1}{16}$  and  $\ell + a \leq \frac{32}{15}(3Q + 7 + \log n)/n$ . The one-round theorem [27, Theorem 5.5] gives  $Q + \frac{1}{8} \geq S(\rho) + \frac{1}{8} \geq c_*(\ell + a)^{-\beta_H}$ . If  $Q \geq 7 + \log n$ , then  $\ell + a \leq \frac{128}{15}Q/n$  and  $Q^{1+\beta_H} \geq c n^{\beta_H}$ . If  $Q < 7 + \log n$ , then  $Q + \frac{1}{8} \geq c_*(15n/(128(7 + \log n)))^{\beta_H}$ , which is impossible for large  $n$ . So  $Q_{\Phi_H}^*(n, \epsilon) \geq c n^{\beta_H/(1+\beta_H)}$  for  $n \geq n_0$ .

## O Source conditions in the Houghton smoothed family

The main text's cube-root service theorem concerns genuine group devices. This appendix records the separate static results for sources in the fixed smoothed geometry. Their full proofs are in Appendices N, P, Q and R.

### O.1 The smoothed geometry with symmetric sources

Genuine actions are not the only class for which the static exponent can be fixed. Consider the existing constant-field smoothed corner models of [27, Section 4.2], with an  $M \times M$  clock and  $k$  exterior-algebra flavours on  $L = 4M + 4$  one-particle modes. Write  $\Pi(W) = (\bigwedge W)^{\otimes k}$  for the collective action of  $U(L)$  on all flavours. This symmetry is a source hypothesis, not a twirl that may be imposed without cost.

**Theorem O.1** (Collectively invariant smoothed sources). *Let  $M \geq 12$ ,  $k \geq 1$ . Suppose every normalized clock-diagonal source block  $\rho_c$  commutes with  $\Pi(W)$  for every  $W \in U(L)$ . Flavours and multiplicity spaces may be arbitrarily entangled; off-diagonal clock blocks are arbitrary. Finite direct sums of these models with arbitrary coherence between flags are also allowed. For the resulting round,  $u \leq 1/16$  and  $\ell \leq 2/(15n)$  imply*

$$S(\rho) + na(\rho) \geq c_H n^{1/3}.$$

*A different zero-leakage exact-support round has target error at most  $1/16$  and cost at most  $c_H n^{1/3}$ . The optimized leaky cost in this class is  $\Theta(n^{1/3})$ , and concentrated replacement costs at most a constant factor.*

The number-sector sources give a useful concrete subclass. Write  $\tau_j = I_{\Lambda^j(\mathbb{C}^L)}/\binom{L}{j}$ .

**Corollary O.2** (Sector-isotropic smoothed sources). *Let  $M \geq 12$ ,  $k \geq 1$  and let a source's clock pinching have the form*

$$\Delta_c \rho = \sum_{c, \mathbf{j}} p(c, \mathbf{j}) |c\rangle\langle c| \otimes \bigotimes_{l=1}^k \tau_{j_l}.$$

*The law  $p$  may correlate clocks and all flavour degrees; off-diagonal clock blocks and finite model-flag coherence are arbitrary. These sources satisfy Theorem O.1.*

This is a static class comparison, not a lower bound on arbitrary services. Entropy and production are measured before pinching. Arbitrary noninvariant internal states, including diagonal ones, and different round geometries remain unexcluded. The arbitrary-field flux theorem for the fully flat fixed-region model already appears in [26, Section 6.2]; this theorem concerns the source extension of its constant-field geometry. Appendix N gives both the concrete sector proof and the symmetry extension.

Full orbital invariance can be weakened, and the exponent is stable under a small trace-distance perturbation without any lower bound on source eigenvalues. Every one-particle link in this geometry fixes  $e_+ = (1, \dots, 1)/\sqrt{L}$ . Let  $G_0 = U(e_+^\perp)$ , acting trivially on  $e_+$ , and let  $\mathcal{E}_0$  twirl each clock-diagonal block over its collective action. Put

$$\delta_{\text{orb}}(\rho) = \frac{1}{2} \|\Delta_c \rho - \mathcal{E}_0(\Delta_c \rho)\|_1.$$

Clock coherence is not constrained by this definition.

**Theorem O.3** (Trace-stable symmetric-source exponent). *In the same constant-field model, let  $M \geq 48$  and  $k \geq 1$ . For every source, possibly singular, the conditions*

$$u \leq 1/16, \quad \ell \leq 2/(15n), \quad \delta_{\text{orb}}(\rho) \leq \frac{\pi^2}{65536 \cdot 288}$$

*imply  $S(\rho) + na(\rho) \geq 4 \cdot 10^{-6} n^{1/3}$ .*

Appendix P proves the entropy transfer and the capped flux estimate needed here. The smaller group leaves an inert mode in which pure-state coherence is allowed, so the class is genuinely larger than full  $U(L)$  invariance. The bound concerns this fixed geometry, not arbitrary rounds. In particular low production has not been shown to imply the required proximity uniformly as  $M$  grows.

## O.2 Literal constant-field sources without imposed symmetry

For the literal rational constant-field round, source entropy and leakage can also be bounded without imposing orbital symmetry. Every source below may be singular, coherent across clocks and entangled across flavours. Entropy production is always the actual gain  $a = S(T\rho) - S(\rho)$  of the physical round, with  $T\rho = \sum_s w_s V_s \rho V_s^*$ . Proposition Q.1 compares this gain with the Frobenius root energy of the actual slots.

The entropy estimate is uniform in flavour count:

$$S + 1 \geq c \min\{M, a^{-1/2}\}, \quad S + na + 1 \geq c \min\{M, n^{1/3}\}.$$

It follows from exact chains of interior adjacent transpositions (Theorem Q.3). The estimate saturates at the clock size; a leakage lower bound must also force that size to grow. The available conclusions are summarized in Table 3.

The all-flavour bound uses a rational power  $O(M^2)$  to expose a fixed local  $SU(4)$  group. With moderately many flavours, many disjoint blocks and a small-angle contraction reduce

| Source or flavour condition                                               | Bound in the literal family                                              |
|---------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Arbitrary $k \geq 1$                                                      | $M^2a + M^4\ell \geq c_0$ ; cost at least $c_K n^{1/4}$<br>(Theorem Q.4) |
| $k \leq c_f \sqrt{M}$                                                     | $M^2a + M^3\ell \geq c_0$ ; cost at least $c_K n^{1/3}$<br>(Theorem R.1) |
| Actual $a = 0$ , arbitrary $k$                                            | $\ell \geq cM^{-3}$ , $S + 1 \geq cM$ (Corollary Q.5)                    |
| Fixed trace neighbourhood of the actual stationary algebra, arbitrary $k$ | $\ell \geq cM^{-3}$ ; cost at least $c_K n^{1/3}$ (Corollary Q.5)        |

Table 3: All rows assume  $M \geq 48$  and  $u \leq 1/16$ . The displayed horizon costs mean  $S + na + 1$  at  $\ell \leq K/n$ , with fixed  $K > 0$ . Constants are independent of  $M, k$  within each row. The moderate-flavour row is asymptotic with an extremely small  $c_f$ ; the stationary tolerance is also small.

that power to  $O(M^{3/2})$ . Fixed and polylogarithmically growing  $k$  meet the moderate-flavour condition eventually, including the usual  $k = 2$  approximation family. At zero actual production, exact invariance under round words allows the overlapping local groups to generate a larger symmetry, with no bandwidth restriction. A bounded capped flux then survives a fixed trace perturbation. Small gain alone has not been shown to imply the required trace proximity.

The complete statements and proofs are in Appendices Q and R. At each fixed rational clock the cost is in fact  $\Omega_M(n)$  (Theorem Q.2); that argument alone gives no uniform rate in  $M$ . For unrestricted flavours the growing-clock cost remains between powers  $1/4$  and  $1/3$ , up to logarithms. Optimized real fields, alternative realizations and positive-production mixtures over clock sizes are outside these conclusions. They are static source bounds, not operational memory lower bounds over every device.

## P Stability near the symmetry that fixes the inert mode

We prove Theorem O.3. The round, clock torus, constant fields and exterior-algebra representation are exactly those of Appendix N; no change of geometry is made. Write  $\Pi(g) = (\bigwedge g)^{\otimes k}$ ,  $L = 4M + 4$  and  $G_0 = U(e_+^\perp) \cong U(L - 1)$ . All link permutations fix  $e_+$ , as do the phase rotations on  $(e_x - e_y)/\sqrt{2}$ . Thus every literal slot is a clock translation followed by a collective  $G_0$  unitary.

Put  $\sigma = \Delta_c \rho$ ,  $\eta = \mathcal{E}_0 \sigma$  and  $\delta = \frac{1}{2} \|\sigma - \eta\|_1$ . For clock-diagonal states,  $\mathcal{E}_0$  commutes with each literal conjugation: Haar twirling is unchanged by the internal link on either side, and the same twirl is used at every clock. Clock pinching also commutes with these conjugations. The actual bath map is their weighted mixture. Applying Holevo data processing to the ensemble of translated sources therefore gives

$$a(\eta) \leq a(\sigma) \leq a(\rho). \quad (35)$$

This comparison does not assert that  $\eta$  implements the full target channel. We transfer only its order-three anchor charge.

### P.1 Entropy transferred from large irreducible sectors

The one-particle three-cycle  $\gamma$  fixes the orthogonal complement of a complex two-plane in  $e_+^\perp$  and has eigenvalues  $e^{2\pi i/3}, e^{-2\pi i/3}$  on that plane. It is consequently an element of the plane's  $SU(2)$ . In every representation, the projection  $Q_\gamma$  onto its nontrivial eigenvalues is bounded above by the projection onto nontrivial  $SU(2)$  summands.

For an irreducible  $G_0$  representation  $\lambda$ , let  $p_\lambda$  be the flat-state mass of these nontrivial  $SU(2)$  summands and  $q_\lambda$  the flat-state mass of  $Q_\gamma$ . Planes are conjugate in  $G_0$ , so  $p_\lambda$  is common to every plane and  $q_\lambda \leq p_\lambda$ . Restrict to  $t = \lfloor (L-1)/2 \rfloor = 2M+1$  disjoint two-planes. Their commuting  $SU(2)$  groups decompose the flat irrep into flat tensor-product irreducibles and multiplicities. A nontrivial spin factor has dimension at least two. The entropy decomposition, including its nonnegative mixing and multiplicity terms, gives

$$\log_2 \dim \lambda \geq \sum_{j=1}^t \Pr(m_j > 0) = tp_\lambda \geq tq_\lambda. \quad (36)$$

This is the same disjoint-plane argument as in Appendix N, applied inside the smaller group.

The anchor calculation (40) gives  $\text{Tr } \rho Q_\gamma \geq 7/12$ . It is unchanged by clock pinching. Since  $Q_\gamma$  is a projection, trace distance implies  $\text{Tr } \eta Q_\gamma \geq 7/12 - \delta \geq 1/2$  when  $\delta \leq 1/32$ . The clock-dependent three-cycles are conjugate in  $G_0$ , so this is the average of the same charges  $q_\lambda$  over the clock and irrep weights of  $\eta$ . Irreps with  $q_\lambda \geq 1/4$  carry total mass  $w \geq 1/3$ , since the total charge is at most  $w + (1-w)/4$ . By (36), each such irrep has dimension at least  $2^{M/2}$ .

We use a dimension-free entropy transfer rather than continuity in the ambient bath dimension.

**Lemma P.1** (A large flat subdensity survives trace proximity). *Let  $\zeta, \xi$  be states,  $\frac{1}{2}\|\zeta - \xi\|_1 \leq e$ , and  $0 \leq \xi_H \leq \xi$  with  $\text{Tr } \xi_H = w$  and  $\xi_H \leq 2^{-B}I$ . For every  $s > 0$ ,*

$$S(\zeta) \geq s(w - e - 2^{s-B}).$$

*Proof.* Let  $P$  be the spectral projection of  $\zeta$  for eigenvalues at least  $2^{-s}$ . Its rank is at most  $2^s$ . Hence  $\text{Tr } \xi_H P \leq 2^{s-B}$  and  $\text{Tr } \zeta(I - P) \geq \text{Tr } \xi(I - P) - e \geq w - 2^{s-B} - e$ . Every eigenvalue of  $\zeta$  on  $I - P$  contributes at least  $s$  times its mass to the entropy. This also proves the inequality when the right side is negative.  $\square$

Write  $\sigma = \bigoplus_c p_c \sigma_c$  and  $\eta = \bigoplus_c p_c \eta_c$ . Twirling preserves  $p_c$ . Schur decomposition of a normalized invariant block gives

$$\eta_c = \bigoplus_\lambda \frac{I_{\dim \lambda}}{\dim \lambda} \otimes A_{c,\lambda}, \quad A_{c,\lambda} \succeq 0, \quad \sum_\lambda \text{Tr } A_{c,\lambda} = 1.$$

The high-charge subdensity is bounded above by  $2^{-M/2}I$  even when multiplicities are entangled or singular. Apply the lemma with  $B = M/2$ ,  $s = M/4$  to each block, and average. The average high-charge mass is  $w \geq 1/3$  and the average block trace distance is  $\delta$ . Coherently copying the clock and applying Araki–Lieb, as in Appendix N, gives

$$S(\rho) \geq \sum_c p_c S(\sigma_c) \geq \frac{M}{4} (1/3 - \delta - 2^{-M/4}) \geq \frac{M}{16} \quad (M \geq 48, \delta \leq 1/32). \quad (37)$$

No physical clock measurement, full-rank hypothesis or eigenvalue domination was used.

## P.2 Capped flux at its natural scale

For an invariant internal state  $\xi$ , let  $p_\xi$  be its common nontrivial  $SU(2)$ -plane mass. The following estimate applies in  $G_0$  as well as in the full orbital group.

**Lemma P.2** (Capped invariant flux). *Let  $W$  have phases  $\theta_j$  on disjoint one-particle modes in  $e_+^\perp$ , and suppose these can be paired with distinct unused modes in that space. If  $|\theta_j| \leq \pi/4$  and  $\sum_j \theta_j^2 \leq h \leq 1$ , then a collectively  $G_0$ -invariant state satisfies*

$$\mathrm{Tr} \xi \min(|\Pi(W) - I|^2, hI) \geq \frac{p_\xi}{2048} \sum_j \theta_j^2.$$

The minimum denotes spectral functional calculus.

*Proof.* Let  $C \in G_0$  swap the paired modes and set  $H = WCW^*C^*$ . Its paired rotations belong to commuting  $SU(2)$  groups. Conditional on irreducible spins  $m_j \geq 0$ , invariance makes their weights independent and uniform:

$$X_j = (m_j - 2r_j)\theta_j, \quad 0 \leq r_j \leq m_j, \quad v_j = \mathbb{E}X_j^2 = m_j(m_j + 2)\theta_j^2/3.$$

They are symmetric, and their fourth moments are  $m_j(m_j + 2)(3m_j^2 + 6m_j - 4)\theta_j^4/15 \leq 3v_j^2$ . For  $Z = \sum_j X_j$  and  $v = \sum_j v_j$ , independence gives  $\mathbb{E}Z^4 \leq 3v^2$ .

If  $0 < v \leq 1$ , Paley–Zygmund gives  $\Pr(Z^2 \geq v/2) \geq 1/12$ , while  $\Pr(|Z| > \pi) \leq 3/\pi^4$ . On the remaining event,  $4\sin^2(Z/2) \geq 2v/\pi^2$ . Thus

$$\mathbb{E} \min(4\sin^2(Z/2), h) \geq (1/12 - 3/\pi^4)(2/\pi^2) \min(v, h) > \min(v, h)/256.$$

If  $v \geq 1$ , each character  $\varphi_j = \mathbb{E}e^{iX_j}$  satisfies  $|\varphi_j| \leq \exp[-\min(v_j, 1)/64]$ . Indeed when  $(m_j + 1)|\theta_j| \leq 1$ ,  $|X_j| \leq 1$  and the cosine inequality gives  $0 \leq \varphi_j \leq e^{-2v_j/\pi^2}$ . Otherwise the spin character formula gives

$$|\varphi_j| = \left| \frac{\sin((m_j + 1)\theta_j)}{(m_j + 1)\theta_j} \right| \left| \frac{\theta_j}{\sin \theta_j} \right| \leq (101/120)(9/8) = 909/960 < e^{-1/64}.$$

Here  $|\sin z|/z \leq \sin 1 \leq 101/120$  for  $z \geq 1$ , and  $|\theta/\sin \theta| < 9/8$  for  $|\theta| \leq \pi/4$ ; zero angles follow by continuity. Since  $\sum_j \min(v_j, 1) \geq 1$ , the variable  $Y = 4\sin^2(Z/2) \leq 4$  has  $\mathbb{E}Y \geq 2(1 - e^{-1/64})$ . The inequality  $\min(Y, h) \geq hY/4$  gives  $\mathbb{E} \min(Y, h) > h/256$ . The case  $v = 0$  is immediate. In all cases the lower is  $\min(v, h)/256$ . As  $v \geq \sum_{m_j > 0} \theta_j^2$  and this sum is at most  $h$ , averaging the spin sectors yields

$$\mathrm{Tr} \xi \min(|\Pi(H) - I|^2, hI) \geq p_\xi \sum_j \theta_j^2/256.$$

On the finite algebra generated by  $\Pi(W), \Pi(C)$ ,  $\omega(A) = \mathrm{Tr} \xi A$  is tracial. For singular-value counting  $N(A; s) = \omega(1_{(s, \infty)}(|A|))$ , the intersection/min–max argument gives  $N(A + B; s) \leq N(A; s/2) + N(B; s/2)$ . One can first remove zero-weight blocks if  $\omega$  is not faithful. Integration of this inequality gives

$$\omega \min(|A + B|^2, hI) \leq 4[\omega \min(|A|^2, hI) + \omega \min(|B|^2, hI)].$$

Use

$$\Pi(H) - I = [(\Pi(W) - I)\Pi(C) - \Pi(C)(\Pi(W) - I)]\Pi(W)^*\Pi(C)^*.$$

Tracial unitary invariance makes each summand's capped flux equal to that of  $\Pi(W) - I$ , giving a factor eight. Combining the two bounds proves the lemma, including arbitrary multiplicity states.  $\square$

In the actual geometry there are at most  $M$  active modes. They lie in  $e_+^\perp$  and are orthogonal because their label pairs are disjoint. Its dimension  $4M + 3$  leaves room for  $M$  partners. With  $m_M = 1 + (M - 1)(M - 2)/2$ , their constant-field angles satisfy

$$|\theta_j| \leq \pi/m_M \leq \pi/4, \quad \sum_j \theta_j^2 \leq M\pi^2/m_M^2 \leq 72/M^3 =: h_M \leq 1.$$

The last estimate holds at  $M = 12$  and improves thereafter. On  $B_0 = \{0, \dots, \lfloor M/4 \rfloor\}^2$ , all  $M$  regions occur, with areas at most  $M^2$ , so  $\sum_j \theta_j^2 \geq \pi^2/M^3$ ;  $B_0$  occupies at least  $1/16$  of the clock torus.

Write  $\eta_0 = (I/M^2) \otimes \bar{\eta}$  for the uniform-clock state with the same internal marginal as  $\eta$ . Its plane mass satisfies  $p_{\bar{\eta}} \geq \text{Tr } \eta Q_\gamma \geq 1/2$ . Set

$$X = \|\Pi(W) - I\|^2, \quad F_M = \min(X, h_M I), \quad C_0 = \pi^2/65536,$$

where the clock-diagonal word  $W = r_4$  uses the holonomy at each clock. The capped lemma gives  $\text{Tr } \eta_0 F_M \geq C_0/M^3$ . The operator-valued clock law (32) uses only invariance under the literal internal slots. It therefore applies to  $\eta$  with  $G_0$  in place of  $U(L)$ , and (35) gives

$$\frac{1}{2} \|\eta - \eta_0\|_1 \leq M \sqrt{104(\ln 2)a(\rho)}.$$

Since  $0 \leq F_M \leq h_M I$  and its clock pinching is unchanged, positivity and trace distance imply

$$\text{Tr } \rho X \geq \text{Tr } \rho F_M \geq M^{-3} \left[ C_0 - 72\delta - 72M \sqrt{104(\ln 2)a(\rho)} \right]. \quad (38)$$

Capping is important: the unbounded relative spectral contrast of  $\rho$  does not multiply the error by the ambient bath dimension.

### P.3 The cost bound and its limitation

Put  $\delta_* = C_0/288$  and  $A_0 = \delta_*^2/(104 \ln 2)$ . Assume  $\delta \leq \delta_*$ , which is smaller than  $1/32$ . If  $a > A_0/M^2$ , (37) gives

$$S + na \geq M/16 + nA_0/M^2 \geq \frac{3}{2^{2/3}} (A_0/16^2)^{1/3} n^{1/3} > 4.64 \cdot 10^{-6} n^{1/3}.$$

Otherwise (38) gives  $\text{Tr } \rho X \geq C_0/(2M^3)$ . The weaker consequence  $\ell \geq \text{Tr } \rho X/(3 \cdot 104)$  of (39), together with  $\ell \leq 2/(15n)$ , implies

$$M^3 \geq \frac{15C_0}{12 \cdot 104} n, \quad S + na \geq \frac{1}{16} \left( \frac{15C_0}{12 \cdot 104} \right)^{1/3} n^{1/3} > 7.61 \cdot 10^{-4} n^{1/3}.$$

Both branches prove the stated  $4 \cdot 10^{-6}$  constant.

The hypothesis is not density domination in disguise. Under the canonical exterior-algebra splitting  $\Lambda(\mathbb{C}^L) \simeq \Lambda(\mathbb{C}e_+) \otimes \Lambda(e_+^\perp)$ , the first factor is inert for every literal slot. It can be in a pure coherent state while the second carries an invariant charged source. Such states can satisfy the theorem's round tests, have zero eigenvalues, and fail full  $U(L)$  invariance. For example, a flat state on  $\Lambda(e_+^\perp)^{\otimes k}$  has the same holonomy characters as the flat full-orbital source; replacing its inert factor by a pure coherent state changes no channel or leakage test.

More generally let  $\tau$  have invariant clock-diagonal blocks and let  $\rho = P\tau P/\text{Tr}(P\tau)$ , where  $\text{Tr}(P\tau) = 1 - r$ . The purification proof of gentle filtering gives  $\frac{1}{2} \|\rho - \tau\|_1 \leq \sqrt{r}$ . Pinching and twirling are trace-distance contractions, so  $\delta_{\text{orb}}(\rho) \leq 2\sqrt{r}$ . Whenever the original round tests still hold, a vanishing filtering perturbation cannot beat the exponent in this geometry. The missing all-source step is to deduce a suitably uniform symmetry proximity from low actual production as  $M$  grows. No such deduction, or comparison with all round geometries, is proved here.

## Q Actual production and Houghton coercivity

### Q.1 The root-energy dictionary

**Proposition Q.1** (Actual-production dictionary). *For the literal smoothed slots  $T\rho = \sum_s w_s V_s \rho V_s^*$ , with  $m \leq 174$ ,  $V_0 = I$  and  $w_0 \geq 1/d$ , put  $\mathfrak{D} = \sum_s w_s \|V_s \sqrt{\rho} V_s^* - \sqrt{\rho}\|_F^2$ . For every source, with  $a = S(T\rho) - S(\rho)$ ,*

$$\mathfrak{D} \leq (d+1) \ln 2 a, \quad a \leq h_2(t) + t \log_2(m-1), \quad t = \min\{\mathfrak{D}, (m-1)/m\}.$$

*On the same feasible physical rounds, optimizing  $S + n\mathfrak{D}$  and  $S + na$  gives equivalent costs up to  $O(\log(en))$  and an additive constant. Error and leakage constraints are unchanged.*

*Proof of Proposition Q.1.* Put  $\xi = \sqrt{\rho}$ ,  $\xi_s = V_s \xi V_s^*$ ,  $\bar{\xi} = \sum_s w_s \xi_s$ , and  $\sigma = T\rho$ . The Holevo identity gives  $a = \sum_s w_s D_2(\rho_s \| \sigma)$ . For states  $P, Q$ ,

$$D_{\text{nat}}(P \| Q) \geq \|\sqrt{P} - \sqrt{Q}\|_F^2.$$

Indeed spectral overlap probabilities reduce the left side to a classical relative entropy, and Jensen gives  $D_{\text{nat}} \geq -2 \ln \text{Tr} \sqrt{P} \sqrt{Q} \geq 2 - 2 \text{Tr} \sqrt{P} \sqrt{Q}$ ; zero eigenvalues follow by limits. The variance identity and minimization over a common centre give

$$v := \sum_s w_s \|\xi_s - \bar{\xi}\|_F^2 = 1 - \|\bar{\xi}\|_F^2 \leq (\ln 2)a.$$

The identity slot implies  $w_0 \|\xi - \bar{\xi}\|_F^2 \leq v$ , hence  $\mathfrak{D} = v + \|\bar{\xi} - \xi\|_F^2 \leq (d+1)(\ln 2)a$ .

For the converse, form the pure-state ensemble  $\Omega = \sum_s w_s |\text{vec } \xi_s\rangle \langle \text{vec } \xi_s|$ . Partial trace maps it to the ensemble  $\rho_s$ , so Holevo data processing gives  $a \leq S(\Omega)$ . Its rank is at most  $m$  and  $\Omega \geq |\text{vec } \bar{\xi}\rangle \langle \text{vec } \bar{\xi}|$ , so  $\lambda_{\max}(\Omega) \geq 1 - v \geq 1 - \mathfrak{D}$ . Maximizing entropy with this eigenvalue constraint gives the stated binary-entropy bound. Zero weights can be omitted;  $m = 1$  is trivial.

For completeness put  $y = \log_2(en)$  and  $C_0 = (d+1) \ln 2$ . The first inequality gives  $S + n\mathfrak{D} \leq \max(1, C_0)(S + na)$ . The second, using  $h_2(t) \leq t \log_2(e/t)$ , gives  $S + na \leq C_m y(1 + S + n\mathfrak{D})$  for a fixed  $C_m$ . If  $\mathfrak{D} \leq 1/n$ , set  $x = n\mathfrak{D} \leq 1$  and use  $x \log_2(en/x) \leq y + 1/(e \ln 2)$ . If  $1/n < \mathfrak{D} < (m-1)/m$ , use  $\log_2(e/\mathfrak{D}) \leq y$ ; above this threshold use  $a \leq \log_2 m \leq 2(\log_2 m)\mathfrak{D}$ . Taking infima over the same feasible family proves the profile claim.  $\square$

### Q.2 The exact leakage coefficient

Here and below  $\Pi_k$  denotes the smoothed model's bath representation,  $W = r_4$ , and  $X = (\Pi_k(W) - I)^*(\Pi_k(W) - I)$ . Inspecting the canonical slots gives, for every source,

$$\ell = \frac{488}{225d} \text{Tr } \rho X. \quad (39)$$

The calculation is weighted least squares, not a defect inequality. The identity class has total squared coefficient  $45/2$ : one exceptional closing slot of weight  $1/2$  versus weight  $22$  with the usual endpoint. The  $\alpha$  class has total  $25/2$ : exceptional weight  $2$  versus usual weight  $21/2$ . The other exceptional word  $\text{aba}^{-1} = r_4 \alpha b$  is alone in its target label class and contributes no residual. The flux phases act on coordinates numbered at least five, so they commute with  $\alpha$ . Thus the two variances multiply the same  $X$ :

$$\Lambda = \left( \frac{(1/2)22}{45/2} + \frac{2(21/2)}{25/2} \right) X = \frac{488}{225} X.$$

All other slot classes are consistent. Let  $Q_\gamma$  be the projection onto the nontrivial eigenvalues of the order-three operator  $\Pi_k(\gamma)$ . The anchor gives  $|\text{Tr } \rho \Pi_k(\gamma)| \leq 2u \leq 1/8$ , so

$$\text{Tr } \rho Q_\gamma \geq 7/12, \quad (40)$$

since  $\text{Re } \text{Tr } \rho \Pi_k(\gamma) = 1 - \frac{3}{2} \text{Tr } \rho Q_\gamma$ .

### Q.3 A representation-uniform gap at fixed clock size

**Theorem Q.2** (Fixed-clock coercivity). *For each fixed  $M \geq 48$  in the literal rational constant-field smoothed model there is  $c_M > 0$ , independent of the flavour count  $k$ , such that every source with  $u \leq 1/16$  satisfies  $a + \ell \geq c_M$ . Consequently  $\ell = O(1/n)$  forces  $S + na = \Omega_M(n)$ .*

We give the compact-group argument for Theorem Q.2. Its only geometric inputs are the actual unitary model of [26, Section 6], specialized to its rational constant fields. At fixed  $M \geq 48$ , put  $L = 4M + 4$ . The one-particle generators lie in

$$G = U(L)^{M^2} \rtimes \mathbb{Z}_M^2 \subset U(\mathbb{C}^{M^2} \otimes \mathbb{C}^L).$$

Their links are permutations times  $e^{i\phi Q_{\text{pair}}}$ . The tree curl construction with fields  $\pm\pi/|R|$  makes  $\phi$  a rational multiple of  $\pi$ , so all their entries are algebraic. The words  $r_1, r_2, r_3, r_5$  are exact,  $\gamma$  is a three-cycle, and  $W = r_4$  has disjoint-plane phases. In particular

$$W^q = I, \quad q = 2 \operatorname{lcm}_R |R|, \quad \|W - I\|_{\text{op}} \leq \pi/m_M < 1/2, \quad m_M = 1 + (M-1)(M-2)/2.$$

For each  $k$ , the model on  $\mathbb{C}^{M^2} \otimes \Lambda(\mathbb{C}^L)^{\otimes k}$  is a continuous unitary representation of this same  $G$ . It is not the exterior power of the full one-particle clock space.

Let  $\Gamma = \langle \mathbf{a}, \mathbf{b}, \alpha \rangle$ ,  $K = \overline{\Gamma}$ , and let  $\Gamma_N$  be the abstract normal closure of  $W$  in  $\Gamma$ . Then  $N = \overline{\Gamma_N}$  is a closed normal subgroup of  $K$ . We first prove that  $N^0$  is semisimple. The torus  $Z = Z(N^0)^0$  is characteristic and hence  $K$ -invariant. Every  $K$ -conjugate of  $W$  acts on its Lie algebra with

$$\|\operatorname{Ad}(W) - I\|_{F \rightarrow F} \leq 2\|W - I\|_{\text{op}} < 1.$$

This is an order- $q$  torus-lattice automorphism. A nonidentity finite-order integral automorphism cannot have every eigenvalue within distance strictly less than one from 1: a nontrivial root of unity  $\zeta$  brings all its Galois conjugates in the integer characteristic polynomial, while the nonzero integer  $\prod_{\zeta'} (1 - \zeta')$  would then have absolute value less than one. Thus every conjugate centralizes  $Z$ , and density shows that  $N$  does too.

The compact abelian Lie quotient  $N/[N, N]$  has exponent dividing  $q$ , since it is generated densely by these order- $q$  conjugates; it is therefore finite. Put  $D = [N^0, N^0]$ , a closed semisimple subgroup. In  $N/D$ , the connected component is the image of  $Z$ , hence is central and has finite index. Schur's theorem gives a finite commutator subgroup [87]. Its abelianization is also finite, as a quotient of the preceding exponent- $q$  abelian group. Thus  $N/D$  is finite, and its connected component is trivial. This proves  $N^0 = D$ .

Next  $\gamma \in N$ . The quotient  $K/N$  satisfies the  $H_3$  presentation. Every finite-dimensional unitary representation of  $H_3$  kills its subgroup  $\operatorname{Alt}_{\text{fin}}$ : simplicity would otherwise give an injection, whereas this group contains elementary abelian 2-groups of arbitrarily large rank and commuting involutions in  $U(p)$  have rank at most  $p$ . Continuous finite-dimensional representations of a compact group separate its points. They therefore kill the image of  $\gamma$  in  $K/N$ , proving the assertion.

The dense subgroup  $\Gamma_N \cap N^0$  consists of algebraic matrices;  $N^0$  is open in  $N$ . If  $N^0$  is nontrivial, Breuillard–Gelder's theorem supplies two elements of this subgroup generating a dense subgroup of  $N^0$  [12, Theorem 1.1]. Their adjoint matrices have algebraic entries in an algebraic Lie-algebra basis. Here is the basis justification, to exclude an implicit rationality assumption on a closed subgroup. A compact matrix group is real Zariski closed: polynomial approximation followed by averaging over the group separates any exterior point from the group by an invariant polynomial. At each bounded degree its vanishing polynomials are the kernel of evaluations on the dense algebraic subgroup. Finite-rank linear algebra selects finitely many algebraic evaluation rows, so this kernel has an algebraic coefficient basis. Noetherianity selects finitely many of these defining equations. Taking their tangent

equations at the identity yields an algebraic basis of the Lie algebra; compact real algebraic groups are smooth. Conjugation by the selected algebraic matrices is algebraic in this basis.

Apply He–de Saxce’s theorem to a symmetric lazy probability measure on these generators and their inverses [46, Theorem 1.4]. It has a spectral gap on  $L_0^2(N^0)$ . Peter–Weyl then gives the same gap in every continuous unitary representation after removing  $N^0$ -invariants. Add finitely many representatives in  $\Gamma_N$  of  $N/N^0$ . The projection onto  $N^0$ -invariants commutes with  $N$ ; on its range the finite quotient has a uniform gap. Combining the two gaps gives a finite set  $F_M \subset \Gamma_N$  and a representation-uniform constant. If  $N^0$  is trivial, the finite-group gap gives the same conclusion directly. Every member of  $F_M$  is a finite word in  $\Gamma$ -conjugates of  $W$  and  $W^{-1}$ . All choices and constants depend on  $M$ , not on a representation or its dimension.

On Hilbert–Schmidt matrices use  $\zeta_k(x, y)A = \Pi_k(x)A\Pi_k(y)^*$ . Left multiplication by a conjugate of  $W$  is  $(g, g)(W, e)(g^{-1}, g^{-1})$  in this representation; there is an analogous right formula. Telescoping each fixed word bounds its squared displacement by its length times the sum of the squared elementary displacements. Diagonal  $\mathbf{a}, \mathbf{b}, \alpha$  displacements are bounded by  $d$  times the actual slot Dirichlet energy, since their singleton weights are at least  $1/d$ ; inverse displacements have the same norm. The remaining elementary energies are left and right  $W - I$ . Apply the finite-set gap separately to the left and right representations of  $N$ . Their invariant projections  $P_L, P_R$  commute, and distance to  $P_L P_R$  is at most the sum of the two squared distances. Since  $\gamma \in N$ , its left and right nontrivial spectral projections annihilate  $P_L P_R$ . We obtain

$$\begin{aligned} \frac{\|Q_{\gamma,k}A\|_F^2 + \|AQ_{\gamma,k}\|_F^2}{2} &\leq C_M \left[ \sum_s w_s \|V_s A V_s^* - A\|_F^2 \right. \\ &\quad \left. + \frac{\|(\Pi_k(W) - I)A\|_F^2 + \|A(\Pi_k(W) - I)\|_F^2}{2} \right]. \end{aligned} \quad (41)$$

Set  $A = \sqrt{\rho}$ . The two symmetrized terms become  $\text{Tr } \rho Q_\gamma$  and  $\text{Tr } \rho X$ , respectively. Equations (39)–(40) and Proposition Q.1 give

$$7/12 \leq C_M \left[ (d+1)(\ln 2)a + \frac{225d}{488}\ell \right],$$

proving the fixed-clock theorem. No invariance or entropy condition on  $\rho$  has been introduced.

#### Q.4 Entropy without source symmetry

**Theorem Q.3** (Entropy without source symmetry). *In the literal rational constant-field smoothed model, every  $M \geq 48$ ,  $k \geq 1$  and arbitrary coherent source with  $u \leq 1/16$  satisfies, with a universal  $c > 0$ ,*

$$S(\rho) + 1 \geq c \min\{M, a(\rho)^{-1/2}\}, \quad S(\rho) + na(\rho) + 1 \geq c \min\{M, n^{1/3}\} \quad (n \geq 1).$$

At  $a = 0$  the first minimum is  $M$ .

*Proof of Theorem Q.3.* Write  $\pi$  for the round’s usual free-group representation and put  $c_g = 2(104 \ln 2)^{1/2}$ . Lemma 8.4 and word telescoping give  $\chi(\pi(w)) \leq |w|c_g\sqrt{a}$  and  $\frac{1}{2}\|\rho - \pi(w)\rho\pi(w)^*\|_1 \leq \chi(\pi(w))$ . The same Holevo argument after tracing to either clock coordinate bounds its cyclic root energy  $E_i$  by  $c_g^2 a$ . Cauchy–Schwarz gives  $\max p_i \leq 1/M + 2\sqrt{E_i}$ . Hence  $G_R = \{0 \leq i, j \leq M - R - 1\}$  has  $p(G_R^c) \leq 2R/M + 4Rc_g\sqrt{a}$ .

We first verify the exact structure on  $G_R$ , for  $5 \leq R < M/4$ . At  $c = (i_0, j_0)$ , a forward horizontal prefix of length at most  $R - 1$  has edge starts at most  $M - 3$  and endpoints at

most  $M - 2$ . It meets neither the seam nor the preliminary exact horizontal move of [26, Section 6.2]. Its base links are  $c_{2M+2-i-j_0}$ . A horizontal edge is interior to a region only when both incident plaquettes belong to it. For  $j_0 = 0$  none of these edges carries a rotation; for  $j_0 \geq 1$  the bulk condition is  $i + j_0 \leq t + M - 8$ , and the  $T$  condition is  $i + j_0 \leq M - 3$ . Thus a region eligible later was eligible earlier. Actual potentials may be sparse; only eligibility is monotone. Pulling each phase plane back through the base prefix gives its fixed starting pair  $(x, x + M + i_0)$ , where  $x = t + M - 1 - i_0 - j_0$  for  $5 \leq t \leq M + 3$ , and  $x = M + 3 - i_0 - j_0$  for  $T$ . These pairs are disjoint.

In column-operator notation the prefix factors as  $P_j = B_j R_j$ , with  $R_j$  a product of commuting rotations on these pairs. The usual representation takes adjoints of the right-action operators, so the one-particle holonomy  $J_j = P_j^* \tau P_j$  of  $\pi(a^j \alpha a^{-j})$  has normal  $R_j^*(e_{j+1} - e_{j+2})/\sqrt{2}$ . The far partners exceed  $R+1$ . Nonadjacent normals therefore have disjoint support; adjacent overlaps have modulus at most  $1/2$ . The exact conjugates of  $r_2$  force that modulus to be  $1/2$ . Phase adjustment gives the standard  $A_R$  root Gram matrix, proving an exact  $S_{R+1}$  representation. Quantization preserves it for every flavour count.

Partition this chain into  $t_R = \lfloor (R+1)/3 \rfloor$  commuting  $S_3$  blocks. Their three-cycles are  $\gamma_b = a^{3b} \gamma a^{-3b}$ . The original anchor gives  $\text{Tr } \rho Q_\gamma \geq 7/12$ . Transport and restriction to  $G_R$  leave each block weighted charge at least  $7/12 - 2R/M - 5Rc_g \sqrt{a}$ . Six representatives for each block have word lengths at most  $2R + 5 \leq 3R$ . Their canonical root pure-state ensemble has rank at most six and an eigenvalue at least  $1 - x$ ,  $x = 9R^2 c_g^2 a$ . Partial trace bounds the mixed-state Holevo gain by  $f_6(x) = h_2(x) + x \log_2 5$  for  $x \leq 5/6$ .

Clock pinching and Holevo data processing bound the weighted conditional gains on  $G_R$  by  $f_6(x)$ . On these clocks only, each previous block twirl commutes with every conjugation in a later block, so it cannot increase that block's gain. The final state is flat on the irreducible factors of  $S_3^{t_R}$ , with arbitrary multiplicities. Only its two-dimensional standard factors have nontrivial three-cycle charge; those charge projections are central and preserved. Final conditional entropy is therefore at least the sum of charges. Coherent clock copying and Araki–Lieb give  $S(\rho) \geq \sum_c p_c S(\rho_c)$ , hence

$$S(\rho) \geq t_R \left[ \frac{7}{12} - \frac{2R}{M} - 5Rc_g \sqrt{a} - f_6(9R^2 c_g^2 a) \right].$$

No global commutation on the excluded clocks is required.

Put  $K = \min\{M, 1/(c_g \sqrt{a})\}$  and  $R = \lfloor K/128 \rfloor$ . For  $R \geq 5$ , the bracket exceeds  $7/16$ , since  $7/12 - 7/128 - f_6(9/16384) > 0.5206$ ; also  $t_R \geq R/4$ . Thus  $S > R/10$  and  $S+1 \geq K/1280$ . For  $R < 5$ ,  $K < 640$  and  $S+1 \geq 1$  suffices. This proves the first inequality with  $c = 1/(1280c_g)$ . For the second,  $a \geq n^{-2/3}$  gives  $na \geq n^{1/3}$ ; otherwise the first inequality applies with  $a^{-1/2} > n^{1/3}$ .  $\square$

## Q.5 A fixed local group and the fourth-power leakage bound

**Theorem Q.4** (Leakage without source symmetry). *For the literal rational constant-field smoothed round, all  $M \geq 48$ ,  $k \geq 1$  and arbitrary coherent sources with  $u \leq 1/16$  satisfy, with constants  $c_0, c > 0$  independent of  $M, k, \rho$ ,*

$$M^2 a + M^4 \ell \geq c_0, \quad S + na + 1 \geq c \min\{n^{1/3}, \ell^{-1/4}\} \quad (n \geq 1).$$

*At zero leakage the second argument is infinite.*

*Proof of Theorem Q.4.* Put  $d = 104$ ,  $c_g = 2\sqrt{104 \ln 2}$  and  $\sigma = \Delta_{\text{clock}} \rho = \bigoplus_c p_c \rho_c$ . Clock covariance and Holevo data processing give  $a(\sigma) \leq a(\rho)$ ; pinching preserves  $X$  and  $Q_\gamma$ . For an actual word  $C$  the identity and generator weights give

$$\chi(C, \sigma) := \|C \sqrt{\sigma} C^* - \sqrt{\sigma}\|_F \leq |C| c_g \sqrt{a}.$$

Tracing to the clock bounds each cyclic root energy by  $c_g^2 a$ . The torus Poincaré inequality and differences of squares imply  $\text{TV}(p, \text{unif}) \leq c_g M \sqrt{a}$ . For  $M \geq 4096$  put  $x = \lfloor M/4 \rfloor$ ,  $R = x + 8$ , and

$$G = \{(i, j) : 0 \leq i \leq M - x - 12, \quad 1 \leq j \leq M - 3, \quad i + j \geq M - x + 16\}.$$

Its size is  $(M - x - 11)(M + x - 48)/2$ , so its density is at least  $15/32 - 40/M \geq 0.45$ . Restricting the original anchor charge gives

$$q_G := \sum_{c \in G} p_c \text{Tr } \rho_c Q_\gamma \geq 1/30 - c_g M \sqrt{a}. \quad (42)$$

Thus  $M^2 a \leq \varepsilon_0 := 1/(120c_g)^2$  implies  $q_G \geq 1/40$ .

We need a stronger version of the preceding interior-chain geometry. At  $c = (i, j) \in G$ , horizontal prefixes of length at most  $R - 1$  have edge starts at most  $M - 6$  and endpoints at most  $M - 5$ . They meet neither the seam nor the preliminary exact  $A$  move. The first reverse cycle has length  $r_0 \geq x + 17 = R + 9$ . In inverse traversal the tracked indices and cycle lengths increase together, so there is no wrap, even though later cycles can be shorter than  $R + 4$ ; the last cycle has length at least eleven. The row bound  $j \leq M - 3$  excludes the exceptional  $T$ -source row. As before, bulk and  $T$  eligibility require both incident plaquettes, and are respectively  $i + j \leq t + M - 8$  and  $i + j \leq M - 3$ . Pulled-back planes are disjoint fixed pairs  $(s, s + M + i)$ . A bulk pair can rotate at prefix step  $q$  only for  $q \leq s - 7$ , and a  $T$  pair only for  $q \leq s - 6$ . Its rotation is therefore stable before a reflection normal uses  $e_s$ . One common product  $D_c$  of these commuting rotations gives orthonormal points  $f_s = D_c^* e_s$  and reflection normals  $(f_{q+1} - f_{q+2})/\sqrt{2}$ , for  $0 \leq q < R$ . Thus the chain is exactly  $S_{R+1}$ , with  $f_1, f_2, f_3 = e_1, e_2, e_3$ . All the same pairs support the field  $w_c$ , so  $D_c$  commutes with it.

Choose  $r \in \{0, \dots, 7\}$ , set  $s = x + r$  and  $t = x + r - M + 1 + i + j$ . Then  $17 \leq t \leq M - 7$  and this bulk region is active, with pair  $(s, s + M + i)$ . Its untruncated triangle count gives

$$A_t = (M - 1)^2 - h(h + 1)/2, \quad h = M + 3 - t.$$

Triangular residues modulo four repeat 0, 1, 3, 2, 2, 3, 1, 0 over eight consecutive values. Since  $(M - 1)^2$  is zero or one modulo four, one choice has  $4 \mid A_t$ . Put  $p = A_t/4 \leq M^2/4$ , relabel  $f_s$  as point four, and take  $f_5 = D_c^* e_{s+M+i}$ . On these five points  $W_c^p$  is  $g_4 = \exp(\pm i\pi P_{f_4-f_5}/4)$ , where  $P$  projects onto the normalized difference. All other phases are outside the five points and commute with  $S_4$  on the first four. The swap of low point  $i_0 \leq 3$  with  $s$  is the actual word

$$C_{i_0, s} = \mathbf{a}^{i_0-1} (\alpha \mathbf{a})^{s-i_0-1} \alpha (\mathbf{a}^{-1} \alpha)^{s-i_0-1} \mathbf{a}^{-(i_0-1)}.$$

Adjacent products collapse to this word of length at most  $4s \leq M + 28 \leq 2M$ . With  $C_4 = I$ , there are only 24 nontrivial global choices. Hence conditional selection costs

$$\sum_{c \in G} p_c \sum_{i=1}^3 \chi(C_i^{r(c)}, \rho_c)^2 \leq 96 c_g^2 M^2 a. \quad (43)$$

Here is the fixed-group gap input. In the canonical five-point basis put  $v_i = (e_i - e_5)/\sqrt{2}$ ,  $g_i = e^{i\pi P_{v_i}/4}$ , and take the twelve matrices  $U_{ij} = g_i g_j^*$  and  $g_i U_{ij} g_i^*$ ,  $i < j$ . They fix the all-ones vector and lie in  $SU(H_0)$ ,  $H_0 = \{z : \sum z_i = 0\}$ , of dimension four. On  $E_{ij} = \text{span}(v_i, v_j)$  the nontrivial eigenvalues of  $U_{ij}$  are  $e^{\pm i\phi}$ , with  $2 \cos \phi = (2 + 3\sqrt{2})/4$ . Its minimal polynomial is  $z^2 - z - 7/8$ , so it is not an algebraic integer. Thus  $\phi/\pi$  is irrational and powers give a continuous  $SU(2)$  circle on  $E_{ij}$ . Its  $g_i$ -conjugate is nonparallel:  $[g_i, U_{ij}] = g_i [g_i, g_j^*] \neq 0$ ; an antiparallel axis is impossible since  $\text{Ad } g_i$  has no eigenvalue  $-1$ . The two circles generate  $SU(2)(E_{ij})$ . The six real skew matrices  $v_i v_j^T - v_j v_i^T$  span  $\mathfrak{so}(4)$ . The same plane algebras contain the

nonzero imaginary symmetric traceless  $i(P_{v_1} - P_{v_2})$ . Orthogonal conjugation diagonalizes it to a nonzero multiple of  $i \operatorname{diag}(1, -1, 0, 0)$ ; permutations and plane rotations span all nine imaginary symmetric traceless directions. Together these span  $\mathfrak{su}(4)$ , proving density. An algebraic orthonormal basis of  $H_0$  makes all entries algebraic. Bourgain–Gamburd’s theorem [11, Theorem 1] and Peter–Weyl give a fixed  $\delta > 0$  such that, in every continuous representation,

$$\|(I - P_{\text{inv}})\xi\|^2 \leq \delta^{-1} \sum_h \|(h - I)\xi\|^2.$$

We use the unnormalized symmetric gap. The negative-angle list is the complex conjugate and has the same gap.

For the actual gates  $G_i = C_i W_c^p C_i^*$  the common outside factor cancels in  $G_i G_j^*$  and  $G_i (G_i G_j^*) G_i^*$ . Quantization gives continuous representations of this same  $SU(4)$  for every  $k$ , with arbitrary spectators. The original three-cycle  $\gamma$  belongs to it, so  $Q_\gamma$  kills its invariant vectors. Apply the gap to left multiplication on Hilbert–Schmidt matrices, not to conjugation. For  $\xi = \sqrt{\rho_c}$ ,  $E = \|(W_c - I)\xi\|_F$  and  $\chi_i = \|C_i \xi C_i^* - \xi\|_F$ , moving  $C_i^*$  across  $\xi$  gives  $\|(G_i - I)\xi\|_F \leq pE + 2\chi_i$ . Products telescope in left-action norm and inverses have the same energy. The two listed gates have respective bounds  $2pE + 2\chi_i + 2\chi_j$  and  $4pE + 6\chi_i + 2\chi_j$ . Three-term Cauchy and summation give

$$\sum_h \|(h - I)\xi\|_F^2 \leq 360p^2 E^2 + 360 \sum_{i=1}^3 \chi_i^2.$$

Since the selected power obeys  $p(c)^2 \leq M^4/16$  pointwise, (43) yields

$$q_G \leq \delta^{-1} \left[ (45/2)M^4 \operatorname{Tr} \rho X + 34560c_g^2 M^2 a \right].$$

With  $A = 34560c_g^2$ ,  $B = (45/2)(225d/488)$ , (42) proves the bound at  $M \geq 4096$  with  $c_{\text{large}} = \min\{\varepsilon_0, \delta/[40 \max(A, B)]\}$ . For  $48 \leq M < 4096$ , Theorem Q.2 gives  $M^2 a + M^4 \ell \geq M^2 c_M$ . The minimum of these finitely many positive constants and  $c_{\text{large}}$  is  $c_0 > 0$ . No numerical gap constant is asserted.

Finally put  $t_0 = \min\{n^{1/3}, \ell^{-1/4}\}$  and  $\beta = \min\{1, (c_0/2)^{1/4}\}$ . If  $M \geq \beta t_0$ , Theorem Q.3 gives  $S + na + 1 \geq c_{\text{ent}} \beta t_0$ . Otherwise  $M^4 \ell < c_0/2$ , so  $na \geq c_0 n/(2M^2) > [c_0/(2\beta^2)]n/t_0^2 \geq [c_0/(2\beta^2)]t_0$ , since  $t_0^3 \leq n$ . This proves the second inequality. It gives the stated  $n^{1/4}$  floor at  $\ell \leq K/n$ , and a cube-root lower at  $\ell \leq n^{-4/3}$ , without asserting a matching zero-leakage upper.  $\square$

## Q.6 Zero production and the actual stationary algebra

Let  $\mathcal{E}_M$  be Haar averaging over the closed group generated by the *actual* slot unitaries on the counted bath. Its range is their commutant. It commutes with clock pinching  $\Delta_c$ , since each slot has a fixed clock translation and a blockwise internal action. This averaging is an analytical conditional expectation.

**Corollary Q.5** (Stationary sources at arbitrary flavour count). *There are constants  $b, \delta_*, c > 0$ , independent of  $M \geq 48$  and  $k \geq 1$ , with the following properties. Put  $h = M^{-3}$ ,  $F = \min(X, hI)$  and  $q_\eta = \operatorname{Tr} \eta Q_\gamma$ . Every actual stationary state  $\eta = \mathcal{E}_M \eta$  obeys*

$$\operatorname{Tr} \eta F / h \geq b(q_\eta - 11/20). \quad (44)$$

*If a physical source has  $u \leq 1/16$  and actual  $a = 0$ , then*

$$\ell \geq cM^{-3}, \quad S(\rho) + 1 \geq cM, \quad S(\rho) + 1 \geq c\ell^{-1/3}.$$

More generally,  $u \leq 1/16$  and

$$\frac{1}{2} \|\Delta_c \rho - \mathcal{E}_M(\Delta_c \rho)\|_1 \leq \delta_*$$

imply  $\ell \geq cM^{-3}$ . If also  $\ell \leq K/n$  for fixed  $K > 0$ , then  $S(\rho) + na(\rho) + 1 \geq c_K n^{1/3}$ . Entropy and production always refer to the original source.

*Proof.* The Holevo identity and the identity slot show that  $a = 0$  holds exactly when every positive-weight slot fixes  $\rho$  by conjugation. Thus  $\rho$  commutes with every actual word. Clock pinching retains stationarity,  $X$  and  $Q_\gamma$ . The two clock translations force each pinched clock mass to be  $1/M^2$ . For any stationary  $\eta$ , write  $\Delta_c \eta = M^{-2} \oplus_c \eta_c$ .

For  $M \geq 4096$ , use the exact stabilized logical-point geometry above, restricted to the smaller wedge of Appendix R. Put  $x = \lfloor M/4 \rfloor$ ,  $z = \lfloor M/256 \rfloor$  and

$$G = \{0 \leq i \leq M - x - 12, 1 \leq j \leq M - 3, i + j \geq M - x + 16 + 2z\}.$$

Its density is at least 0.45. Choose even  $b_0 \in [x - z, x - z + 1]$ ,  $J = \lfloor (z - 1)/8 \rfloor$  and  $b_j = b_0 + 8j$ ,  $0 \leq j < J$ . Then  $M/8192 \leq J \leq M/2048$ . In each eight-index interval choose  $s_j$  so that the actual region  $t = s_j - M + 1 + i + j$  has  $4 \mid A_t$ . The triangular residues used above guarantee this choice. These active regions have  $M^2/3 \leq A_t \leq M^2$ . Their pairs  $(f_{s_j}, D_c^* e_{s_j + M + i})$  are disjoint from one another and from the common anchor modes  $f_1, f_2, f_3$ . The usual-representation angles have absolute value  $\pi/A_t$ ; taking adjoints reverses the written right-action sign.

The rational power  $A_t/4$  and the three low-to-selected swaps give the fixed local  $SU(4)$  of the preceding proof on the sum-zero space  $H_j$  of those five modes. Outside phases cancel in its generators. Stationarity gives exact commutation with these groups; no word-length or congestion estimate is needed. They generate  $SU(H)$  on the sum-zero space of all  $3+2J$  modes, of dimension  $2J + 2$ . To verify this, choose a unit vector  $e$  in the common two-dimensional anchor sum-zero plane. Each local Lie algebra contains both off-diagonal skew-Hermitian generators between  $e$  and every vector in  $H_j \cap e^\perp$ . The spaces  $H_j$  span  $H$ : the two common anchor differences and the two independent differences from each selected pair to an anchor give a spanning set. Thus these local perpendicular spaces span  $H \cap e^\perp$ . Linearity and commutators give every off-diagonal direction and every traceless diagonal direction of  $\mathfrak{su}(H)$ .

On  $H$ , the field has  $J$  orthogonal active difference modes  $v_j$  with angles  $\theta_j$ , and fixes a complementary  $(J + 2)$ -space. Pair the  $v_j$  with  $J$  orthogonal fixed modes. The resulting product  $SU(2)$  groups lie in  $SU(H)$ . An invariant conditional state has the same nontrivial spin mass  $p_c$  on every such plane, by conjugacy. The anchor three-cycle lies in an anchor  $SU(2)$ , so  $q_c \leq p_c$ . Arbitrary flavour and multiplicity correlations are allowed in its flat product-spin decomposition.

Apply the capped spin estimate of Appendix P to the commutator of the full field with the product of the paired quarter turns  $v_j \mapsto w_j$ ,  $w_j \mapsto -v_j$ . Each quarter turn lies in  $SU(2) \subset SU(H)$ , unlike a swap, whose determinant on  $H$  is  $-1$ , and it produces the same commutator. Outside phases cancel. Its transfer to  $F$  is valid because the stationary conditional state commutes with both the full field and the quarter turns, making the corresponding functional tracial. Here  $\sum_j \theta_j^2 \leq 9\pi^2/(2048M^3) < h$  and  $|\theta_j| \leq \pi/4$ . Consequently

$$\text{Tr } \eta_c F \geq \frac{p_c}{2048} \sum_j \theta_j^2 \geq \frac{q_c J \pi^2}{2048 M^4}.$$

Uniform clock mass and positivity allow discarding clocks outside  $G$ . Their lost charge is at most  $11/20$ , giving (44) with  $b_{\text{large}} = \pi^2/(2048 \cdot 8192)$ . For  $48 \leq M < 4096$ ,  $X \leq 4I$  gives

$F \geq hX/4$ . Equation (41) at stationarity gives  $q_\eta \leq C_M \text{Tr } \eta X$ . Choose

$$b = \min \left\{ b_{\text{large}}, \min_{48 \leq M < 4096} (4C_M)^{-1} \right\} > 0.$$

This is an existence constant; the finite-clock gaps are not evaluated.

Put  $\beta = b/80$  and  $\delta_* = \min\{1/48, \beta/2\}$ . For  $\sigma = \Delta_c \rho$  and  $\eta = \mathcal{E}_M \sigma$ , the stated trace distance transfers only the anchor charge:  $q_\eta \geq 7/12 - 1/48 = 9/16$ . Hence  $\text{Tr } \eta F \geq \beta h$  and, since  $0 \leq F \leq hI$ ,

$$\text{Tr } \rho X = \text{Tr } \sigma X \geq \text{Tr } \sigma F \geq (\beta - \delta_*)h \geq \beta h/2.$$

Equation (39) proves the leakage lower. At  $a = 0$  the entropy theorem gives  $S + 1 \geq c_{\text{ent}} M$ ; combining with  $\ell \geq cM^{-3}$  gives its third inequality. For  $\ell \leq K/n$ , the leakage bound forces  $M \geq (cn/K)^{1/3}$ , and Theorem Q.3 gives the final cost bound. No entropy of  $\sigma$  or  $\eta$  is substituted for  $S(\rho)$ .  $\square$

The condition concerns the actual stationary algebra, which need not coincide with the collective symmetry algebra. Small entropy production does not automatically supply its required trace proximity uniformly as  $M$  grows. The corollary is about this literal family and does not give an unrestricted-source or all-realization lower bound.

## Q.7 The unresolved sharp quantitative estimate

Write  $\mathcal{D} = \sum_s w_s (\text{Ad } V_s - I)^* (\text{Ad } V_s - I)$ ,  $\mathcal{X} = (L_X + R_X)/2$  and  $\mathcal{Q} = (L_{Q_\gamma} + R_{Q_\gamma})/2$  on Hilbert–Schmidt space. The concrete sufficient estimate

$$M^2 \mathcal{D} + M^3 \text{polylog}(M) \mathcal{X} \geq \kappa \mathcal{Q} \quad (45)$$

uniformly in  $k$ , with  $\kappa > 0$  independent of  $M$ , is open. Theorem Q.3 supplies the entropy half for all coherent sources in this literal geometry at growing  $M$ . Theorem Q.4 gives a weaker physical-source coercivity with  $M^4 \ell$ , not this sharper operator inequality. Its fixed rational power  $p = O(M^2)$  explains the extra factor of  $M$ . Theorem R.1 removes that factor for physical sources with  $k \leq c_f \sqrt{M}$ , using many small-angle blocks. It does not prove the displayed operator inequality uniformly in  $k$ . Unrestricted flavour growth, all-round scope and optimized universal comparison remain unproved.

There is a useful exact reduction for testing the operator estimate. For  $c_D, c_X \geq 0$ , positivity of  $\mathcal{H} = c_D \mathcal{D} + c_X \mathcal{X} - \lambda \mathcal{Q}$  on all complex Hilbert–Schmidt matrices is equivalent to positivity on positive matrices. Indeed  $\mathcal{H}$  commutes with adjoint, so a negative witness splits into a negative Hermitian witness. For Hermitian  $A$ , replacing  $A$  by  $|A|$  keeps  $A^2$  and hence both potential energies. It lowers the Dirichlet energy because

$$\| |A| - |B| \|_F \leq \| A - B \|_F \quad (A = A^*, B = B^*).$$

To check this inequality, expand in eigenbases: each spectral overlap term has  $|a||b| \geq ab$ . Apply it with  $B = V_s A V_s^*$ . Thus a negative positive witness exists.

The operator commutes with clock pinching and with each flavour’s degree pinching. A positive ground witness survives pinching (its trace is nonzero) and stays in the ground eigenspace. A nonzero degree-tuple block is therefore a clock-diagonal positive witness. This is a spectral-test reduction, not an entropy-preserving source twirl: pinching can increase source entropy, and the operator witness need not satisfy all the channel’s coarse-correctness constraints.

## R Sharp Houghton cost with moderate flavour growth

The proof keeps the original source and actual entropy production. An analytical average is used only to test a bounded observable; its entropy is never substituted for the original source entropy.

**Theorem R.1** (Sharp cost with moderate flavour growth). *There are universal constants  $c_f, c_0, c > 0$  such that, in the literal rational constant-field round, every  $M \geq 48$ ,  $1 \leq k \leq c_f \sqrt{M}$  and arbitrary coherent source with  $u \leq 1/16$  satisfy*

$$M^2 a + M^3 \ell \geq c_0, \quad S + na + 1 \geq c \min\{n^{1/3}, \ell^{-1/3}\} \quad (n \geq 1).$$

*At zero leakage the second argument is infinite. For fixed  $K > 0$ ,  $\ell \leq K/n$  therefore forces  $S + na + 1 \geq c_K n^{1/3}$ . A different exact concentrated corner frame has cost at most  $C_K(1 + S + na)$  at the fixed coarse-error allowance of that construction.*

### R.1 Disjoint logical blocks and sweep words

*Proof of Theorem R.1.* Put  $c_g = 2\sqrt{104 \ln 2}$  and  $\sigma = \Delta_{\text{clock}} \rho = \bigoplus_c p_c \rho_c$ . The actual-weight dictionary of Appendix Q gives, with  $X = (W - I)^*(W - I)$  and  $W = \Pi_k(r_4)$ ,

$$\ell = \kappa_\ell \text{Tr } \rho X, \quad \kappa_\ell = 488/(225d), \quad \text{Tr } \sigma Q_\gamma \geq 7/12, \quad \chi(C, \sigma) \leq |C| c_g \sqrt{a}.$$

Pinching preserves  $X, Q_\gamma$  and decreases production. Its clock law satisfies  $\text{TV}(p, \text{unif}) \leq c_g M \sqrt{a}$ .

For  $M \geq 4096$  let  $x = \lfloor M/4 \rfloor$ ,  $R = x + 8$ ,  $z = \lfloor M/256 \rfloor$ , and retain

$$G = \{(i, j) : 0 \leq i \leq M - x - 12, \quad 1 \leq j \leq M - 3, \quad i + j \geq M - x + 16 + 2z\}.$$

Direct counting gives  $|G| = (M - x - 11)[(M + x - 48)/2 - 2z]$  and density at least  $15/32 - 40/M - 1/128 \geq 0.45$ . This is a subset of the wedge in the proof of Theorem Q.4. The exact stable logical points there are  $f_s = D_c^* e_s$ , with adjacent swaps given by  $\pi(a^q \alpha a^{-q})$ ,  $0 \leq q < R$ . The same disjoint pairs support the field, which commutes with  $D_c$ .

Choose even  $b_0 \in [x - z, x - z + 1]$ , set  $J = \lfloor (z - 1)/8 \rfloor$  and  $b_j = b_0 + 8j$ . Block  $j$  uses  $f_{b_j}, f_{b_j+2}, f_{b_j+4}, f_{b_j+6}$  and their four partners  $D_c^* e_{s+M+i}$ . All eight-mode blocks are disjoint, and

$$M/8192 \leq J \leq M/2048, \quad b_{J-1} + 7 \leq x.$$

For a selected point  $s$ , the active bulk region is  $t = s - M + 1 + i + j$ , with  $z + 17 \leq t \leq M - 16$ . Its area and signed angle are

$$A_t = (M - 1)^2 - h(h + 1)/2, \quad h = M + 3 - t, \quad \theta_t = -(-1)^{t-5} \pi / A_t.$$

The minus sign comes from the adjoint defining the usual representation; the right-action field in the companion has the opposite sign. The four angles in a block have the same sign,  $M^2/3 \leq A_t \leq M^2$ , and  $|\theta_t - \theta_{t'}| \leq 54\pi/M^3$  within that block: adjacent areas differ by at most  $M$ , and  $|t - t'| \leq 6$ .

Let  $G_j = S_4$  permute the four selected points, and put  $G_{\text{prod}} = \prod_j G_j$ . A local element is a permutation of eight points, hence has at most 28 adjacent swaps. Each swap in the first eight logical points has literal length at most 15; a local representative  $w_j$  therefore has length at most 420. Shifting to successive blocks cancels the intervening powers:

$$a^{b_0-1} w_0 a^8 w_1 \cdots a^8 w_{J-1} a^{-b_{J-1}+1}.$$

Every product-group element has a clock-preserving representative of length at most  $2R + 420J \leq M$ . No group identity on excluded clocks is required. The three-cycle  $\Gamma_j$  on the first three selected points is conjugate to  $\gamma$  by  $C_j = \mathbf{a}^{b_j-1}w$ , where one fixed eight-point permutation sends 1, 2, 3 to 1, 3, 5. Its length is at most  $M$ .

Put  $e = c_g M \sqrt{a}$  and  $A = \sqrt{\sigma_G}$ , without normalizing the restriction to  $G$ . Transporting charge and restricting clocks give  $\text{Tr } \sigma_G Q_{\Gamma_j} \geq 1/30 - 2e$ . Average the conjugations of  $A$  by  $G_{\text{prod}}$  on  $G$ , obtaining  $B \geq 0$ . Short-word centrality implies

$$\|A - B\|_F \leq e, \quad \|A^2 - B^2\|_1 \leq 2e.$$

Write  $\eta = B^2$ . Let  $P_j$  be the complement of the  $A_4$ -invariant subspace in block  $j$ . It commutes with  $S_4$ , and  $Q_{\Gamma_j} \leq P_j$ . Consequently, if  $e \leq 1/480$ ,

$$\frac{1}{J} \sum_j \text{Tr } \eta P_j \geq 1/30 - 4e \geq 1/40. \quad (46)$$

## R.2 A six-mode infinitesimal gap

In one block the equal-angle reference is  $e^{i\theta H}$ , with  $H = \frac{1}{2} \begin{pmatrix} I_4 & -I_4 \\ -I_4 & I_4 \end{pmatrix}$ . Split each four-point space into its all-ones line and standard three-dimensional real  $S_4$  space. On the standard six modes, the conjugated Hamiltonians are  $I_6/2 + K_g$ , where

$$K_g = -\frac{1}{2} \begin{pmatrix} 0 & R_g \\ R_g^T & 0 \end{pmatrix}.$$

Here  $R_g$  is the real standard representation. The two remaining modes and  $I_6/2$  give a common commuting unitary factor in every exterior quantization. Conjugation by  $\text{diag}(I_3, iI_3)$  identifies  $iK_g$  with real skew cross-space generators of  $SO(6)$ . Matrix-coefficient orthogonality gives  $\mathbb{E}_g(R_g)_{ab}(R_g)_{cd} = \delta_{ac}\delta_{bd}/3$  and  $\mathbb{E}_g R_g = 0$ . If  $X_{ab}$  is the Hermitian generator of a unit-speed cross-space coordinate rotation, then

$$X_{ab} = -6\mathbb{E}_g(R_g)_{ab}K_g, \quad \|d\pi(X_{ab})\xi\|^2 \leq 12\mathbb{E}_g\|d\pi(K_g)\xi\|^2.$$

Every  $SO(6)$  matrix is a product of at most 15 coordinate Givens rotations with angles at most  $\pi$ . A within-space rotation is a cross-space rotation conjugated by a cross-space  $\pi/2$  rotation. Thus at most 45 cross-space flows, of total absolute angle at most  $45\pi$ , suffice. The spectral theorem and product telescoping bound every group displacement squared by  $12(45\pi)^2\mathbb{E}_g\|d\pi(K_g)\xi\|^2$ . Haar averaging gives, in every continuous finite-dimensional unitary representation,

$$\mathbb{E}_g d\pi(K_g)^2 \geq \delta(I - P_{\text{inv}}), \quad \delta = \frac{2}{12(45\pi)^2}. \quad (47)$$

In  $k$  exterior flavours the generators have mean zero and norm at most  $h = 3k/2$ : each one-particle  $K_g$  has three eigenvalues  $1/2$  and three  $-1/2$ , exterior eigenvalues are subset sums, and flavour generators add.

## R.3 Contraction and the actual unequal fields

For Hermitian  $H_g$  of mean zero and norm at most  $h$ , put  $T(t) = \mathbb{E}_g e^{itH_g}$  and  $A_2 = \mathbb{E}_g H_g^2$ . The scalar spectral bounds  $|e^{iy} - 1 - iy| \leq y^2/2$  and  $2(1 - \cos y) \geq y^2 - y^4/12$  imply

$$\|(T - I)\xi\|^2 \leq (t^4 h^2/4)\langle \xi, A_2 \xi \rangle,$$

and  $2 \operatorname{Re}\langle \xi, (I - T)\xi \rangle \geq t^2(1 - t^2 h^2/12) \langle \xi, A_2 \xi \rangle$ . Subtracting proves the operator inequality

$$I - T^*T \geq t^2(1 - t^2 h^2/3)A_2. \quad (48)$$

Thus  $|t|h \leq 1$  and (47) give norm at most  $1 - \delta t^2/3$  on the  $SO(6)$ -noninvariant sector. Even permutations of the four selected points embed in this  $SO(6)$ , so its invariant vectors are  $A_4$ -invariant. The same norm bound holds on  $P_j$ , including the common commuting factor. No normality of the average is assumed.

Take  $p = \lfloor M^{3/2} \rfloor$  and  $c_f = \delta\pi/5184$ . A reference angle satisfies

$$\frac{\pi}{2\sqrt{M}} \leq |p\theta| \leq \frac{3\pi}{\sqrt{M}}.$$

For  $k \leq c_f\sqrt{M}$ ,  $|p\theta|h \leq 1$ , and the reference average has norm at most  $1 - \delta\pi^2/(12M)$  on  $P_j$ . The four disjoint number operators have norm at most  $k$ ; the actual-field perturbation is therefore bounded by

$$216\pi pk/M^3 \leq 216\pi c_f/M = \delta\pi^2/(24M).$$

The actual block average  $B_j = \mathbb{E}_{g \in G_j} \Pi_k(gW_j^p g^{-1})$  commutes with  $S_4$ , hence with  $P_j$ , and obeys

$$\|B_j|_{P_j}\| \leq 1 - b/M, \quad b = \delta\pi^2/24. \quad (49)$$

Its norm on the complement is at most one.

#### R.4 Global correlations and the cost bound

The blocks have disjoint one-particle modes. Their exterior quantizations commute, as do all outside phases with the block groups. The full average is  $K = \mathbb{E}_{g \in G_{\text{prod}}} gW^p g^{-1} = W_{\text{out}}^p \prod_j B_j$ . It commutes with the  $P_j$  and has norm at most  $(1 - b/M)^N$  on a sector with  $N$  charged blocks. Since  $\eta$  commutes with the product group,  $\operatorname{Tr} \eta W^p = \operatorname{Tr} \eta K$ . Sector decomposition and (46) give

$$\operatorname{Tr} \eta(I - \operatorname{Re} W^p) \geq \frac{b}{2M} \sum_j \operatorname{Tr} \eta P_j \geq \frac{b}{655360} =: \kappa.$$

We used  $N \leq J \leq M$ ,  $b \leq 1$  and  $1 - (1 - b/M)^N \geq bN/(2M)$ . This norm estimate permits arbitrary interblock correlations.

The positive observable  $F_p = I - \operatorname{Re} W^p$  has norm at most two. Root transfer costs at most  $4e$ . If  $e \leq e_0 = \min\{1/480, \kappa/8\}$ , it follows that  $\operatorname{Tr} \rho F_p \geq \kappa/2$ . For integer  $p$ ,  $F_p \leq p^2 X/2$ ; since  $p^2 \leq M^3$ ,  $M^3 \ell \geq \kappa_\ell \kappa$ . Otherwise  $M^2 a > e_0^2/c_g^2$ . Thus  $c_0 = \min\{e_0^2/c_g^2, \kappa_\ell \kappa\} > 0$  suffices. Because  $c_f < 1/64$ , the conditions  $k \geq 1$  and  $k \leq c_f\sqrt{M}$  already imply  $M > 4096$ ; the smaller-clock range has no admissible flavour count.

Finally put  $t_0 = \min\{n^{1/3}, \ell^{-1/3}\}$  and  $\beta = \min\{1, (c_0/2)^{1/3}\}$ . If  $M \geq \beta t_0$ , Theorem Q.3 gives  $S + na + 1 \geq c_{\text{ent}}\beta t_0$ . Otherwise  $M^3 \ell < c_0/2$  and

$$na \geq c_0 n/(2M^2) > [c_0/(2\beta^2)]n/t_0^2 \geq [c_0/(2\beta^2)]t_0,$$

since  $t_0^3 \leq n$ . The different exact corner frames used in Appendix N have cost  $O(n^{1/3})$  and fixed coarse correctness. This proves the stated static replacement comparison, without any proximity requirement on the original source.  $\square$

## S Proofs for Section I

*Proof of Theorem I.1.* (a)  $B_a^* B_a = \cos^2 \vartheta U^* U + \sin^2 \vartheta U^* U = I$ , and likewise for the others.  $B_1^* B_{a-1} = \cos \vartheta U^* = B_a^* B_1$  is the identity of the triangle  $(a, a^{-1}, \emptyset)$ .  $\text{Tr}(B_1^* B_{a\pm 1}) = \cos \vartheta \text{Tr} U^{\pm 1} = 0$  and  $\text{Tr}(B_a^* B_{a-1}) = \cos 2\vartheta \text{Tr} U^{-2} = 0$  for  $k \geq 3$ , so the traces are exact, and the span of the images is  $\mathbb{C}^{2k}$  since  $\sin \vartheta \neq 0$ . For the flat round,  $T(I/k) = \sum_g \mu_g B_g B_g^*/k$ , since the  $A_g$  have disjoint supports, and the off-diagonal blocks  $\pm 0.3 \cos \vartheta \sin \vartheta I$  cancel:  $T(I/k) = \text{diag}((1 - 0.6 \sin^2 \vartheta)I_k, 0.6 \sin^2 \vartheta I_k)/k$ , so  $a = h_2(0.6 \sin^2 \vartheta)$ .

(b) Let  $J$  be an isometry onto  $\mathcal{A}$ . The cosines of the principal angles between  $Y_1$  and  $Y_a$  are the singular values of  $J^*(B_1 \otimes I)^*(B_a \otimes I)J = \cos \vartheta J^*(U \otimes I)J$ , a compression of a unitary, so all principal angles are at least  $\vartheta$ . On  $Y_1 + Y_a$  the operator  $X = (P_{Y_1} + P_{Y_a})/2$  has the eigenvalues  $(1 \pm \cos \varphi_i)/2$ ,  $\varphi_i \geq \vartheta$ , and the two eigenvalues of each pair sum to 1. Write  $c = \cos \vartheta$ . The sum of the  $D + m$  largest eigenvalues ( $0 \leq m \leq D$ ) is at most  $m + (D - m)(1 + c)/2 = D(1 + c)/2 + m(1 - c)/2$ , and the sum of the  $D'$  largest ( $D' \leq D$ ) is at most  $D'(1 + c)/2$ . The hypothesis gives  $\text{Tr}(\Pi X) \geq (1 - \eta)D$ , and Ky Fan's maximum principle [34, Theorem 1] bounds  $\text{Tr}(\Pi X)$  by the sum of the rank  $\Pi$  largest eigenvalues of  $X$ . If  $\text{rank } \Pi \geq 2D$  there is nothing to prove. If  $\text{rank } \Pi = D + m$  with  $0 \leq m \leq D$ , then  $m \geq D(1 - 2\eta/(1 - c))$ . If  $\text{rank } \Pi = D' \leq D$ , then  $D' \geq 2(1 - \eta)D/(1 + c)$  and  $\eta \geq (1 - c)/2$ ; and  $2(1 - \eta)/(1 + c) - (2 - 2\eta/(1 - c)) = 2c(c - 1 + 2\eta)/(1 - c^2) \geq 0$ , so again  $\text{rank } \Pi \geq D(2 - 2\eta/(1 - c))$ . In the device of [28, Theorem 12.2] the round-one memory is  $I/D_0$  on the active space  $\mathcal{A}_0$ , unchanged by  $V_1$ , and by (F1) the anchor input of  $g$  leaves the memory in  $(B_g \otimes I)(I/D_0)(B_g \otimes I)^* = P_{Y_g}/D_0$ .

(c) For a gadget channel  $\text{Tr}(A_h^* A_g) = d\mu_g \delta_{gh}$ , so with maximally mixed input a round maps  $\sigma$  to  $\sum_g \mu_g (B_g \otimes I) V \sigma V^* (B_g \otimes I)^*$ , and  $\text{Tr} \sigma'^2 = \sum_{g,h} \mu_g \mu_h \text{Tr}(V \sigma V^* Y_{gh} V \sigma V^* Y_{gh}^*)$  with  $Y_{gh} = B_g^* B_h \otimes I$ . For  $g = h$  the term is  $\text{Tr} \sigma^2$ . For Haar  $V$  on dimension  $N$ ,  $\mathbb{E}[(V \sigma V^*)^{\otimes 2}] = \alpha I + \beta F$  with  $F$  the swap,  $\alpha = (1 - \text{Tr} \sigma^2/N)/(N^2 - 1)$  and  $\beta = (\text{Tr} \sigma^2 - 1/N)/(N^2 - 1)$ , so  $\mathbb{E} \text{Tr}(V \sigma V^* Y V \sigma V^* Y^*) = \alpha \text{Tr}(Y Y^*) + \beta |\text{Tr} Y|^2$ . Exact traces give  $\text{Tr} Y_{gh} = 0$  for  $g \neq h$ , and  $\text{Tr}(Y Y^*) \leq N$ , so the cross terms are at most  $N/(N^2 - 1) \leq 2/N$ . This proves the recursion, since the unitaries after the round do not change the spectrum and an export followed by a flat import cannot increase the purity:  $\text{Tr}((\text{Tr}_e \sigma)^2)/e \leq \text{Tr} \sigma^2$ . Unrolling with  $N_s \geq D_0 q^s$  and  $\text{Tr} \sigma_1^2 \leq \text{Tr} \sigma_0^2 = 1/D_0$ ,  $\mathbb{E} \text{Tr} \sigma_t^2 \leq s_2^{t-1}/D_0 + \sum_{u=1}^{t-1} 2s_2^{t-1-u}/(D_0 q^u) \leq 3q^{-(t-1)}/(D_0(1 - qs_2))$ . By Markov's inequality, with probability at least  $\frac{1}{2}$ ,  $\text{Tr} \sigma_t^2 \leq 6q^{-(t-1)}/(D_0(1 - qs_2))$ . For  $\Pi$  with  $\text{Tr}(\Pi \sigma_t) \geq 1 - \eta$ , Cauchy-Schwarz gives  $(1 - \eta)^2 \leq (\text{Tr} \Pi \sigma_t \Pi)^2 \leq \text{rank } \Pi \cdot \text{Tr} \sigma_t^2$ , and  $D_t \leq 2q^t D_0$  [28, proof of Theorem 12.2] finishes.  $\square$

*Proof of Corollary I.2.* The thin rounds of Theorem G.12(b) have distance at most  $2\pi F\sqrt{d}/N^2 \leq \frac{1}{16}$  for large Fibonacci  $N$ . A rounding with  $S' \leq S + C_0$  and  $a' \leq a + C_1 \ell \log(1/\ell)$  is the case  $\lambda = \gamma = m = 1$  of Theorem G.12(b), which requires  $\lambda \geq 2\gamma$ .  $\square$

*Proof of Proposition I.3.* (a) The rounds of Lemma L.1 are square. A zero-leakage square round has  $B_g K \subseteq K$ , so its  $B_g$  are unitaries of  $K$ ; replacing each  $B_g$  by  $B_1^* B_g$ , which leaves every  $B_g^* B_h$  unchanged, gives  $B_1 = I$ , and Lemma 2.10, with the phases of  $\Phi_\omega$ , gives  $B_{[y]} = \zeta_t B_{[x]}^* B_{[z]}$  on every triangle. As in the proof of Theorem G.12(a), now with exact identities, the commutator  $C = B_b B_a^* B_b^* B_a$  equals  $\bar{\omega} I$ , and  $\det C = 1$  gives  $\bar{\omega}^k = 1$ , which is impossible for irrational  $\theta$ .

(b) The smoothed rounds of [27, Proposition 4.14] are gadget unitaries of unitary assignments on a finite space, hence square, with flat baths, so  $a = 0$ ; their distance is at most  $\frac{1}{16}$  and their leakage tends to 0. A zero-leakage square round at distance below  $\frac{1}{2}$  would, by Lemma H.5, give a finite-dimensional representation of  $P_\Phi$  that separates the labels. Composed with the homomorphism  $H_3 \rightarrow P_\Phi$  that sends each generator to its label (the relators map to 1 by chaining the triangles along them, and the word  $\alpha a \alpha a^{-1}$  maps

to the label  $\gamma$ ), it would separate  $\gamma$  from 1 in a finite-dimensional representation of  $H_3$ , which Remark 8.1 excludes. The same argument excludes zero-leakage square rounds at distance below  $\frac{1}{2}$  for every gadget channel whose labels no finite-dimensional representation separates.  $\square$

*Proof of Proposition I.4.* By [27, Proposition 4.4],  $A^*(3N) \geq (3N)^2/16$  when  $3N \geq 23$ , so for  $N \geq 8$  condition (25) forces  $10^8 \cdot 2^{17} \cdot \frac{9}{16} N^{5/2} \sqrt{\nu} \leq 1$ , that is  $N \leq (16/(9 \cdot 10^8 \cdot 2^{17}))^{2/5} \nu^{-1/5} = 7.13 \cdot 10^{-6} \nu^{-1/5}$ . At the rank rate, Theorem 7.1 with  $\lambda = 2$  and Proposition 7.2 give a round with  $\nu \leq \frac{32}{15}(3Q + 7 + \log n)/n$ , so the method certifies at most  $Q \geq S(\rho) \geq N/16$  with  $N \lesssim ((Q + \log n)/n)^{-1/5}$ , that is  $Q^{6/5} \lesssim n^{1/5}$  up to logarithms: at most  $Q$  of order  $n^{1/6}$ .  $\square$

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